1D/2D modelling suite for integral water solutions

SOBEK Suite

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Hydrodynamics

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Technical Reference Manual

D-Flow 1D (SOBEK 3)

Hydrodynamics

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List of Symbols

Symbol	Unit	Description
A_F	m^2	Cross sectional flow area
A_R	m^2	Rainfall runoff area
A_S	m^2	Cross sectional storage area
A_T	m^2	Cross sectional total area (i.e. $A_F + A_S$)
A_z	m^2	Surface area
$A_{\zeta,m}$	m^2	Total free surface area for location m
C	$m^{1/2}/s$	Chézy coefficient
C_d	-	Drag coefficient
C_{wind}	-	The wind friction coefficient.
f_d	-	Global design factor for overall extreme high or low load (rainfall)
g	m/s^2	Acceleration due to gravity (≈ 9.81)
h	m	Total water depth
$h(x, y, t)$	m	Total water depth, $h = \zeta - z_b$
h_0	m	average water depth in a branch
i_R	mm/s	Intensity of rainfall
i_s	mm/s	Intensity of seepage
O	m	Wetted perimeter
Q	m^3/s	Discharge
q_{lat}	m^2/s	Lateral discharge per unit length
R	m	Hydraulic radius
t	s	Time
w_f	m	Cross sectional width at water level
W_T	m	Total width, including flow and storage area
x	m	Cartesian coordinate along channel
y	m	Cartesian coordinate across channel
z_b	m	Bed level, positive up
$z_b(x)$	m	Bed level, positive up
ϕ_{wind}	deg	Angle between the wind direction and the local channel direction
ρ_w	kg/m^3	Water density
ρ_{air}	kg/m^3	Density of air
τ_{wind}	N/m^2	Wind shear stress
$\zeta(x, t)$	m	Water level

1 Conceptual description

1.1 Hydrodynamics D-Flow 1D

1.1.1 Model equations

The water flow is computed by solving the complete De Saint Venant (1871) equations for unsteady flow and are based upon the the following series of assumptions (Cunge *et al.*, 1980):

- The flow is one-dimensional i.e. the velocity can be represented by a uniform flow over the cross-section and the water level can be assumed to be horizontal across the section.
- The streamline curvature is small and the vertical accelerations are negligible, hence the pressure is hydrostatic.
- The effects of boundary friction and turbulence can be accounted for through resistance laws analogous to those used for steady flow.
- The average channel bed slope is small so that the cosine of the angle it makes with the horizontal may be replace by unity.

For one dimensional flow (Channel Flow and Sewer Flow modules) the following equations are solved

- continuity equation 1D
- momentum equation 1D

These equations are solved numerically using the Delft-scheme.

1.1.1.1 Continuity equation (1D)

The 1D continuity equation reads:

$$\frac{\partial A_T}{\partial t} + \frac{\partial Q}{\partial x} = q_{lat} \tag{1.1}$$

where:

A_T	Total area (sum of flow area and storage area) [m^2]
Q	Discharge [m^3/s]
q_{lat}	Lateral discharge per unit length [m^2/s]. Positive value refers to inflow. Negative value refers to outflow.

1.1.1.2 Momentum equation (1D)

The 1D momentum equation reads:

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A_F}\right) + gA_F\frac{\partial \zeta}{\partial x} + \frac{gQ|Q|}{C^2RA_F} - w_f\frac{\tau_{wind}}{\rho_w} + gA_F\frac{\xi Q|Q|}{L_x} = 0 \tag{1.2}$$

The first term describes the inertia
The second term describes the convection
The third term describes the water level gradient
The fourth term describes the bed friction
The fifth term describes the influence of the wind force
The sixth term describes the influence of extra resistance

where:

A_F	Flow area [m^2]
C	Chézy value [$m^{1/2}/s$]
g	Acceleration due to gravity [m/s^2]
ζ	Water level [m]
L_x	Length of branchsegment, accomodating an Extra Resistance Node [m]
Q	Discharge [m^3/s]
R	Hydraulic radius [m]
t	Time [s]
w_f	Water surface width [m]
x	Distance along the channel axis [m]
ρ_w	Density of fresh water [kg/m^3]
τ_{wind}	Wind shear stress [N/m^2] (see section 1.1.7)
ξ	Extra Resistance coefficient [s^2/m^5]

1.1.2 Hydrodynamic definitions

1.1.2.1 Model datum/reference level

The Model datum and reference level both refer to a horizontal plane from which elevations are defined (positive in upward direction). By default the model datum and reference level are equal to zero.

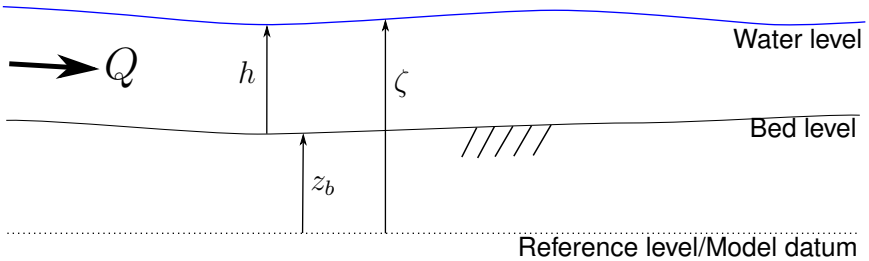

Figure 1.1: *Definition of model datum/reference level*

Note: All levels (quantities with a vertical coordinate) in SOBEK are defined with respect to the model datum or reference level.

1.1.2.2 Bed level

The bed level is defined as the lowest point in the cross section. In the definition of the cross section an example is given of the interpolation and extrapolation of the bed levels over a branch. The used symbol in the D-Flow 1D module is z_b.

It is given relative to a reference level, for example Mean Sea Level .

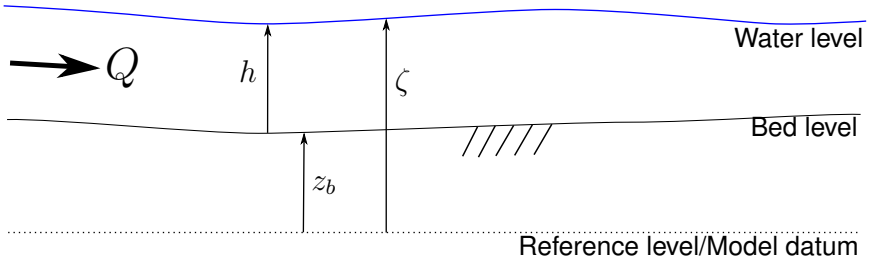

Figure 1.2: *Definition of bed level*

If the water level is lower than the bed level, drying occurs.

1.1.2.3 Water depth

In the D-Flow 1D module the water depth is the distance between the water level and the bed level. The symbol used for the water depth is h.

$$h = \zeta - z_b \tag{1.3}$$

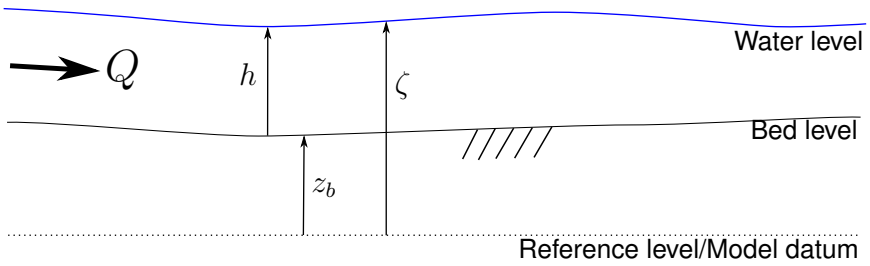

Figure 1.3: *Definition of water depth* $h = \zeta - z_b$

1.1.2.4 Water level

The water level (ζ) is the level of the water surface relative to the reference level or Model datum (= reference level).

In the one-dimensional D-Flow 1D module the water level perpendicular to the flow direction is assumed to be horizontal.

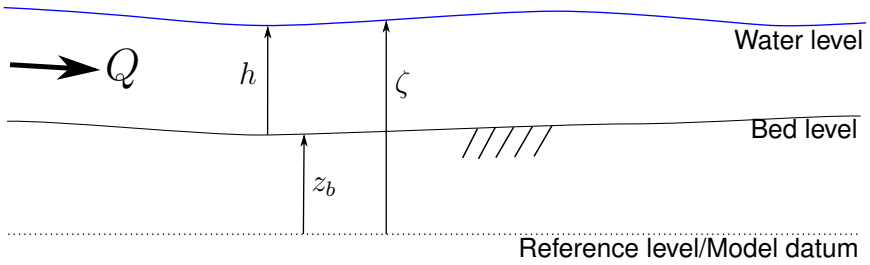

Figure 1.4: *Definition of water level*

The water level together with the discharge form the result of the D-Flow 1D module computations. Water levels are calculated at the connection nodes and the ζ-calculation points.

Note: In each D-Flow 1D, all levels are relative to the reference level

1.1.2.5 Flow area

The flow area A_F of a cross-section is the area through which water is actually flowing.

In a cross-section, distinction can be made between the flow area and the storage area. In the latter part, water is stored only, see Figure 1.5.

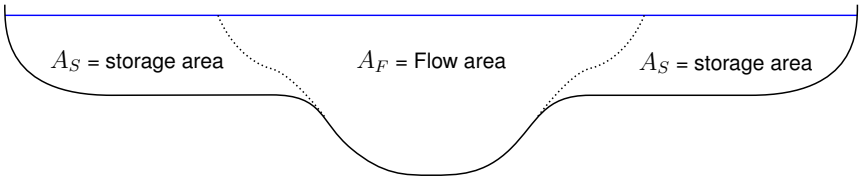

Figure 1.5: *Definiton of flow area (A_F) and storage area (A_S)*

The shape of the cross-section, and the distinction between the flow area and storage area are defined for each cross-section by user input.

1.1.2.6 Storage area

The storage area A_S of a cross-section is the area in which only water is stored i.e. the non-conveying part of the cross-section.

In a cross-section, distinction can be made between the flow area and the storage area, see Figure 1.5.

The shape of the cross-section, and the distinction between the flow area and storage area is defined for each cross-section by user input.

The total area A_T is defined as the flow area plus the storage area.

1.1.2.7 Wetted area

The wetted area (A_F [m^2]) is the part of the cross section that is filled with water. See Figure 1.6

1.1.2.8 Wetted perimeter

The wetted perimeter (O [m]) is the interface between the soil and the flowing water.

See Figure 1.6.

In the case of a main channel and one or two floodplains in the cross-section, a wetted perimeter is computed for each section. In that case the water interface between the sections is not included in the wetted perimeter. See also Figure 1.6 and Figure 1.7

The wetted perimeter is used to compute the hydraulic radius, which is important for the computation of the bed friction.

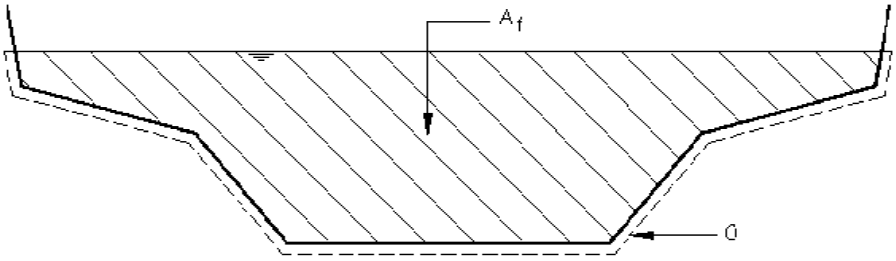

Figure 1.6: *Wetted perimeter*

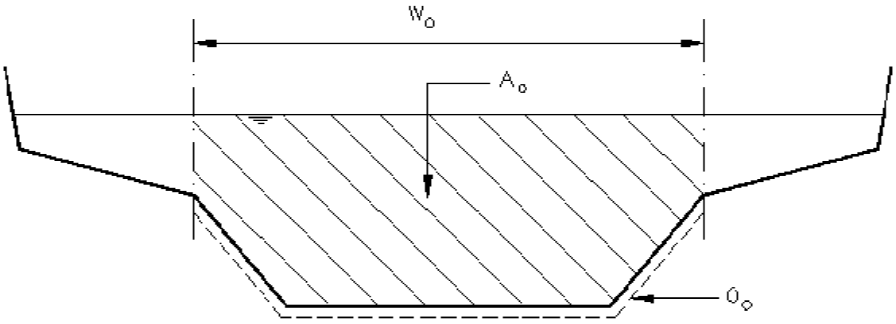

Figure 1.7: *Wetted perimeter for the main channel*

1.1.2.9 Flow velocity

The flow velocity (u) is defined as the average flow velocity in the flow section of the cross-section. It is by default given in [m/s].

The average flow velocity is derived by dividing the discharge [m^3/s] by the flow area [m^2]. Output of the overall average flow velocity, the flow velocity in the main channel, the flow velocity in floodplain 1 and the flow velocity in floodplain 2 is possible.

$$u = \frac{Q}{A_F}, \qquad u_0 = \frac{Q}{A_0}, \qquad u_1 = \frac{Q}{A_1}, \qquad u_2 = \frac{Q}{A_2}. \tag{1.4}$$

The indices 0, 1, 2 indicate the main channel, floodplain 1 and floodplain 2 respectively.

1.1.2.10 Velocity

The velocity (u) is defined as the average flow velocity in the wetted area of the cross section. It is given in [m/s].

$$u = \frac{Q}{A_F} \tag{1.5}$$

Note: Discharges and velocities are defined in branch segments, whereas cross sections are defined in ζ-calculation points. For the computation of the velocities according to the above formula. The D-Flow 1D module uses the upstream cross section

1.1.2.11 Hydraulic radius

The hydraulic radius is defined as the wetted area of the cross section divided by the wetted perimeter.

$$R = \frac{A_F}{O} \tag{1.6}$$

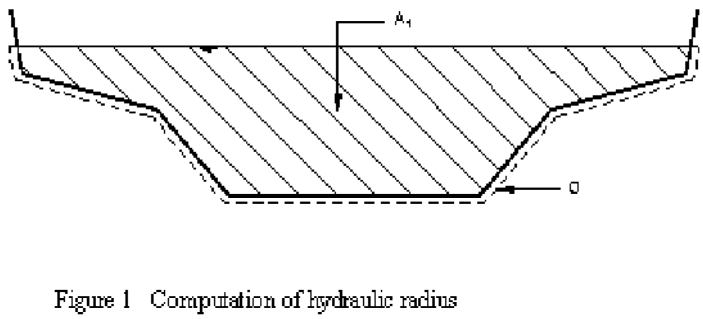

Figure 1.8: *Computation of hydraulic radius*

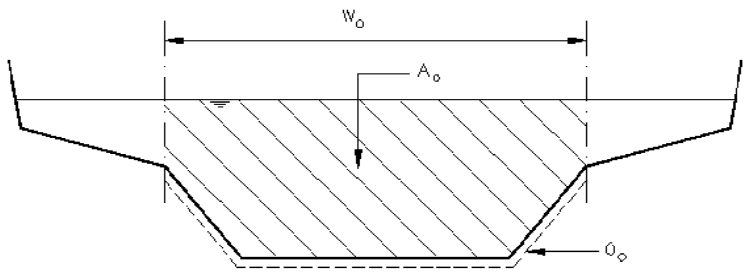

Figure 1.9: *Wetted perimeter for the main channel (Subdivision in main channel and floodplains only in SOBEK-River)*

1.1.3 Inertia

The flow in one dimension is described by two equations: the momentum equation and the continuity equation. The first term in the momentum equation is the inertia term:

$$\text{inertia} = \frac{\partial Q}{\partial t} \tag{1.7}$$

Q	Discharge [m^3/s]
t	Time [s]

1.1.4 Convection

In the D-Flow 1D module an convection term is used to guarantee the conservation of momentum in a branch (between branch segments). The same holds for a connection node when two branches are connected to this node. It that case the connection nodes could also have been modelled as a ζ-calculation point. If there are more than two branches connected to a connection node the convection term is set to zero.

When a branch has subsections, all subsections have an independent solution for the velocity. The velocity that is used for the conservation of momentum over the branch is the average of the velocities u_i of the subsections in the cross section.

1.1.5 Convection (1D)

The flow in one dimension is described by two equations: the momentum equation and the continuity equation. One of the terms in the momentum equation is the convection term:

$$\text{convection} = \frac{\partial}{\partial x}\left(\frac{Q^2}{A_F}\right) \tag{1.8}$$

Q	Discharge [m^3/s]
A_F	Flow area [m^2]
x	Distance [m]

1.1.6 Water level gradient

The third term in the momentum equation is the water level gradient.

$$\text{water level gradient} = gA_F\frac{\partial \zeta}{\partial x} \tag{1.9}$$

x	Distance [m]
A_F	Flow area [m^2]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
ζ	Water level [m] (with respect to the reference level)

This force tries to achieve a flat water surface under influence of the gravitational acceleration. This force together with the bed friction have the greatest effect on the water movements.

1.1.7 Wind friction

The wind friction term in the momentum equation is expressed as:

$$\tau_{wind} = -\rho_{air} C_{wind} u_{wind}^2 \cos(\varphi_{wind} - \varphi_{channel}) \tag{1.10}$$

τ_{wind}	Wind shear stress (positive if acting along the positive x-axis of an open channel) [N m^{-2}]
ρ_{air}	Density of air, 1205 kg m^{-3}
C_{wind}	$\alpha_{wind,1} + \alpha_{wind,2} u_{wind}$, the wind friction coefficient. The used values are:

$$a_{wind,1} = 0.50 \times 10^{-3} \qquad [-] \tag{1.11}$$

$$a_{wind,2} = 0.06 \times 10^{-3} \qquad [\text{s m}^{-1}] \tag{1.12}$$

u_{wind}	Wind velocity [m s^{-1}]
$\varphi_{channel}$	Clock-wise angle between the positive x-axis of the 1D open channel and the North (Nautical convention) [degrees]
φ_{wind}	Clock-wise angle between the wind direction and the North (Nautical convention) [degrees]

The wind direction that is specified in the D-Flow 1D module is the direction from which the wind is blowing with respect to the north. So, if the wind is blowing from the north, the wind direction is 0 degrees. Wind from the southeast has a wind direction of 135 degrees.

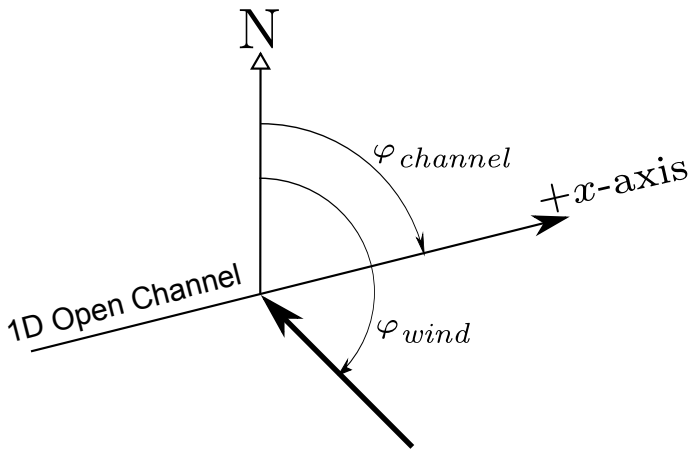

Figure 1.10: *Wind direction*

When a cross section has a top width smaller than 20 mm (circular, egg-shaped, or nearly closed tabulated cross section) than the cross section is considered closed. In that case the wind is neglected in the momentum equation

1.1.8 Initial conditions

The initial conditions are the water levels or depths and the discharges at the beginning of the simulation. The initial conditions are defined over a branch (water level and depth at the ζ-calculation points, discharges at the branch segments). Therefore the water level that initially should be taken at connection nodes is not strictly defined. This happens when, for example, branch 1 with initial water level 0.5 is connected by a node to branch 2 with initial water level 0.6. In that case the water level at the connection node is set to the lowest value of the connected branches (0.5). Because the ζ-calculation point at the end of branch 1, the connection node and the ζ-calculation point at the beginning of branch 2 have the same location, the water levels of these three points are set to 0.5.

1.1.9 Boundary

A boundary can be applied at the locations where the model network ends with a boundary node.

In order to solve the water flow equations (continuity equation and momentum equation), information about the water flow at the model boundaries must be supplied.

At each boundary node, one condition for the water flow must be specified. The following options are available:

- ◇ discharge (constant, tabulated function of time, tabulated function of the water level)
- ◇ water level (constant, tabulated function of time)

Because in the staggered grid, used by D-Flow 1D module, the discharges are defined for the branch segments, a discharge boundary is imposed on the first branch segment next to the boundary. Therefore the ζ-calculation point just before this first branch segment (first one in a branch) is undefined and will not be taken into consideration during a calculation.

A water level boundary is defined in the first calculation point next to the boundary. This ζ-calculation point actually has the same coordinates as the boundary node.

Note: In channel flow, usually, the discharge will be specified at boundaries where water is flowing into the model, and the water level where water is flowing out of the model. In both sewer and channel systems a dead end (or beginning) of a branch can be a connection node at the end of a pipe or channel. In contrast to a boundary node, this node has storage.

1.1.10 Discharge

The discharge Q is the amount of water passing a branch segment per unit of time. It is given in [m^3/s].

The discharges at the branch segments together with the water levels at the ζ-calculation points, form the results of a water flow simulation.

The D-Flow 1D module gives a positive discharge when the water flows in positive direction with respect to the defined direction of a branch. If the water is flowing from branch end to branch beginning, the discharge is negative.

1.1.11 Lateral discharges

A distinction is to be made between a *Point* and *Diffusive* lateral discharge.

- **Point lateral discharge**: A point lateral discharge (Q_{lat}, [m^3/s]) refers to the inflow or outflow of water at a specific point/location in a model schematization.
- **Diffusive lateral discharge**: A diffusive lateral discharge (q_{lat}, [m^2/s]) refers to the inflow or outflow of water along a specific branchsegment in a model schematization. A branch segment is the distance (l_i, [m]) between two ζ-calculation points. A branch may consists of several branchsegments. The total diffusive lateral discharge at a branch segment equals $q_{lat} \times l_i$. A diffusive lateral discharge is uniform distributed along the axis of a branch (see Figure 1.11).

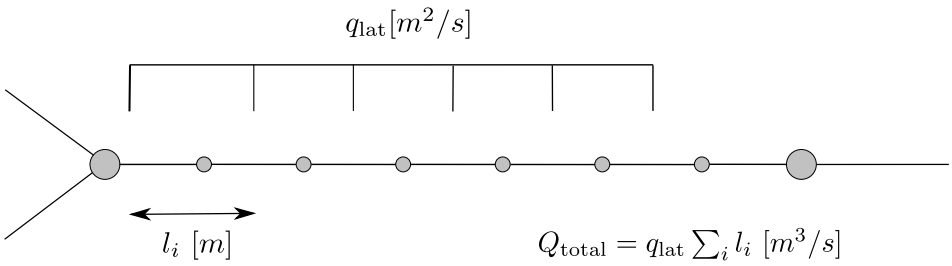

Figure 1.11: *Diffuse lateral discharge*

For both point lateral discharges and diffuse lateral discharges yields that they are assigned to a ζ*-calculation point(s)*. More precisely, their lateral inflow or outflow volume is accounted for in the local water balance of such ζ-calculation point (e.g. local discretization of the continuity equation, see section 1.1.11.1).

Two options are available for determining to which adjacent ζ-calculation point a lateral discharges is assigned (see section 1.1.11.2).

An overview of lateral discharges, available in SOBEK, is given in section 1.1.11.3. In general lateral discharges can be defined as a constant or as a function of time. Exceptions are:

- The Area Based Point lateral inflow (see section 1.1.11.4)

1.1.11.1 Incorporating Lateral discharges in the Continuity Equation

Both point lateral discharges and diffusive lateral discharges (see Figure 1.12) are included in the third term (q_{lat}) of the continuity equation (see Equation (1.13)).

$$\frac{\partial A_F}{\partial t} + \frac{\partial Q}{\partial x} = q_{lat} \tag{1.13}$$

A_F	Flow area [m^2]
t	Time [s]
Q	Discharge [m^3/s]
x	Distance [m]
q_{lat}	Lateral discharge per unit length [m^2/s]

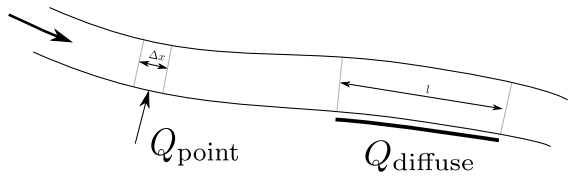

Figure 1.12: *Lateral discharges:*
Point lateral discharge: $q_{lat} = Q_{point}/\Delta x$*; and*
Diffuse lateral discharge: $q_{lat} = Q_{diffuse}/l$

Note: A large withdrawal of water (negative lateral discharge) can cause problems when the amount of outflowing water is larger than the storage available in a ζ-calculation point. As a result negative volumes could be computed in the continuity equation. For that reason the time step is temporarily reduced when the outflowing water volume is larger than half the available volume (safety factor) in the ζ-calculation point. The time step is set to a value that allows a withdrawal of half the available volume in one time step. If the time step resulting from this action is smaller than 0.01 second, the lateral withdrawal is set to zero.

1.1.11.2 Options for Assigning Lateral Discharges to ζ-calculation point(s)

Two different *lateral discharge assignment options* are available in Settings (see 1D Flow, Simulation Settings Tab), that determine to which ζ-calculation (or water level) point(s) a particular lateral discharge is to be assigned. The two different lateral discharge assignment options are:

Option 1: Lateral Assigned to Lowest Water Level Point,
Option 2: Lateral Assigned to Nearest Water Level Point.

Hereafter, the consequences of the two different lateral discharge assignment options are discussed for each type of lateral discharge (see section 1.1.11.3) separately.

- Lateral discharge on a Node [m^3/s]:
  Irrespective of the selected lateral discharge assignment option, a lateral discharge on a Node is assigned to its concerning Node (e.g. ζ-calculation point, Connection Node or Manhole).
- Point-lateral discharge on a Branch [m^3/s]:
  A point-lateral discharge on a Branch is either assigned to one or equally distributed over both of the ζ-calculation points, that are respectively located at the left-side and the right-side of the point-lateral discharge.
  - If **option 1** (Lateral Assigned to Lowest Water Level Point) is selected, the point-lateral discharge is assigned to the ζ-calculation point having the lowest bed level. If both

calculation points have the same bed level and the point-lateral discharge is located at a u-velocity point (e.g. centre-point of the two adjacent ζ-calculation points), the point-lateral discharge is equally distributed over both adjacent calculation points. Else the point-lateral discharge is assigned to the nearest ζ-calculation point.

- If **option 2** (Lateral Assigned to Nearest Water Level Point) is selected and the point-lateral discharge is located at a u-velocity point (e.g. centre-point of the two adjacent ζ-calculation points), the point-lateral discharge is equally distributed over both adjacent ζ-calculation points. Else the point-lateral discharge is assigned to the nearest ζ-calculation point.

◇ Diffusive lateral discharge along a Pipe or Branch [m^2/s]:
A diffusive lateral discharge along a Pipe or Branch is either assigned to one or equally distributed over both of the ζ-calculation points, that are respectively located at the beginning and the end of a pipe or branchsegment, receiving a diffusive lateral discharge.

- If **option 1** (Lateral Assigned to Lowest Water Level Point) is selected, the diffusive lateral discharge is assigned to the ζ-calculation point having the lowest bed level. Else the diffusive lateral discharge is equally distributed over both adjacent calculation points.
- If **option 2** (Lateral Assigned to Nearest Water Level Point) is selected, the diffusive lateral discharge is equally distributed over both adjacent ζ-calculation points.

1.1.11.3 Examples of Lateral Discharges

Three different types of lateral discharges are discerned:

1. Lateral Discharge on a Node [m^3/s]:
   Examples of lateral discharges on a Node are:
   - *Point-Lateral Discharge, that is exactly located on a ζ-calculation point*,
   - *Connection Node with Storage and Lateral Flow*,
   - *Manhole with Lateral Flow*,
   - *Manhole with Lateral Disch. and Runoff*
   - *Manhole with Runoff*
   - *Flow - RR Connection on Flow Connection Node*.
2. Point-Lateral Discharge on a Branch [m^3/s]:
   Examples of a point-lateral discharge on a Branch are:
   - *Flow - Lateral Flow Node*
   - *Flow - RR Connection on Channel*.
3. Diffusive Lateral Discharge along a Pipe or Branch [m^2/s]:
   Examples of a diffusive lateral discharge along a Pipe or Branch are:
   - *Flow - Pipe with Runoff*,
   - *Flow - Dry Weather Pipe*,
   - *Flow - Rain Pipe*,
   - *Flow - Pipe with Infiltration*
   - *Flow - Channel with Lateral Discharge*.

1.1.11.4 Area Based Point Lateral Flow

Area based Point lateral flow is computed as:

$$Q_{lat} = A(f_d R + S) \tag{1.14}$$

Q_{lat} Lateral discharge [m^3/s]

f_d	Design factor for area based lateral flow [$-$].
R	Rainfall [m/s].
A	The catchment runoff area [m^2]
S	Seepage (positive, inflow of groundwater) or Infiltration (negative, outflow to groundwater reservoir) [m/s].

1.1.12 Bed friction

The bed friction is the friction between the flowing water and the channel bed. As such, it exerts a force on the flowing water always in the direction opposite the water flow.

In water courses, this force together with the force caused by earth gravity usually determines the flow conditions: the other forces are far less important.

The fourth term of the momentum equation is the bed-friction term:

$$\text{bed friction} = \frac{gQ|Q|}{C^2 R A_F} \tag{1.15}$$

where:

g	Acceleration due to gravity [m/s^2] (≈ 9.81)
Q	Discharge [m^3/s]
C	Chézy coefficient [$m^{1/2}/s$]
R	Hydraulic radius [m]
A_F	Wetted area [m^2]

Note:
If the specified roughness parameter is equal to zero (Chézy=0, Manning=0, etc.), then the bed friction term is not taken into account in the momentum equation.

The following roughness definitions can be used:

1.1.12.1, Bos-Bijkerk;
1.1.12.2, Chézy;
1.1.12.3, Manning;
1.1.12.4, Nikuradse;
1.1.12.5, Strickler;
1.1.12.6, White-Colebrook.

1.1.12.1 Bos-Bijkerk

The bed friction formulation according to de Bos-Bijkerk describes the Manning coefficient as a function of the water depth and a parameter. This parameter can be used to shape the function to a certain empirical curve:

$$k_m = \gamma h^{1/3} \tag{1.16}$$

k_m	Manning roughness coefficient [$m^{1/3}/s$]
h	Water depth [m]
γ	Parameter [$1/s$], normally between 20 and 40

This formula can be rewritten to a formulation for the Chézy coefficient as a function of the water depth

$$C = \gamma h^{1/3} R^{1/6} \tag{1.17}$$

C	Chézy coefficient [$m^{1/2}/s$]
R	Hydraulic radius [m]

For every computational time step the new friction value is calculated according to the Bos-Bijkerk formula and sequentially applied to the hydrodynamics calculations.

1.1.12.2 Chézy

The D-Flow 1D module uses the Chézy bed friction value in solving the water flow equations.

The following roughness formulations are possible:

- Chézy coefficient
- Bos-Bijkerk friction shape parameter g, resulting in a Chézy value according to:

$$C = \gamma h^{1/3} R^{1/6} \tag{1.18}$$

h	Water depth [m]
γ	Parameter [$1/s$], normally between 20 and 40
C	Chézy coefficient [$m^{1/2}/s$]
R	Hydraulic radius [m]

- White-Colebrook, using the Nikuradse roughness coefficient k_n, results in a Chézy value according to:

$$C = 18\ {}^{10}\log\left(\frac{12R}{k_n}\right) \tag{1.19}$$

- Manning coefficient, n_m, resulting in a Chézy coefficient according to:

$$C = \frac{R^{1/6}}{n_m} \tag{1.20}$$

- Strickler, using the Nikuradse k_n roughness coefficient, results in a Chézy value according to:

$$C = 25\left(\frac{R}{k_n}\right)^{1/6} \tag{1.21}$$

- Strickler, using the Strickler k_s roughness coefficient, results in a Chézy value according to:

$$C = k_s R^{1/6} \tag{1.22}$$

- Engelund-like roughness predictor (for main sections of SOBEK-River profiles only) resulting in a Chézy value according to:

$$C = C_{90}\sqrt{\frac{\theta_{90}}{\theta_s}} \tag{1.23}$$

For SOBEK-River, the Chézy, Nikuradse, Manning or Strickler coefficients may be

- a constant
- spatially varying
- a tabulated function of the water level (h) or the total discharge (Q)

Different values or tables are possible for positive as well as negative flow. See also the chapter 1D hydraulic friction concepts for information on the different friction concepts present in the user interface.

1.1.12.3 Manning

One of the methods to define the bed roughness is using the Manning coefficient, symbol n_m. In the D-Flow 1D module, the Manning coefficient is used to compute the actual value of the Chézy coefficient, by:

$$C = \frac{R^{1/6}}{n_m} \tag{1.24}$$

C	Chézy coefficient [$m^{1/2}/s$]
R	Hydraulic radius [m]
n_m	Manning coefficient [$s/m^{1/3}$]

1.1.12.4 Nikuradse

One of the methods to define the bed friction is by entering an equivalent roughness according to Nikuradse, represented by the symbol k_n [m].

Values of k_n for open channels with bed forms are in the same order of magnitude as the height of the bed forms.

Using this option, the actual value of the Chézy coefficient will be computed according to the White-Colebrook formula.

1.1.12.5 Strickler

One of the methods to define the bed roughness is by using the Strickler formula. The actual value of the Chézy coefficient is computed using:

$$C = 25 \left(\frac{R}{k_n} \right)^{1/6} \tag{1.25}$$

or:

$$C = k_s R^{1/6} \tag{1.26}$$

C	Chézy coefficient [$m^{1/2}/s$]
R	Hydraulic radius [m]
k_s	Strickler roughness coefficient [$m^{1/3}/s$]

in which k_n is the Nikuradse equivalent roughness, and k_s is the Strickler roughness coefficient. You may select either ks or kn as input value.

The value of the hydraulic radius R is taken from the last iteration loop.

It is possible to enter the coefficients varying in space and depending on the local flow direction.

1.1.12.6 White-Colebrook

One of the methods to define the bed friction is by specifying an equivalent roughness according to Nikuradse. Using this option, the value of the Chézy coefficient will be computed according to the White-Colebrook formula:

$$C = 18\ {}^{10}\log\left(\frac{12R}{k_n}\right) \tag{1.27}$$

C	Chézy coefficient
R	Hydraulic radius
k_n	Nikuradse equivalent roughness

Note:
Actually, this formula is a simplification of the complete White-Colebrook formula.

1.1.13 Froude number

The Froude number is defined by u/c, with u the average flow velocity and c the wave celerity.

Herein is c as follows:

$$c = \sqrt{\frac{gA_F}{W_f}} \tag{1.28}$$

Thus

$$\text{Froude number} = \frac{u}{\sqrt{gA_F/W_f}} \tag{1.29}$$

u	Velocity ($u = Q/A_F$) [m/s]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
A_F	Wetted area [m^2]
W_f	Flow width [m]

Remarks:
- ◇ When the Froude number is less than 1, the flow is sub-critical, when it is larger than 1, the flow is super-critical
- ◇ Transitions from sub-critical to super-critical flow causes an hydraulic jump.

1.1.14 Boussinesq

The Boussinesq coefficient is a parameter in the momentum equation for the water flow. It accounts for the non-uniform velocity distribution in a cross-section.

The Boussinesq constant α_B is multiplier to the convective term:

$$\frac{\partial}{\partial x}\left(\alpha_B \frac{Q^2}{A_F}\right) \tag{1.30}$$

The definition of the coefficient α_B is:

$$\alpha_B = \frac{A_F}{Q^2}\int_0^{W_f} \frac{q(y)^2}{h(y)}\,dy \tag{1.31}$$

The constant of Boussinesq is computed by D-Flow 1D. The computation is based upon the Engelund approach (Engelund and Hansen, 1967). In this approach the water level gradient and bed-friction term are assumed to be an order of magnitude larger than the other terms in the momentum equation. The resulting equation is:

$$gA_F \frac{\partial h}{\partial x} = \frac{gQ^2}{C^2 R A_F} \tag{1.32}$$

in which locally a constant water level gradient is assumed. In an arbitrary cross-section the discharge can be expressed as:

$$Q = \int_0^{W_f} C(y)h(y)\sqrt{R(y)\frac{\partial h}{\partial x}}\, dy \tag{1.33}$$

in which h is the water depth. Furthermore, it is assumed that the water level and water level slope are the same for main and floodplains. Combining Equation (1.32) and Equation (1.33) leads to a more accurate estimate of the Chézy coefficient based upon the average of the local Chézy coefficients.

$$C = \frac{1}{A_F\sqrt{R}} \int_0^{W_f} C(y)h(y)\sqrt{R(y)}\, dy \tag{1.34}$$

Now Equation (1.31) gives combined with Equation (1.32) and Equation (1.33) the following expression for the Boussinesq coefficient:

$$\alpha_B = \frac{1}{A_F R C^2} \int_0^{W_f} C^2(y)h(y)R(y)\, dy \tag{1.35}$$

Cross-sections in D-Flow 1D may be divided in two or three parts. So the following three options are available.

option 1:
The cross-section is not divided into sections with different roughness (only a main channel is present). This gives for the Boussinesq coefficient:

$$\alpha_B = 1 \tag{1.36}$$

option 2:
The cross-section is divided into two sections with different roughness for the modeling of, for example, a cross-section with a main channel and a floodplain. During a computation this gives two possibilities:

possibility 1:
Actual water flow is in the main channel only:

$$\alpha_B = 1 \tag{1.37}$$

possibility 2:
Actual water flow is in the main channel and the floodplain:

$$\alpha_B = \frac{C_0^2 A_{f_0} R_0 + C_1^2 A_{f_1} R_1}{C^2 R A_F} \tag{1.38}$$

The indices 0 and 1 respectively indicate the main channel and floodplain.

option 3:
The cross-section is divided into three sections with different roughness (main channel, floodplain 1 and floodplain 2). During a computation this gives three possibilities:

possibility 1:
Actual water flow is in the main channel only:

$$\alpha_B = 1 \tag{1.39}$$

possibility 2:
Actual water flow is in the main channel and floodplain 1:

$$\alpha_B = \frac{C_0^2 A_{f_0} R_0 + C_1^2 A_{f_1} R_1}{C^2 R A_F(\zeta)} \tag{1.40}$$

The indices 0 and 1 respectively indicate the main channel and floodplain.
possibility 3:
Water is actually flowing in all sections, main channel, floodplain 1 and floodplain 2:

$$\alpha_B = \frac{C_0^2 A_{f_0} R_0 + C_1^2 A_{f_1} R_1 + C_2^2 A_{f_2} R_2}{C^2 R A_F} \tag{1.41}$$

The indices 0, 1 and 2 respectively indicate the main channel, floodplain 1 and floodplain 2.

1.1.15 Accuracy

Whether you are measuring in a prototype, studying water flow in a scale model or modelling with a software system, you should always make some considerations about the accuracy of your activities and of the results of your study.

As for the results of a mathematical model study, the following elements play a role in establishing the overall accuracy:

- ◇ The reliability of the available data describing the prototype;
- ◇ The accuracy of the available data describing the prototype;
- ◇ The violation of certain assumptions underlying the mathematical modelling concept being used;
- ◇ The experience and skill of the modeller(s);
- ◇ The overall accuracy that is required (perhaps more accurate results could be obtained but the aim of the study may not require such accuracy or due to a lack of time you may have to settle for less);
- ◇ The accuracy of the applied numerical modelling technique.

In general practice the first five items are the most important when you are using a one-dimensional mathematical model for hydrodynamic flow. The numerical accuracy is in that case normally of minor importance.

The D-Flow 1D module uses the Delft-scheme to solve the water flow equations. This scheme is developed with robustness as the most important design aspect. It can deal with phenomena such as drying/flooding, super-critical flow and it guarantees a solution for every time step.

The accuracy of the solution depends on the grid size; the smaller the grid sizes, the more accurate the solution is. The time step can be chosen arbitrarily. It is reduced internally

when this is necessary to guarantee stability by means of a time step estimation procedure. Obviously small grid sizes result in a network with more elements and therefore in a longer simulation time.

The Delft-scheme is designed to produce a closed water balance.

1.1.16 Structures

In the D-Flow 1D-modules the following structure types are available:

1.1.16.1 Advanced weir
1.1.16.2 Bridge
1.1.16.3 Compound structure
1.1.16.4 Culvert
1.1.16.5 Database structure
1.1.16.6 General structure
1.1.16.7 Inverted siphon
1.1.16.8 Orifice
1.1.16.9 Pump station and Internal Pump station
1.1.16.10 External Pump station
1.1.16.11 River Pump
1.1.16.12 River Weir
1.1.16.13 Siphon
1.1.16.14 Universal Weir
1.1.16.15 Vertical obstacle friction
1.1.16.16 Weir

The flow through structures is computed based on:

- Upstream water level or energy level (River weir, Advanced weir and General structure only)
- Downstream water level or energy level (River weir, Advanced weir and General structure only)
- Structure dimensions (some can be controlled: i.e. crest level, opening height, pump capacity, opening of valve)
- A number of user-defined parameters, depending on the structure type (contraction coefficient, reduction factor, etc.).

The discharges and wetted areas that are computed in the structure formulas are imposed in the branch segment where the structure is located. The formulas use the water levels of the ζ-calculation points on either side of the branch segment. For the River weir, Advanced weir, General structure, River pump, Database structure and Compound structure yields that SOBEK by default places a computational point 0.5 m upstream and 0.5 m downstream of the structure location. Hence such structure is located in a branch having a default length of 1 m.

Note:
The dimensions of the structure do not contribute to the storage in the water system.

1.1.16.1 Advanced weir

Please note that the discharge through an Advanced weir is computed on basis of upstream and downstream energy levels. Further on please note that default a computational point is located 0.5 m in front and 0.5 m behind an Advanced weir.

Notation
The flow direction can be either positive or negative. A positive flow direction is a flow in the direction that the branches have been specified, i.e. with increasing x-coordinates. The upstream side, facing the beginning of the branch is denoted by the subscript 1, whereas the downstream location facing the end of the branch is denoted with 2. The subscript s is used for locations at the sill.

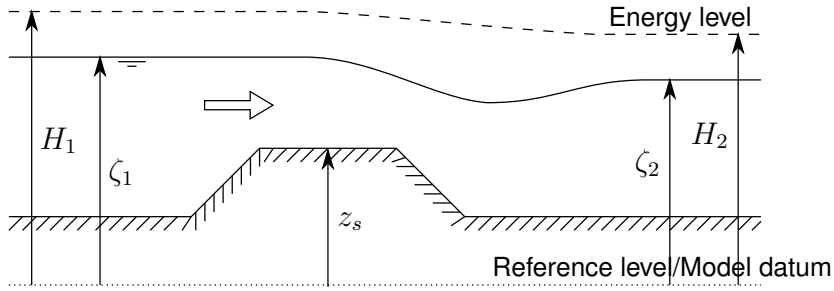

Figure 1.13: *Definition of energy and water level*

and the energy level H is defined as:

$$H = \zeta + \frac{u^2}{2g} \tag{1.42}$$

The user must enter the following input parameters for this type of hydraulic structure:

z_s	Level of crest
W_n	Total net width
N	Number of piers
P	Level of upstream face
H_0	Design head of the weir
K_p	Pier contraction coefficient
K_a	Abutment contraction coefficient

The following parameters can be controlled by a hydraulic structure controller:

W_s	Width across flow section
z_s	Crest level of weir

The discharge through the structure is computed with:

$$Q = C_1 C_2 W \sqrt{2g(H_1 - z_s)^3} \tag{1.43}$$

In which W is the active width and C_1 and C_2 are factors computed in the following way.

Note:
if $\zeta_1 < z_s$ then $Q = 0$.

$$W = W_k - 2(NK_p + K_a)(H_1 - z_s) \tag{1.44}$$

$$C_1 = C_0 C_k \tag{1.45}$$

The value of C_0 is computed depending on the value of the ratio between the upstream face and the design head:

$$\chi = \frac{P}{H_0} \tag{1.46}$$

if $\chi < 2$ then:

$$C_0 = (0.1256\chi^5 - 1.0178\chi^4 + 3\chi^3 - 3.94\chi^2 + 2.28\chi + 1.66)\frac{\xi}{\sqrt{2g}} \tag{1.47}$$

where ξ is a correction factor computed with:

$$\xi = -0.052\chi^3 + 0.145\chi^2 - 0.096\chi + 1.01 \tag{1.48}$$

if $\chi > 2$ then:

$$C_0 = \frac{2.1549008}{\sqrt{2g}} \tag{1.49}$$

The value of χ is computed depending on the value of the ratio between the energy level minus the sill level and the design head:

$$\chi_1 = \frac{H_1 - z_s}{H_0} \tag{1.50}$$

$$C_k = \begin{cases} 0.1394\chi_1^3 - 0.416\chi_1^2 + 0.488\chi_1 + 0.785 & \chi_1 < 1.6 \\ 1.0718224 & \chi_1 \geq 1.6 \end{cases} \tag{1.51}$$

The value of C_t is computed depending on the value of the ratio between the energy level minus the downstream water level and the energy level minus the sill level:

$$\chi_2 = \frac{H_1 - h_2}{H_1 - z_s} \tag{1.52}$$

$$C_t = \begin{cases} \sqrt{(1 - \frac{(\chi_2 - 0.7)^2}{0.49})} + 27(0.7 - \chi_2)^4\sqrt{\chi_2^3} & \chi_2 < 0.7 \\ 1 & \chi_2 \geq 0.7 \end{cases} \tag{1.53}$$

1.1.16.2 Bridge

A bridge is one of the structure types that can be included in the SOBEK-Flow-module. The following types of bridges can be modelled:

- Pillar bridge
- Abutment bridge
- Fixed bed bridge
- Soil bed bridge

The general description of the discharge through a bridge is given by:

$$Q = \mu A_f \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.54}$$

Q	Discharge through bridge [m^3/s]
μ	Discharge coefficient derived from loss-coefficients [$-$]
A_f	Wetted area [m^2] of flow through bridge at upstream side
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
ζ_1	Upstream water level [m]

ζ_2 Downstream water level [m]

Note: For numerical reasons (e.g. validity of structure Equation (1.54)) the discharge coefficient (μ) is limited to a maximum of 1.0. In other words if the discharge coefficient (μ) according to Equation (1.55) (Pilar bridge) or Equation (1.57) (Abutment bridge, Fixed bed bridge or Soil bed bridge) becomes larger than 1.0, the actual applied discharge coefficient in Equation (1.54) is limited to 1.0. This means that for a given discharge, the water level difference over such bridge might be larger than anticipated, when considering the defined friction loss coefficients. In such case it is advised to apply the Extra Resistance Node for modelling this particular bridge. An Extra Resistance Node (see **??** and section 1.1.1.2) adds additional(extra) resistance to the St. Venant momemtum equation. Hence no structure equation is solved at an Extra Resistance Node.

Pillar bridge

A pillar bridge has one or more pillars that affect the discharge through the bridge with the following definition for the discharge coefficient (μ)

$$\mu = \frac{1}{\sqrt{\xi_y}} \tag{1.55}$$

ξ_y Pillar loss coefficient, defined as

$$\xi_y = \beta \frac{\alpha_y}{A_f} \tag{1.56}$$

β Parameter [$-$] depending on shape of pillar[s] (shape factor). Normally between 0.22 and 1.56

A_f Wetted area [m^2] of flow through bridge at upstream side

α_y Area [m^2] of wetted part of pillar [s] perpendicular to the flow direction, considered at upstream side

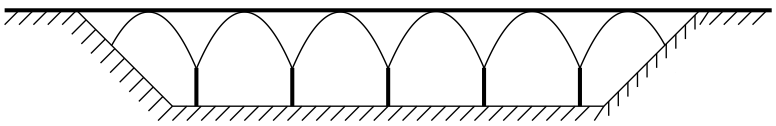

Figure 1.14: *Pillar bridge*

Note: For numerical reasons the discharge coefficient (μ) is limited to a maximum of 1.0. For more information, see the note below Equation (1.54).

The plate of the bridge is always so high that it does not effect the flow through the bridge. So the cross section is considered as open.

Note:
Pillar bridges are not allowed in closed cross-sections.

Abutment bridge

For the abutment bridge the overall loss coefficient is defined as:

$$\mu = \frac{1}{\sqrt{\xi_i + \xi_f + \xi_o}} \tag{1.57}$$

ξ_i	Entrance loss coefficient (in). Constant
ξ_f	Friction loss coefficient
ξ_o	Exit loss coefficient (out).

Note: For numerical reasons the discharge coefficient (μ) is limited to a maximum of 1.0. For more information, see the note below Equation (1.54).

The ξ_o coefficient is defined as:

$$\xi_o = k\left(1 - \frac{A_f}{A_{f_2}}\right)^2 \tag{1.58}$$

k	Constant exit loss coefficient
A_f	Wetted area [m^2] of flow through bridge at upstream side
A_{f_2}	Wetted area [m^2] of flow in branch at downstream side of bridge

The ξ_f coefficient is defined as:

$$\xi_f = \frac{2gL}{C^2R} \tag{1.59}$$

g	Acceleration due to gravity [m/s^2] (≈ 9.81)
L	Length of bridge [m]
C	Chézy coefficient [$m^{1/2}/s$]
R	Hydraulic radius [m]

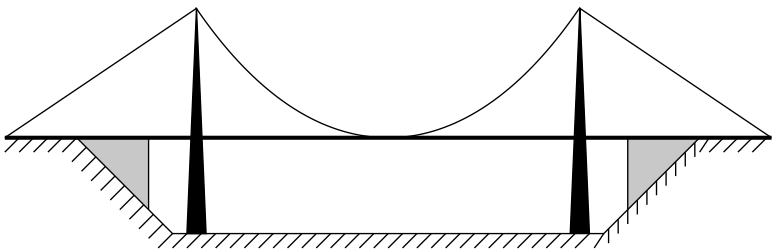

Figure 1.15: *A suspension bridge with abutments*

Here, the plate of the bridge can effect the flow through the bridge. The cross section is closed

Note:
The definition of this bridge is similar to the definition of a culvert.

Fixed bed bridge

This bridge has the same formulation as the abutment bridge except it has a rectangular profile.

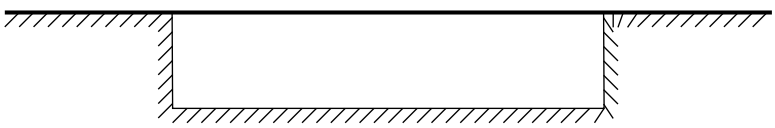

Figure 1.16: *Fixed bed bridge*

Soil bed bridge

This bridge has the same formulation as the fixed bed bridge including the rectangular profile. In addition to this it has a ground layer with a different friction formulation. This ground layer can have a zero thickness.

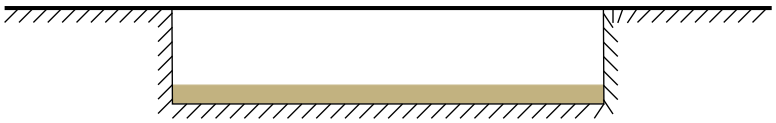

Figure 1.17: *Soil bed bridge*

1.1.16.3 Compound structure

A compound structure consists of several hydraulic structures parallel to each other at one location. These hydraulic structures may be of the same type or of different types. Presently following structure types can be placed as a member in a compound structure, viz.: General structure, Database structure, Advanced weir, River weir and River pump.

Please note that a value for structure inertia damping factor can be defined for each individual member of the compound structure.

Each member of the compound structure has its own triggers and controllers.

1.1.16.4 Culvert

A culvert is one of the structure types that can be included in the SOBEK-Flow-module. A culvert is an underground structure that normally connects two open channels. The flow through a culvert is affected by its upstream and downstream invert levels, the size and shape of its closed cross section, its ground layer thickness, its entrance loss, its friction loss, its valve loss and its exit loss.

Figure 1.18 shows a side view of a culvert.

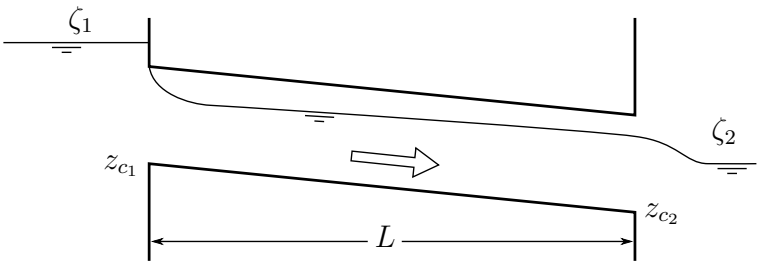

Figure 1.18: *Side view of a culvert*

Two flow conditions can occur:

Free flow when $\zeta_2 < z_{c2} + h_{c2}$

$$Q = \mu A_{fc}\sqrt{2g(\zeta_1 - (z_{c2} + h_{c2}))} \tag{1.60}$$

Submerged flow when $\zeta_2 \geq z_{c2} + h_{c2}$:

$$Q = \mu A_{fc}\sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.61}$$

Q	Discharge through culvert [m^3/s]
μ	Discharge coefficient, derived from loss-coefficients [$-$]
A_{fc}	Discharge culvert flow area $\min(A_{fc1}, A_{fcgate})$ [m^2]
	A_{fc1}: Flow area in the culvert at its upstream side [m^2]
	A_{fcgate}: Flow area under the culvert gate [m^2]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
ζ_1	Upstream water level [m]
ζ_2	Downstream water level [m]
z_{c2}	Downstream culvert invert level [m]
h_{c2}	Critical culvert depth at the downstream side, $\sqrt[3]{Q^2/(gT_2^2)}$ [m]
T_2	Surface width in the culvert at its downstream side [m]

For numerical reasons the discharge coefficient (μ) is limited to a maximum of 1.0. The discharge coefficient (μ) is computed as follows:

$$\mu = \frac{1}{\sqrt{\xi_i + \xi_f + \xi_v + \xi_o}} \tag{1.62}$$

ξ_i	Entrance loss coefficient [$-$]
ξ_f	Friction loss coefficient [$-$]
ξ_v	Valve loss coefficient [$-$]
ξ_o	Exit loss coefficient [$-$]

The entrance loss coefficient (ξ_i) can be defined as a constant value only.

The friction loss coefficient (ξ_f) is computed as follows:

$$\xi_f = \frac{2gL}{C_1^2 R} \tag{1.63}$$

L	Length of the culvert [m]
C_1	Chézy coefficient in the culvert at its upstream side [m$^{1/2}$/s]
R	Hydraulic radius [m]

$$R = \begin{cases} R_{c1} & \zeta_1 \geq z_{c1} + h_{gate} \\ R_{gate} & \zeta_1 < z_{c1} + h_{gate} \end{cases} \tag{1.64}$$

h_{gate}	Gate height opening [m]
R_{c1}	Hydraulic radius in the culvert at its upstream side [m]
R_{gate}	Hydraulic radius based on actual gate height opening [m]
z_{c1}	Upstream culvert invert level [m]

The valve loss coefficient (ξ_v) can be defined as a constant value or as a function of the ratio of the “Gate height opening” and the “maximum inner culvert height”.

Note:

- ◇ In case the valve loss coefficient (ξ_v) is not a constant, in computations the actual valve loss coefficient (ξ_v) is derived from the user defined table, while using the ratio of the “actual gate height opening and the “actual maximum inner culvert height”.
- ◇ In case the ground layer thickness is greater than zero, both the “actual gate height opening” and the “actual maximum inner culvert height” will differ from the values as defined in the user interface (see next paragraph)

The exit loss coefficient (ξ_o) is computed as follows:

Submerged flow ($\zeta_2 = z_{c2} + h_{c2}$):

$$\xi_o = k \left(1 - \frac{A_{fc}}{A_{fr2}}\right)^2 \tag{1.65}$$

k	User defined constant exit loss coefficient [$-$]
A_{fr2}	Flow area in the branch, adjacent to the downstream culvert side [m^2]
A_{fc}	Culvert flow area [m^2]

$$A_{fc} = \begin{cases} A_{f_{cgate}} & h_1 \geq z_{c1} + h_{gate} \\ A_{f_{c1}} & h1 < z_{c1} + h_{gate} \end{cases} \tag{1.66}$$

h_{gate}	Gate height opening [m]
z_{c1}	Upstream culvert invert level [m]
$A_{f_{cgate}}$	Flow area under the culvert gate [m^2]
$A_{f_{c1}}$	Flow area in the culvert at its upstream side [m^2]

Free flow ($\zeta_2 < z_{c2} + h_{c2}$):

$$\xi_o = 0 \tag{1.67}$$

Culvert cross-sections, bed friction and ground layer

For a culvert all available closed cross-section types can be used. In a culvert, a ground layer with constant thickness can be defined. Culvert friction and ground layer friction can be specified, using any of the available bed friction formulations.

Defining a ground layer thickness > 0 implies that in culvert computations:

- Defined invert levels are raised with the ground layer thickness,
- Gate height openings are reduced with the ground layer thickness,
- Maximum inner height of the culvert is reduced with the ground layer thickness,
- Cross-sectional parameters (such as: flow areas, hydraulic radius and so on) are computed based on a cross-sectional profile, that is reduced by the ground layer thickness.

Culvert, Good modelling practice aspects

It is advised to avoid that the bed level of a cross-section in front of a Culvert, Inverted Siphon or Siphon is above the ground-layer level (= invert level + ground-layer thickness), since such situation can result in very small computational time-steps (e.g. long required wall-clock times) or even in a termination of the simulation.

This is explained as follows. Consider the situation depicted in Figure 1.38 were the bed level in front of a Culvert (Inverted Siphon or Siphon) is 0.60 m above the ground-layer level. This means that at small upstream water depths, water will be sucked into the culvert (Inverted Siphon or Siphon), resulting in large flow velocities. For the computational time-step (Δt) yields that $\Delta t \leq \Delta x / U$, where U is the local flow velocity and Δx is distance between two water level computational points (or ζ-points). At very low discharges even negative water depths may be computed, leading to a termination of the simulation.

The situation explained above can be avoided by making the bed level in front of the Culvert (Inverted Siphon or Siphon) equal to the ground-layer level. In other words by defining a

bed level slope from ζ-point ζ_2 to ζ_3 as depicted in Figure 1.38. Providing for the parameter "Maximum Lowering of Cross-section Bed Level at Culvert" a value greater or equal to 0.60 m means that before the computation starts, the bed level at ζ-point ζ_3 is set equal to the ground-layer level. In **??** it is shown how to provide a value for parameter "Maximum Lowering of Cross-section Bed Level at Culvert".

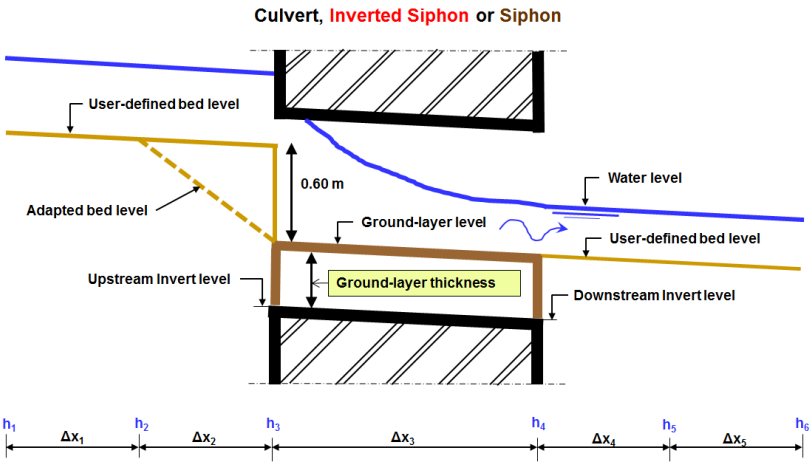

Figure 1.19: *Good modelling practice, Culvert, Inverted Siphon and Siphon*

1.1.16.5 Database structure

Please note that default a computational point is located 0.5 m before and 0.5 m behind a Database structure. In a hydraulic structure the discharge through the structure depends on upstream and downstream water levels and structure parameters that define the dimension of the structure etc. In other words the hydraulic behaviour of a structure can be defined in the structure equation as a relationship between upstream and dowstream water level. In the database structure however, this relationship is defined in a tabulated form which is stored in a database. The user has to define this database.

The database consists in fact of a matrix of discharges. At every point of the matrix a discharge (Q) is defined as a function of two corresponding water levels: one facing the beginning of the branch (ζ_1) and one facing the end (ζ_2), are defined. There are two ways to define the relation:

- ◇ a function of both water levels $Q = Q(\zeta_1 - z_s, \zeta_2 - z_s)$, to be used for structures with relatively high head differences; $z_s = crest\ level$ w.r.t. datum
- ◇ a function of the water level and the water level difference $Q = Q(\zeta_1 - z_s, \Delta\zeta = \zeta_1 - \zeta_2)$, to be used for structures with relatively small water level differences combined with large water level variations; $z_s = crest\ level$ w.r.t. datum

All discharges in a row correspond to the same water level (ζ_1) facing the beginning of the branch segment. All discharges in a column correspond either to the same water level (ζ_2) facing the end of the branch segment or to the water level difference ($\Delta\zeta = \zeta_1 - \zeta_2$). The water levels that are part of the database are defined with respect to a user defined datum.

During a simulation, discharges at any point in the domain of the matrix will be obtained by interpolation. In most cases this will be a linear interpolation in two directions. However when the discharge is requested at a water level that lies within 10 percent of a water level defined in the matrix, cubic interpolation takes place for that water level.

Warning:

- ◇ The crest level can be adjusted by a controller. This implies the assumption that the discharge head relation is independent of the level of the crest. This is not completely correct. The user should be aware of this.

1.1.16.6 General structure

Please note that default a computational point is located 0.5 m in front and 0.5 m behind a general structure. In the general structure type in sobek weir and gate flow is combined in one structure type. In addition, the general structure gives more freedom in defining the dimensions and the geometry of the hydraulic structure. The geometrical shape is given in Figure 1.20 and Figure 1.21. See **??** for the definition of input parameters. Please note that the discharge through a General structure is computed on basis of upstream and downstream energy levels. Please note as well that a structural inertia damping factor can be defined for each individual General structure.

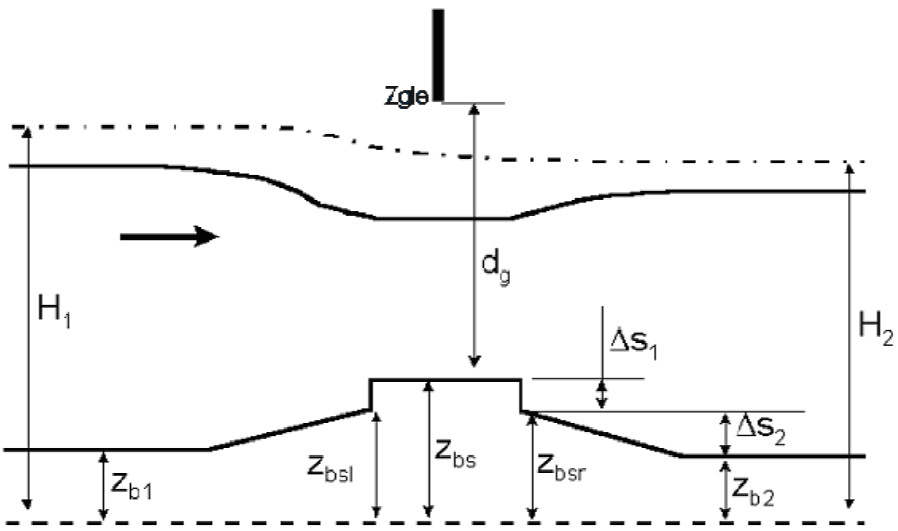

Figure 1.20: *General structure, side view*

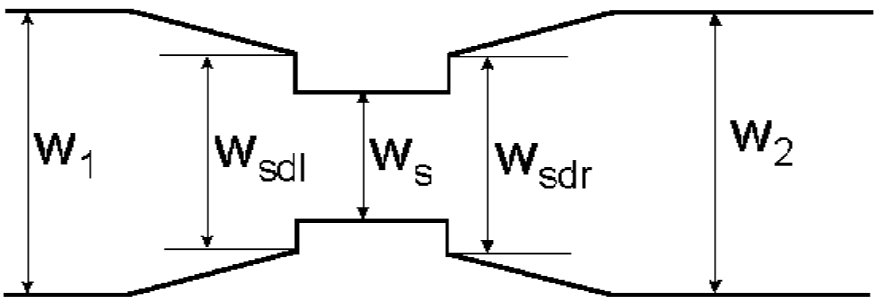

Figure 1.21: *General structure, top view*

Flow across the general structure can be of the following types: drowned weir flow, free weir flow, drowned gate flow, and free gate flow, depending on the dimensions of the structure and

the flow conditions.

When salt intrusion is modelled, the density difference of the water over the structure is incorporated in the impulse balance (this is not the case with the other structure types).

In the solution method for the general structure particular attention has been given to the modelling of the transition between free- and submerged flow.

For this purpose, the water level at the sill (ζ_s) is computed by applying an impulse balance instead of taking it equal to the water level further downstream.

As a result the general structure is especially attractive for those who want to simulate the shifting conditions from free- to submerged gate flow accurately and of course in case of important density-differences over the structure.

Computation of downstream water level

In case of drowned gate flow or drowned weir flow the water level at the sill or downstream of the gate is required. This level is computed by application of the impulse balance.

◇ drowned gate-flow

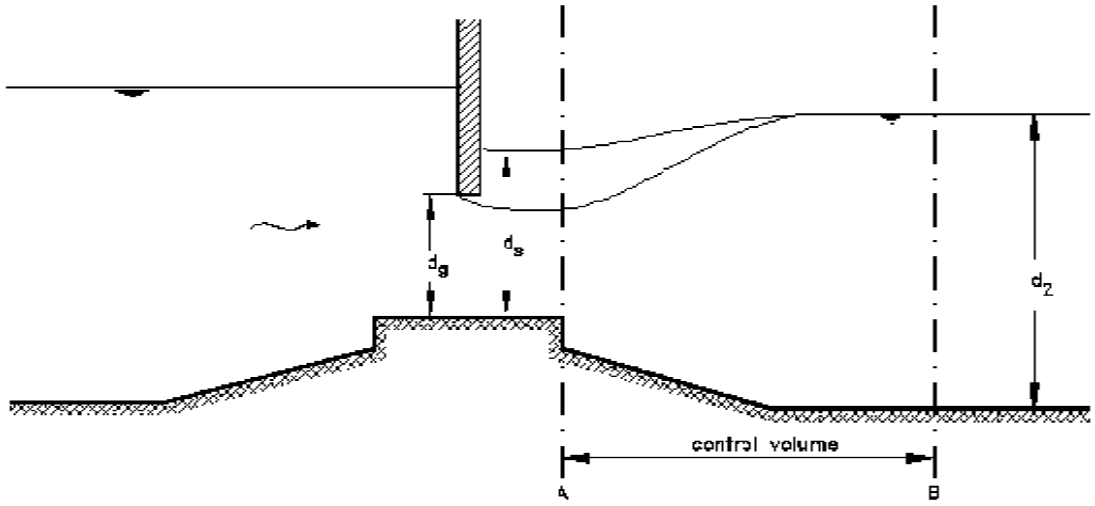

Figure 1.22: *Drowned gate flow*

The water depth h_s can be described by a second order algebraic equation:

$$A_w h_s^2 + B_w h_s + C_w = 0 \tag{1.68}$$

with:

$$A_g = (1+\rho^*)\left(\frac{W_{sd}}{3}+\frac{W_2}{6}\right)+(1-\rho^*)\left(\frac{W_{sd}}{4}+\frac{W_2}{12}\right) \tag{1.69}$$

$$\begin{aligned} B_g = (1+\rho^*)\Bigg(&(\Delta S_1+\Delta S_2-h_2)\left(\frac{W_{sd}}{3}+\frac{W_2}{6}\right)+(2\Delta S_1+h_2)\frac{W_{sd}}{6}+\\ &(\Delta S_1+2h_2)\frac{W_2}{6}\Bigg)+\\ &+(1-\rho^*)\left(\Delta S_1\frac{W_{sd}}{3}+(\Delta S_1+h_2)\frac{W_{sd}+W_2}{6}\right)+\\ &+\frac{4\rho^* c_{gd}^2\mu_{gd}^2 d_g^2 W_s^2}{W_2 h_2}(1+\frac{\lambda}{d_2})-4c_{gd}\mu_{gd}d_g W_s \end{aligned} \tag{1.70}$$

$$\begin{aligned} C_g = &(1+\rho^*)(\Delta S_1+\Delta S_2-h_2)\left(2\Delta S_1+h_2)\frac{W_{sd}}{6}+(\Delta S_1+2h_2)\frac{W_2}{6}\right)+\\ &+(1-\rho^*)\left((\Delta S_1)^2\frac{W_{sd}}{6}+(\Delta S_1+h_2)\frac{W_{sd}+W_2}{12}\right)+\\ &-\frac{4\rho^* c_{gd}^2\mu_{gd}^2 d_g^2 W_s^2 H_{s_1}}{W_2 h_2}(1+\frac{\lambda}{h_2})+4c_{gd}\mu_{gd}d_g W_s H_{s_1} \end{aligned} \tag{1.71}$$

Equation (1.68) leads to:

$$h_s = \frac{-B_g+\sqrt{B_g^2-4A_gC_g}}{2A_g} \tag{1.72}$$

◇ drowned weir-flow

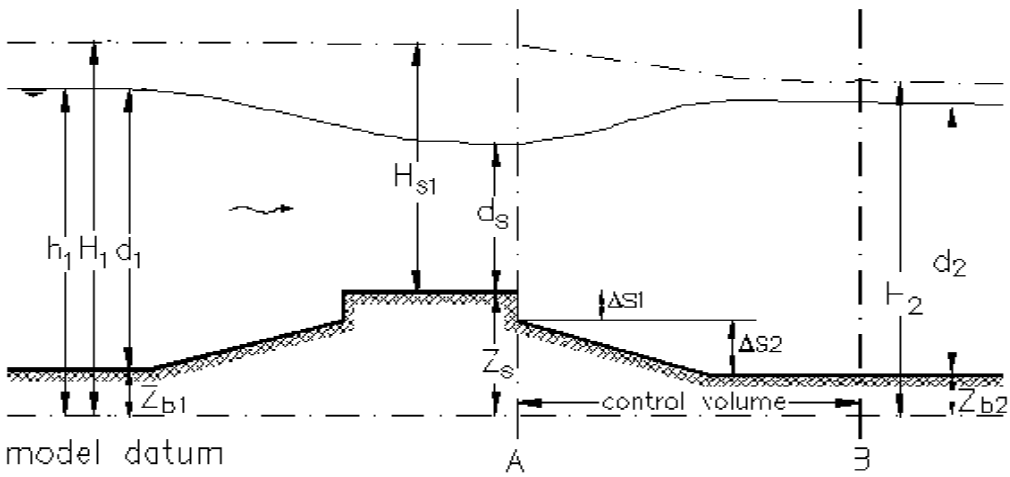

Figure 1.23: *Drowned weir-flow*

The water depth at the sill h_s is described by a third order algebraic equation:

$$D_w h_s^3 + A_w h_s^2 + B_w h_s + C_w = 0 \tag{1.73}$$

with:

$$D_w = \frac{4\rho^* c_{wd}^2 W_s^2}{W_2 h_2}(1 + \frac{\lambda}{h_2}) \tag{1.74}$$

$$A_w = (1+\rho^*)\left(\frac{W_{sd}}{3} + \frac{W_2}{6}\right) + (1-\rho^*)\left(\frac{W_{sd}}{4} + \frac{W_2}{12}\right) + \\ - \frac{4\rho^* c_{wd}^2 W_s^2 H_{s_1}}{W_2 h_2}(1 + \frac{\lambda}{h_2}) - 4c_{wd}W_s \tag{1.75}$$

$$B_w = (1+\rho^*)\left((\Delta S_1 + \Delta S_2 - h_2)\left(\frac{W_{sd}}{3} + \frac{W_2}{6}\right) + (2\Delta S_1 + h_2)\frac{W_{sd}}{6} + \right. \\ \left. (\Delta S_1 + 2h_2)\frac{W_2}{6}\right) + \\ + (1-\rho^*)\left(\Delta S_1 \frac{W_{sd}}{3} + (\Delta S_1 + h_2)\frac{W_{sd} + W_2}{6}\right) + 4c_{wd}W_s H_{s_1} \tag{1.76}$$

$$C_w = (1+\rho^*)(\Delta S_1 + \Delta S_2 - h_2)\left((2\Delta S_1 + h_2)\frac{W_{sd}}{6} + (\Delta S_1 + 2h_2)\frac{W_2}{6}\right) + \\ + (1-\rho^*)\left((\Delta S_1)^2 \ \frac{W_{sd}}{6} + (\Delta S_1 + h_2)^2 \ \frac{W_{sd} + W_2}{12}\right) \tag{1.77}$$

A direct method is applied to calculate h_s from equation Equation (1.73).

Please note that:

The relative density ρ^* is given by the expression:

$$\rho^* = \frac{\rho_2}{\rho_1} \tag{1.78}$$

where:

ρ^*	relative density [$-$]
ρ_1	density of the water at the upstream side of the General structure [kg/m^3]
ρ_2	density of the water at the downstream side of the General structure [kg/m^3]

The extra resistance parameter λ relates to the friction force in the impulse balance, that in case of drowned gate flow or drowned weir flow is solved for the water movement at the downstream side of the General structure. The extra resistance parameter λ is given by the expression:

$$\lambda = \frac{gL}{C^2} \tag{1.79}$$

where:

λ	Extra resistance parameter [m]
L	Length of the expansion zone at the downstream side of the General structure [m]
g	Acceleration due to gravity [m/s^2]
C	Chézy value, representing the friction-roughness at the downstream side of the General structure [$m^{1/2}/s$]

Discharge equations

The following discharge equations are applied during the computations.

◇ Free gate flow:

$$u_s = \mu_{gf} c_{gf} \sqrt{2g(H_1 - (z_s + \mu_{gf} d_g))} \tag{1.80}$$
$$A_f = W_s d_g \tag{1.81}$$
$$Q = u_s A_f = \mu_{gf} c_{gf} W_s d_g \sqrt{2g(H_1 - (z_s + \mu_{gf} d_g))} \tag{1.82}$$

◇ Drowned gate flow:

$$u_s = \mu_{gd} c_{gd} \sqrt{2g(H_1 - (z_s + d_s))} \tag{1.83}$$
$$A_f = W_s d_g \tag{1.84}$$
$$Q = u_s A_f = \mu_{gd} c_{gd} W_s d_g \sqrt{2g(H_1 - (z_s + d_s))} \tag{1.85}$$
$$\tag{1.86}$$

◇ Free weir flow:

$$u_s = c_{wf} \sqrt{\frac{2}{3} g(H_1 - z_s)} \tag{1.87}$$
$$A_f = W_s \ \frac{2}{3}(H_1 - z_s) \tag{1.88}$$
$$Q = u_s A_f = c_{wf} W_s \ \frac{2}{3} \sqrt{\frac{2}{3}} g(H_1 - z_s)^{3/2} \tag{1.89}$$

◇ Drowned weir flow:

$$u_s = c_{wd} \sqrt{2g(H_1 - (z_s + d_s))} \tag{1.90}$$
$$A_f = W_s d_s \tag{1.91}$$
$$Q = u_s A_f = c_{wd} W_s d_s \sqrt{2g(H_1 - (z_s + d_s))} \tag{1.92}$$

where:

μ_{gf}	contraction coefficient for free gate flow
μ_{gd}	contraction coefficient for drowned gate flow ($\mu_{gd} = \mu_{gf}$)
c_{gf}	correction coefficient for free gate flow
c_{gd}	correction coefficient for drowned gate flow
c_{wf}	correction coefficient for free weir flow
c_{wd}	correction coefficient for drowned weir flow

Note:
The contraction coefficient has a maximum value $\mu = 1.0$. In case the user specifies a higher value, a warning will be generated

Criteria for flow types

First it is assumed that it is weir flow. Then the water level at the sill ζ_s and the critical depth $h_c = \frac{2}{3}(H_1 - z_s)$ are calculated.

The criteria are:

- $\zeta_s > h_c$ and $d_g > \zeta_s$ drowned weir flow
- $\zeta_s < h_c$ and $d_g > \zeta_s$ free weir flow
- otherwise gate flow.

In the latter case the water level at the sill ζ_s is recalculated using the gate flow conditions. The critical depth h_c is now defined as $\mu_{gf} d_g$.

The criteria are.

- $\zeta_s > h_c$ drowned gate flow
- $\zeta_s < h_c$ free gate flow
- ζ_s imaginary free gate flow (i.e. downstream water level below crest level)

Note:
In case upstream water level is above gate lower edge level, there can still be **drowned weir flow** if $\zeta_s > h_c$ and $d_g > \zeta_s$ or **free weir flow** if $\zeta_s < h_c$ and $d_g > \zeta_s$.

1.1.16.7 Inverted siphon

An inverted siphon is one of the structure types, that can be included in the D-Flow 1D-module. An inverted siphon is a structure that normally connects two open channels, that are separated by a particular infrastructural work (e.g. dike, railroad). The inverted siphon makes an underground connection through such infrastructural work. An inverted siphon is assumed to be fully filled with water at its deepest point. In case this does not yield for your stucture, you can consider to use a culvert. The flow through an inverted siphon is affected by its upstream and downstream invert level, the size and shape of its closed cross section, its ground layer thickness, its entrance loss, friction loss, its valve loss, losses due to its bends (bend loss), and its exit loss.

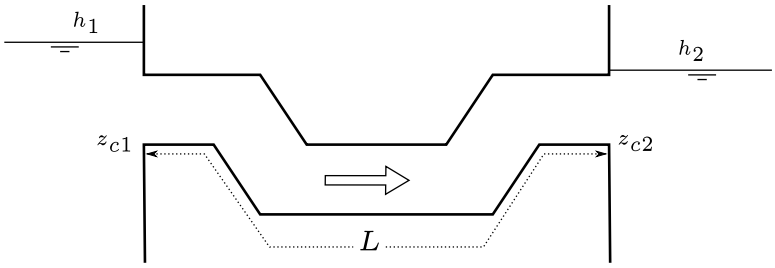

Figure 1.24: *Side view of an inverted siphon*

Two flow conditions can occur:

1 Free flow when $\zeta_2 < z_{c2} + d_{c2}$

$$Q = \mu A_{fis} \sqrt{2g(\zeta_1 - (z_{c2} + d_{c2})} \tag{1.93}$$

2 Submerged flow when $\zeta_2 \geq z_{c2} + d_{c2}$:

$$Q = \mu A_{fis} \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.94}$$

Q	Discharge through inverted siphon [m^3/s]
μ	Discharge coefficient, derived from loss-coefficients [$-$]
A_{fis}	Discharge inverted siphon flow area ($= \min(A_{fcrs}, A_{fisgate})$) [$m^2$]

A_{fcrs}	Cross-sectional area of the inverted siphon. At its deepest point the inverted siphon is considered to be completely filled with water [m^2]
$A_{fisgate}$	Flow area under the inverted siphon gate [m^2]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
ζ_1	Upstream water level [m]
ζ_2	Downstream water level [m]
z_{c2}	Downstream inverted siphon invert level [m]
d_{c2}	Critical inverted siphon depth at the downstream side

$$\left(d_{c2} = \sqrt[3]{Q^2/(gT_2^2)}\right) \qquad [m] \tag{1.95}$$

T_2	Surface width in the inverted siphon at its downstream side [m]

For numerical reasons the discharge coefficient (μ) is limited to a maximum of 1.0. The discharge coefficient (μ) is computed as follows:

$$\mu = \frac{1}{\sqrt{\xi_i + \xi_f + \xi_v + \xi_b + \xi_o}} \tag{1.96}$$

ξ_i	Entrance loss coefficient [$-$]
ξ_f	Friction loss coefficient [$-$]
ξ_v	Valve loss coefficient [$-$]
ξ_b	Bend loss coefficient [$-$]
ξ_o	Exit loss coefficient [$-$]

The entrance loss coefficient (ξ_i) can be defined as a constant value only.

The friction loss coefficient (ξ_f) is computed as follows:

$$\xi_f = \frac{2gL}{C^2R} \tag{1.97}$$

L	Length of the inverted siphon [m]
C	Chézy coefficient for fuly water filled inverted siphon [m$^{1/2}$/s]
R	Hydraulic radius [m]
	If GHO $<$ MIISH; $R = R_{gate}$
	If GHO $\geq$ MIISH; R= $R_{inverted\ siphon}$
GHO	Gate height opening [m]
MIISH	Maximum inner inverted siphon height [m]
$R_{inverted\ siphon}$	Hydraulic radius based on a fully water filled inverted siphon [m]
R_{gate}	Hydraulic radius based on actual gate height opening [m]

The valve loss coefficient (ξ_v) can be defined as a constant value or as a function of the ratio of the “Gate height opening” and the “maximum inner inverted siphon height”.

Note:

- In case the valve loss coefficient (ξ_v) is not a constant, in computations the actual valve loss coefficient (ξ_v) is derived from the user defined table, while using the ratio of the “actual gate height opening and the “actual maximum inner inverted siphon height”.
- In case the ground layer thickness is greater than zero, both the “actual gate height opening” and the “actual maximum inner inverted siphon height” will differ from the values as defined in the user interface.

The exit loss coefficient (ξ_o) is computed as follows:

Submerged flow ($\zeta_2 = z_{c2} + d_{c2}$):

$$\xi_o = k \left(1 - \frac{A_{fis}}{A_{fr2}}\right)^2 \tag{1.98}$$

k	User defined constant exit loss coefficient [$-$]
A_{fr2}	Flow area in the branch, adjacent to the downstream inverted siphon side [m^2]
A_{fis}	Inverted siphon flow area [m^2]
	If GHO $\leq$ MIISH; $A_{fis} = A_{fisgate}$
	If GHO $>$ MIISH; $A_{fis} = A_{inverted\ siphon}$
GHO	Gate height opening [m]
MIISH	Maximum inner inverted siphon height [m]
$A_{fisgate}$	Flow area under the inverted siphon gate [m^2]
$A_{inverted\ siphon}$	Flow area based on a fully water filled inverted siphon [m^2]

Free flow ($\zeta_2 < z_{c2} + d_{c2}$):

$$\xi_o = 0 \tag{1.99}$$

Inverted siphon cross-sections, bed friction and ground layer:
For an inverted siphon all available closed cross-section types can be used. In an inverted siphon, a ground layer width constant thickness can be defined. Inverted siphon friction and ground layer friction can be specified, using any of the available bed friction formulations.

Defining a ground layer thickness $>$ 0 implies that in inverted siphon computations:

- Defined invert levels are raised with the ground layer thickness,
- Gate height openings are reduced with the ground layer thickness,
- Maximum inner height of the inverted siphon is reduced with the ground layer thickness,
- Cross-sectional parameters (such as: flow areas, hydraulic radius and so on) are computed based on a cross-sectional profile, that is reduced by the ground layer thickness.

Inverted Siphon, Good modelling practice aspects

It is advised to avoid that the bed level of a cross-section in front of a Culvert, Inverted Siphon or Siphon is above the ground-layer level (= invert level + ground-layer thickness), since such situation can result in very small computational time-steps (e.g. long required wall-clock times) or even in a termination of the simulation.

This is explained as follows. Consider the situation depicted in Figure 1.38 were the bed level in front of a Culvert (Inverted Siphon or Siphon) is 0.60 m above the ground-layer level. This means that at small upstream water depths, water will be sucked into the culvert (Inverted Siphon or Siphon), resulting in large flow velocities. For the computational time-step (Δt) yields that $\Delta t \leq \Delta x / U$, where U is the local flow velocity and Δx is distance between two water level computational points (or ζ-points). At very low discharges even negative water depths may be computed, leading to a termination of the simulation.

The situation explained above can be avoided by making the bed level in front of the Culvert (Inverted Siphon or Siphon) equal to the ground-layer level. In other words by defining a bed level slope from ζ-point ζ_2 to ζ_3 as depicted in Figure 1.38. Providing for the parameter "Maximum Lowering of Cross-section Bed Level at Culvert" a value greater or equal to 0.60 m

means that before the computation starts, the bed level at ζ-point ζ_3 is set equal to the ground-layer level. In **??** it is shown how to provide a value for parameter "Maximum Lowering of Cross-section Bed Level at Culvert".

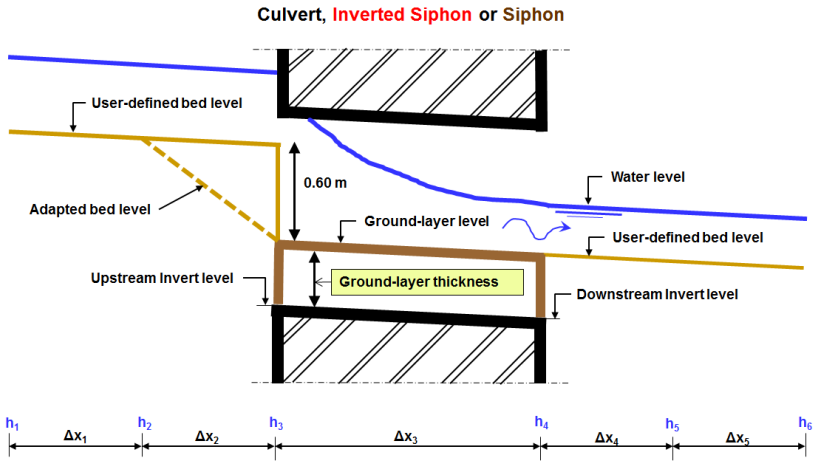

Figure 1.25: *Good modelling practice, Culvert, Inverted Siphon and Siphon*

1.1.16.8 Orifice

The geometrical shape of an orifice is given in Figure 1.26. Four different types of orifice flow can be discerned, viz: free weir flow, submerged weir flow, free gate flow and submerged gate flow. No discharge flows through the orifice if its gate is closed, if the upstream water level equals the downstream water level or if both the upstream water level and the downstream water level are below the crest level.

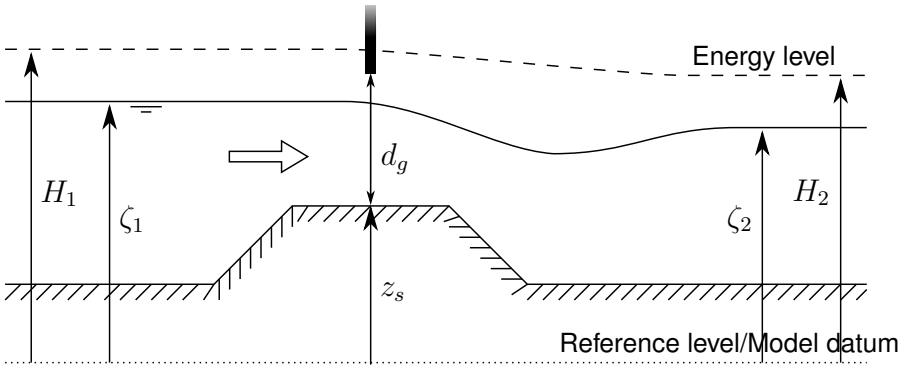

Figure 1.26: *Orifice*

The following four discharge equations are applied:

1 Free weir flow:

$$A_f = W_s \ \frac{2}{3}(\zeta_1 - z_s) \tag{1.100}$$

$$Q = c_w W_s \ \frac{2}{3}\sqrt{\frac{2}{3}g} \ \ (\zeta_1 - z_s)^{3/2} \tag{1.101}$$

2 Submerged weir flow:

$$A_f = W_s \left(\zeta_1 - z_s - \frac{{u_s}^2}{2g} \right) \tag{1.102}$$

$$Q = c_w W_s \left(\zeta_1 - z_s - \frac{{u_s}^2}{2g} \right) \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.103}$$

3 Free gate flow:

$$A_f = W_s \mu \, d_g \tag{1.104}$$

$$Q = c_w \, W_s \, \mu d_g \sqrt{2g(\zeta_1 - (z_s + \mu \, d_g))} \tag{1.105}$$

4 Submerged gate flow:

$$A_f = W_s \mu \, d_g \tag{1.106}$$

$$Q = c_w \, W_s \, \mu \, d_g \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.107}$$

Q	Discharge across orifice [m^3/s]
A_F	Flow area [m^2]
μ	Contraction coefficient [$-$] Normally 0.63
c_w	Lateral contraction coefficient [$-$]
W_s	Crest width [m]
d_g	Opening height [m] (gate lower edge level minus crest level)
g	Gravity acceleration [m/s^2] (≈ 9.81)
ζ_1	Upstream water level [m]
ζ_2	Downstream water level [m]
z_s	Crest level [m]
u_s	Velocity over crest [m/s]

The different formulas are applied when the following conditions are met

◇ Free weir flow:

$$\zeta_1 - z_s < \frac{3}{2} d_g \quad \text{and} \quad \zeta_1 - z_s > 3/2(\zeta_2 - z_s) \tag{1.108}$$

◇ Submerged weir flow:

$$\zeta_1 - z_s < \frac{3}{2} d_g \quad \text{and} \quad \zeta_1 - z_s \leq \frac{3}{2}(\zeta_2 - z_s) \tag{1.109}$$

◇ Free gate flow:

$$\zeta_1 - z_s \geq \frac{3}{2} d_g \quad \text{and} \quad \zeta_2 \leq z_s + d_g \tag{1.110}$$

◇ Submerged gate flow:

$$\zeta_1 - z_s \geq \frac{3}{2} d_g \quad \text{and} \quad \zeta_2 > z_s + d_g \tag{1.111}$$

1.1.16.9 Pump station and Internal Pump station

The functionality of a Pump station and an Internal Pump station is identical. The only difference comprises the fact that:

- Pump station: A Pump station is located on an open channel branch. The pump discharge is determined using the water levels at the nearest ζ-calculation points, respectively located at the upstream-side and downstream-side of the Pump station.
- Internal Pump station: An Internal Pump station is accommodated in a pipe. The pump discharge is determined using the water levels (or hydrostatic pressure heads) at the upstream side of the pipe and at the downstream side of the pipe.

Here after, both the Pump station and the Internal Pump station are referred to as the Pump station. Please note that:

- When activated, water is always pumped from the suction-side towards the delivery-side,
- A Pump station cannot be placed in a compound structure,
- A Time Controller, Hydraulic controller, a PID controller or an Interval controller can overrule the pump capacity of a Pump station.
- Pump station output parameters becomes available by checking the Pump Data check-box on the 1DFLOW/Output options/Structures Tab in Settings.

Pump direction

- Positive pump direction means that the pump discharge is flowing in the positive x-direction along a branch (see Figure 1.27). Hence, water is pumped in downstream direction towards the pump-side with the highest x-coordinate.
- Negative pump direction means that the pump discharge is flowing in the negative x-direction along a branch (see Figure 1.28). Hence, water is pumped in upstream direction towards the pump-side with the lowest x-coordinate.

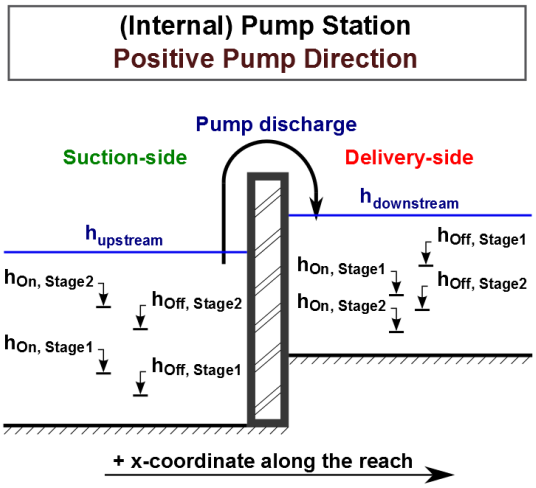

Figure 1.27: *Pump station with positive pump direction and two pump stages*

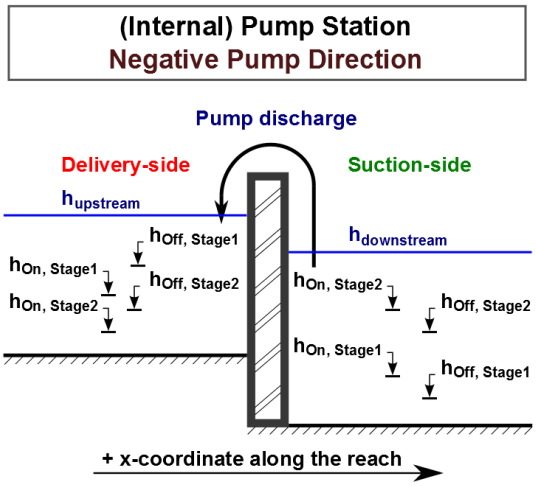

Figure 1.28: *Pump station with negative pump direction and two pump stages*

Six possible pump stages

A Pump station comprises of one (1) pump only, which can have up to six different pump stages (1, 2, 3, 4, 5 or 6). Each pump stage can have a different pump capacity. At each point-in-time, the pump can be in one (1) particular pump stage only.

The actual pump stage

The actual pump stage is determined using dead-band triggers, defined either at the suction-side only, at the delivery-side only or both at the suction-side and the delivery-side of the pump. At the beginning of each time-step for all pump stages, the status [On/Off] of their dead-band trigger at the suction-side (if defined) and their dead-band trigger at the delivery-side (if defined) are evaluated. This evaluation is done on basis of actual water levels and the Switch-on and Switch-off levels, defined for the concerning pump stage. For more details, see dead-band trigger algorithm. For more information on how to define Switch-on and Switch-off levels at the suction-side and the delivery-side of the pump, see Conventions for switch-on and switch-off levels. The actual pump stage, applied in the time-step from $t = t$ to $t = t + \Delta t$, is the number of the highest pump stage (1, 2, 3, 4, 5 or 6) that is triggered at $t = t$, meaning that all its dead-band triggers have the status [On]. If not any pump stage is triggered, the pump becomes inactive (pump discharge is zero) and the actual pump stage is set to 0. If the pump is overruled by a controller, the actual pump stage is set to -1.

Controllers

A Time-controller, a Hydraulic-controller, a PID controller or an Interval controller can be assigned to a pump station. A controller is only active in case the pump is triggered in accordance with the Switch-on and Switch-off levels, defined for stage 1. An active controller overrules the pump capacity of the triggered pump stage, while the actual pump stage is set to -1. Please note that capacity reduction factors are applied to the pump capacities set by a controller. Advice: In using a controller at a pump station, define only one (1) pump stage and take care that its dead-band trigger is always [On].

Capacity reduction table

A capacity reduction table can be defined, which is applied to all pump stages as well as to pump capacities, that are set by a controller. In the capacity reduction table a capacity reduction factor can be given as function of the pump head (e.g. water level at delivery-side minus water level at suction-side). Pump heads (first column) in a capacity reduction table should be in increasing order. Capacity reduction factors should be equal or larger than 0. The pump discharge equals the pump capacity times the capacity reduction factor. If no capacity reduction table is defined in effect a capacity reduction factor equal to 1 is applied. Please note that the pump head at $t = t$ is used to determine the capacity reduction factor to be applied in the time-step from $t = t$ to $t = t + \Delta t$.

Pump station output parameters

Pump station output parameters can be viewed in Result in Charts (Pump.his file) by checking the Pump Data check-box on the 1DFLOW/Output options/Structures Tab in Settings. Available pump station output parameters as function of time are:

- ◇ Suction-Side level: Water level at the suction-side of the pump.
- ◇ Delivery-Side level: Water level at the delivery-side of the pump.
- ◇ Pump Head: The water level at the delivery-side of the pump minus the water level at the suction-side of the pump.
- ◇ Actual Pump Stage: Equal to -1 if pump is overruled by a controller; equal to 0 if pump is inactive; or equal to 1, 2, 3, 4, 5 or 6 depending on the actual pump stage that is triggered (see Dead-band triggering algorithm).
- ◇ Pump Capacity: The pump capacity is either defined by its controller; equal to zero if pump is inactive; or equal to the pump capacity defined for the actual pump stage.
- ◇ Reduction Factor: The reduction factor follows from the pump head and the capacity reduction table.
- ◇ Pump Discharge: The pump discharge is equal to the pump capacity times the reduction factor.

Dead-band triggering algoritm

A distinction is to be made between evaluating the status [On/Off] of dead-band triggers defined at the suction-side of the pump and dead-band triggers defined at the delivery-side of the pump:

Dead-band triggering at the Suction-side of the pump

Parameters used in determining the status [On/Off] of a suction-side dead-band trigger are $h_{Suction}$ (water level at the suction-side), $h_{On,Suction}$ (switch-on-level) and $h_{Off,Suction}$ (switch-off-level), where $h_{On,Suction} > h_{Off,Suction}$. Two booleans are used (e.g. $S_{On,Suction}$ and $S_{Off,Suction}$):

- ◇ If $h_{Suction} > h_{On,Suction} \quad \Rightarrow \quad S_{On,Suction} = True;\ Else\ S_{On,Suction} = False$
- ◇ If $h_{Suction} < h_{Off,suction} \quad \Rightarrow \quad S_{Off,Suction} = True;\ Else\ S_{Off,Suction} = False$

If at the start of a computation yields that $h_{Off,Suction} \leq h_{Suction} \leq h_{On,Suction}$ the dead-band trigger is set [On]. During a computation the dead-band trigger is set [On] or [Off] according to the following rules:

- ◇ If $(S_{On,Suction} = True \quad and \quad S_{Off,Suction} = False) \quad \Rightarrow \quad Trigger = [On]$
- ◇ If $(S_{On,Suction} = False \quad and \quad S_{Off,Suction} = True) \quad \Rightarrow \quad Trigger = [Off]$
- ◇ If $(S_{On,Suction} = False \quad and \quad S_{Off,Suction} = False) \quad \Rightarrow \quad Trigger$ obtains the status determined in the previous computational time step.

Dead-band triggering at the Delivery-side of the pump

Parameters used in determining the status [On/Off] of a delivery-side dead-band trigger are $h_{Delivery}$ (water level at the delivery-side), $h_{On,Delivery}$ (switch-on-level), $h_{Off,Delivery}$ (switch-off-level), where $h_{Off,Delivery} > h_{On,Delivery}$. Two booleans are used (e.g. $S_{On,Delivery}$ and $S_{Off,delivery}$):

- ◇ If $h_{Delivery} < h_{On,Delivery} \Rightarrow S_{On,Delivery} = True;\ Else\ S_{On,Delivery} = False$
- ◇ If $h_{Delivery} > h_{Off,Delivery} \Rightarrow S_{Off,Delivery} = True;\ Else\ S_{Off,Delivery} = False$

If at the start of a computation yields that $h_{On,Delivery} \leq h_{Delivery} \leq h_{Off,Delivery}$ the dead-band trigger is set [On]. During a computation the dead-band trigger is set [On] or [Off] according to the following rules:

- ◇ If $(S_{On,Delivery} = True \quad and \quad S_{Off,Delivery} = False) \quad \Rightarrow \quad Trigger = [On]$
- ◇ If $(S_{On,Delivery} = False \quad and \quad S_{Off,Delivery} = True) \quad \Rightarrow \quad Trigger = [Off]$
- ◇ If $(S_{On,Delivery} = False \quad and \quad S_{Off,Delivery} = False) \quad \Rightarrow \quad Trigger$ obtains the status determined in the previous computational time step

Conventions for switch-on and switch-off levels

A distinction is to be made between switch-on-levels and switch-off-levels, defined at the suction-side and defined at the delivery-side of the pump (see Table 1.1).

- ◇ Suction-side: The switch-on level should increase with increasing stage number. For each stage, its switch-off-level should be lower than its switch-on-level. The switch-off-level should increase with increasing stage number.
- ◇ Delivery-side: The switch-on-level should decrease with increasing stage number. For each stage, its switch-off-level should be higher than its switch-on-level. The switch-off-level should decrease with increasing stage number.

Table 1.1: *Example of switch-on-levels and switch-off-levels at the suction-side and the delivery-side of a pump station*

Pump		**Suction-side**		**Delivery-side**	
Stage No	**Capacity** m^3/s	**Switch-on-level** m	**Switch-off-level** m	**Switch-on-level** m	**Switch-off-level** m
1	0.1	0.80	0.10	0.80	1.90
2	0.2	0.90	0.20	0.70	1.80
3	0.3	1.00	0.30	0.60	1.70
4	0.4	1.10	0.40	0.50	1.60
5	0.5	1.20	0.50	0.40	1.50
6	0.6	1.30	0.60	0.30	1.40

1.1.16.10 External Pump station

An External Pump station is to located at the end of an open channel branch or at the end of a (closed) pipe. The end of an open channel branch can either have the highest x-coordinate or an x-coordinate equal to zero. The same applies for a pipe. The pump discharge is either determined using the boundary water level and the water level at the branch-side of the pump or determined using the boundary water level and the water level (or hydrostatic pressure head) at the pipe-side of the pump. Please note that:

- When activated, water is always pumped from the suction-side towards the delivery-side.
- Water can be stored at an External Pump station, for more information see Storage options at an External Pump station.
- The "External Pump station" cannot be placed in a compound structure,
- A Time Controller, Hydraulic controller, a PID controller or an Interval controller can overrule the pump capacity of an External Pump station.
- External Pump station output parameters becomes available by checking the Pump Data check-box on the 1DFLOW/Output options/Structures Tab in Settings.

External Pump direction:

- Pump direction IN means that (see Figure 1.29):
  - Water is pumped into the model or flows towards the branch/pipe-side (or upstream-side) of the pump,
  - The pump discharge has a negative sign, this is irrespective of the x-direction along the branch (or pipe).
- Pump direction OUT means that (see Figure 1.30):
  - Water is pumped out off the model or flows towards the boundary-side (or downstream-side) of the pump ,
  - The pump discharge has a postive sign, this is irrespective of the x-direction along the branch (or pipe).

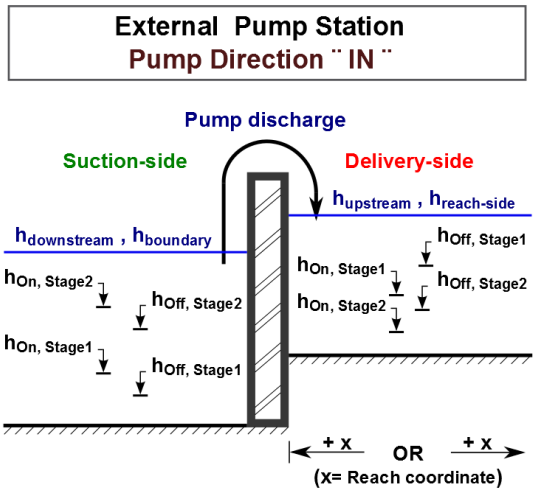

Figure 1.29: *External pump station with pump direction IN and two pump stages*

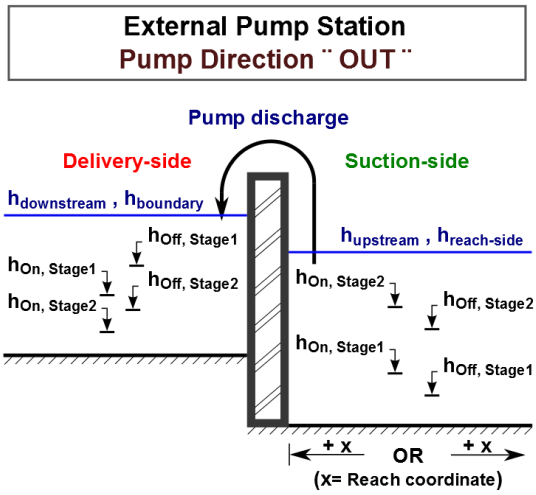

Figure 1.30: *External pump station with pump direction OUT and two pump stages*

Six possible External Pump stages

An External Pump station comprises of one (1) pump only, which can have up to six different pump stages (1, 2, 3, 4, 5 or 6). Each pump stage can have a different pump capacity. At each point-in-time, the pump can be in one (1) particular pump stage only.

The actual pump stage

The actual pump stage is determined using dead-band triggers, defined either at the suction-side only, at the delivery-side only or both at the suction-side and the delivery-side of the pump. At the beginning of each time-step for all pump stages, the status [On/Off] of their dead-band trigger at the suction-side (if defined) and their dead-band trigger at the delivery-side (if defined) are evaluated. This evaluation is done on basis of actual water levels and the Switch-on and Switch-off levels, defined for the concerning pump stage. For more details, see dead-band trigger algorithm. For more information on how to define Switch-on and Switch-off levels at the suction-side and the delivery-side of the External Pump station, see Conventions for switch-on and switch-off levels. The actual pump stage, applied in the time-step from t=t to t=t+dt, is the number of the highest pump stage (1, 2, 3, 4, 5 or 6) that is triggered at t=t, meaning that all its dead-band triggers have the status [On]. If not any pump stage is triggered, the pump becomes inactive (pump discharge is zero) and the actual pump stage is set to 0. If the pump is overruled by a controller, the actual pump stage is set to -1.

Controllers

A Time-controller, a Hydraulic-controller, a PID controller or an Interval controller can be assigned to an External Pump station. A controller is only active in case the pump is triggered in accordance with the Switch-on and Switch-off levels, defined for stage 1. An active controller overrules the pump capacity of the triggered pump stage, while the actual pump stage is set to -1. Please note that capacity reduction factors are applied to the pump capacities set by a controller. Advice: In using a controller at an External Pump station, define only one (1) pump stage and take care that its dead-band trigger is always [On].

Capacity reduction table

A capacity reduction table can be defined, which is applied to all pump stages as well as to pump capacities, that are set by a controller. In the capacity reduction table a capacity reduction factor can be given as function of the pump head (e.g. water level at delivery-side minus water level at suction-side). Pump heads (first column) in a capacity reduction table should be in increasing order. Capacity reduction factors should be equal or larger than 0. The pump discharge equals the pump capacity times the capacity reduction factor. If no capacity reduction table is defined in effect a capacity reduction factor equal to 1 is applied. Please note that the pump head at $t = t$ is used to determine the capacity reduction factor to be applied in the time-step from $t = t$ to $t = t + \Delta t$.

External Pump station output parameters

External Pump station output parameters can be viewed in Result in Charts (Pump.his file) by checking the Pump Data check-box on the 1DFLOW/Output options/Structures Tab in Settings. Available External pump station output parameters as function of time are:

- ◇ Suction-Side level: Water level at the suction-side of the pump.
- ◇ Delivery-Side level: Water level at the delivery-side of the pump.
- ◇ Pump Head: The water level at the delivery-side of the pump minus the water level at the suction-side of the pump.
- ◇ Actual Pump Stage: Equal to -1 if pump is overruled by a controller; equal to 0 if pump is inactive; or equal to 1, 2, 3, 4, 5 or 6 depending on the actual pump stage that is triggered (see Dead-band triggering algorithm).
- ◇ Pump Capacity: The pump capacity is either defined by its controller; equal to zero if pump is inactive; or equal to the pump capacity defined for the actual pump stage.
- ◇ Reduction Factor: The reduction factor follows from the pump head and the capacity reduction table.
- ◇ Pump Discharge: The pump discharge is equal to the pump capacity times the reduction factor.

Dead-band triggering algoritm

A distinction is to be made between evaluating the status [On/Off] of dead-band triggers defined at the suction-side of the External Pump station and dead-band triggers defined at the delivery-side of the External Pump station:

Dead-band triggering at the Suction-side of the External Pump station:

Parameters used in determining the status [On/Off] of a suction-side dead-band trigger are $h_{Suction}$ (water level at the suction-side), $h_{On,Suction}$ (switch-on-level) and $h_{Off,Suction}$ (switch-off-level), where $h_{On,Suction} > h_{Off,Suction}$. Two booleans are used (e.g. $S_{On,Suction}$ and $S_{Off,Suction}$):

- ◇ If $h_{Suction} > h_{On,Suction} \quad \Rightarrow \quad S_{On,Suction} = True;\ Else\ S_{On,Suction} = False$
- ◇ If $h_{Suction} < h_{Off,suction} \quad \Rightarrow \quad S_{Off,Suction} = True;\ Else\ S_{Off,Suction} = False$

If at the start of a computation yields that $h_{Off,Suction} \leq h_{Suction} \leq h_{On,Suction}$ the dead-band trigger is set [On]. During a computation the dead-band trigger is set [On] or [Off] according to the following rules:

- ◇ If $(S_{On,Suction} = True \quad and \quad S_{Off,Suction} = False) \quad \Rightarrow \quad Trigger = [On]$
- ◇ If $(S_{On,Suction} = False \quad and \quad S_{Off,Suction} = True) \quad \Rightarrow \quad Trigger = [Off]$
- ◇ If $(S_{On,Suction} = False \quad and \quad S_{Off,Suction} = False) \quad \Rightarrow \quad Trigger$ obtains the status determined in the previous computational time step.

Dead-band triggering at the Delivery-side of the External Pump station:

Parameters used in determining the status [On/Off] of a delivery-side dead-band trigger are $h_{Delivery}$ (water level at the delivery-side), $h_{On,Delivery}$ (switch-on-level), $h_{Off,Delivery}$ (switch-off-level), where $h_{Off,Delivery} > h_{On,Delivery}$. Two booleans are used (e.g. $S_{On,Delivery}$ and $S_{Off,delivery}$):

- If $h_{Delivery} < h_{On,Delivery} \Rightarrow S_{On,Delivery} = True;\ Else\ S_{On,Delivery} = False$
- If $h_{Delivery} > h_{Off,Delivery} \Rightarrow S_{Off,Delivery} = True;\ Else\ S_{Off,Delivery} = False$

If at the start of a computation yields that $h_{On,Delivery} \leq h_{Delivery} \leq h_{Off,Delivery}$ the dead-band trigger is set [On]. During a computation the dead-band trigger is set [On] or [Off] according to the following rules:

- If $(S_{On,Delivery} = True \quad and \quad S_{Off,Delivery} = False) \quad \Rightarrow \quad Trigger = [On]$
- If $(S_{On,Delivery} = False \quad and \quad S_{Off,Delivery} = True) \quad \Rightarrow \quad Trigger = [Off]$
- If $(S_{On,Delivery} = False \quad and \quad S_{Off,Delivery} = False) \quad \Rightarrow \quad Trigger$ obtains the status determined in the previous computational time step

Conventions for switch-on and switch-off levels

A distinction is to be made between switch-on-levels and switch-off-levels, defined at the suction-side and defined at the delivery-side of the External Pump station (see Table 1.2).

- Suction-side: The switch-on level should increase with increasing stage number. For each stage, its switch-off-level should be lower than its switch-on-level. The switch-off-level should increase with increasing stage number.
- Delivery-side: The switch-on-level should decrease with increasing stage number. For each stage, its switch-off-level should be higher than its switch-on-level. The switch-off-level should decrease with increasing stage number.

Table 1.2: *Example of switch-on-levels and switch-off-levels at the suction-side and the delivery-side of an external pump station*

Pump		**Suction-side**		**Delivery-side**	
Stage No	**Capacity** m^3/s	**Switch-on-level** m	**Switch-off-level** m	**Switch-on-level** m	**Switch-off-level** m
1	1.1	1.80	1.10	2.80	3.90
2	2.2	1.90	1.20	2.70	3.80
3	3.3	2.00	1.30	2.60	3.70
4	4.4	2.10	1.40	2.50	3.60
5	5.5	2.20	1.50	2.40	3.50
6	6.6	2.30	1.60	2.30	3.40

Storage options at an External Pump station

Water can be stored at the site, accommodating an external pump station. In SOBEK a distinction is made between water stored below and above a certain design level, hereafter referred to as street level.

- ◇ Below street level: At water levels below street level, water is considered to be stored in a so-called bottom-storage-reservoir.
- ◇ Above street level: For water levels above street level, there are three options:
  - ▫ Water is stored in a street-level-reservoir, having a bed level equal to street level and walls that cannot be overtopped;
  - ▫ Water cannot be stored (e.g. closed pump station), hence water levels inside the pump station may become pressurized; and
  - ▫ Water leaves the external pump station (e.g. loss of water) by flowing towards low lying areas, while the water level inside the pump station is considered to remain at street level.

1.1.16.11 River Pump

The River Pump is to be located on an open channel branch. Please note that:

- ◇ When activated, water is always pumped from the suction-side towards the delivery-side.
- ◇ The River – Pump can be placed in a compound structure.
- ◇ Controllers cannot be assigned to a River - Pump,
- ◇ Flow - ζ-calculation points are default located 0.5 m in front and 0.5 m behind a River - Pump.
- ◇ River pump output parameters becomes available by checking the Pump Data check-box on the 1DFLOW/Output options/Structures Tab in Settings.

Control direction determines location of trigger-water level (h_{trig}) of a River pump:

The control direction of the River pump defines the location of the so-called trigger-water level (h_{trig}):

- ◇ Upward control direction means that the trigger water level (h_{trig}) is located upward, e.g. at the pump-side with the lowest x-coordinate along the branch.
- ◇ Downward control direction means that the trigger water level (h_{trig}) is located downward, e.g. at the pump-side with the highest x-coordinate along the branch.

Start-level and stop-level determine location of the suction-side and the delivery-side:

The elevation of the start-level (h_{start}) and the elevation of the stop-level (h_{stop}) define the location of the suction-side and the delivery-side of a river pump as follows:

- ◇ If start-level is above stop-level: water is pumped away from the trigger-water level side of the pump (e.g. trigger-water level side coincides with the suction-side of the pump).
- ◇ If stop-level is above start-level: water is pumped towards the trigger-water level side of the pump (e.g. trigger-water level side coincides with the delivery-side of the pump).

Discharge direction of a River pump:

The discharge direction of a river pump with respect to the positive x-direction along the branch is defined by the control direction and the elevations of the start-level and the stop-level. Four different situations can be discerned:

- ◇ Upward control and start-level above stop-level (see Figure 1.31),
- ◇ Upward control and stop-level above start-level (see Figure 1.32),
- ◇ Downward control and start-level above stop-level (see Figure 1.33),
- ◇ Downward control and stop-level below start-level (see Figure 1.34),

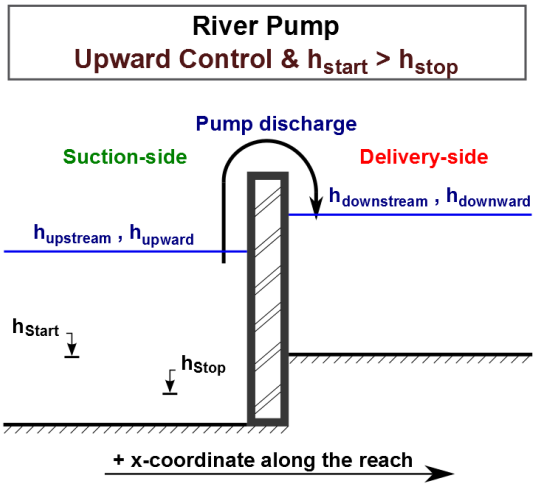

Figure 1.31: *River pump with Upward control and start-level above stop-level*

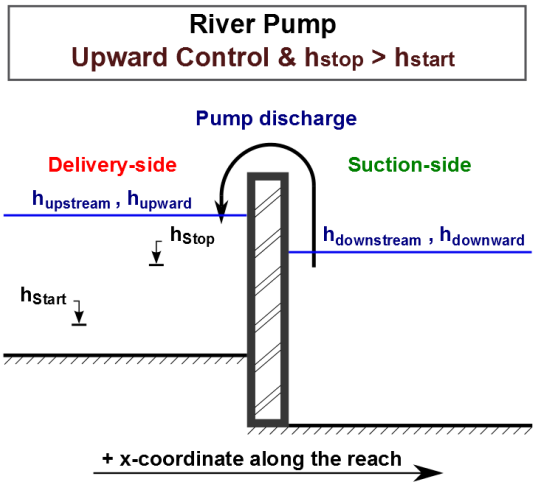

Figure 1.32: *River pump with Upward control and stop-level above start-level*

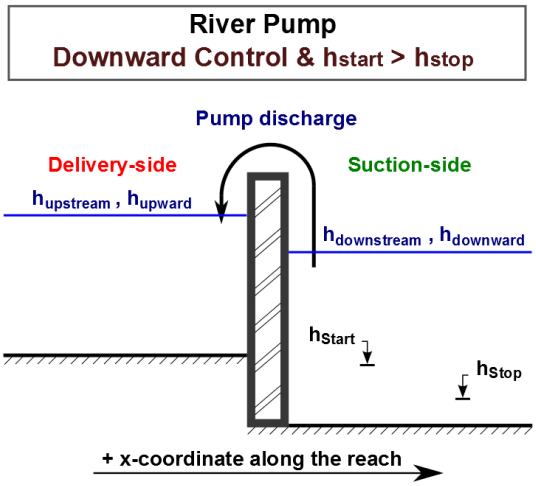

Figure 1.33: *River pump with Downward control and start-level above stop-level*

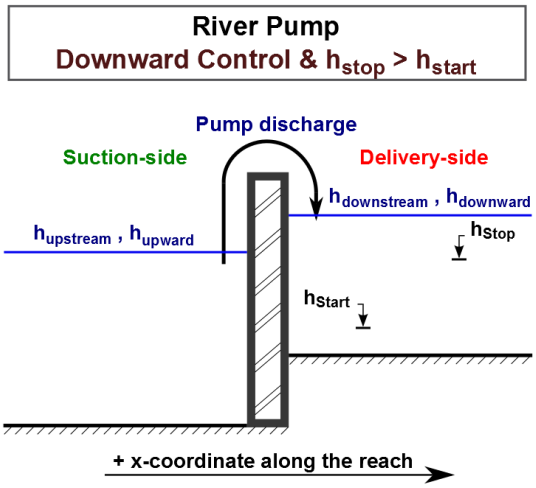

Figure 1.34: *River pump with Downward control and stop-level above start-level*

River pump stages

A River pump has only one pump stage, hence only one pump capacity can be defined.

Constant Reduction Factor or Reduction Factor F(Pump Head)

- Constant Reduction Factor: WARNING: This means that irrespective of the pump head, the actual pump capacity always equals the defined pump capacity times the constant reduction factor. Say you defined a pump capacity of 100 m^3/s and constant reduction factor of 0.7. This means that the pump will have a capacity of 70 (=0.7*100) m^3/s for both positive and negative pump heads.
- Reduction Factor F(Pump Head): With this option you define a capacity reduction factor as function of the pump head (e.g. water level at the delivery-side minus the water level at the suction-side). The capacity reduction factor may vary between 0 and 1. The first row in the $F(PumpHead)$ table should read 0,1 (pump head, capacity reduction factor). For positive pump-heads, the pump discharge equals the pump capacity times the capacity reduction factor. For negative pump-heads a capacity reduction factor equal to 1 is applied. Please note that the pump head at $t = t^n$ is used to determine the capacity reduction factor to be applied in the time-step from $t = t^n$ to $t = t^n + \Delta t$.

River pump output parameters

River pump output parameters can be viewed in Result in Charts (<Pump.his> file) by checking the Pump Data check-box on the 1DFLOW/Output options/Structures Tab in Settings. Available River pump output parameters as function of time are:

- Suction-Side level: Water level at the suction-side of the pump.
- Delivery-Side level: Water level at the delivery-side of the pump.
- Pump Head: The water level at the delivery-side of the pump minus the water level at the suction-side of the pump.
- Actual Pump Stage: Equal to 0 if pump is inactive; or equal to 1 if the pump is triggered (Please note that a river pump has one pump stage only).
- Pump Capacity: The pump capacity is either equal to zero if pump is inactive; or equal to its defined pump capacity.
- Reduction Factor: The reduction factor is either a constant or follows from the F(Pump Head) table
- Pump Discharge: The pump discharge is equal to the pump capacity times the reduction factor.

Dead-band triggering algorithm

A dead-band trigger [On/Off] determines if the River Pump should pump (e.g. actual pump stage = 1) or not pump (e.g. pump discharge is zero; actual pump stage = 0). The actual status [On/Off] of the dead-band trigger is determined as follows. Firstly, the value of two Booleans (S_{on} and S_{off}) is determined. In case of triggering at the suction-side of the pump, or in other words if the start-level (h_{Start}) is above the stop-level (h_{Stop}) yields (see Figure 1.31 and Figure 1.33):

- If $h_{trig} > h_{Start} \quad \Rightarrow S_{on} = True; \quad Else \quad S_{on} = False$
- If $h_{trig} < h_{Stop} \quad \Rightarrow S_{off} = True; \quad Else \quad S_{off} = False$

In case of triggering at the delivery-side of the pump, or in other words if the stop-level (hStop) is above the start-level (hStart) yields (see Figure 1.32 and Figure 1.34):

- If $h_{trig} < h_{Start} \quad \Rightarrow S_{on} = True; \quad Else \quad S_{on} = False$
- If $h_{trig} > h_{Stop} \quad \Rightarrow S_{off} = True; \quad Else \quad S_{off} = False$

If at the start of a computation yields that $h_{stop} \leq h_{trig} \leq h_{off}$, the dead-band trigger is set [On]. During a computation the dead-band trigger is set [On] or [Off] according to the following algorithm

- If $(S_{on} = True \ and \ S_{off} = False) \quad \Rightarrow \quad Trigger = [On]$
- If $(S_{on} = False \ and \ S_{off} = True) \quad \Rightarrow \quad Trigger = [Off]$
- If $(S_{on} = False \ and \ S_{off} = False) \quad \Rightarrow \quad Trigger$ obtains the status determined in the previous computational time step

Comparing a River Pump and a Pump station:

- From a pump direction point of view the Pump station covers the options available at a River Pump (see Table 1.3).
- Advantage of the River Pump over a Pump station is the fact that it can be placed in a compound structure
- Limitations of the River Pump with respect to the Pump station:
  - No controllers can be assigned,
  - Only one (1) pump stage can be defined,

- Only one (1) dead-band trigger can be defined at a River Pump, hence it is not possible to operate the pump on basis of water levels at both its suction-side and its delivery-side.

Table 1.3: *Flow-Pump station covers the options of a River pump with respect to the pump direction*

River pump		**Pump station**	
Control Direction	**Start-level** (h_{start}) **Stop-level** (h_{stop})	**Pump Direction**	**Triggering at**
Upward	$h_{start} > h_{stop}$	Positive	Suction-side
Upward	$h_{stop} > h_{start}$	Negative	Delivery-side
Downward	$h_{start} > h_{stop}$	Negative	Suction-side
Downward	$h_{stop} > h_{start}$	Positive	Delivery-side

1.1.16.12 River Weir

Please note that discharges are computed on basis of upstream and downstream energy levels. Further on please note that default a computational point is located 0.5 m in front and 0.5 m behind a River weir. A structure inertia damping factor can be defined for each individual River weir.

Two types of flow conditions can occur in the case of weir flow. These are free (modular) flow and drowned (submerged) flow. If high tail water conditions do affect the flow, the weir is said to be drowned. These conditions may then be computed by applying a reduction factor to the modular function, i.e. to the equation applied if the weir or flume is not drowned (Ackers *et al.*, 1978; Schmidt, 1955; Bos, 1989).

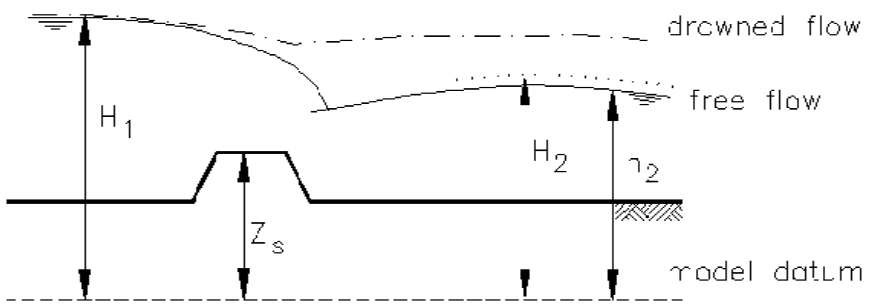

Figure 1.35: *Free and drowned weir flow*

The discharge through the weir is computed with:

$$Q = c_w f W_s \frac{2}{3}\sqrt{\frac{2}{3}g} \;\; (H_1 - z_s)^{3/2} \tag{1.112}$$

in which:

c_w	correction coefficient
W_s	width across flow section
H_1	upstream energy level

f_s	drowned flow reduction factor
z_s	crest or sill level of weir

The submergence factor of the weir is defined as:

$$S_f = \frac{H_2 - z_s}{H_1 - z_s} \tag{1.113}$$

in which:

S_f	submergence factor
H_2	downstream energy level

If the actual submergence factor exceeds the submergence limit, drowned flow occurs. In all other cases modular flow occurs and a drowned flow reduction factor equal to 1.0 should be applied. The submergence limit depends on the crest shape (see Table 1.4).

For a weir the user must specify:

- level of crest or crest height
- crest width across flow section
- crest shape (broad, triangular, round, or sharp)

Depending on the crest shape the following parameters are used (Table 1.4), which can be adjusted by the user:

- correction coefficient (default see Table 1.4)
- submergence limit (default see Table 1.4)
- drowned flow reduction curve (default see Table 1.4)

Table 1.4: *Crest shape and coefficients for simple weir structure (default values)*

Crest shape	**Correction coef.**	**Submergence limit**	**Reduction curve (Figure 1.36)**
broad	1.0	0.82	1
triangular	1.05	0.67	2
round	1.1	0.3	4
sharp	1.2	0.01	5

Notes:

- If only one flow direction has been specified, the coefficients for that flow direction are used as default coefficients for the other flow direction.
- The default drowned flow reduction curve depends on the submergence limit.

The drowned flow reduction curve is stored in tabulated format.

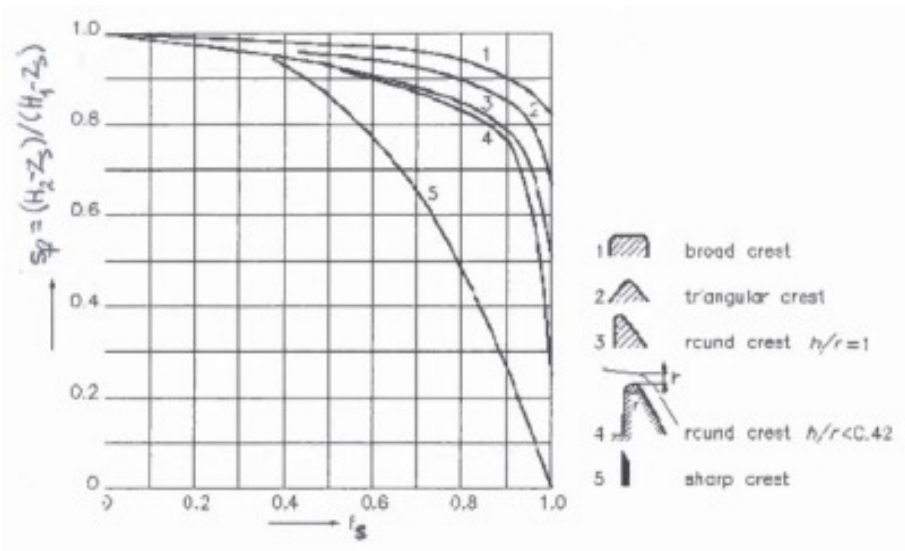

Figure 1.36: *Drowned flow reduction curves*

1.1.16.13 Siphon

A siphon (see Figure 1.37) can only transport water from its upstream defined side to its downstream defined side. ***Hence, only positive flow is possible.***

A switch-on-level and a switch-off-level at its downstream side determine if the siphon may or may not discharge water. If the downstream water level drops below the switch-on-level, the siphon start discharging water. If the downstream water level rises above the switch-off-level, the siphon stops discharging water. If the downstream water level is in between the switch-on-level and the switch-off-level (e.g. in between its dead-band), the siphon remains in its present state of operation. In case the downstream water level is in its dead-band at the onstart of a computation, the siphon will start discharging water.

Discharge can only flow through a siphon if its upstream cross-sectional area is fully filled with water, meaning that the upstream water level is above the obvert level (invert level + inner cross-sectional height) of the siphon. If the upstream cross-sectional area is not fully filled with water, air will be contained in het upper part of the siphon, this air will prevent the siphon from discharging water.

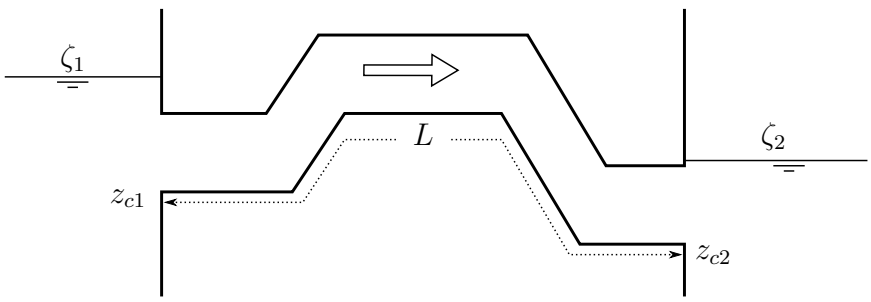

Figure 1.37: *Siphon*

Two flow conditions can occur:

Free flow when $\zeta_2 < z_{c2} + d_{c2}$

$$Q = \mu A_{fs} \sqrt{2g(\zeta_1 - (z_{c2} + d_{c2})} \tag{1.114}$$

Submerged flow when $\zeta_2 \geq z_{c2} + d_{c2}$:

$$Q = \mu A_{fs} \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.115}$$

Q	Discharge through siphon [m^3/s]
μ	Discharge coefficient, derived from loss-coefficients [$-$]
A_{fs}	Discharge siphon flow area ($= \min(A_{fcrs}, A_{fsgate})$) [$m^2$]
A_{fcrs}	Cross-sectional area of the siphon. At its highest point the siphon is considered to be completely filled with water [m^2]
A_{fsgate}	Flow area under the siphon gate [m^2]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
ζ_1	Upstream water level [m]
ζ_2	Downstream water level [m]
z_{c2}	Downstream siphon invert level [m]
d_{c2}	Critical siphon depth at the downstream side

$$(= \sqrt[3]{Q^2/(gT_2^2)}) \qquad [m] \tag{1.116}$$

T_2	Surface width in the siphon at its downstream side [m]

For numerical reasons the discharge coefficient (μ) is limited to a maximum of 1.0. The discharge coefficient (μ) is computed as follows:

$$\mu = \frac{1}{\sqrt{\xi_i + \xi_f + \xi_v + \xi_b + \xi_o}} \tag{1.117}$$

ξ_i	Entrance loss coefficient [$-$]
ξ_f	Friction loss coefficient [$-$]
ξ_v	Valve loss coefficient [$-$]
ξ_b	Bend loss coefficient [$-$]
ξ_o	Exit loss coefficient [$-$]

The entrance loss coefficient (ξ_i) can be defined as a constant value only.

The friction loss coefficient (ξ_f) is computed as follows:

$$\xi_f = \frac{2gL}{C^2 R} \tag{1.118}$$

L	Length of the siphon [m]
C	Chézy coefficient for fuly water filled siphon [m$^{1/2}$/s]
R	Hydraulic radius [m] If GHO $<$ MIISH; $R = R_{gate}$ If GHO $\geq$ MIISH; R$= R_{siphon}$
GHO	Gate height opening [m]
MIISH	Maximum inner inverted siphon height [m]
R_{siphon}	Hydraulic radius based on a fully water filled siphon [m]
R_{gate}	Hydraulic radius based on actual gate height opening [m]

The valve loss coefficient (ξ_v) can be defined as a constant value or as a function of the ratio of the “Gate height opening” and the “maximum inner siphon height”.

Note:

- In case the valve loss coefficient (ξ_v) is not a constant, in computations the actual valve loss coefficient (ξ_v) is derived from the user defined table, while using the ratio of the “actual gate height opening and the “actual maximum inner siphon height”.
- In case the ground layer thickness is greater than zero, both the “actual gate height opening” and the “actual maximum inner siphon height” will differ from the values as defined in the user interface (see the paragraph "Siphon cross-sections, bed friction and ground layer" below)

The exit loss coefficient (ξ_o) is computed as follows:

Submerged flow ($\zeta_2 = z_{c2} + d_{c2}$):

$$\xi_o = k \left(1 - \frac{A_{fs}}{A_{fr2}}\right)^2 \tag{1.119}$$

k	User defined constant exit loss coefficient [$-$]
A_{fr2}	Flow area in the branch, adjacent to the downstream siphon side [m^2]
A_{fis}	Siphon flow area [m^2]
	If GHO $\leq$ MISH; $A_{fs} = A_{fsgate}$
	If GHO $>$ MISH; $A_{fs} = A_{siphon}$
GHO	Gate height opening [m]
MISH	Maximum inner siphon height [m]
A_{fsgate}	Flow area under the siphon gate [m^2]
A_{siphon}	Flow area based on a fully water filled siphon [m^2]

Free flow ($\zeta_2 < z_{c2} + d_{c2}$):

$$\xi_o = 0 \tag{1.120}$$

Siphon cross-sections, bed friction and ground layer:

For a siphon all available closed cross-section types can be used. In a siphon, a ground layer with constant thickness can be defined. Siphon friction and ground layer friction can be specified, using any of the available bed friction formulations.

Defining a ground layer thickness $>$ 0 implies that in siphon computations:

- Defined invert levels are raised with the ground layer thickness,
- Gate height openings are reduced with the ground layer thickness,
- Maximum inner height of the siphon is reduced with the ground layer thickness,
- Cross-sectional parameters (such as: flow areas, hydraulic radius and so on) are computed based on a cross-sectional profile, that is reduced by the ground layer thickness.

Siphon, Good modelling practice aspects

It is advised to avoid that the bed level of a cross-section in front of a Culvert, Inverted Siphon or Siphon is above the ground-layer level (= invert level + ground-layer thickness), since such

situation can result in very small computational time-steps (e.g. long required wall-clock times) or even in a termination of the simulation.

This is explained as follows. Consider the situation depicted in Figure 1.38 were the bed level in front of a Culvert (Inverted Siphon or Siphon) is 0.60 m above the ground-layer level. This means that at small upstream water depths, water will be sucked into the culvert (Inverted Siphon or Siphon), resulting in large flow velocities. For the computational time-step (Δt) yields that $\Delta t \leq \Delta x/U$, where U is the local flow velocity and Δx is distance between two water level computational points (or ζ-points). At very low discharges even negative water depths may be computed, leading to a termination of the simulation.

The situation explained above can be avoided by making the bed level in front of the Culvert (Inverted Siphon or Siphon) equal to the ground-layer level. In other words by defining a bed level slope from ζ-point ζ_2 to ζ_3 as depicted in Figure 1.38. Providing for the parameter "Maximum Lowering of Cross-section Bed Level at Culvert" a value greater or equal to 0.60 m means that before the computation starts, the bed level at ζ-point ζ_3 is set equal to the ground-layer level. In **??** it is shown how to provide a value for parameter "Maximum Lowering of Cross-section Bed Level at Culvert".

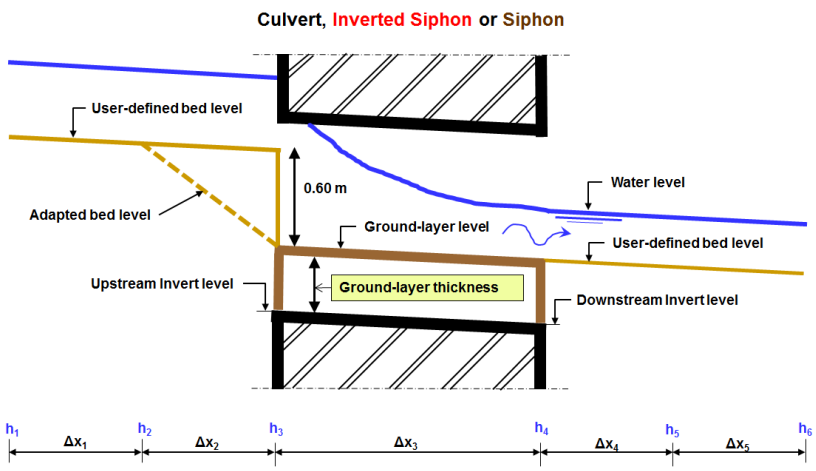

Figure 1.38: *Good modelling practice, Culvert, Inverted Siphon and Siphon*

1.1.16.14 Universal Weir

Crest levels of a universal weir (see Figure 1.39) can be defined as a Y-Z profile. The crest level profile of a universal weir is divided into *rectangular weir sections* having a horizontal bed and *triangular weir sections* having a sloping bed. It is assumed that the discharge over an universal weir is the summation of the discharges over each individual weir section. Rectangular weir sections are considered as a broad-crested weir. Triangular weir sections are considered as (the half of) a broad-crested weir with truncated triangular control section (Bos, 1989).

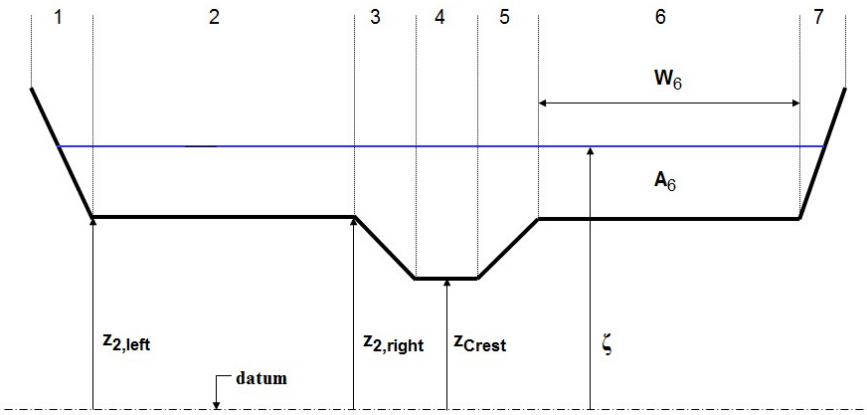

Figure 1.39: *Crest level (Y-Z) profile of a Universal weir, divided into three rectangular weir sections (2, 4 and 6) and four triangular weir sections (1, 3, 5 and 7)*

Parameters depicted in Figure 1.39 are:

A_i	Flow area of section i [m^2]
W_i	Width of weir section i [m]
$z_{i,left}$	Elevation at the left side of weir section i [mAD]
$z_{i,right}$	Elevation at the right side of weir section i [mAD]
z_{Crest}	Crest level of the Universal weir (output parameter only), equal to the lowest elevation of its crest level profile [mAD]
ζ	Water level [mAD]

Following equations yield for a universal weir:

$$Q = \sum_i^N (u_i A_i) = \sum_i^N Q_i \tag{1.121}$$

$$A = \sum_i^N A_i \tag{1.122}$$

$$U_{structure} = Q/A \tag{1.123}$$

where:

A	Flow area of the universal weir [m^2]
A_i	Flow area of weir section i [m^2]
N	Number of weir sections i [$-$]
Q	Discharge over the universal weir [m^3/s]
Q_i	Discharge over weir section i [m^3/s]
u_i	Flow velocity in weir section i [m/s]
$U_{structure}$	Average flow velocity over the universal weir [m/s]

Following equations yield for a rectangular weir section i:

$$z_{i,crest} = min(z_{i,left}\ ,\ z_{i,right}) \tag{1.124}$$

$$ml_{rectangular,i} = 2/3 \tag{1.125}$$

◇ Free rectangular weir flow if $\left(\frac{\zeta_2 - z_{i,crest}}{\zeta_1 - z_{i,crest}}\right) \leq ml_{rectangular,i}$

$$u_i = c_e \sqrt{\frac{2}{3} g} \sqrt{\zeta_1 - z_{i,crest}} \tag{1.126}$$

$$A_i = \frac{2}{3} W_i (\zeta_1 - z_{i,crest}) \tag{1.127}$$

$$Q_i = u_i \, A_i = c_e \, \frac{2}{3} \sqrt{\frac{2}{3} g} \quad W_i \, (\zeta_1 - z_{i,crest})^{3/2} \tag{1.128}$$

◇ Submerged rectangular weir flow if $\left(\frac{\zeta_2 - z_{i,crest}}{\zeta_1 - z_{i,crest}}\right) > ml_{rectangular,i}$

$$u_i = c_e \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.129}$$

$$A_i = W_i (\zeta_2 - z_{i,crest}) \tag{1.130}$$

$$Q_i = u_i \, A_i = c_e \, W_i \, (\zeta_2 - z_{i,crest}) \, \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.131}$$

Following equations yield for a triangular weir section i:

$$z_{i,crest} = min(z_{i,left} \, , \, z_{i,right}) \tag{1.132}$$

$$dz_i = |z_{i,left} - z_{i,right}| \tag{1.133}$$

$$ml_{triangular,i} = \begin{cases} \frac{4}{5} & \text{if } (\zeta_1 - z_{i,crest})) \leq 1.25 \, dz_i \\ \frac{2}{3} + \frac{1}{6} \frac{dz_i}{(\zeta_1 - z_{i,crest})} & \text{if } (\zeta_1 - z_{i,crest})) > 1.25 \, dz_i \end{cases} \tag{1.134}$$

◇ Free triangular weir flow if $\left(\frac{\zeta_2 - z_{i,crest}}{\zeta_1 - z_{i,crest}}\right) \leq ml_{triangular,i}$

$$u_i = c_e \sqrt{2g(1 - ml_{triangular,i})(\zeta_1 - z_{i,crest})} \tag{1.135}$$

$$A_i = \begin{cases} W_i \left(\frac{(ml_{triangular,i} \, (\zeta_1 - z_{i,crest}))^2}{2 \, dz_i} \right) & \text{if } \frac{\zeta_1 - z_{i,crest}}{1/ml_{triangular,i}} \leq dz_i \\ W_i \left(\frac{\zeta_1 - z_{i,crest}}{1/ml_{triangular,i}} - \frac{dz_i}{2} \right) & \text{if } \frac{\zeta_1 - z_{i,crest}}{1/ml_{triangular,i}} > dz_i \end{cases} \tag{1.136}$$

$$Q_i = u_i \, A_i \tag{1.137}$$

◇ Submerged triangular weir flow if $\left(\frac{\zeta_2 - z_{i,crest}}{\zeta_1 - z_{i,crest}}\right) > ml_{triangular,i}$

$$u_i = c_e \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.138}$$

$$A_i = \begin{cases} W_i \left(\frac{(\zeta_2 - z_{i,crest})^2}{2 \, dz_i} \right) & \text{if } \zeta_2 - z_{i,crest} \leq dz_i \\ W_i \, (\zeta_2 - z_{i,crest} - \frac{dz_i}{2}) & \text{if } \zeta_2 - z_{i,crest} > dz_i \end{cases} \tag{1.139}$$

$$Q_i = u_i \, A_i \tag{1.140}$$

where:

A_i	Flow area of weir section i [m^2]
c_e	Discharge coefficient, applicable to all rectangular weir sections as well as to all triangular weir sections of a universal weir [$-$]
dz_i	Vertical distance between the elevation at the left side and the right side of a triangular weir section i [m]
g	Acceleration due to gravity [m/s^2] (≈ 9.814)
$ml_{rectangular,i}$	*Water-level-based* modular limit of a rectangular weir section i. Its value is always equal to 2/3 [$-$]
$ml_{triangular,i}$	*Water-level-based* modular limit of a triangular weir section i. Its value depends on the actual water depth. [$-$]
Q_i	Discharge over weir section i [m^3/s]
u_i	Velocity in weir section i [m/s]
W_i	Width of weir section i [m]
$z_{i,crest}$	Crest level of weir section i [mAD]
$z_{i,left}$	Elevation at the left side of weir section i [mAD]
$z_{i,right}$	Elevation at the right side of weir section i [mAD]
ζ_1	Water level at the upstream side of the universal weir [mAD]
ζ_2	Water level at the downstream side of the universal weir [mAD]

1.1.16.15 Vertical obstacle friction

The vertical obstacle friction terms have been introduced to account for the added resistance that is caused by vertical obstacles, like houses or trees. The vertical obstacle friction coefficient is based on the number of obstacles, their diameter and drag coefficient. This spatial coefficient can be prescribed in a ARC-INFO grid file, like the bathymetry.

The vertical obstacle friction coefficient a is specified as the summation of the product of obstacle width and drag coefficient over the number of obstacles per unit area.

$$a = \frac{1}{A} \sum_{i=1}^{N} D_i C_{d_i} \tag{1.141}$$

where:

a	Vertical obstacle friction coefficient [$1/m$]
N	Number of obstacles in unit area [$-$]
A	Unit area [m^2]
D_i	Diameter of obstacle i [m]
C_{d_i}	Drag coefficient of obstacle i [$-$]

For round pillars, Cd is usually 1.0. In this formulation it is assumed that the obstacle height will always be larger than the waterdepth. In case the obstacles do become submerged, a more advanced option is the Depth-Dependent Vegetation Roughness.

1.1.16.16 Weir

Three types of flow conditions can occur in the case of weir flow. These are free (modular) flow, submerged flow and no flow (water levels below crest level). If high tail water conditions do affect the flow, the weir is said to be submerged.

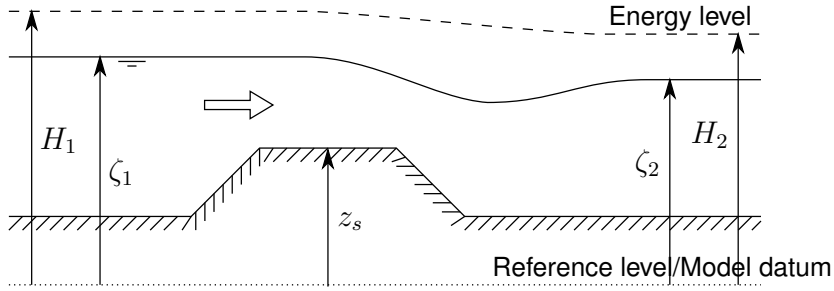

Figure 1.40: *Weir*

The discharge and wetted area through the weir is computed with the following formulas:

Free weir flow:

$$u_s = c_e \sqrt{\frac{2}{3} g} \sqrt{\zeta_1 - z_s} \tag{1.142}$$

$$A_f = c_w W_s \frac{2}{3} (\zeta_1 - z_s) \tag{1.143}$$

$$Q = u_s A_f = c_e c_w W_s \frac{2}{3} \sqrt{\frac{2}{3} g} (\zeta_1 - z_s)^{3/2} \tag{1.144}$$

Submerged weir flow:

$$u_s = c_e \sqrt{2g(\zeta_1 - \zeta_2)} \tag{1.145}$$

$$A_f = c_w W_s (\zeta_2 - z_s) \tag{1.146}$$

$$Q = u_s A_f = c_e c_w W_s \sqrt{2g(\zeta_1 - \zeta_2)} (\zeta_2 - z_s) \tag{1.147}$$

Q	Discharge across weir [m^3/s]
A_f	Wetted area [m^2]
c_e	Discharge coefficient [$-$]
c_w	Lateral contraction coefficient [$-$]
W_s	Crest width [m]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
ζ_1	Upstream water level [m]
ζ_2	Downstream water level [m]
z_s	Crest level [m]
u_s	Velocity over crest [m/s]

The different formulas are applied when the following conditions occur

Free weir flow:

$$\zeta_1 - z_s > \frac{3}{2} (\zeta_2 - z_s) \tag{1.148}$$

Submerged weir flow:

$$\zeta_1 - z_s \leq \frac{3}{2}(\zeta_2 - z_s) \tag{1.149}$$

Note:
Broad weirs can cause oscillations, because the discharge calculated with the broad crest move a lot of water at a time step. This large discharge can lower the upstream water level significantly resulting in a discharge with a reverse flow direction at the next time step. The following (rather conservative) rule-of-thumb can be used to avoid oscillations:

$$A_s = \frac{3}{2} W_s \tag{1.150}$$

A_s	Storage area upstream of the structure [m^2]
W_s	Crest width [m]

1.1.17 Staggered grid; ζ- and u-calculation points at different locations

A SOBEK-Flow-model consists of a network of branches connected to each other at connection nodes. The SOBEK-Flow-model uses a staggered grid. On a staggered grid the water level points (e.g. ζ-calculation points) and velocity points (e.g. u-calculation points) are at different locations (see Figure 1.41). Water levels are computed at the ζ-calculation points, while velocities are computed half way between the ζ-calculation points, on the so called u-calculation points. Please note that Connection Nodes and Boundary Nodes are ζ-calculation points.

In each branch a number of ζ-calculation points can be defined. These ζ-calculation points represent the spatial numerical grid used in the simulation. The momentum equation is solved between ζ-calculation points, while the continuity equation is solved between the u-calculation points.

The location of each ζ-calculation point should be selected on various criteria:

- the distance between two neighbouring ζ-calculation points should not be too large (for accuracy and proper representation of the physical processes)
- the distance between two neighbouring ζ-calculation points should not be too small because of increasing simulation time. The default minimum distance SOBEK uses during a simulation is 1 meter.
- the location of the ζ-calculation points may be non-equidistant.

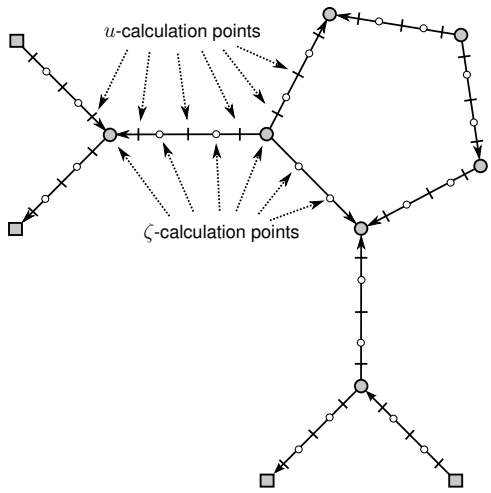

Figure 1.41: *Staggered grid, ζ- and u-calculation points at different locations*

1.1.18 Construction of the numerical bathymetry on basis of user-defined cross-sections

Cross-sections (e.g. bathymetrical data) may be defined at arbitrary locations in a model, provided that for each branch at least one cross-section is specified. An overview of the available type of cross-sections is given in **??**.

This section explains how the numerical bathymetry of a SOBEK model is constructed on basis of the user-defined cross-sections. With numerical bathymetry is referred to the bathymetrical information, that depending on the cross-section type is available at ζ-calculation points only or available at both ζ- and u-calcuation points. Water levels and discharges are computed using this numerical bathymetrical information. For more information on ζ- and u-calculation points, reference is made to section 1.1.17.

In constructing the numerical bathymetry a distinction is to be made between"'

- ◇ The Y-Z type of profiles (e.g. Open Vertical Segmented Y-Z profile, Asymmetrical Trapezium profile, Open Lumped Y-Z profile and Closed Lumped Y-Z profile).
- ◇ All cross-section types except for the Y-Z type of profiles.

1.1.18.1 The Y-Z type of profiles

With the Y-Z type of profiles is referred to the Open Vertical Segmented Y-Z profile, the Asymmetrical Trapezium profile, the Open Lumped Y-Z profile and the Closed Lumped Y-Z profile (for more information, see **??**).

For the Y-Z type of profiles yields that numerical bathymetrical information is made available at both ζ- and u-calculation points by interpolating (see section 1.1.19) between the user-defined cross-sectional profiles. The construction of the numerical bathymetrical information for Y-Z type of profiles is depicted in Figure 1.42.

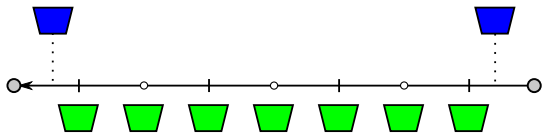

Figure 1.42: *Construction of the numerical bathymetrical information for Y-Z type of profiles; Blue coloured objects are the user-defined cross-sections; Green coloured objects are the interpolated numerical bathymetrical information; Open circle are ζ-calculation points; Small vertical lines are u-calculation points*

Note: If only one cross-section is specified for a branch, the bathymetrical information of each ζ-calculation point and each u-calculation point located on this branch will be identical to this specified cross-section.

1.1.18.2 All cross-section types except for the Y-Z type of profiles

With the Y-Z type of profiles is referred to the Open Vertical Segmented Y-Z profile, the Asymmetrical Trapezium profile, the Open Lumped Y-Z profile and the Closed Lumped Y-Z profile (for more information, see **??**).

For all cross-section types except for the Y-Z type of profiles yields that numerical bathymetrical information is made available at ζ-calculation points only by interpolating (see section 1.1.19) between user-defined cross-sectional profiles. The construction of the numerical bathymetrical information for all cross-section types except the Y-Z type of profiles is depicted in Figure 1.43.

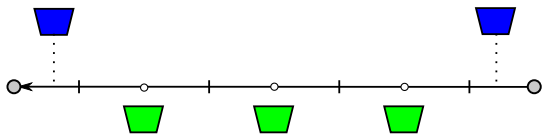

Figure 1.43: *Construction of the numerical bathymetrical information for all cross-section types except for the Y-Z type of profiles; Blue coloured objects are the user-defined cross-sections; Green coloured objects are the interpolated numerical bathymetrical information; Open circle are ζ-calculation points; Small vertical lines are u-calculation points*

Note: If only one cross-section is specified for a branch, the numerical bathymetrical information for each ζ-calculation point located on this branch will be identical to this specified cross-section.

1.1.19 Method of interpolating between user-defined cross-sections

This section explains the method of interpolating between user-defined cross-sections. An overview of the available type of cross-sections is given in **??**. Following distinction between cross-sections types is to be made:

- Round and Egg-shape (see section 1.1.19.1).
- Open Vertical Segmented Y-Z profile and Asymmetrical Trapezium profile (see section 1.1.19.2).
- Cross-sections not being a Round, Egg-shape, Open Vertical Segmented Y-Z profile or Asymmetrical Trapezium profile (see section 1.1.19.3).

1.1.19.1 Method of Interpolating between Round cross-sections and between Egg-shape cross-sections

A Round is specified by an invert level, a diameter and the elevation of a ground-layer surface. Interpolating between two Round cross-sections refers to the linear interpolation along the branch of invert level, diameter and the elevation of the ground-layer surface.

An Egg-shape is specified by an invert level, a height, a width and the elevation of a ground-layer surface. Interpolating between two Egg-shape cross-sections refers to the linear interpolation along the branch of invert level, height, width and the elevation of the ground-layer surface.

1.1.19.2 Method of Interpolating between Open Vertical Segmented Y-Z profiles and between Asymmetrical Trapezium profiles

For both an Open Vertical Segmented Y-Z profile and an Asymmetrical Trapezium profile yields that first from the user-defined cross-sectional information so-called conveyance tables are constructed. In a conveyance table the width, area and conveyance as function of water level are given. Interpolation between two of these type of profile refers to the linear interpolation along the branch of their respective conveyance tables.

1.1.19.3 Method of Interpolating between Cross-sections not being a Round, Egg-shape, Open Vertical Segmented Y-Z profile or Asymmetrical Trapezium profile

For all cross-sections not being a Round, Egg-shape, Open Vertical Segmented Y-Z profile or Asymmetrical Trapezium profile yields that the cross-sectional profile is defined in a Table.

Before interpolation starts it is taken care of that the concerning two cross-sections contain exactly the same vertical spacing for flow and total widths. In this context vertical spacing means the level at which a certain width in one of the two cross-sections is defined minus the bed level (lowest point) of the concerning cross-section. Afterwards, for each vertical spacing the flow width and total width of an intermediate constructed cross-section is determined by linear interpolation along the branch. The ground-layer surface of an intermediate constructed cross-section is determined by linear interpolation along the branch as well.

Note: In case of a River profile, the properties of a summer dike (e.g. Flow area, Total area, Dike crest level and Flood plain base level) are linear interpolated along the branch. The same applies for the width of the main section, the width of floodplain 1 and the width of flood plain 2.

Note: The height of a cross-section is the maximum level at which a width is defined minus the bed level (lowest point) of this cross-section. For open cross-sectional profiles yields that in case two cross-sections have different heights, the height of the lowest cross-section will be vertically extended with a width equal to the width defined at its maximum level. This is done to obtain in the lowest cross-section exactly the same vertical spacing as in the highest cross-section. Please note that the same procedure is applied for closed cross-sections, with the exception that the width of the lowest cross-section above its obvert level (maximum defined level) becomes equal to zero.

1.1.20 Methods for computing conveyance

Conveyance is computed for each u-calculation point (or velocity calculation point). A u-calculation point is located in the centre point of its two adjacent ζ-calculation points (or water level calculation points). A distinction is to be made between:

- ◇ The lumped conveyance approach (see section 1.1.20.1),
- ◇ Analytical formulae for computing lumped conveyance, and
- ◇ The vertically segmented conveyance approach (see section 1.1.20.2).

The vertically segmented conveyance approach is used for Y-Z profiles and Asymmetrical Trapezium cross-sections only. Analytical formulae for computing lumped conveyance are used for the Egg-shaped cross-section and the Round cross-section. For the remaining cross-section types, the lumped conveyance approach is used.

1.1.20.1 Lumped conveyance approach

In the lumped conveyance approach, it is assumed that there is an uniform flow velocity ($\overline{U}$) in the cross-sectional profile for a given water level (see Figure 1.44). Hence, it is assumed that differences in flow velocities across the cross-sectional profile can be neglected. This assumption generally does not yield for rivers. Therefore, in modelling rivers Y-Z profiles (or Asymmetrical Trapeziums) are preferred, where conveyance is computed using the vertically segmented conveyance approach (see section 1.1.20.2).

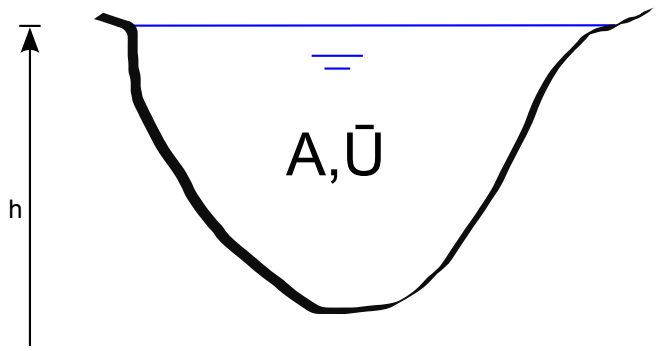

Figure 1.44: *Concept of the lumped conveyance approach*

First an asymmetrical cross-sectional profile as depicted in Figure 1.44 is transferred into a symmetrical cross-sectional profile. Thereafter the lumped conveyance is computed as:

$$K(h) = A(h)C(h)\sqrt{R(h)} \tag{1.151}$$

where:

$K(h)$	Lumped conveyance at water level h
$A(h)$	Cross-sectional area at water level h
$C(h)$	Chézy friction value at water level h
$P(h)$	Wetted Perimeter at water level h
$R(h)$	Hydraulic radius at water level h ($R(h) = A(h)/P(h)$)

If the water level in an open tabulated rises above the highest level in a cross-sectional profile, the lumped conveyance is computed taking the flow width at this water level equal to the flow width, defined at the highest cross-sectional level.

Note: Please note that in case of a River profile, the lumped conveyance of its main section, floodplain 1 and floodplain 2 are computed separately, and thereafter added to obtain the total lumped conveyance of a river profile at a particular water level.

1.1.20.2 Vertically segmented conveyance approach

The vertically segmented conveyance approach is suited for modelling flow in rivers, since in this approach it is assumed that flow velocities vary across the cross-sectional profile for a given water level. In line with this assumption, the cross-section is divided in vertical segments (see Figure 1.45).

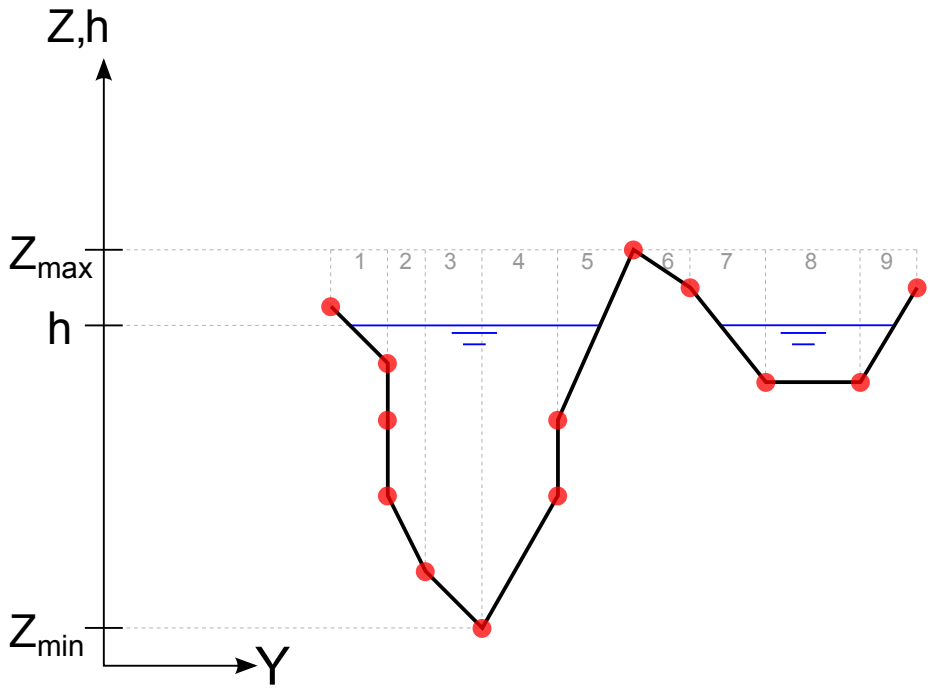

Figure 1.45: *Example of constructed vertical segments in a Y-Z profile*
13 (y_i, z_i) points red dots
9 vertical segments

Note: Conveyance table constructed from $Z_{\min}$ to $Z_{\max}$.

A Y-Z profile is defined by a number (n) of (y_i, z_i) points. For the total number $(nseg)$ of vertical segments yields: $n_{seg} = n - a - 1$, where a = number of succeeding (y_i, z_i) points for which $y_i = y_{i+1}$. So, *please note* that vertical walls in a Y-Z profile do not induce a vertical segment. *Moreover*, the length of vertical walls is not considered in computing the conveyance of a Y-Z profile. Omitting vertical walls in computing the conveyance of a Y-Z profile, means that for a canal having a rectangular profile with a small ratio of width and water depth only, substantial difference in discharge capacity can occur if such canal is modelled as a rectangular cross-section (i.e. lumped conveyance approach) or as a rectangle shaped Y-Z profile (i.e. vertically segmented conveyance approach). The surface level of a Y-Z profile is equal to its highest (y_i, z_i) point. In analogy, the same yields for an Asymmetrical Trapezium cross-section.

The conveyance (K_i) of each vertical segment is computed separately (for formulae, see

conveyance per vertical segment further below). The conveyance of each inclined vertical segment is determined by solving an integral from y_i to y_{i+1}. The advantage of using an integral is that irrespective of the number of user-defined intermediate points, that lie on a straight line between (y_0, z_0) to (y_1, z_1), the same conveyance for the cross-sectional part from y_0 to y_1 is computed. The total conveyance (K_{tot}) at a particular water level is the summation of the conveyance of all the vertical segments ($K_{tot} = \sum K_i$).

Conveyance tables, applied in the vertically segmented conveyance approach

A conveyance table gives the conveyance, flow width and flow area of a cross-section as function of water level. Conveyance tables are used for Y-Z profiles and Asymmetrical Trapeziums only. These conveyance tables are made by a pre-processor before the onstart of a hydrodynamic calculation. In constructing the conveyance table of each user-defined cross-section, it is ensured that the difference in conveyance between two succeeding water levels is less than 1 %. Conveyance tables are used for reducing computational efforts, hence increasing SOBEK's computational performance.

Conveyance tables are made from the bed-level up to the highest (y_i, z_i) point, irrespective of the actual location of the highest (y_i, z_i) point within the cross-sectional profile. This implies that no distinction is made between the left and the right embankment height.

Conveyance tables for computational points located in between two user-defined cross-sections are obtained by linear interpolating the conveyance tables, constructed at these two user-defined cross-sections (see also cross-section).

Extrapolation of conveyance tables

Say the water level (h) rises above the highest level ($n_{\max}$) of a conveyance table. In this case the conveyance (K_h) at water level (h) is computed as

$$K_h = a(h - z_{\min})^b, \tag{1.152}$$

where: $z_{\min}$ is the lowest level in the cross-sectional profile; and a, b are constants derived from the conveyances at levels $n_{\max}$ and $n_{\max} - 1$.

Lumped hydraulic output for Y-Z profiles and Asymmetical Trapezium

As a courtesy at subsection branch output, lumped hydraulic output (Q, C_{lumped}, A_{flow}, W_{flow} and R) are provided for Y-Z profiles and Asymmetrical Trapeziums. The discharge (Q) is the discharge through the profile, having a flow area (A_{flow}) and flow width (W_{flow}). The hydraulic radius (R) is computed on basis of A_{flow} and W_{flow} assuming a symmetrical profile as function of water level. The latter assumption is made, since different Y-Z profiles may have different shapes, that might have to be interpolated. The lumped Chézy value (C_{lumped}) is computed as

$$C_{lumped} = \frac{K}{A_{flow}\sqrt{R}}, \tag{1.153}$$

where K is the conveyance.

Conveyance per vertical segment

A vertical segment is defined by two (y_0, z_0) and (y_1, z_1) points for which yields that y_0 is not equal to y_1 (see Figure 1.46).

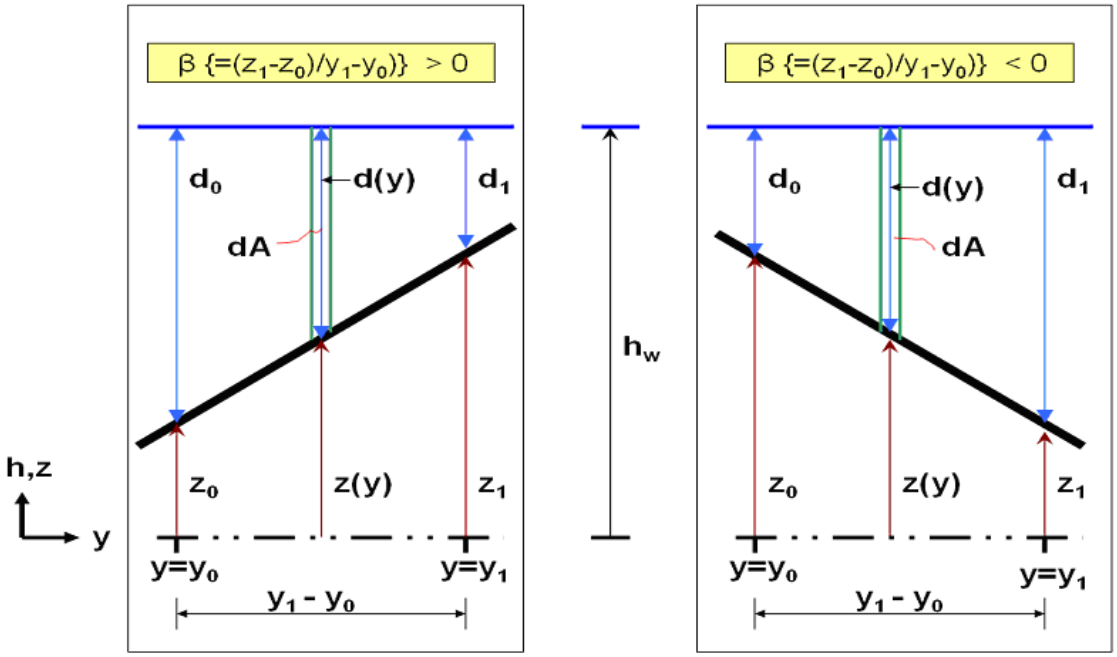

Figure 1.46: *Definition sketch of a vertical segment, considered in computing conveyance for a Y-Z profile and an Asymmetrical Trapezium cross-section*

Hereunder, the conveyance formulae for a vertical segment are given in case of:

- ◇ Chézy roughness,
- ◇ Manning roughness,
- ◇ Strickler (k_n) roughness,
- ◇ Strickler (k_s) roughness,
- ◇ White-Colebrook (k_n) roughness, and
- ◇ Bos&Bijkerk (γ) roughness.

Chézy roughness conveyance for a vertical segment (see Figure 1.46):

$d = \max(d_0, d_1)$
if $d \leq 0$ then
    $k = 0$
else
    if $d < |z_1 - z_0|$ then
        if $|\beta| \leq 0.01$ then
            $k_1 = \frac{2C}{|\beta|(1+\beta^2)^{1/4}} \left(\frac{d}{2}\right)^{5/2}$
        elseif $|\beta| > 0.01$ then
            $k_2 = \frac{2C}{5|\beta|(1+\beta^2)^{1/4}} d^{5/2}$
    elseif $d \geq |z_1 - z_0|$ then
        if $|\beta| \leq 0.01$ then
            $k_3 = \frac{C}{(1+\beta^2)^{1/4}} \left(\frac{d_0+d_1}{2}\right)^{3/2} (y_1 - y_0)$
        elseif $|\beta| > 0.01$ then
            $k_4 = \frac{2C}{5|\beta|(1+\beta^2)^{1/4}} \left| d_0^{5/2} - d_1^{5/2} \right|$
        endif
    endif

endif

Manning roughness conveyance for a vertical segment (see Figure 1.46):

$d = \max(d_0, d_1)$
if $d \leq 0$ then
  $k = 0$
else
  if $d < |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      $k_1 = \frac{2}{n|\beta|(1+\beta^2)^{1/4}} \left(\frac{d}{2}\right)^{8/3}$
    elseif $|\beta| > 0.01$ then
      $k_2 = \frac{3}{8n|\beta|(1+\beta^2)^{1/4}} d^{8/3}$
  elseif $d \geq |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      $k_3 = \frac{1}{n(1+\beta^2)^{1/4}} \left(\frac{d_0+d_1}{2}\right)^{5/3} (y_1 - y_0)$
    elseif $|\beta| > 0.01$ then
      $k_4 = \frac{3}{8n|\beta|(1+\beta^2)^{1/4}} \left|d_0^{8/3} - d_1^{8/3}\right|$
    endif
  endif
endif

Strickler (k_n) roughness conveyance for a vertical segment (see Figure 1.46):

$d = \max(d_0, d_1)$
if $d \leq 0$ then
  $k = 0$
else
  if $d < |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      $k_1 = \frac{50k_n^{-1/6}}{|\beta|(1+\beta^2)^{1/4}} \left(\frac{d}{2}\right)^{8/3}$
    elseif $|\beta| > 0.01$ then
      $k_2 = \frac{75k_n^{-1/6}}{8|\beta|(1+\beta^2)^{1/4}} d^{8/3}$
  elseif $d \geq |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      $k_3 = \frac{25k_n^{-1/6}}{(1+\beta^2)^{1/4}} \left(\frac{d_0+d_1}{2}\right)^{5/3} (y_1 - y_0)$
    elseif $|\beta| > 0.01$ then
      $k_4 = \frac{75k_n^{-1/6}}{8|\beta|(1+\beta^2)^{1/4}} \left|d_0^{8/3} - d_1^{8/3}\right|$
    endif
  endif
endif

Strickler (k_s) roughness conveyance for a vertical segment (see Figure 1.46):

$d = \max(d_0, d_1)$
if $d \leq 0$ then
  $k = 0$

else
  if $d < |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      $k_1 = \frac{2k_s}{|\beta|(1+\beta^2)^{1/4}} \left(\frac{d}{2}\right)^{8/3}$
    elseif $|\beta| > 0.01$ then
      $k_2 = \frac{3k_s}{8|\beta|(1+\beta^2)^{1/4}} d^{8/3}$
  elseif $d \geq |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      $k_3 = \frac{k_s}{(1+\beta^2)^{1/4}} \left(\frac{d_0+d_1}{2}\right)^{5/3} (y_1 - y_0)$
    elseif $|\beta| > 0.01$ then
      $k_4 = \frac{3k_s}{8|\beta|(1+\beta^2)^{1/4}} \left|d_0^{8/3} - d_1^{8/3}\right|$
    endif
  endif
endif

White-Colebrook (k_n) roughness conveyance for a vertical segment (see Figure 1.46):

$d = \max(d_0, d_1)$
$c_1 = (1+\beta^2)^{1/4} \ln(10)$
$c_2 = \frac{k_n}{12}$
if $d \leq 0$ then
  $k = 0$
else
  if $d < |z_1 - z_0|$ then
    if $|\beta| \leq 0.01$ then
      if $\frac{6d}{k_n} \leq 1.0005$ then
        $k_1 = \frac{36}{|\beta|(1+\beta^2)^{1/4}} 0.00022 \left(\frac{d}{2}\right)^{5/2}$
      else
        $k_1 = \frac{36}{|\beta|(1+\beta^2)^{1/4}} \, {}^{10}\log\left\{\frac{6d}{k_n}\right\} \left(\frac{d}{2}\right)^{5/2}$
      endif
    elseif $|\beta| > 0.01$ then
      if $\frac{d}{c_2} \leq 1.495$ then
        $k_2 = \frac{36}{5|\beta|c_1} d^{5/2} 0.00213$
      else
        $k_2 = \frac{36}{5|\beta|c_1} d^{5/2} \left(\ln\left\{\frac{d}{c_2}\right\} - \frac{2}{5}\right)$
      endif
    elseif $d \geq |z_1 - z_0|$ then
      if $|\beta| \leq 0.01$ then
        if $\frac{6(d_0+d_1)}{k_n} \leq 1.0005$ then
          $k_3 = \frac{18}{(1+\beta^2)^{1/4}} 0.00022 (y_1 - y_0) \left(\frac{d_0+d_1}{2}\right)^{3/2}$
        else
          $k_3 = \frac{18}{(1+\beta^2)^{1/4}} \, {}^{10}\log\left\{\frac{6(d_0+d_1)}{k_n}\right\} (y_1 - y_0) \left(\frac{d_0+d_1}{2}\right)^{3/2}$
        endif
      elseif $|\beta| > 0.01$ then
        if $\frac{d_0}{c_2} \leq 1.495$ and $\frac{d_1}{c_2} \leq 1.495$ then
          $k_4 = \frac{36}{5|\beta|c_1} \left|d_0^{5/2} 0.00213 - d_1^{5/2} 0.00213\right|$

elseif $\frac{d_0}{c_2} \leq 1.495$ and $\frac{d_1}{c_2} > 1.495$ then

$$k_4 = \frac{36}{5|\beta|c_1}\left|d_0^{5/2}0.00213 - d_1^{5/2}\left(\ln\left\{\frac{d_1}{c_2}\right\} - \frac{2}{5}\right)\right|$$

elseif $\frac{d_0}{c_2} > 1.495$ and $\frac{d_1}{c_2} \leq 1.495$ then

$$k_4 = \frac{36}{5|\beta|c_1}\left|d_0^{5/2}\left(\ln\left\{\frac{d_0}{c_2}\right\} - \frac{2}{5}\right) - d_1^{5/2}0.00213\right|$$

else

$$k_4 = \frac{36}{5|\beta|c_1}\left|d_0^{5/2}\left(\ln\left\{\frac{d_0}{c_2}\right\} - \frac{2}{5}\right) - d_1^{5/2}\left(\ln\left\{\frac{d_1}{c_2}\right\} - \frac{2}{5}\right)\right|$$

endif

endif

endif

endif

endif

Bos&Bijkerk (γ) roughness conveyance for a vertical segment (see Figure 1.46):

$d = \max(d_0, d_1)$

if $d \leq 0$ then

$k = 0$

else

if $d < |z_1 - z_0|$ then

if $|\beta| \leq 0.01$ then

$$k_1 = \frac{2\gamma}{|\beta|(1+\beta^2)^{1/4}}\left(\frac{d}{2}\right)^3$$

elseif $|\beta| > 0.01$ then

$$k_2 = \frac{\gamma}{3|\beta|(1+\beta^2)^{1/4}}d^3$$

elseif $d \geq |z_1 - z_0|$ then

if $|\beta| \leq 0.01$ then

$$k_3 = \frac{\gamma}{(1+\beta^2)^{1/4}}\left(\frac{d_0+d_1}{2}\right)^2(y_1 - y_0)$$

elseif $|\beta| > 0.01$ then

$$k_4 = \frac{\gamma}{3|\beta|(1+\beta^2)^{1/4}}\left|d_0^3 - d_1^3\right|$$

endif

endif

endif

1.1.21 Drying/flooding

When a water level upstream of a branch segment is lower than 5 mm above the bed level of the ζ-calculation point at this upstream side, the branch segment is assumed to be dry; the discharge through this branch segment is said to be zero. When a branch segment is dry and upstream or downstream of the segment the water level rises higher than 10 mm above the bed level of the ζ-calculation point at that side, the branch segment starts flooding again.

The same deadband is used for the check on drying and flooding of structures. In case of a weir and orifice, the upstream water level is compared to the crest level. In case of culverts, siphons and inverted siphons, the water level is compared to the maximum of the bed level and the bottom of the structure at the upstream side.

1.1.22 Free board

Freeboard is the difference between surface level and water level at some location in the system. It is defined from surface level to water level, so the freeboard is positive when the water level is lower than the surface level. When the freeboard is negative, there is a flood at that location.

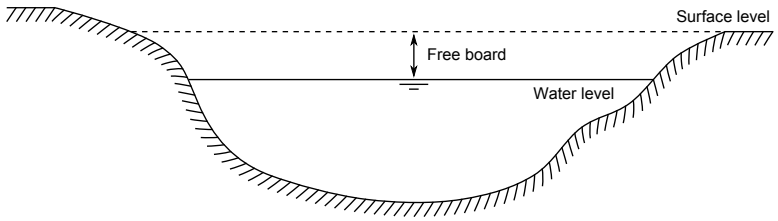

Figure 1.47: *Free board*

1.1.23 Ground layer

Except for a Y-Z profile, cross sections can have a fixed horizontal ground layer (or sediment layer). The ground layer level (top level of the ground layer) equals the bottom level (or invert level) plus the ground layer thickness. The part of the cross-section located beneath the ground layer level is omitted in the hydraulic computations. Hence, a ground layer reduces both the wetted area and the wetted perimeter for a given water level. In Figure 1.48 a circular cross section is shown with a ground layer with thickness D_{gr}

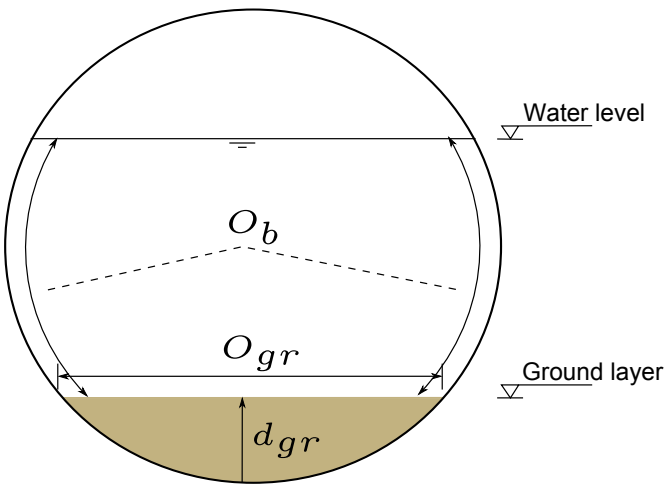

Figure 1.48: *Ground layer in circular cross section*

For a ground layer a friction value can be specified that differs from the bed friction value. The ground layer friction value is applied for the cross-sectional width at the ground layer level only. For the remaining wetted perimeter of the cross-section, the bed friction value is applied. The same friction formulation (i.e. Chézy, Manning etc.) is to be applied for both the ground layer and the remaining wetted perimeter (i.e. the bed) of the cross-sectional profile. The combined friction that is used is computed according to:

$$k_s = \sqrt{\frac{O_b k_{s_b}^2 + O_{gr} k_{s_{gr}}^2}{O_b + O_{gr}}} \tag{1.154}$$

k_s	Combined Strickler friction value
k_{s_b}	Strickler bed friction value
$k_{s_{gr}}$	Strickler friction value of the ground layer
O_b	Wetted perimeter of the bed
O_{gr}	Wetted perimeter of the ground layer

1.1.24 Measurement station

In channel flow a measurement station is placed on a branch. From a measurement station the actual (calculated) value of a water level or a discharge can be retrieved. This value can then be used in an algorithm of a controller.

The water level that a measurement station retrieves is the water level at the nearest calculation point. The retrieved discharge is the discharge at the branch segment where the measurement station is located.

In sewer flow the measurement station can be placed on a connection node or on a branch Water levels are taken at the nodes, discharges at the branches.

1.1.25 Network

The schematic basis of each model is the network of branches. These branches are connected to each other at connection nodes.

One can say that the network of branches and nodes is the modelled representation of the real geographic data of the modelled area.

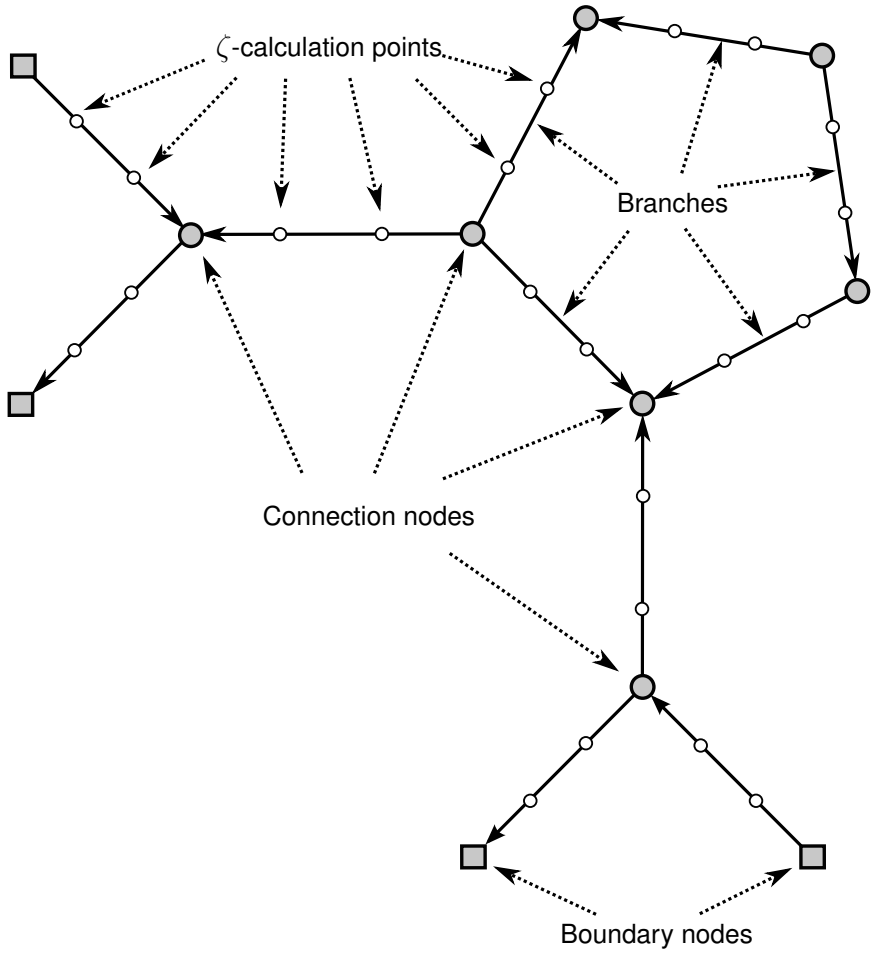

Figure 1.49: *Network*

In the Delft-scheme used by the D-Flow 1D module, the network is split up into a staggered grid. The water levels are defined at the connection nodes and the ζ-calculation points of this grid, while the discharges are defined at the branch segments.

1.1.25.1 Branch

A branch in a SOBEK-Flow-model is one of the basic elements in the schematisation. A branch connects two nodes, and has the following attributes:

- ◇ Begin node
- ◇ End node
- ◇ Branch length
- ◇ A geographical orientation (used only in case wind friction effect is included in your model).

The direction of the branch (positive x-direction) is from begin node to end node.

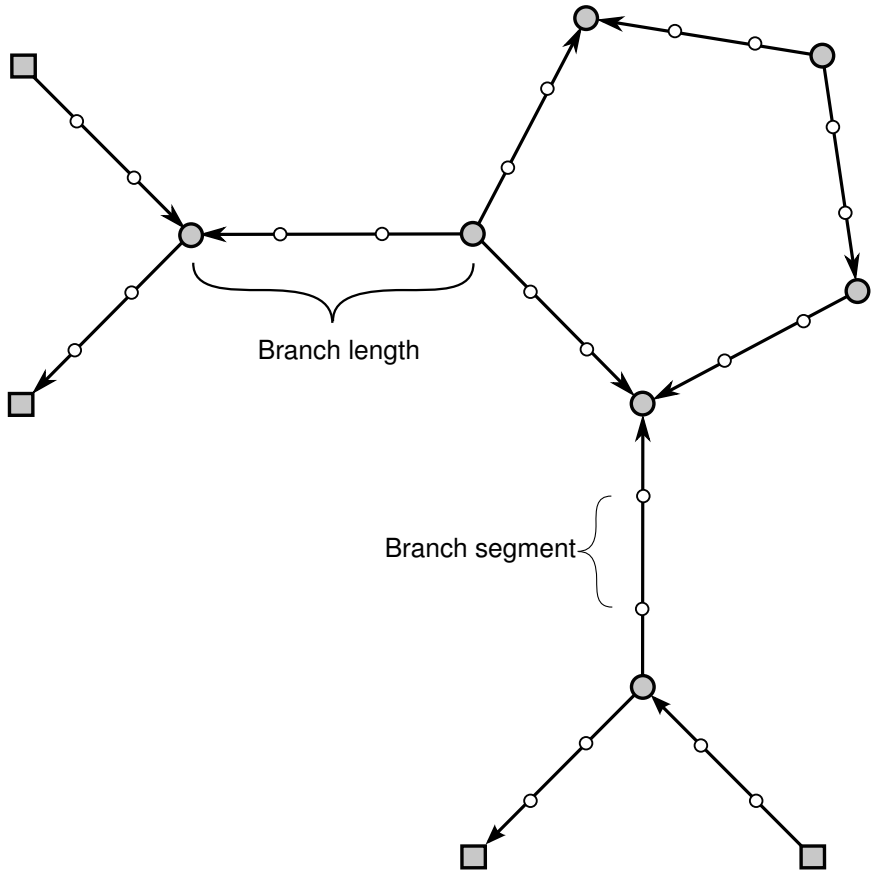

Figure 1.50: *Branch length in model network*

When using channel flow, ζ-calculation points can be specified on a branch. If this is done, the branch is split up into branch segments. The numerical simulation will be carried out based on these ζ-calculation points and branch segments in a staggered grid. The discharges are calculated at the branch or branch segments Water levels are calculated at the connection nodes and ζ-calculation points.

In case of sewer flow, each branch has a maximum of two ζ-calculation points at the beginning and end of that branch. The branch can be interpreted as one branch segment

1.1.25.2 Branch length

In sewer systems the length of a branch is the length from the begin connection node to the end connection node. This length is referred to as the co-ordinate length. In open channel systems, a branch can be made longer than this minimum co-ordinate length, with the use of vector points on the branch. This length is referred to as vector length. With this vector length the actual length of a meandering river or channel can be modelled. If the vector length is defined on a branch, this length is used in the computation of the water flow equations. If this length is not defined, the co-ordinate length is used.

Note: The vector length is always equal to or greater than the co-ordinate length

1.1.25.3 Branch segment

A SOBEK-Flow-network consists of branches that are connected in connection nodes. These branches can be divided by ζ-calculation points into segments; the so-called branch seg-

ments. The discharges that result from the water flow equations are defined in these branch segments

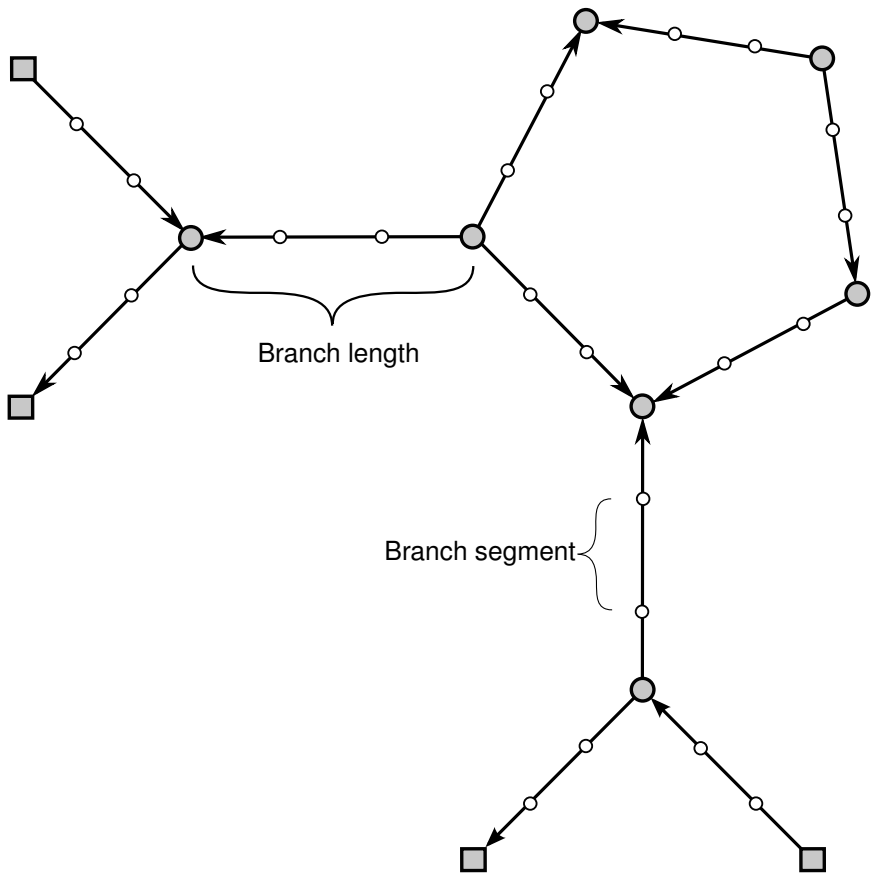

Figure 1.51: *Branch segment*

In sewer networks no ζ-calculation points can be used. So each branch consists of one branch segment.

1.1.25.4 Connection node

A connection node in a SOBEK-Flow-model is a location where branches can be joined to other branches. Water levels are defined at the connection nodes and the ζ-calculation points.

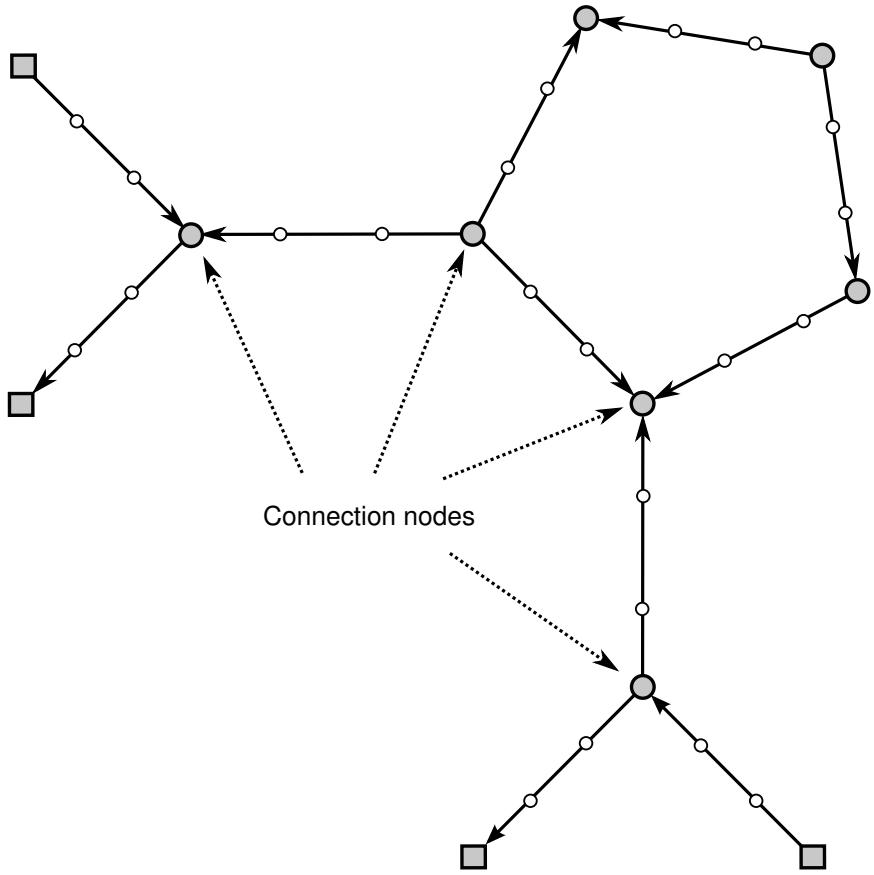

Figure 1.52: *Connection nodes*

Note: A node located at the bounds of the model can be a connection node (with storage) or a boundary node (no storage).

1.1.26 Robustness

The Delft-scheme is developed with robustness as its most important design aspect. This scheme can deal with phenomena such as drying/flooding, supercritical flow and flow transitions (hydraulic jumps) and guarantees a solution for every time step, though the actual computational time step can be reduced internally under certain flow conditions (large discharges in combination with little storage).

The computational time step reduction is done by a time step estimation procedure before a new time step or at the end of a time step when negative depths are computed. In the latter case the step is repeated with a time step reduced by a factor 2 until a correct solution is achieved.

1.1.27 Simulation output parameters at branch segments

For more background information on time step reductions, see section Time step reductions during the simulation. The simulation output parameters available at each branch segment (see option "Simulation Info at the Branch Segments" in the file <flowanal.his> or in Results in Charts and Results in Maps) are:

- **Time step estimation**: This refers to the number of times that the time step is reduced, since the allowable Courant time step of this branch segment is smaller than the user-defined time step.
- **No iteration up and No iteration down**: Parameters "No iteration up" and "No iteration down", respectively refer to the number of times that convergence in the Newton iteration process is not satisfied at the water level point with the lowest x-coordinate ("No iteration up") and the water level point with highest x-coordinate ("No iteration down") of the branch segment.
- **Negative depth up and Negative depth down**: Parameters "Negative depth up" and "Negative depth down", respectively refer to the number of times that a time step reduction is needed for avoiding negative depths at the upstream water level point (lowest x-coordinate) and the downstream water level point (highest x-coordinate) of the branch segment.
- **Total**: The number of times that the allowable Courant time step is smaller than the user-defined time step and the number of times that the time step is reduced for either satisfying the convergence condition or for avoiding negative depths.

1.1.28 Time step reductions during the simulation

The user can define a time step in the Settings Task block. During the computation this user-defined time step may be reduced for stability reasons (e.g. to prevent the simulation from crashing). D-Flow 1D performs the following checks:

- At the start ($t = t^k$) of each computational time-step, the allowable Courant time-step is determined. This is done as follows:
  - For each 1D branch segment (i) its 1D Courant time step is computed as:

$$\Delta t_i = \sigma \frac{Vol_{i,tot}}{Q_{i,t}} \qquad \text{1D Courant} \tag{1.155}$$

    where:

σ	the user-defined value for "maximum flow (1D) and velocity (2D) Courant number" in Settings
$Vol_{i,tot}$	the total volume of water at the upwind (or upstream) water level point
Q_{i,t^k}	the discharge at $t = t^k$

    Then the 1D branch segment having the smallest 1D Courant time step (e.g. the allowable 1D Courant time step) is determined. The applicable 1D time step at $t = t^k$ is the minimum of the user-defined time step and the allowable 1D Courant time step. The branch segment with the smallest 1D Courant time step is logged in case its 1D Courant time step is smaller than the user-defined time step.
  - The 2D Courant time step for all 2D links in U-direction and for all 2D links in V-direction are respectively computed as:

$$\Delta t = \sigma \frac{\Delta x_m}{U_{m,t^k}} \qquad \text{2D Courant, U-direction} \tag{1.156}$$

    and

$$\Delta t = \sigma \frac{\Delta y_n}{V_{n,t^k}} \qquad \text{2D Courant, V-direction} \tag{1.157}$$

where:

σ	the user-defined value for "maximum flow (1D) and velocity (2D) Courant number" in Settings
Δx_m	the grid spacing in x-direction
Δy_n	the grid spacing in y-direction
U_{m,t^k}	the depth-averaged flow velocity in x-direction at $t = t^k$
V_{n,t^k}	the depth-averaged flow velocity in y-direction at $t = t^k$

The allowable 2D Courant time step is the minimum of the 2D Courant time steps of all 2D U-links and 2D V-links. The applicable 2D time step is the minimum of the user-defined time step and the allowable 2D Courant time step.

- In case of a combined 1D-2D hydraulic computation, the applied time step (Δt) is the minimum of the applicable 1D time step and the applicable 2D time step.

- In each computational time step (from $t = t^k$ to $t^{k+1} = t^k + \Delta t$), it is checked that for each water level point yields that the lateral inflow volume is not larger than its volume of water at $t = t^k$. If this condition is not satisfied, the user defined time step is reduced to such extend that this condition is met. The reduced time step, however, is not made smaller than the user-defined "minimum time step" in Settings.
- Due to various circumstances (discontinuities in model schematization, unrealistic boundary conditions, etc.), the above determined actual time step (Δt) may still be too large for finding a valid solution at $t^{k+1} = t^k + \Delta t$. Two conditions are to be fulfilled in finding a valid solution:
  - The first condition is that the Newton iteration process converges in terms of water levels and volumes (both computed at water level points). In other words differences in water levels and volumes between the last and previous Newton solution are to be within the user-defined values for "epsilon for water depth" and "epsilon for volume". In Settings, the user can define the "maximum number of Newton iterations". If the convergence condition is not satisfied within the maximum number of Newton iterations, the time step is reduced by a factor 2. If this does not satisfy the convergence condition, the time step is again reduced by a factor 2. The simulation is terminated if a time step smaller than the user-defined "minimum time step" (see Settings) is needed for satisfying the convergence condition. All water level points, that after the maximum number of Newton iterations are conducted, still having a water level convergence error are logged during a computational time step.
  - The second condition is that no negative depths (e.g. water level is below bed level) may occur. Negative depths at a particular water level point may occur if the outflow during a time step is larger than the volume stored at $t = t^k$. If negative depths occur, the time step is reduced up to a time step for which yields that the outflow during the reduced time step is smaller than the volume stored at $t = t^k$ and hence no negative depth occurs. The simulation is terminated if a reduced time step smaller than the user-defined "minimum time step" (see Settings) is needed to avoid negative depths. All water level points for which the time step needs to be reduced in order to avoid negative depths are logged during a computational time step.

Note: If a model run requires a lot of computational effort (e.g. an unexpected long simulation time is needed), often a large number of time step reductions during simulation is performed. These time step reductions are needed to find a valid solution and/or to prevent the model from crashing. Defining a smaller calculation time step in Settings may result in shorter simulation times. The reason for this, the smaller user-defined time step allows more moderated intermediate solutions to be found. More moderate solutions reduce the number of circumstances in which the model encounters large discontinuities, which require a considerable

amount of time step reductions before finding a valid solution. The final actual time step can be much smaller than the user-defined time step. In other words: A smaller user-defined time step can result in significant less total number of time step reductions leading to a decrease of total computational time. This is because the total required simulation time is directly related to the total number of computational time steps.

1.1.29 Slope

In the D-Flow 1D module slope can refer to the slope of the water level or the bed slope. If the word "slope" is used without further indication, then it pertains to the slope of the water level in the model. The symbol used is i:

$$i = \frac{\partial \zeta}{\partial x} \tag{1.158}$$

where

ζ	Water level [m]
x	Distance [m]

For bed slope, the symbol used is i_b:

Bed slope:

$$i_b = \frac{\partial z_b}{\partial x} \tag{1.159}$$

with z_b as the bed level.

1.1.30 Stationary computation

A stationary computation is performed in the D-Flow 1D module by running a non-stationary simulation until a steady state is branched. During this simulation, all time depending parameters, such as lateral and boundary conditions, are set to their initial value and kept constant.

In some cases a steady state cannot be branched, for example, when a pump is switching on and off. In that case the computation stops after 1 000 simulation steps (not time steps, see Time step estimation) and the maximum difference of water level compared to the last time step in the network is given. If this value is low enough ($<$1 mm for example) the solution can be considered as steady.

The results of a stationary computation can be used as an initial condition for a non-stationary simulation.

1.1.31 Summer dike

A flood plain might be separated from a river by means of a small dike, having a crest level lower than the crest level of the main dike (see Figure 1.53). In the Netherlands large floods usually occur in winter. The main dike and the small dike are, therefore, respectively referred to as the “winter dike” and the “summer dike”.

The presence of a summer dike implies that the cross-sectional profile as measured in situ (see shadowed line in Figure 1.53) does not increase monotonously with rising water levels. When the water level becomes slightly higher than the crest level of the summer dike, first the area behind the summer dike is to be filled, before water levels can become any higher. This implies a local attenuation of the flood wave at water levels just above the crest level of

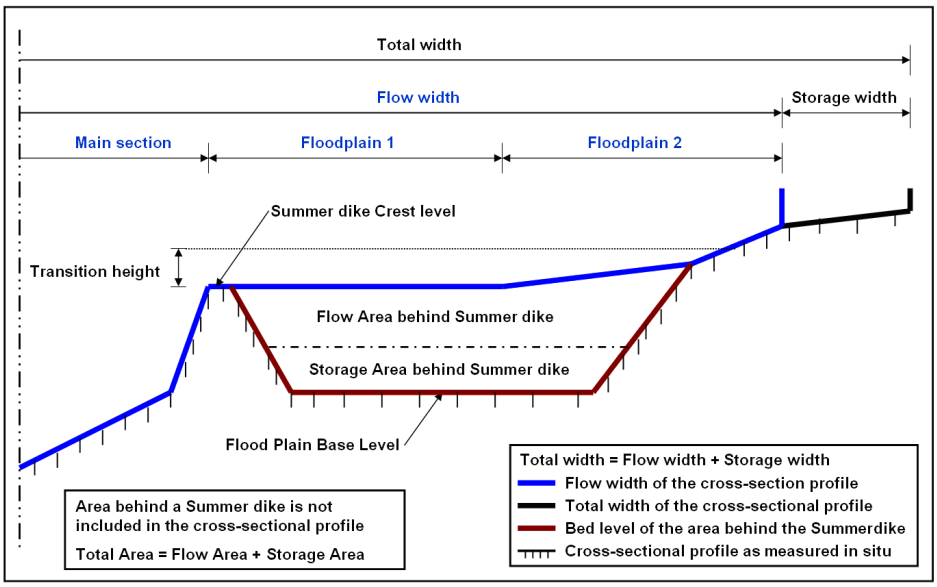

Figure 1.53: *Summer dike option, available in a river profile*

the summer dike. This hydraulic phenomenon can be modelled using the summer dike option available at a river profile.

In defining a summer dike in a river profile a distinction is to be made between:

- **The cross-sectional profile**
  The cross-sectional profile comprises of a “flow area” and a “storage area”, which in Figure 1.53 are respectively located above the blue lines and the black lines. The summation of flow area and storage area is called the “total area”. Both flow area and total area are defined by its width as function of elevation. At each elevation in the cross-sectional profile yields that storage width equals total width minus flow width. Hence, no storage is defined if total width equals flow width. The flow area of the cross-sectional profile can be divided in three separate flow sections, respectively called the (i) main section, (ii) floodplain 1 and (iii) floodplain 2 . Each such flow section can have a different roughness value and a different roughness formulation.
- **The area behind the summer-dike**
  In analogy with the cross-sectional profile, the “total area behind a summer dike” is divided in a “flow area” and a “storage area”. The “Flood Plain Base Level” coincides with the lowest part of the area behind the summer dike (see Figure 1.53).

Remark:

- Both the cross-sectional profile and the area behind the summer dike should be mutually consistent with the cross-sectional profile as measured in situ (see shadowed line in Figure 1.53). Hence, both the flow area and the storage area behind the summer dike should not be included in the cross-sectional profile.

The so-called “transition height” is used for avoiding numerical oscillations. More precisely, the transition height ensures that the flow area and storage area behind the summer dike are gradually taken into computation. Transition height for summer dikes is a model-wide parameter, that is defined in the Settings Taskblock (*Edit* button next to 1DFLOW (River); Advanced settings Tab).

Lets consider the following summer dike properties:

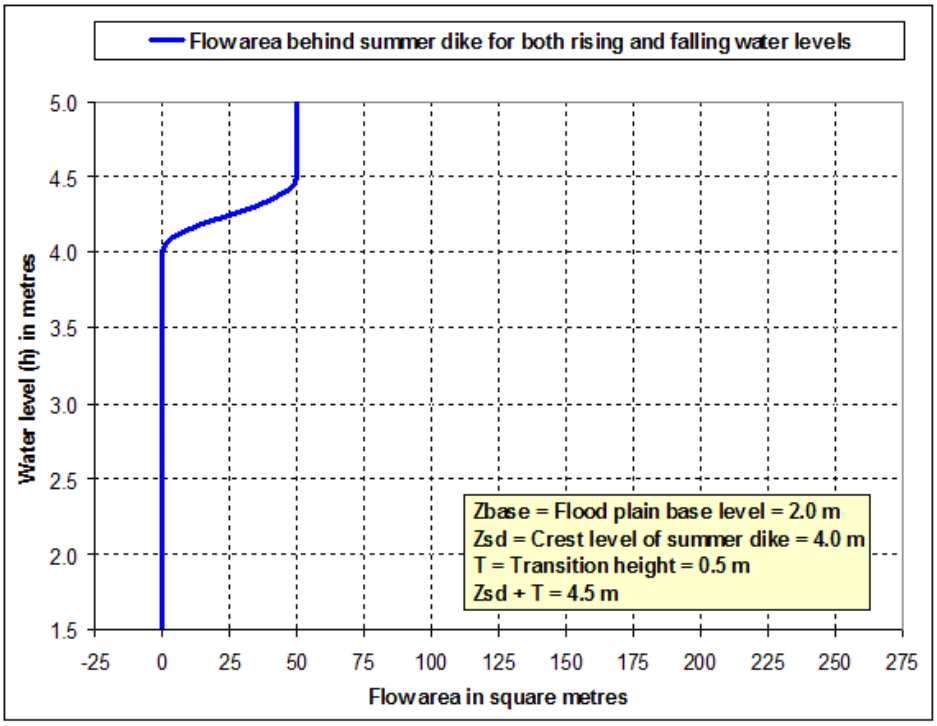

Figure 1.54: *Flow area behind the summer dike as function of local water levels.*

- flood plain base level (z_{base}) of 2 m,
- summer dike crest level (z_{sd}) of 4 m,
- transition height (T) of 0.5 m,
- storage area behind the summer dike of 200 m^2,
- flow area behind the summer dike (ExtraFlowArea) of 50 m^2, and
- total area behind the summer dike (ExtraTotalArea) of 250 m^2.

In explaining how a summer dike is taken into computation, we make a distinction between:

- The actual value of the (extra) flow area behind the summer dike,
- The flow velocity applied to the (extra) flow area behind the summer dike, and
- The actual value of the (extra) total area behind the summer dike.

The actual (extra) flow area is a unique function of the local water level (h) in the cross-sectional profile (see Figure 1.54). More precisely:

- If $\zeta < z_{\text{sd}}$; Flow area behind the summer dike is zero.
- If $z_{\text{sd}} \leq \zeta \leq z_{\text{sd}} + T$; Flow area behind the summer dike varies as a quadratic function of water level (ζ) between zero and ExtraFlowArea (= 50 m^2).
- If $\zeta > z_{\text{sd}} + T$; Flow area behind the summer dike is equal to ExtraFlowArea.

The flow velocity applied to the (extra) flow area behind the summer dike is the flow velocity in one of the three available flow sections. Hence, either the flow velocity in the main section, in floodplain 1 or in floodplain 2. The actual flow velocity applied to the (extra) flow area behind the summer dike is determined as follows:

1. Default the flow velocity in floodplain 2 is used.
2. However, if the wetted area of floodplain 2 becomes less than 0.4 m^2, the flow velocity in floodplain 1 is used,

3 Furthermore, if the wetted area of floodplain 1 becomes less than 0.4 m^2, the flow velocity of the main section is used.

Remarks:

- Please contact Deltares if you like to use a different threshold value (i.e. not 0.4 m^2) for discerning which flow velocity is to be applied to the (extra) flow area behind the summer dike.
- The flow velocity applied to the (extra) flow area behind the summer dike is computed as follows

$$U_i = C_i\sqrt{R_i S} \qquad i \in \{\text{main section}, \text{floodplain 1}, \text{floodplain 2}\} \tag{1.160}$$

where:

U_i	average flow velocity
C_i	Chézy coefficient of the flow section
R_i	$(A_i + A_{\text{sd}})/P_i$: hydraulic radius of the flow section
A_i	wetted area of the flow section
A_{sd}	extra flow area behind the summer dike
P_i	wetted perimeter of the flow section
S	water level slope

- The discharge including the (extra) flow area behind the summer dike is computed as follows

$$Q_i = U_i(A_i + A_{\text{sd}}) \tag{1.161}$$

Figure 1.55 depicts the actual (extra) total area behind the summer dike as function of the local water level (ζ) in the cross-sectional profile. The (extra) total area behind the summer dike is added to the total area of the cross-sectional profile. There is a hysteresis in the extra total area for water levels varying from flood plain base level ($z_{\text{base}} = 2.0\ m$) to crest level of summer dike plus transition height ($z_{\text{sd}} + T = 4.5\ m$). We call line ABC the rising limp and line CA the falling limp of the hysteresis. The so-called "hysteresis flag" determines which limp of the hysteris is to be followed. In case the hysteresis `flag=1` the rising limp of the hysteresis is followed, else the falling limp is followed. At the onstart of a computation yields that the hysteresis `flag=1`. The actual applied extra total area behind the summer dike is computed as follows:

1 If $\zeta < z_{\text{base}}$; Hysteresis flag is set to 1. Total area behind the summer dike is zero.
2 If $z_{\text{base}} \leq \zeta \leq z_{\text{sd}} + T$ and hysteresis `flag=1`;
   - For $z_{\text{base}} \leq \zeta < z_{\text{sd}}$; Total area behind the summer dike is zero.
   - For $z_{\text{sd}} \leq \zeta \leq z_{\text{sd}} + T$; Total area behind the summer dike varies as a quadratic function of water level (ζ) between zero and ExtraTotalArea (= 250 m^2)
3 If $z_{\text{base}} \leq \zeta \leq z_{\text{sd}} + T$ and hysteresis `flag=0`; Total area behind the summer dike varies as a quadratic function of water level (ζ) between zero and ExtraTotalArea.
4 If $\zeta > z_{\text{sd}} + T$; Hysteresis flag is set to 0. Total area behind the summer dike is equal to ExtraTotalArea.

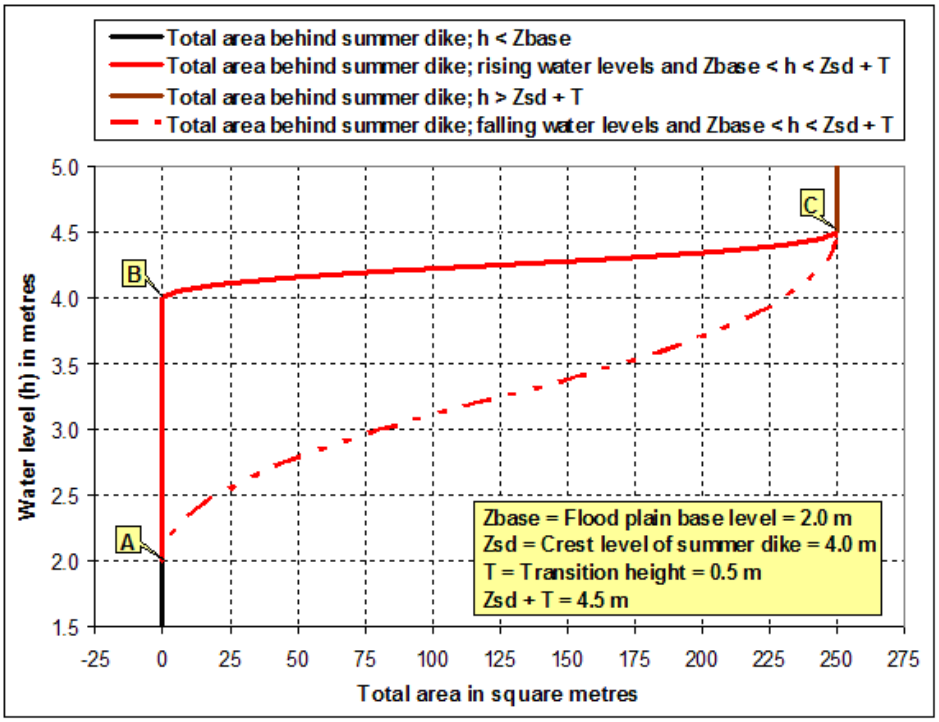

Figure 1.55: *Total area behind the summer dike as function of local water levels.*

1.1.32 Super-critical flow

The D-Flow 1D module can deal with super-critical flow. The flow in a branch or branch segment is super-critical if the Froude number [$-$] is higher than 1:

$$\text{Froude} = \frac{u}{\sqrt{gA_f/W_f}} \tag{1.162}$$

u	Velocity [m/s]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
A_f	Wetted area [m^2]
W_f	Flow width [m]

Remarks:

- ◇ When the Froude number is less than 1, the flow is sub-critical.
- ◇ Transitions from super-critical to sub-critical flow causes an hydraulic jump.

1.1.33 Surface level

The surface level is the level of the soil surface i.e. the embankment level of the cross section. When the water level is higher than this level there is a flood. Normally water is stored above this level on a certain storage area. When the water level drops, this storage is emptying back into the branch or node.

1.2 Transport equation

The transport of salt, temperature and sediment in estuaries, tidal rivers and alluvial rivers can be considered as transport of conservative substance in water. These transports are described by an advection-diffusion equation, which is called the transport equation.

Next to the additional transport equation, density differences have to be accounted for in the momentum equation of the water flow module. The water flow module is therefore coupled with the salt intrusion module and temperature by the density and the flow field. The concentrations are denoted by C.

The transport equation is described by an advection-diffusion equation including source term and read:

$$\frac{\partial A_T C}{\partial t} + \frac{\partial}{\partial x}\left(Q\,C - A_F D \frac{\partial C}{\partial x}\right) = S_s \tag{1.163}$$

in which

C	concentration of salt or chloride, averaged over the total cross-sectional area [kg/m^3]
D	dispersion coefficient [m^2/s]
S_s	source term [$kg/(ms)$]
Q	discharge (water) [m^3/s]
A_T	total cross-sectional area [m^2]
A_F	flow area [m^2]

1.2.1 Temperature: heat flux model

1.2.1.1 General

In D-Flow 1D the heat exchange at the free surface is modelled by taking into account the separate effects of solar (short wave) and atmospheric (long wave) radiation, and heat loss due to back radiation, evaporation and convection. In literature there is a great variety of empirical formulations to calculate these heat fluxes across the sea surface. Most formulations differ in the dependency of the exchange on the meteorological parameters such as wind speed, cloudiness and humidity. Some formulations were calibrated for coastal seas others for lakes.

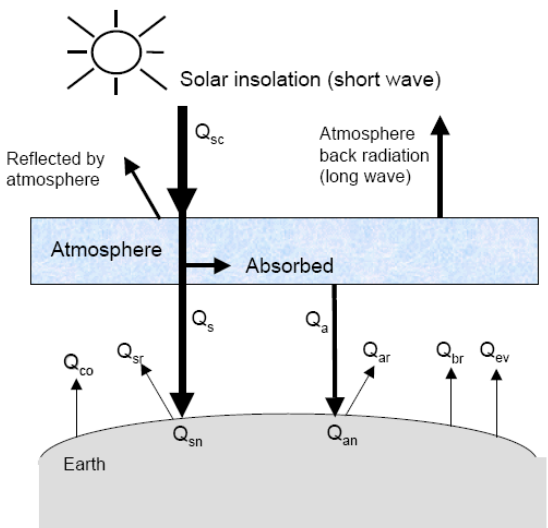

Figure 1.56: *Overview of the heat exchange mechanisms at the surface*

Legend for Figure 1.56:

Q_{sc}	radiation (flux) for clear sky condition in [J/m^{2}s]
Q_{co}	heat loss due to convection (sensible) in [J/m^{2}s]
Q_{sr}	reflected solar radiation in [J/m^{2}s]
Q_s	solar radiation (short wave radiation) in [J/m^{2}s]
Q_{sn}	net incident solar radiation (short wave), $= Q_s - Q_{sr}$
Q_a	atmospheric radiation (long wave radiation) in [J/m^{2}s]
Q_{an}	net incident atmospheric radiation (long wave)
Q_{ar}	reflected atmospheric radiation in [J/m^{2}s]
Q_{br}	back radiation (long wave radiation) in [J/m^{2}s]
Q_{ev}	heat loss due to evaporation (latent) in [J/m^{2}s]

In D-Flow 1D the heat exchange at the free surface is modelled by taking into account the separate effects of solar (short wave) and atmospheric (long wave) radiation, and heat loss due to background radiation, evaporation and convection. Two heat flux models have been implemented:

1. *Excess temperature model*
   The heat exchange flux at the air-water interface is computed. Only the background temperature is required.
2. *Composite temperature model*
   The fraction of the sky covered by clouds is prescribed (in %). The effective back radiation and the heat losses due to evaporation and convection are computed by the model. Additionally, when air and water densities and/or temperatures are such that free convection occurs, free convection of latent and sensible heat is computed by the model. This model formulation typically applies for large water bodies.

For the physical background of the heat exchange at the air-water interface and the definitions, we refer to Octavio *et al.* (1977) for the excess model and Lane (1989) for the composite model.

1.2.1.2 Heat balance

The change in temperature T_s (in °C) is given by:

$$\frac{\partial T_s}{\partial t} = \frac{Q_{tot}}{\rho_w c_p \Delta z_s} \tag{1.164}$$

with:

Q_{tot}	total heat flux through the air-water surface (in J/(m^{2}s))
ρ_w	specific density of water (in kg/m^3)
c_p	specific heat capacity of sea water (= 3930 J/(kg K))
Δz_s	water depth

In D-Flow 1D, the heat exchange at the bed is assumed to be zero. This may lead to overprediction of the water temperature in shallow areas.

Remarks:

- The temperature T is by default expressed in °C. However, in some formulas the absolute temperature $\bar{T}$ in K is used. They are related by:

$$\bar{T} = T + 273.15. \tag{1.165}$$

- In Equation (1.164) the total incoming heat flux is assumed to be absorbed in the water column. This may result in an unrealistically high surface temperature when the top

layer is thin. This can be prevented by absorbing the incoming radiation as a function of depth. Currently, this is only implemented in heat flux models 4 and 5.

◇ When using the composite heat flux model, the free convection of latent and sensible heat is also determined.

1.2.1.3 Excess temperature model

The excess temperature model computes the heat fluxes through the water surface in such a way that the temperature of the surface layer relaxates to the natural background temperature specified by you. The heat transfer coefficient mainly depends on the water temperature and the wind speed.

In the excess temperature model the solar radiation does not play an explicit role, but is part of the background temperature.

$$Q_{tot} = -\lambda \left(T_s - T_a\right), \tag{1.166}$$

with:

λ Heat exchange coefficient, is defined by (Sweers, 1976):

$$\lambda = 4.48 + 0.049 T_s + f\left(U_{10}\right)\left(1.12 + 0.018 T_s + 0.00158 T_s^2\right) \tag{1.167}$$

T_a air temperature
T_s water surface temperature
$f\left(U_{10}\right)$ wind function:

$$f\left(U_{10}\right) = \left(3.5 + 2.0 U_{10}\right)\left(\frac{5.0 \times 10^6}{S_{area}}\right)^{0.05} \tag{1.168}$$

U_{10} wind speed
S_{area} exposed water surface in m^2

1.2.1.4 Composite temperature model

In the Composite temperature model, the heat fluxes through the water surface by incoming radiation, back radiation, evaporation and convection are computed separately. Evaporation and convection depend on the air temperature, the water temperature near the free surface, relative humidity, and wind speed.

The total heat flux through the free surface reads:

$$Q_{tot} = Q_{sn} + Q_{an} - Q_{br} - Q_{ev} - Q_{co}, \tag{1.169}$$

with:

Q_{sn} net incident solar radiation (short wave)
Q_{an} net incident atmospheric radiation (long wave)
Q_{br} back radiation (long wave)
Q_{ev} evaporative heat flux (latent heat)
Q_{co} convective heat flux (sensible heat).

$$Q_{tot} = Q_{sn} - Q_{eb} - Q_{ev} - Q_{co}, \tag{1.170}$$

with:

$$Q_{eb} \qquad = Q_{br} - Q_{an}$$

The subscript n refers to a net contribution.

1.2.1.5 Solar radiation

The short-wave radiation emitted by the sun that reaches the earth surface under a clear sky condition is computed by means of an emperical formula

The solar radiation is computed by D-Flow 1D and is dependent on the geographical position at the earth and the local time.

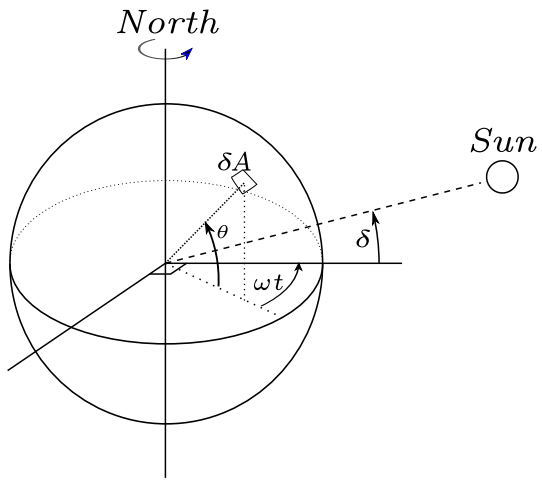

Figure 1.57: *Co-ordinate system position Sun*
δ: declination; θ: latitude; ωt: angular speed

Not all of the radiation is absorbed at the water surface. A part is transmitted to deeper water. Short waves can penetrate over a distance of 3 to 30 meters, depending on the clarity of the water, while the relatively longer waves are absorbed at the surface. Therefore, it is convenient to separate the incoming solar insolation into two portions:

1. βQ_{sn}, the longer wave portion, which is absorbed at the surface and
2. $(1-\beta)\, Q_{sn}$, the remainder part, which is absorbed in deeper water.

The absorption of heat in the water column is an exponential function of the distance H from the water surface:

$$(1-\beta)\, Q_{sn} = \int_0^H e^{-\gamma z}\, dz \Rightarrow \tag{1.171}$$

$$Q_{sn}\,(h) = \frac{\gamma e^{-\gamma h}}{1-e^{-\gamma H}}(1-\beta)Q_{sn}, \tag{1.172}$$

with:

β	part of Q_{sn} absorbed at the water surface which is a function of the wavelength. The default value of β in Delft3D-FLOW is 0.06.
γ	extinction coefficient (measured) in m^{-1}, also related to the so-called Secchi-depth $\gamma = \frac{1.7}{H_{Secchi}}$
h	distance to the water surface in meters.
H	total water depth.

Remark:

- ◇ The exponential decay function, Equation (1.172) has only been implemented in 3D-computations. Since D-Flow 1D is a depth averaged model, the distinction between long and short wave radiation is not relevant.

The incoming energy flux at the water surface depends on the angle (declination) between the incoming radiation and the Earth's surface. This declination depends on the geographical position on the Earth and the local time. The Earth axis is not perpendicular to the line connecting the Sun with Earth. This tilting (angle δ) varies with the time of the year and it leads to a seasonal variation of the radiation flux. At June 21, the declination is maximal, 23.5 degrees. Of course, by the rotation of the Earth the solar radiation also varies during the day. Near twelve o'clock local time, the sun elevation above the horizon is maximal. For an overview of the angles used to determine the solar elevation angle γ, see Figure 1.57.

The temporal and latitude-dependent solar elevation angle γ is estimated by:

$$\sin(\gamma) = \sin(\delta)\sin\left(\frac{\pi\phi}{180}\right) - \cos(\delta)\cos\left(\frac{\pi\phi}{180}\right)\cos(\omega_1 t) \tag{1.173}$$

with:

δ	$= \frac{23.5\pi}{180}\cos(\omega_0 t - 2.95)$
ω_0	frequency of the annual variation ($= 2\pi/(365.24 \times 24)$)
ω_1	frequency of the diurnal variation ($= 2\pi/24$)
ϕ	latitude
t	number of hours since 1st of january

The incoming short-wave solar radiation through a clear sky at ground level Q_{sc} is about 0.76 of the flux incident at the top of the atmosphere (Gill, 1982):

$$Q_{sc} = \begin{cases} 0.76S\sin(\gamma), & \sin(\gamma) \geq 0, \\ 0.0, & \sin(\gamma) < 0. \end{cases} \tag{1.174}$$

The solar constant $S = 1\,368$ J/(m^{2}s) or W/m^2. This is the average energy flux at the mean radius of the Earth.

A part of the radiation that reaches the water surface is reflected. The fraction reflected or scattered (surface albedo) is dependent on latitude and season. Cloud cover will reduce the magnitude of the radiation flux that reaches the sea surface. The cloudiness is expressed by a cloud cover fraction F_c, the fraction of the sky covered by clouds. The correction factor for cloud cover is an empirical formula. The absorption of solar radiation is calculated (Gill, 1982) as the product of the net downward flux of short wave-radiation in cloudless conditions and factors correcting for reflection and cloud cover:

$$Q_{sn} = Q_s - Q_{sr} = (1-\alpha)\,Q_{sc} f(F_c)\,, \tag{1.175}$$

with:

Q_s	solar radiation (short wave radiation) in [$J/(m^2 s)$]
Q_{sr}	reflected solar radiation in [$J/(m^2 s)$]
Q_{sc}	see Equation (1.174)
$\sin(\gamma)$	see Equation (1.173)
$f(F_c)$	$= 1.0 - 0.4F_c - 0.38F_c^2$

α	albedo (reflection) coefficient (= 0.06)
F_c	fraction of sky covered by clouds (user-defined input)
S	solar constant (=1368 J m^{-2} s^{-1})

1.2.1.6 Effective back radiation

The total net long wave radiation flux is called the effective back radiation:

$$Q_{eb} = Q_{br} - Q_{an}. \tag{1.176}$$

The atmospheric radiation depends on the vapour pressure e_a, the air temperature T_a and the cloud cover F_c. The back radiation depends on the surface temperature T_s.

$$Q_{eb} = \varepsilon\sigma\bar{T}_s^4 \left(0.39 - 0.05\sqrt{e_a}\right)\left(1.0 - 0.6F_c^2\right), \tag{1.177}$$

with:

ε	emissivity factor (0.985)
σ	Stefan-Boltzmann's constant = 5.67×10^{-8} in [$J/(m^2sK^4$]
$\bar{T}_s$	the (absolute) water surface temperature in [K].
e_a	actural vapour pressure:

$$e_a = r_{hum} 10^{\frac{0.7859+0.03477T_a}{1.0+0.00412T_a}} \tag{1.178}$$

F_c	fraction of sky covered by clouds (user-defined input)

1.2.1.7 Evaporative heat flux

Evaporation is an exchange process that takes place at the interface between water and air and depends on the conditions both in the water near the surface and the air above it. The evaporation depends on meteorological factors (wind-driven convection) and vapour pressures.

The evaporative heat flux Q_{ev} is defined by:

$$Q_{ev} = L_v E, \tag{1.179}$$

with L_v the latent heat of vaporisation in J/kg water:

$$L_v = 2.5 \times 10^6 - 2.3 \times 10^3 T_s. \tag{1.180}$$

The evaporation rate E is defined as the mass of water evaporated per unit area per unit time [kg/(m^{2}s)]. It is computed by using a form of Dalton's law of mass transfer:

$$E = f\left(U_{10}\right)\left(e_s - e_a\right), \tag{1.181}$$

where the actual vapour pressure e_a and the saturated vapour pressure e_s, is defined as:

$$e_s = 10^{\frac{0.7859+0.03477T_s}{1.0+0.00412T_s}}, \tag{1.182}$$

$$e_a = r_{hum} 10^{\frac{0.7859+0.03477T_a}{1.0+0.00412T_a}}. \tag{1.183}$$

Here r_{hum} is the relative humidity in [-].

Remarks:

- ◇ The relative humidity r_{hum} is specified as a function of time.
- ◇ The vapour pressures are calculated using air and water surface temperature in °C.
- ◇ When the computed E is negative, it is replaced by zero, assuming that it is caused by modelling misfit and not by the actual physical process of water condensation out ot the air into the water.

The evaporation rate is computed from the difference in relative humidity, rather than from the difference in vapour pressure. The contribution to the evaporative heat flux is split in a contribution by forced convection and a contribution by free convection. These are discussed separately.

Forced convection of latent heat

The latent heat flux due to forced convection reads:

$$Q_{ev,\text{forced}} = L_V \rho_a f\left(U_{10}\right)\left\{q_s\left(T_s\right) - q_a\left(T_a\right)\right\}, \tag{1.184}$$

with q_s and q_a the specific humidity of respectively saturated air and remote air (10 m above water level):

$$q_s\left(T_s\right) = \frac{0.62 e_s}{P_{atm} - 0.38 e_s}, \tag{1.185}$$

$$q_a\left(T_a\right) = \frac{0.62 e_a}{P_{atm} - 0.38 e_a}. \tag{1.186}$$

The saturated and remote vapour pressures e_s and e_a are given by Eqs. (1.182) and (1.183). The latent heat L_v is given by equation (1.180)

The wind function in Equation (1.184) is defined as:

$$f\left(U_{10}\right) = c_e U_{10}, \tag{1.187}$$

Free convection of latent heat

Loss of heat due to evaporation occurs not only by forced convection, wind driven, but also by free convection. Free convection is driven by buoyant forces due to density differences (by temperature and/or water vapour content) creating unstable conditions in the atmospheric boundary layer (ABL). Evaporation due to free convection is important in circumstances where inverse temperature/density gradients are present and wind speeds are almost negligible so that the amount of forced convection is small. Neglecting free convection in this situation will lead to underestimating the heat loss. (Ryan *et al.*, 1974) developed a correction to the wind function, accounting for free convection. The derivation of evaporation by just free convection is based on the analogy of heat and mass transfer.

The latent heat flux due to free convection reads:

$$Q_{ev,\text{free}} = k_s L_V \overline{\rho}_a \left(q_s - q_a\right), \tag{1.188}$$

with the average air density:

$$\overline{\rho}_a = \frac{\rho_{a0} + \rho_{a10}}{2}, \tag{1.189}$$

and with the heat transfer coefficient defined as:

$$k_s = \begin{cases} 0 & \text{if} \quad \rho_{a10} - \rho_{a0} \leq 0 \\ c_{fr.\text{conv}} \left\{ \frac{g\alpha^2}{\nu_{air}\overline{\rho}_a} \left(\rho_{a10} - \rho_{a0}\right) \right\}^{1/3} & \text{if} \quad \rho_{a10} - \rho_{a0} > 0 \end{cases} \tag{1.190}$$

where the coefficient of free convection $c_{fr.\text{conv}}$ was calibrated to be 0.14, see (Ryan *et al.*, 1974). The viscosity of air ν_{air} is assumed to have the constant value 16.0×10^{-6} m^2/s. The molecular diffusivity of air α m^2/s is defined as

$$\alpha = \frac{\nu_{air}}{\sigma}, \tag{1.191}$$

with $\sigma = 0.7$ (for air) the Prandtl number. In Equation (1.188), the saturated air density is given by:

$$\rho_{a0} = \frac{\frac{100P_{atm} - 100e_s}{R_{dry}} + \frac{100e_s}{R_{vap}}}{T_s + 273.15}, \tag{1.192}$$

the remote air density (10 m above the water level):

$$\rho_{a10} = \frac{\frac{100P_{atm} - 100e_a}{R_{dry}} + \frac{100e_a}{R_{vap}}}{T_{air} + 273.15}, \tag{1.193}$$

where R_{dry} is the gas constant for dry air: 287.05 J/(kg K) and R_{vap} is the gas constant for water vapour: 461.495 J/(kg K). The specific humidity of respectively saturated air and remote air (10 m above the water level), q_s and q_a are given by Eqs. (1.185) and (1.196). The saturated and remote vapour pressure e_s and e_a are defined in Eqs. (1.182) and (1.183).

Q_{ev} *evaporative heat flux summarized*

The total heat flux due to evaporation then results from adding the forced convection of latent heat in Equation (1.184) and the free convection of latent heat in Equation (1.188):

$$Q_{ev} = Q_{ev,\text{forced}} + Q_{ev,\text{free}} \tag{1.194}$$

$$Q_{ev,\text{free}} = k_s L_V \overline{\rho}_a \left\{ q_s\left(T_s\right) - q_a\left(T_a\right) \right\} \tag{1.195}$$

$$Q_{ev,\text{forced}} = L_V \overline{\rho}_a f\left(U_{10}\right) \left\{ q_s\left(T_s\right) - q_a\left(T_a\right) \right\} \tag{1.196}$$

with:

k_s heat transfer coefficient:

$$k_s = \begin{cases} 0 & \text{if} \quad \rho_{a10} - \rho_{a0} \leq 0 \\ c_{fr.\text{conv}} \left\{ \frac{g\alpha^2}{\nu_{air}\overline{\rho}_a} \left(\rho_{a10} - \rho_{a0}\right) \right\}^{1/3} & \text{if} \quad \rho_{a10} - \rho_{a0} > 0 \end{cases} \tag{1.197}$$

L_V L_v the latent heat of vaporisation

$$L_V = 2.5 \times 10^6 - 2.3 \times 10^3 T_s \tag{1.198}$$

$q_s\left(T_s\right)$ The specific humidity of saturated air

$$q_s\left(T_s\right) = \frac{0.62e_s}{P_{atm} - 0.38e_s} \tag{1.199}$$

$q_a\left(T_a\right)$ The specific humidity of remote air

$$q_a\left(T_a\right) = \frac{0.62e_a}{P_{atm} - 0.38e_a} \tag{1.200}$$

$f(U_{10})$ wind function: $c_e U_{10}$
$\overline{\rho}_a$ average air density:

$$\overline{\rho}_a = \frac{\rho_{a0} + \rho_{a10}}{2} \tag{1.201}$$

ρ_{a0} The saturated air density is given by:

$$\rho_{a0} = \frac{\frac{100P_{atm} - 100e_s}{R_{dry}} + \frac{100e_s}{R_{vap}}}{T_s + 273.15} \tag{1.202}$$

ρ_{a10} remote air density (10 m above the water level):

$$\rho_{a10} = \frac{\frac{100P_{atm} - 100e_a}{R_{dry}} + \frac{100e_a}{R_{vap}}}{T_a + 273.15} \tag{1.203}$$

e_s saturated pressure:

$$e_s = 10^{\frac{0.7859 + 0.03477T_s}{1.0 + 0.00412T_s}} \tag{1.204}$$

e_a remote vapour pressure:

$$e_a = r_{hum} 10^{\frac{0.7859 + 0.03477T_a}{1.0 + 0.00412T_a}} \tag{1.205}$$

R_{dry} gas constant for dry air: 287.05 J/(kg K) is the
R_{vap} gas constant for water vapour: 461.495 J/(kg K)
P_{atm} Atmospheric pressure
U_{10} Wind velocity at 10 m
c_e Dalton number (=0.0015)
r_{hum} relative humidity
T_a air temperature
T_s water surface temperature
$c_{fr.\text{conv}}$ coefficient of free convection (= 0.14)
g gravity
α molecular diffusivity (= ν_{air}/σ)
ν_{air} viscosity of air (=16.0×10^{-6} m^2/s)
σ Prandtl number (= 0.7)

1.2.1.8 Convective heat flux

The convective heat flux is split into two parts, just as the evaporative heat flux. The convective heat flux is divided into a contribution by forced convection and a contribution by free convection.

$$Q_{co} = Q_{co,\text{forced}} + Q_{co,\text{free}} \tag{1.206}$$

Forced convection of sensible heat

The sensible heat flux due to forced convection is computed by:

$$Q_{co,\text{forced}} = \rho_a c_p g(U_{10})(T_s - T_a), \tag{1.207}$$

with:

ρ_a air density as specified in input
$g(U_{10})$ wind speed function is defined following Gill (1982): $c_H U_{10}$

c_H	Stanton number (= 0.0013)
T_a	air temperature
T_s	water surface temperature
c_{pa}	specific heat capacity of air; = 1004.0 J/(kg K)

Free convection of sensible heat

The sensible heat flux due to free convection is computed by:

$$Q_{co,\text{free}} = k_s \overline{\rho}_a c_{pa} \left(T_s - T_a\right) \tag{1.208}$$

with:

k_s heat transfer coefficient:

$$k_s = \begin{cases} 0 & \text{if} \quad \rho_{a10} - \rho_{a0} \leq 0 \\ c_{fr.\text{conv}} \left\{ \frac{g\alpha^2}{\nu_{air}\overline{\rho}_a} \left(\rho_{a10} - \rho_{a0}\right) \right\}^{1/3} & \text{if} \quad \rho_{a10} - \rho_{a0} > 0 \end{cases} \tag{1.209}$$

$\overline{\rho}_a$ average air density:

$$\overline{\rho}_a = \frac{\rho_{a0} + \rho_{a10}}{2} \tag{1.210}$$

ρ_{a0} The saturated air density is given by:

$$\rho_{a0} = \frac{\frac{100P_{atm} - 100e_s}{R_{dry}} + \frac{100e_s}{R_{vap}}}{T_s + 273.15} \tag{1.211}$$

ρ_{a10} remote air density (10 m above the water level):

$$\rho_{a10} = \frac{\frac{100P_{atm} - 100e_a}{R_{dry}} + \frac{100e_a}{R_{vap}}}{T_a + 273.15} \tag{1.212}$$

c_H	Stanton number (= 0.0013)
T_a	air temperature
T_s	water surface temperature
c_{pa}	specific heat capacity of air; = 1004.0 J/(kg K)

1.2.1.9 Input parameters for composite model

The input parameters for the composite model are:

F_c	percentage of sky covered by clouds (time dependent, Meteo data)
T_a	air temperature in °C (time dependent, Meteo data)
r_{hum}	percentage of relative humidity (time dependent, Meteo data)
c_e	Dalton number (default value: 0.0013)
c_H	Stanton number (default value: 0.0013)
c_p	specific heat capacity of water (default value: 3930 J/(kg K) for sea water)

1.2.2 Salinity dispersion

When one of the boundaries of a river or estuary is in contact with the sea, salt water may enter the system. Salt intrusion into a river or estuary is determined by the fresh water discharge on the river side and mixing of the water column by the tides. Mixing occurs due to tidal shear between streamlines of different velocities, tidal trapping of water in tidal flats or side channels, residual currents within the cross section (tidal pumping) and turbulent mixing at small scales. The dominant physical process is the gravitational circulation: it transports salt and thus more dense water near the bed in landward direction and fresh water near the surface in seaward direction. Mixing of the water column by the tide reduces the gravitational circulation.

For the one-dimensional D-Flow 1D the three-dimensional mixing processes are parametrized into a dispersion term. This dispersion term represents the processes that cannot be described in a cross section averaged model. The advection term in D-Flow 1D only represents the transport of salt with the cross sectional averaged flow. In D-Flow 1D a dispersion coefficient is used in the advection-diffusion equation SOBEK3_TRM (2013, §1.7). The advection-diffusion equation is coupled via a state equation, relating salinity and density, to the momentum equation.

For estuaries several formulations for the dispersion coefficient are available. Thatcher and Harleman (1972a) defined a dispersion formula based on field observations in three estuaries in the USA and on a tidal flume experiment. Within their formula they made distinction between shear dispersion which is also present in the fresh water region of an estuary (first part) and tidal dispersion (second part):

$$D = 77\, u\, n\, R^{5/6} + k L_e^2 \frac{u_0}{s_{0,max}} N_r^{1/4} \left| \frac{\partial s}{\partial x} \right| \tag{1.213}$$

in which:

u	flow velocity [m s^{-1}],
n	Manning coefficient [s m$^{-\frac{1}{3}}$],
R	hydraulic radius [m],
k	parameter for tidal dispersion to be defined by the user [-],
L_e	the estuary length [m],
u_0	absolute maximum flood flow velocity [m s^{-1}],
$s_{0,max}$	the maximal salt concentration in the estuary mouth [kg m^{-3}],
N_r	estuarine Richardson number [-],
s	salt concentration [kg m^{-3}],
x	Coordinate along branch [m].

The estuarine Richardson number (N_r) expresses the balance between the stratifying factors (fresh water input) and the mixing factors due to the tide. Therewith it is a measure for the degree of stratification. It is used in all formulations. For low numbers of N_r the estuary is well mixed, while for high numbers it is highly stratified. According Fischer *et al.* (1979) the transition for N_r occurs between 0.08 and 0.8. The estuarine Richardson number is defined as:

$$N_r = \frac{\Delta\rho}{\rho_w} \frac{g h Q_f T}{u_0^2 P_e} \tag{1.214}$$

in which:

$\Delta\rho$	density of sea water minus density of inflowing fresh water [kg/m^3]
ρ_w	density of inflowing fresh water [kg/m^3]
Q_f	fresh water discharge [m^3/s]

T	tidal period [s]
P_e	tidal prism: the water entering the estuary from the sea side during flood [m^3]
g	acceleration due to gravity [m s^{-2}],
h	Total water depth [m]

Various newer dispersion formulations exist. Kuijper and Van Rijn (2011) modified a formulation for the tidal dispersion term described by Savenije (2012). Based on Ippen and Harleman (1961) they added a friction parameter to account for vertical mixing related to the ratio of dissipated energy by means of bed friction and gained potential energy of the fresh river water due to an increase in density. Furthermore, they added depth explicitly in the formulation. They distinguished between convergent (trumpet shaped) and prismatic estuaries (having straight banks). The formulation for prismatic channels is defined at high water slack (HWS):

$$D^{HWS} = 6\alpha_c u_0 d_0 N_r^{1/2} \frac{C}{\sqrt{g}} e^{\phi x} \frac{s^{HWS}}{s_0^{HWS}} \tag{1.215}$$

and for convergent estuaries:

$$D^{HWS} = 60\alpha_c u_0 \frac{h_0}{a} E_0 N_r^{1/2} \frac{C}{\sqrt{g}} e^{\phi x} \frac{s^{HWS}}{s_0^{HWS}} \tag{1.216}$$

in which:

$$\phi = \frac{\delta_u}{2} + \frac{3(a-b)}{2ab} + \frac{1}{2a} \tag{1.217}$$

with:

α_c	calibration factor with default 1, and $0.7 \leq \alpha_c \leq 1.3$ [-],
s_0	salt concentration in the mouth [kg m^{-3}],
ϕ	damping term for the damping of tidal amplitude [m^{-1}],
x	Location increasing into the estuary [m],
δ_u	damping coefficient [m^{-1}],
a	cross-sectional convergence length [m],
b	width convergence length [m].
h_0	average water depth in a branch [m], ,
E_0	tidal excursion length [m],
C	Chézy coefficeint [m$^{0.5}$ s^{-1}],

1.2.3 Sediment transport capacity

The D-Flow 1D module computes the sediment transport capacity according to the following two formulations.

Frijlink

$$T_v = D_{50}\sqrt{g\mu R|S_e|}5\exp\left(\frac{-0.27D_{90}}{\mu R|S_e|}\right) \tag{1.218}$$

$$\mu = \left(\frac{C}{18\ {}^{10}\log\left(\frac{12R}{D_{90}+\frac{3\nu}{\sqrt{gR|S_e|}}}\right)}\right)^{3/2} \tag{1.219}$$

$$S_E = \frac{\Delta\left(h + \frac{u^2}{2g}\right)}{\Delta x} \tag{1.220}$$

T_v	Sediment transport capacity per m bed width [m^2/s]
μ	Bottom roughness factor (ripple factor)
D_{50}	Grain diameter at which 50 % of the grains is smaller [m]
D_{90}	Grain diameter at which 90 % of the grains is smaller [m]
R	Hydraulic radius [m]
S_e	Energy slope [$-$]
C	Chézy coefficient [$m^{1/2}/s$]
ν	Kinematic viscosity [m^2/s]. Normally 1.002×10^{-6}
h	Water level [m]
u	Velocity [m/s]
g	Acceleration due to gravity [m/s^2] (≈ 9.81)
x	Length of branch segment [m]

Van Rijn

$$T_v = 0.053 \frac{T_{gr}^{2.1}}{D_{gr}^{0.3}} \sqrt{(s-1)g} D_{50}^{1.5} \tag{1.221}$$

$$T_{gr} = \frac{u_*^2 - u_{*c}^2}{u_{*c}^2} \text{ if } u_*^2 > u_{*c}^2 \tag{1.222}$$

$$T_{gr} = 0 \text{ if } u_*^2 \leq u_{*c}^2 \tag{1.223}$$

$$u_* = \frac{u\sqrt{g}}{C} \tag{1.224}$$

$$C = 18\ {}^{10}\log\left(\frac{12R}{3D_{90}}\right) \tag{1.225}$$

C	Chézy coefficient related to bed material
u	Average flow velocity [m/s]
u_{*c}	Critical shear stress velocity according to Shields [m/s]

The grain size under water is defined dimensionless by D_gr:

$$D_{gr} = D_{50} \sqrt[3]{\frac{(s-1)g}{\nu^2}} \tag{1.226}$$

$(s-1)$	Relative density
s	s = s/w
r_s	Density of sediment [kg/m^3]. Normally 2650 kg/m^3
r_w	Density of water [kg/m^3]. Normally 1000 kg/m^3
ν	Kinematic viscosity [m^2/s]

The critical shear stress velocity according to Shields u_{*c} can be found with

$$u_{*c} = \sqrt{\theta_{cr}(s-1)gD_{50}} \tag{1.227}$$

$$\theta_{cr} = \begin{cases} 0.24D_{gr}^{-1} & \text{if } D_{gr} \leq 4 \\ 0.14D_{gr}^{-0.64} & \text{if } 4 < D_{gr} \leq 10 \\ 0.04D_{gr}^{-0.1} & \text{if } 10 < D_{gr} \leq 20 \\ 0.013D_{gr}^{-0.29} & \text{if } 20 < D_{gr} \leq 150 \\ 0.055 & \text{if } D_{gr} > 150 \end{cases} \tag{1.228}$$

Note: The sediment transport capacity computed with these formulas is always positive. To give an indication of the direction of the sediment transport in the network, the result is multiplied by the sign of the velocity.

The mentioned formulas are defined in sediment transport capacity per meter bed width. To arrive at a more comprehensible definition the results are multiplied with a certain characteristic width. For closed cross sections, the maximum width of the profile is taken. For open cross sections the wetted perimeter at a certain level (normally 0.30 m) above the bed level is taken.

Note: The parameters that are used in the formulations, like grain diameters, are constant over the whole network

1.3 Hydrodynamics Overland 2DFLOW

1.3.1 Introduction

This section is about the modeling of the lateral coupling between 1D horizontal network flow models (obtained upon averaging over width and depth) and 2D (depth-averaged) horizontal overland flow models in between the network branches, for the modeling of flooding in urban and rural areas:

- flash flooding (or pluvial flooding);
- fluvial flooding;
- coastal flooding;
- groundwater flooding.

Both the physical modeling of such a lateral coupling and its numerical implementation will be considered.

This section assumes the availability of model components that provide, or that are required to compute, the incoming discharge:

- rainfall;
- (upstream) discharges;
- (tidal) water levels;
- groundwater fluxes.

The focus lies on the modeling of the response to these discharges by the flow through an urban or rural area with drainage channels and rivers:

- inundation patterns;
- (maximum) water levels;
- local discharges and flow velocities.

Point of departure is to model the flow through networks of channels and rivers by means of efficient 1D shallow-water network models, and to include a 2D shallow-water model for the flow across any urban or rural area when the discharges or water levels are high enough for such an area to flood. This requires couplings between the two models that become active everywhere in the domain where there is a flow from a 1D network to an adjacent 2D area or vice versa.

A 1D network flow description assumes that the flow dynamics in crosswise direction can be neglected, i.e., the effect of any crosswise flow can be ignored and the water level in crosswise direction can be assumed uniform. For example, 2D/3D dynamic effects in (sharp) bends are not taken into account, i.e., their effect on the global flow behavior is assumed negligible.

When a 1D-modeled channel section is flooded, strong crosswise flow may occur, in which case the assumptions of negligible crosswise flow dynamics and uniform crosswise water level may not be valid anymore. If that is the case, then the only solution is to model that stretch of channel 2D or even 3D.

An alternative solution seems to be to model the flow in channel sections 1D for water levels below the flooding level and to apply 2D modeling for the flood flow above the channel's flooding level when the water level rises above that level.

In this approach a 2D model for the flow in flood plains and other flooded areas lies on top of the 1D channel flow models, with a vertical coupling in between: a *1D-2D modeling with vertical coupling*. This approach poses several problems:

- ◇ Since the water level in crosswise direction is assumed constant in the 1D model and can have a gradient in the 2D model, which water level to use in the 1D model to compute the lengthwise hydrostatic pressure gradient if the 2D layer is active?
- ◇ If the 2D layer is active, crossflow effects are still neglected in the 1D part and therefore only partially included.
- ◇ The flow in lengthwise direction inside the channel and in the flood layer above the channel are not independent. It is not clear how the 1D model and 2D model should be coupled vertically to take this dependency into account properly. Some sort of friction condition between the two layers?
- ◇ The vertical coupling should ensure that the 2D model above the channel bed feels the bed friction. Bed friction is included in the 1D model, but it is not clear how to include it in the 2D model on top of the 1D model. It is also not clear how to include the effect of the flow in the 2D layer in the friction formulation used in the 1D model.

The first two items do not pose a problem if in the, usually small, area covered by the channel network the effect of crossflow momentum can also be neglected in case of flooding. In that case a horizontal coupling between a network of channels modeled 1D and the flood plains between the channels modeled 2D may just as well be considered. This may even be a better approach. Describing the flow in the channels at any water level with one and only one 1D model makes the modeling of this part of the flow problem more transparent and easier to calibrate.

Notice that when a relatively small water layer in flooded areas is present, a flow in the flood plains will cause an at most small crossflow in the channels where the water depth is (much) larger. When the water layer in flooded areas is rather large, the velocity variations and water level gradients are usually moderate to small. In other words, in both cases the effect of any momentum transport across the channels caused by flooding may generally be expected to be negligible, indicating that a *1D-2D modeling with a horizontal coupling* in between may often be adequate. Obviously, the assumption is only valid if the channels in the 1D network are sufficiently deep. In particular, the applied 1D channel models should *not* include any shallow areas like river flood plains.

Summarizing, there are basically two ways to establish a 1D-2D coupling:

- ◇ **vertical 1D-2D coupling:** the 2D area is defined in the entire domain and lies 'on top of' the 1D network where it coincides with channels and rivers. **Note:** in this approach, *the 1D channels and rivers have limited height*. They end at the bottom of the 2D area, where in vertical direction the 2D area starts.
- ◇ **horizontal 1D-2D coupling:** separate 2D areas are defined in between the branches of the 1D network of channels and rivers (domain decomposition approach). NOTE: in this approach, *the 1D channels and rivers have limited width*. They end at the boundaries of

the 2D areas, where in horizontal direction the 2D areas start.

Because of the close connection between the 1D model and the 2D model, the shape of the 2D model affects that of the 1D model or vice verse. With the vertical coupling, the 1D model is limited from above; with the horizontal coupling, the boundaries of the 2D models limit the width of the 1D model, or, equivalently, the 1D model determines the position of the boundaries of the 2D models in between the branches. This is not the case when the 1D model and the 2D model are strictly separated. Seyoum *et al.* (2012) present a vertical coupling by means of the flow through manholes between an otherwise separated 1D sewer network model and 2D overland flow model. Chen *et al.* (2012) and Chen *et al.* (2013) present a horizontal coupling connecting independently designed 1D network models and 2D or 3D shallow-water models through the exchange of information at the boundaries.

An overview[1] of the properties of both types of couplings is presented in Table 1.5. Based on these properties, and on experiences with the vertical coupling technique in SOBEK 2 it was decided to implement a new horizontal coupling for SOBEK 3. **Note:** A question mark has been added to some properties, because the experiences with the new horizontal 1D-2D coupling are of course still limited.

	vertical 1D-2D coupling	horizontal 1D-2D coupling
accurate	$-$	$+$
efficient	$+$	$?$
robust	$+$	$+?$
extendable, easy maintenance	$-$	$+$
easy to set up model	$+$	$+$
modeling restrictions	$+$	$-$
insight in physical modeling	$-$	$+$

Table 1.5: *Estimation of the properties of the vertical 1D-2D coupling versus the horizontal 1D-2D coupling.*

Explanation to the properties of the new horizontal 1D-2D coupling:

- *accuracy* and *insight in physical modeling* is obtained because each 1D-2D interface is a separate weir-type model component that can be set up independently for each 1D channel bank (*no modeling restrictions*);
- tests have so far shown that the *efficiency* is reasonable; further improvement is possible and probably required;
- tests have so far shown that the *robustness* is quite good;
- there are several ways to *extend* the currently implemented 1D-2D coupling; the possibility to use non-matching grids and non-matching time steps at 1D-2D interfaces, to be realized in the future, is particularly interesting because that will improve the efficiency, flexibility and ease of use of the new 1D-2D modeling approach; the algorithm has been set up such that this is fairly easy to realize (*easy maintenance*);
- the developed preprocessing tools make it *easy to set up a 1D-2D model*.

From here on, the focus of this document will lie on the conceptual description of the horizontal 1D-2D coupling. In section 1.3.2 the technique currently applied in D-Flow 1D is described: a technique based on an implicit coupling between the 1D model (as described in section 1.1) and the 2D model (as described in the D-Flow FM technical reference manual) that is solved iteratively by means of an optimized explicit exchange of flow information.

[1] This overview is indicative and for a typical complex 1D-2D application. Depending on the model and the effort spent in its optimization, the comparison of both types of couplings may be different for some of the properties.

Note: The 1D network modeling capabilities of D-Flow FM and a technique to couple a 1D D-Flow FM model to a 2D D-Flow FM model are currently also under development for use in future versions.

A horizontal 1D-2D coupling as mentioned above has never been developed. The principle is:

- a network grid for the 1D channel and river network;
- boundary-fitted (unstructured) grids for the 2D areas in between the branches and in between branches and the domain boundaries;
- horizontal couplings to connect at the interfaces the solution of the 1D network and the solution of the 2D areas.

An example is shown in Figure 1.58.

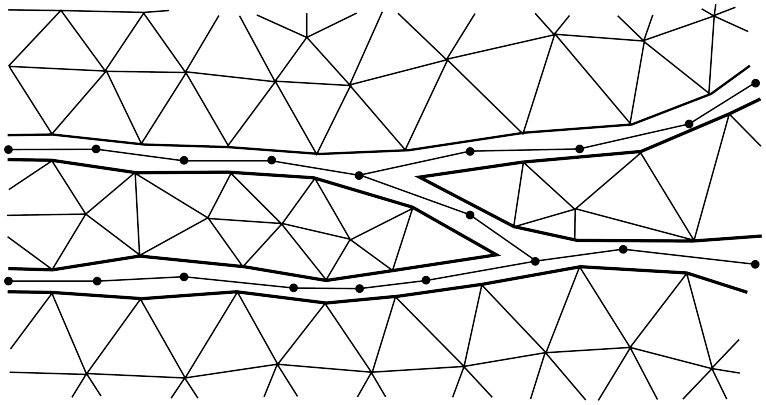

Figure 1.58: *The proposed 1D-2D modeling with horizontal coupling.*

Flooding in urban and rural areas can also be modeled by the use of just a 2D model, provided that a sufficiently high resolution is available locally to capture the details of the flow through the 1D network. An example of a fully 2D D-Flow FM river-overland flow model is given in Figure 1.59.

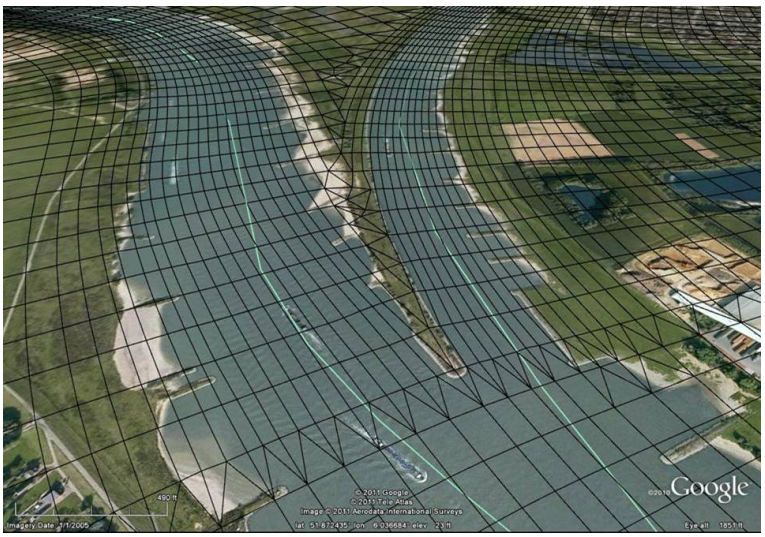

Figure 1.59: *Example of a D-Flow FM grid with sufficient resolution across channels to capture the details in that direction.*

Because of the (much) larger number of grid cells inside the channels, an excessive amount of computational time will then be required to simulate large-scale network flows. Moreover,

since in a 2D model the momentum equation in crosswise channel direction is solved as well, a large time-step restriction and hence very long computational times may result, because of the partially explicit nature of the applied numerical scheme in combination with the usually small grid size across channels. This may even be the case when narrow channels are discretized with a single row of long but thin grid cells with high aspect ratio.

The conclusion reads that the large-scale modeling of flooding in urban and rural areas with just a 2D model is not convenient.

Kim *et al.* (2012) describe a model consisting of a coupling between a hydrological model and a 2D hydrodynamic model using unstructured grids of regular grid cells with refinements in channels. The locally fine grids (also in streamwise direction, which is a waste of resolution in elongated channels) require the use of small time steps. By applying local time stepping the efficiency has been improved by 50–70 %. See also, e.g., Bradford and Sanders (2002), Begnudelli *et al.* (2008), Gallegos *et al.* (2009), and Sanders *et al.* (2010).

Note: The coupling between 1D and 2D in *longitudinal direction*, i.e., the coupling of 1D models to 2D models at the boundaries of 1D branches and in the direction of the 1D flows in those branches (instead of through 1D lateral discharges along the branches that are normal to the direction of the 1D flows, as in the lateral 1D-2D coupling), is under development and not yet available in this version. Because an efficient longitudinal 1D-2D coupling can be realized using standard open-boundary formulations, its realization in the future is considered fairly straightforward. With regard to its accuracy, integration of the discharges and averaging of the water levels across the (parts of) the outer 2D cell faces connected to a 1D cross-sectional area suffices for a longitudinal coupling from 2D to 1D. For the longitudinal coupling in the other direction, from 1D to 2D, the average discharge and water level at a 1D coupling boundary need to be distributed across the faces of the 2D grid cells where a 2D domain is connected to an outer 1D channel cross-section. For the discharge a distribution based on the conveyance across the 1D width could be used, but it may also be possible to use the distribution as it is computed at the interface in the 2D domain. These aspects are currently under consideration in the development of a longitudinal horizontal 1D-2D coupling compatible with the lateral horizontal 1D-2D coupling described below.

1.3.2 Lateral horizontal 1D-2D coupling

In this approach, 1D network flow models are combined with 2D overland flow models that lie *in between* (and *not* across, as in vertical 1D-2D couplings) the branches, as well as in between branches and the boundaries of the domain. A schematic of this modeling approach is depicted in Figure 1.60.

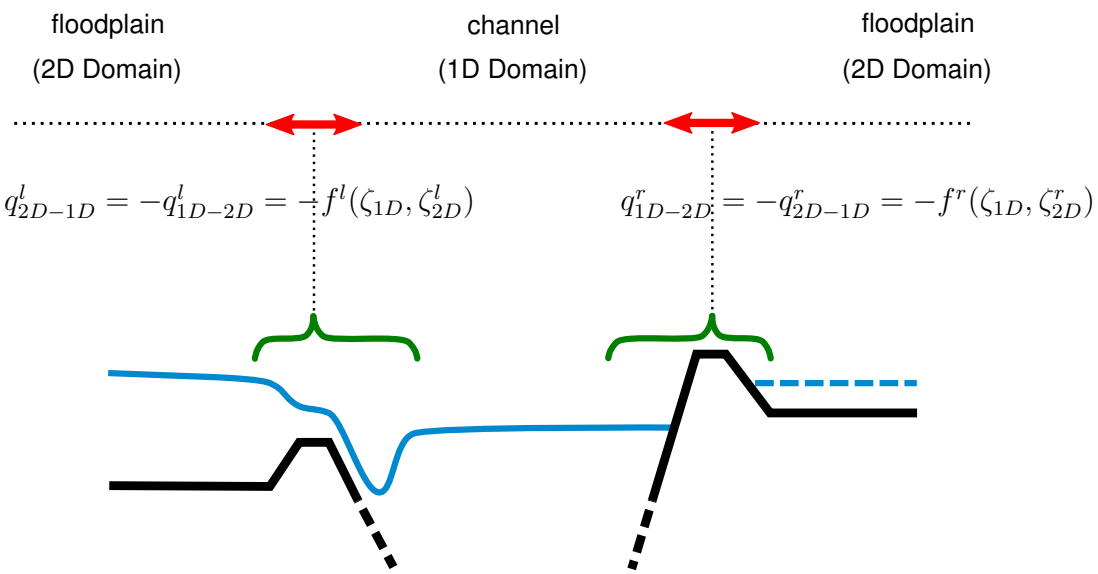

Figure 1.60: *Principle of the lateral horizontal 1D-2D coupling, with 2D modeled areas lying next to 1D modeled waterways.*
The two red horizontal two-sided arrows indicate the couplings that independently model the flow across two separate waterway banks. That flow exchange, (usually) not modeled in detail, occurs in the strips along the banks left and right of the waterway indicated by the green braces. For clarity, the size of the dikes and the free-surface variation across the dike at the left interface have been exaggerated in the figure.
The lateral discharge per unit length $q_{2D\text{-}1D}$ from a 2D domain to a 1D domain is a function of the water levels $\zeta_{2D,I}$ and ζ_{1D} of those domains at the interface. Because of mass conservation, the lateral discharge per unit length in the other direction, $q_{1D\text{-}2D}$, equals $-q_{2D\text{-}1D}$.

Note: Two ways of horizontal 1D-2D coupling are envisioned: embedded (in future D-Flow FM) and with strict model separation (in D-Flow 1D and presented here).

The principle of the **embedded lateral horizontal 1D-2D coupling** is shown in Figure 1.61.

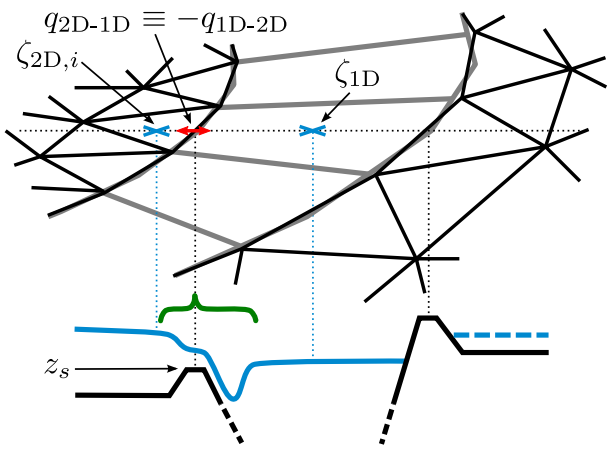

Figure 1.61: *Principle of the embedded lateral horizontal 1D-2D coupling, more or less applying and solving flow equations across the 1D-2D interface.*

The principle of the **lateral horizontal 1D-2D coupling with strict model separation** is

shown in Figure 1.62.

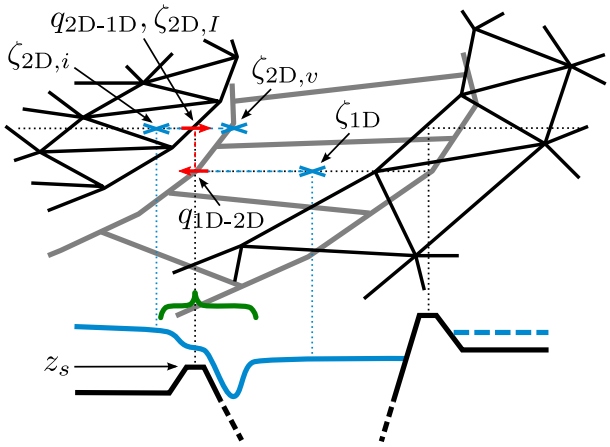

Figure 1.62: *Principle of the lateral horizontal 1D-2D coupling with strict model separation, using separate models for the 2D and the 1D areas coupled at the interfaces by coupling conditions.*

In the Figure 1.61 and Figure 1.62, the 2D areas are discretized using unstructured grids, but the use of other type of grids may be feasible as well. The figures illustrate that it will generally not be possible to have the 2D grids coincide exactly with the boundaries of the 1D channels. Note, however, that the construction of those boundaries is an approximation anyway and that the gap can be made small compared to the 2D grid size in which case its effect will be negligible.

Description and properties of both lateral horizontal 1D-2D couplings:

- ◇ In D-Flow FM at lateral horizontal 1D-2D interfaces: mass conservation (single discharge $q_{\text{2D-1D}} \equiv -q_{\text{1D-2D}}$ in Figure 1.61) and a flow link, i.e., some discretized normal momentum equation using the nearest 1D and 2D water level (ζ_{1D} and $\zeta_{\text{2D},i}$ in Figure 1.61) and some distance between them to determine a water level gradient (not necessarily normal to the interface). If an (explicit) discretization of the convection terms is included (not clear which velocity variables to use here), there may be a severe time-step stability restriction in case of a narrow 1D channel and (hence) small 2D grid cells.
  Advantage: fully implicit coupling of water levels.
  Disadvantages: not flexible: 1D and 2D grids must match at interfaces, 1D and 2D time steps must be equal, and intertwining of 1D implementation and 2D implementation due to a fairly strong coupling between 1D modeled areas and 2D modeled areas (at each next time level one large overall system of equations is to be constructed for the determination of the next water level).
- ◇ In the coupling technique of SOBEK 3 at lateral horizontal 1D-2D interfaces: mass conservation ($q_{\text{1D-2D}} = q_{\text{2D-1D}}$ in Figure 1.62) and a q-h relation, i.e., a reduced normal momentum equation using the 1D and 2D water level at the interface at the same location (ζ_{1D} and $\zeta_{\text{2D},I}$ in Figure 1.62). This defines an implicit coupling at interfaces that is solved iteratively using explicit updates of flow information crossing the interfaces.
  Advantages: flexible: allows in principle non-matching 1D and 2D grids as well as different 1D and 2D time steps. Because of the explicit iterative updates, interpolations are feasible and of moderate complexity. Inherent parallelism: due to the explicit update of coupling information across the 1D-2D interfaces, the update of the solution per 2D area consists of independent computations. Strict separation of 1D modeled areas and 2D modeled areas, hence a strict separation between the 1D implementation and the 2D implementation is possible, with an explicit (iterative) exchange of information through a generalized

implementation of lateral discharges (1D) and boundary conditions (2D).
Disadvantage: requires an iterative solution procedure to be optimized, which is likely to require more advanced 1D lateral discharge implementations and 2D boundary condition implementations than currently available, especially when going for fast convergence.

Because of the advantages, and especially because it is the only technique that allows to combine two different computational models with each their own software implementation (in this case D-Flow 1D and 2D D-Flow FM), it has been decided to equip D-Flow 1D with a lateral horizontal 1D-2D coupling with strict model separation.

1.3.3 The D-Flow 1D lateral horizontal 1D-2D coupling with strict model separation

The SOBEK 3 lateral horizontal coupling technique works as follows:

- Discretize the 1D network using a 1D network model. **Note:** It is important in this step to clearly define the lateral boundaries of the channels, as they define the position of the interfaces between the 1D network at one side and the 2D areas where an overland model will be used at the other side.
- Discretize the 2D areas in between the channels of the 1D network model and the 2D areas in between the 1D channels and the outer boundaries of the (2D) domain of simulation by D-Flow FM-type unstructured grids.
  Matching grids at the interfaces (like in the Figure 1.58 and Figure 1.62) is undoubtedly the easiest option to realize. This may require the subdivision of grid cells of the 1D model in stretches where the orientation of the channel changes significantly, to ensure a proper representation of curved channel boundaries by the straight cell faces of the 2D grid. It will also require a redesign of the grid of the 1D model in stretches where the 1D grid size varies significantly, to avoid excessively large grid-cell variations in the 2D grid.
  Since the coupling between the 1D network model and the 2D overland models will be explicit per iteration per time step (implicit per time step upon convergence), *non-matching grids* are not much more complicated. With suitable integration/interpolation procedures of discharges and water levels at or near the interfaces (easy to apply because the coupling is explicit), the grids of the 1D network model and the 2D overland models could be designed fully independently. The subdivision of certain 1D grid cells may then still be useful, to avoid a loss of resolution in the case of flow across a 1D channel. This is because information of a 2D model will be integrated if the 1D grid has a coarser resolution at the interface than the 2D grid, while information of a 1D model to an adjacent 2D model will then be interpolated. So with a fine-grid 2D model on both sides of a coarse-grid 1D model section, 2D flow details along the interface may get lost when crossing the channel, because the interpolation of 1D model data applied after crossing the channel will not fully restore the details present in the 2D solution prior to crossing and lost in the integration. The possibility of non-matching grids is especially useful to avoid that an irregular grid size distribution in the 1D model affects the quality of the grid of the 2D models, thereby eliminating the need for a redesign of the 1D grid.
- Define the couplings between the 1D network model and the 2D overland models, by applying mass conservation across the interfaces (set the lateral discharge of the 1D model equal to the discharge at the boundary of a 2D model) and by a dynamic condition. The latter describes the discharge throught the interface as a function of the water level slope normal to the interface: a weir-type q-h relation that depends on dike height, dike shape, that includes resistance terms, etc.
- Solve the couplings (iteratively, if necessary) by solving alternately per time step the 1D network flow model and the 2D overland flow models using optimized updates of coupling information (optimized multiplicative Schwarz, a DD algorithm).
  At present, the best procedure seems to be to first do an inexpensive 1D network computation with an optimized combination of the two coupling conditions (using the solution of

the 2D models of the previous time step at the interfaces), to determine a first estimate of the lateral discharges at the next time level. Then compute the 2D models with an optimized (non-reflecting or Riemann) combination of the two coupling conditions as boundary condition at the interfaces, using the updated water levels of the 1D model. Repeat those two steps if necessary. The final step is a 1D network computation with imposed lateral discharge, i.e., the discharge at the interface computed in the last executed 2D model runs. This final step is to ensure mass conservation.

Note:

- ◇ In the above approach, the assumption is that the 1D channels are narrow enough for the lateral momentum changes across the channel width to be negligible. In case of flooding, the flow (lateral discharge) is simply passed from one bank to the other. Friction losses across the channel banks (due to the presence of levees or dikes) are to be modeled by including them in the q-h relations applied at the interfaces. Remark that such a modeling is also required when the flow is modeled full 2D everywhere (i.e., also inside the channels) and a 2D grid is used that is too coarse to capture 'small-scale' features like levees and dikes in detail.
- ◇ The application of optimized coupling conditions required an extension of the available boundary-condition implementation of D-Flow FM and an extension of the implementation of lateral discharges in SOBEK, as expected.
- ◇ Supercritical flow at the interfaces into or out of the 2D areas may require special measures. In the former case, two boundary conditions should normally be applied; in the latter, no boundary conditions should be applied. Some experience is available in changing the number of applied boundary conditions as a function of the flow regime in a 1D shallow-water model.
  For the time being the flow at the 2D side of 1D-2D interfaces is assumed to be subcritical. The flow *across* a 1D-2D interface, however, may be both subcritical and supercritical, i.e., both drowned-weir flow and free-weir flow are supported.

1.3.3.1 Properties of the D-Flow 1D lateral horizontal 1D-2D coupling

Advantages:

- ◇ Modular approach: the 1D model and the 2D models can be separate codes, loosely connected by the exchange of 'external' data determining the lateral discharges and the boundary values to be applied. This includes the possibility of using the implementation of different model equations, of using different (implementations of) numerical algorithms, and of using different (parameter) settings in different 1D and 2D areas, which gives much room for optimization.
- ◇ Modular approach: independent development of the 1D models and the 2D models remains possible.
  Note: Of particular interest is the extension of D-Flow FM (and D-Flow 1D) with a more implicit time integration that may significantly reduce the current time-step stability restriction. This is expected to improve the efficiency of D-Flow FM (and D-Flow 1D) with at most a marginal loss in accuracy. The latter is especially true if in addition a variable time step will be implemented: small time steps when transient phenomena like flooding are to be modeled; large time steps when the flow is nearly stationary.
- ◇ Modular approach: fully independent design of the 1D schematization and of each 2D schematization.
  Note: This advantage will become fully available when in the future the coupling will be extended to non-matching grids and the use of different 1D and 2D time steps at the interfaces.
- ◇ Inherent parallelism: models of all 2D areas are independent and can be computed in

parallel.

Disadvantages:

- Non-proven technology.
  Note: The results obtained so far with the current implementation confirm the feasibility of the method.
- Several iterations are generally required per time step.
  Note: Further study and development is required to check and improve the efficiency of this 1D-2D coupling method. For example, it may be possible to combine the 1D-2D coupling iterations with the existing iterative solution procedures that solve for the water level. Further optimization of the 1D-2D iterative solver is also possible.
- 2D grid generation required per 2D area, which may not be obvious when there are many of such areas.
  Note: Algorithms have been designed to fully automate this procedure, using the D-Flow FM grid generation tools already developed.

1.4 The D-Flow 1D lateral horizontal 1D-2D coupling — the details

The flow in the domain is either modeled 1D or it is modeled 2D. The flow is modeled 1D in the branches of the network(s) of channels and rivers. In the areas in between the flow is modeled 2D. A horizontal 1D-2D coupling is applied to connect the 1D and 2D flow models.

The elements to be discerned in the development of a horizontal 1D-2D coupling are:

- 1D network flow model,
- 2D flood flow model,
- horizontal 1D-2D coupling.

The issues to be considered in the coupling are the physical modeling of the coupling ($\rightarrow$ the coupling equations) and the numerical implementation of the coupling ($\rightarrow$ discretization and solution procedure). Applying the coupling equations is trivial when they are discretized explicitly in time. However, since the 1D and 2D flow models that will be used (SOBEK, D-Flow FM) are partially implicit in time, the coupling equations must be implemented implicitly in time as well, to avoid a loss in stability and accuracy. In fact, the space and time discretization of the flow models dictates the numerical implementation of the coupling equations. For this reason, the discretization of the flow models as far as relevant for the implementation of the coupling is first presented, following a memo by Kernkamp (Kernkamp, 2008) and the D-Flow FM technical reference manual (D-Flow FM TRM, 2014). Next, two ways of lateral horizontal 1D-2D coupling are presented: the one under development for D-Flow FM (embedded coupling) and the one implemented in SOBEK 3 (coupling with strict model separation).

1.4.1 1D and 2D flow modeling

The flow in the 2D parts of the domain is modeled by the 2DH shallow-water equations. Because of the numerical implementation that is used both in SOBEK and in D-Flow FM, the 2DH continuity equation and 2DH momentum equation may be written as:

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0 \ , \tag{1.229)a}$$

$$\frac{\partial \mathbf{u}}{\partial t} + g\nabla\zeta + b\mathbf{u} = \mathbf{e} \ , \tag{1.229)b}$$

with $\mathbf{e}$ all terms in the momentum equation that are discretized explicitly (convection, viscosity,

Coriolis, wind force) and with $b\mathbf{u}$ the bottom-friction term in quasi-linear form that is discretized implicitly, where b is the linearization coefficient (e.g., $g|\mathbf{u}|/(C^2h)$) that is evaluated explicitly (Picard linearization). Furthermore, $h = h(\mathbf{x}, t) = \zeta - z_b$ is the total water depth, with $\zeta = \zeta(\mathbf{x}, t)$ and $z_b = z_b(\mathbf{x})$ respectively surface elevation and stationary depth relative to a horizontal plane of reference, $\mathbf{u} = \mathbf{u}(\mathbf{x}, t) = (u, v)^{\mathrm{T}}$ is the horizontal velocity vector, and $\nabla = (\partial/\partial x, \partial/\partial y)^{\mathrm{T}}$ is the horizontal gradient operator.

The numerical implementation of the 1D depth- and width-averaged shallow-water equations in D-Flow 1D and D-Flow FM is similar, hence these equations are written similarly:

$$\frac{\partial A}{\partial t} + \frac{\partial (Au)}{\partial x} = q \ , \tag{1.230)a}$$

$$\frac{\partial u}{\partial t} + g\frac{\partial \zeta}{\partial x} + bu = e \ , \tag{1.230)b}$$

with A the wetted cross-sectional area, u the average flow velocity over the wetted cross-sectional area in the longitudinal direction x of the 1D channel, and with q the incoming lateral discharge per unit length.

Because of the applied staggered-grid technique, momentum equation Equation (1.229)b is discretized per normal velocity component u that are defined at the cell faces:

$$\frac{u_f^{n+1} - u_f^n}{\Delta t} + \frac{g}{\Delta x_u}\big(\theta(\zeta_2^{n+1} - \zeta_1^{n+1}) + (1-\theta)(\zeta_2^n - \zeta_1^n)\big) + b^n u_f^{n+1} = e^n \ , \tag{1.231}$$

with ζ_2 and ζ_1 the water level at the circumcenter downstream (in positive u_f-direction) and upstream (in negative u_f-direction) of the cell face, with Δx_u the distance between these two circumcenters, and with θ the parameter that determines the level of implicitness of the terms taken implicitly.

The discretization of Equation (1.230)b has exactly the same form as Equation (1.231).

The mass-conservative finite volume discretization of Equation (1.229)a is written in a generic form that also represents the discretization of Equation (1.230)a:

$$\frac{V_c^{n+1} - V_c^n}{\Delta t} + \sum_{\text{cell faces}} A_f^n s_{c,f}\big(\theta u_f^{n+1} + (1-\theta)u_f^n\big) = \Delta x_q q_c^* \ , \tag{1.232}$$

with $V_c = V_c(\zeta_c)$ the water level dependent volume of water that is contained in a control volume (the grid cells) over which Equation (1.229)a is integrated and subsequently discretized, with the A_f the surfaces of the *vertical* cell faces (where the velocity unknowns u_f are located), with q_c^* the average value over the time step[2] of some incoming lateral discharge per unit length q_c for the grid cell (mass control volume) under consideration, and with Δx_q the horizontal length of the part of the cell where depth-integrated mass flow q_c is specified. If both ζ and z_{b_c} are approximated piecewise constant per grid cell, this reads $V_c = S_\zeta(\zeta_c - z_{b_c})$, with S_ζ the constant *horizontal* surface of the control volume (the surface of a grid cell). In that case the first term in the left-hand side of Equation (1.232) can be written as $(V_c^{n+1} - V_c^n)/\Delta t = S_\zeta(\zeta_c^{n+1} - \zeta_c^n)/\Delta t$.

The sum in Equation (1.232) is taken over the mass fluxes through the faces of the grid cell under consideration (the faces of the mass control volume). They are approximated per cell

[2]The level of implicitness of q_c turns out to be irrelevant in the horizontal 1D-2D coupling, hence no θ-weighing here.

face by the product of the wetted cell-face surface $A_f = A_f(\zeta)$ and normal velocity u_f[3]. The direction of the mass flux depends on the direction of u_f relative to the grid cell (outward or inward) and is taken care of by the switches $s_{c,f}$ for the faces of the cell:

$$s_{c,f} = \begin{cases} 1 & \text{if } u_f > 0 \text{ is directed } \textit{outward} \text{ of cell } c \ , \\ -1 & \text{if } u_f > 0 \text{ is directed } \textit{inward} \text{ of cell } c \ . \end{cases} \tag{1.233}$$

An external discharge q_c has been added in the right-hand side of Equation (1.232) to have this discretization in a form that can also represent the discretization of 1D continuity Equation (1.230)a. In that case this reads $V_c(\zeta) = A_\zeta(\zeta)\Delta x_\zeta$, with A_ζ the water level dependent wetted cross-sectional area at the location of the ζ-point and with Δx_ζ the length of a 1D control volume, i.e., the distance between two adjacent u-points where the two mass fluxes in the sum of Equation (1.232) are computed. The generalization to ζ-points where several 1D branches are connected to (connection nodes) is straightforward. The A_f in the 1D mass fluxes are the wetted vertical faces of the 1D mass control volumes that are normal to the channel axis, i.e., the wetted cross-sectional areas at the location of the u-points.

The discretized time derivative in Equation (1.232) is generally nonlinear in the unknown ζ_c^{n+1}, the water level at the cell center/control volume. Its determination therefore requires linearization in combination with an iterative solution procedure:

$$\frac{V_c^{n+1} - V_c^n}{\Delta t} \approx \frac{V_c^{p-1} + (\zeta_c^p - \zeta_c^{p-1})(\partial V/\partial \zeta)^{p-1} - V_c^n}{\Delta t} = \frac{V_c^{p-1} + (\zeta_c^p - \zeta_c^{p-1})S_\zeta^{p-1} - V_c^n}{\Delta t} \ , \tag{1.234}$$

where the superscripts $p-1$ and p indicate respectively the previous and the next iterative approximation of variables at the next time level $n+1$.

Variable S_ζ in Equation (1.234) represents the (*horizontal*) surface of the mass control volume at the free surface. In 1D $S_\zeta = W_\zeta \Delta x_\zeta$, with $W_\zeta = W_\zeta(\zeta_c)$ the channel width at the free surface at the location of the ζ-point. In 1D S_ζ is a constant per mass control volume if channel width W_ζ is constant over the depth. In 2D S_ζ is constant per grid cell (and equal to the surface of the grid cell) as soon as the water level is above the shallowest bottom level inside a grid cell, and hence a constant per mass control volume if, as mentioned above, both ζ and z_b are approximated piecewise constant per grid cell.

Substituting Equation (1.234), Equation (1.232) can be written as (replacing u_f^{n+1} by u_f^p and q_c^* by q_c^p, since when the ζ_c^{n+1} are determined iteratively, the u_f^{n+1} and q_c^* are as well):

$$\begin{aligned} &\frac{S_\zeta^{p-1}\zeta_c^p}{\Delta t} + \sum_{\text{cell faces}} A_f^n s_{c,f} \theta u_f^p - \Delta x_q q_c^p \\ &= \frac{S_\zeta^{p-1}\zeta_c^{p-1} - V_c^{p-1}}{\Delta t} + \frac{V_c^n}{\Delta t} - \sum_{\text{cell faces}} A_f^n s_{c,f}(1-\theta) u_f^n \ . \end{aligned} \tag{1.235}$$

Equation (1.231) can be rewritten as:

$$u_f^p = R_u^n - F_u^n(\zeta_2^p - \zeta_1^p) \ , \tag{1.236}$$

[3]Please recall that often an upwind approximation is used for the wetted face surface A_f in Equation (1.232). Since A_f is evaluated explicitly, the upwind direction is determined by u_f^n and *not* by the velocity approximation that is actually used per time step, i.e., $\theta u_f^{n+1} + (1-\theta)u_f^n$. As a result, the evaluation of A_f may be *downwind* at locations and time steps where the flow normal to a face changes direction. Since this normally occurs at at most a small number of faces while u_f at these faces will then be small, the destabilizing effect of this downwind discretization is expected to be negligible.

with:

$$F_u^n = \frac{1}{1+b^n\Delta t}\frac{g\Delta t}{\Delta x_u}\theta,\ R_u^n = \frac{1}{1+b^n\Delta t}\left(u_f^n + \Delta t e^n - \frac{g\Delta t}{\Delta x_u}(1-\theta)(\zeta_2^n - \zeta_1^n)\right)\ . \quad (1.237)$$

Substitution of Equation (1.236) in Equation (1.235) gives:

$$\begin{aligned}&\left(\frac{S_\zeta^{p-1}}{\Delta t} + \sum_{\text{cell faces}} A_f^n \theta F_u^n\right)\zeta_c^p - \sum_{\text{cell faces}} A_f^n \theta F_u^n \zeta_a^p - \Delta x_q q_c^p \\ &= \frac{S_\zeta^{p-1}\zeta_c^{p-1} - V_c^{p-1}}{\Delta t} + \frac{V_c^n}{\Delta t} - \sum_{\text{cell faces}} A_f^n s_{c,f}(1-\theta)u_f^n - \sum_{\text{cell faces}} A_f^n s_{c,f}\theta R_u^n\ ,\end{aligned} \quad (1.238)$$

with ζ_c^p the value of ζ^p at the grid cell/control volume under consideration, and with the ζ_a^p the value of ζ^p at the adjacent grid cells/control volumes.

Please recall that the above equations (Equation (1.235), Equation (1.236), Equation (1.237), and Equation (1.238)) have been formulated such that they represent the equations used in 2D simulations as well as in 1D simulations.

The equations Equation (1.238) per control volume, together with the discretized boundary conditions, form the system of equations to be solved for the determination of the next iterative estimation ζ^p of the solution ζ^{n+1} that is sought.

Note: A couple of remarks on solving Equation (1.238):

Linearization coefficient S_ζ^{p-1} in the left-hand side of Equation (1.238) as well as the first term in the right-hand side are to be updated at each nonlinear iteration. The first means that the matrix of the system of equations to be solved changes and hence has to be rebuilt at each nonlinear iteration. Keeping S_ζ^{p-1} fixed at its first value $S_\zeta^0 = S_\zeta^n$ inside each nonlinear iteration loop would prevent that, and would allow the reuse of the first matrix decomposition at the remaining iterations. This is not applied in D-Flow FM or SOBEK as it may be at the expense of quite some convergence speed. By keeping the value of S_ζ^{p-1} fixed, the convergence speed changes from quadratic (Newton linearization) to linear, which is generally considerably slower. This applies in particular to typical SOBEK applications where $S_\zeta^{p-1} = W_\zeta^{p-1}\Delta x_\zeta$. Because W_ζ can be a strongly varying function of ζ, the value of W_ζ^{p-1} may change significantly within a nonlinear iteration loop. On the other hand, since both D-Flow FM and SOBEK use rather small time steps, S_ζ^{p-1} will probably not vary much within the nonlinear iteration process per time step. It may be worth investigating if by keeping the linearization fixed, or by updating the matrix only every two or three iterations (or only the first couple of iterations, when the nonlinear changes of S_ζ^{p-1} are the largest), a substantial decrease in computational time could be obtained.

In typical D-Flow FM applications, the water level will be largely above the bottom, so S_ζ^{p-1} will generally be constant nearly everywhere anyway and the nonlinear iteration loop will converge very rapidly. However, in drying and flooding areas, typical for flooding applications, keeping S_ζ^{p-1} fixed in D-Flow FM within each nonlinear iteration loop may have a negative effect on the nonlinear convergence speed. This *only* applies when the discrete approximation of the bathymetry is different from that of the water level, i.e., when it is not piecewise constant. Else, as mentioned before, $S_\zeta^{p-1} = S_\zeta$ is always constant and hence system of equations Equation (1.238) linear.

It needs to be mentioned here that investigations are going on to enhance the efficiency of D-Flow FM by a modification of the algorithm that allows the use of larger time steps.

Efficiency issues related to any nonlinear behavior of $V(\zeta)$ (variable S_ζ^{p-1}) will then become more pronounced. These issues are avoided by having V linear in ζ, i.e., by approximating the bathymetry piecewise constant per grid cell.

When the S_ζ^{p-1} and hence the matrix are updated, the A_f^n could be updated as well, making it effectively A_f^{p-1}. In particular, their upwind value could be adapted to the direction of the velocity approximation at the previous iteration $\theta u_f^{p-1} + (1-\theta)u_f^n$. While upon convergence an A_f^n may have become a downwind value (because the sign of $\theta u_f^{n+1} + (1-\theta)u_f^n$ may be different from that of u_f^n, cf. Footnote 3), the A_f^{p-1} will always be upwind. This may improve both the accuracy and the stability of the discretization. However, it introduces a switch in the solution algorithm (a small change in some u_f^{p-1} may result in a large change in the A_f^{p-1} value at that location) that may hamper convergence. On the other hand, by definition that large change can only occur at a change in velocity direction and hence is always multiplied by a small velocity value. Just because this issue is only relevant at changes in the velocity direction normal to a cell face and hence small normal velocities, the gain in accuracy and stability that can be expected of this modification is likely to be small. That gain may *not* be small if u_f^{n+1} in Equation (1.232) is multiplied by A_f^{n+1} instead of by A_f^n, with the A_f^{n+1} a function of both the u_f^{n+1} (for their upwind location) and the ζ^{n+1} (for their value). NB, note that the iterative determination of the A_f^{n+1} by subsequent updates of their approximations A_f^{p-1} can conveniently be combined with the updates S_ζ^{p-1} of that other geometry-dependent variable.

At each nonlinear iteration, linear system of equations Equation (1.238) is to be solved. An iterative solver is used for this. As long as the outer nonlinear iteration loop has not converged (ζ^p not yet a close approximation of ζ^{n+1}), full convergence of the inner linear iteration loop is not required. Exploiting this by making the convergence criterion of the inner iteration loop a small fraction of the convergence error of the outer iteration loop (inexact solvers) may give a nice reduction in computational time while having virtually no effect on the quality of the solution.

Strict mass conservation, independent of the number of nonlinear iterations, is obtained by taking V_c^{n+1} equal to $V_c^{p-1} + S_\zeta^{p-1}(\zeta_c^p - \zeta_c^{p-1})$. This follows directly from Equation (1.235). Strict mass conservation, independent of the number of nonlinear *and* linear iterations, is obtained by computing V_c^{n+1} directly from Equation (1.232), using the most recent approximations of u_f^{n+1} and q_c^*:

$$V_c^{n+1} = V_c^n + \Delta t\Big(\Delta x_q q_c^p - \sum_{\text{cell faces}} A_f^n s_{c,f}\big(\theta u_f^p + (1-\theta)u_f^n\big)\Big) \ . \tag{1.239}$$

This may be a rather inaccurate approximation of the genuine solution of Equation (1.232) if the nonlinear (and linear) iteration loop has branched an insufficient level of convergence. Moreover, solving Equation (1.232) with limited accuracy weakens the implicit character of the time integration scheme and may have a negative effect on stability.

1.4.2 Embedded lateral horizontal 1D-2D coupling

The embedded horizontal 1D-2D coupling is currently under development and will become available within stand-alone D-Flow FM. For completeness sake a short and preliminary description of this coupling is given.

Across the interface additional momentum equations like Equation (1.236) are applied (cf. Figure 1.61):

$$s_{c,I} u^p_{\text{2D},I} = s_{c,I} R^n_I - F^n_I (\zeta^p_{\text{1D}} - \zeta^p_{\text{2D},i}) \ , \tag{1.240}$$

with the $u_{\text{2D},I}$ the values of additional 2D velocity components u_f along the interface, and with the R^n_I and F^n_I as in Equation (1.237).

Similar to Equation (1.233), directional switch $s_{c,I}$ is set to 1 or -1, depending on whether $u_{\text{2D},I} > 0$ is directed from the 2D domain to the 1D domain or vice versa. The switch is included in Equation (1.240) since here, unlike in Equation (1.236) and Equation (1.231) where the water level difference $\zeta_2 - \zeta_1$ is set in positive u_f-direction, the water level difference is set from the 1D domain to the 2D domain, which may be either in positive $u_{\text{2D},I}$-direction or in negative $u_{\text{2D},I}$-direction.

The Δx_u in the R^n_I and F^n_I are taken equal to the local effective thickness of the interface. The b^n used in the R^n_I and F^n_I contain additional terms to model the losses across the interface due to, e.g., the presence of a bank or the sudden change in water depth. How to deal with the e^n in the R^n_I is a subject of investigation. They may contain special discretizations modeling the convection of normal momentum across the interface. Alternatively, those e^n may be set to zero. Since the complex flow across interfaces cannot be modeled in detail anyway (the variation of velocity and water level across interfaces is not included in the discretization), the effect of convection (and viscosity, Coriolis, wind force) across a 1D-2D interface may just as well be ignored. Since the 1D-2D interfaces form only a small part of the entire modeling area, this is a viable approach, especially when the net effect of the omitted terms across the interface is included in the b^n (subgrid modeling).

The mass flux through the 1D-2D interface is $A^n_I s_{c,I}(\theta u^p_{\text{2D},I} + (1-\theta) u^n_{\text{2D},I})$, cf. the second term in the left-hand side of Equation (1.232), with A_I the surface of the relevant part of the flow area across the interface. So we have:

$$Q^p_{\text{2D-1D}} = A^n_I s_{c,I}(\theta u^p_{\text{2D},I} + (1-\theta) u^n_{\text{2D},I}) \text{ and } Q^p_{\text{1D-2D}} = -A^n_I s_{c,I}(\theta u^p_{\text{2D},I} + (1-\theta) u^n_{\text{2D},I}) \ , \tag{1.241}$$

with $Q^p_{\text{2D-1D}}$ and $Q^p_{\text{1D-2D}}$ respectively the discharge from the 2D domain to the 1D domain and from the 1D domain to the 2D domain. These mass fluxes are added to the 2D and to the 1D model equations by setting $\Delta x_q q^p_c$ in Equation (1.235) or Equation (1.238) equal to $Q_{\text{1D-2D}}$ (mass flux from 1D to 2D) for the *2D* grid cell where this flux is applied, and by setting $\Delta x_q q^p_c$ in Equation (1.235) or Equation (1.238) equal to $Q_{\text{2D-1D}}$ (mass flux from 2D to 1D) for the *1D* grid cell where this flux is applied. In both cases the flux represents a (sort of) lateral discharge.

Mass conservation Equation (1.241) and normal momentum approximation Equation (1.240) together define the embedded horizontal 1D-2D coupling. Its properties have been briefly discussed in section 1.3.2.

Note that in this approach it is not required that the 2D domains and the 1D domains are adjacent. Equation (1.240) and Equation (1.241) can also be applied when there is a gap in

between the domains, connecting some 1D grid cell to some 2D grid cell that is at the boundary of a 2D domain. The orientation of the boundaries of both domains does not have to be the same either. Because of this flexibility, it is even possible to have the 2D domains overlapping the 1D domains. That latter way of modeling 1D-2D interfaces actually corresponds with an embedded vertical coupling (not presented in this document).

1.4.3 Lateral horizontal 1D-2D coupling with strict model separation

The approach that is used in SOBEK 3 is very similar to the one presented in Kuiry *et al.* (2010). The difference is their use of a quasi-2D horizontal flood inundation model, using a steady-state friction formulation instead of the full 2D horizontal shallow-water momentum equations, and solving the coupling between the 1D horizontal network flow model and the quasi-2D flood model implicitly. The applied 1D-2D coupling equations, however, are the same as the ones that will be used here.

The coupling equations are expressed in terms of the variables at the 1D-2D interface (cf. Figure 1.62): z_s is the height of the bank that separates the 1D and 2D domain, $q_{\text{2D-1D}}$ is the lateral discharge per unit length from the 2D domain to the 1D domain, $q_{\text{1D-2D}}$ the lateral discharge per unit length in the other direction, and $\zeta_{\text{2D},I}$ and ζ_{1D} are the water level of respectively the 2D domain and the 1D domain at the interface. NB, no subscript I in ζ_{1D}, because in 1D domains the water level is taken constant across the channel width. The water level at the banks (and at a 1D-2D interface) is the same as the one at the center axis of the channel and are all denoted by ζ_{1D} (without subscript I).

1D-2D interfaces are not modeled in detail[4] (by simulation) but globally (by parameterization) by modeling the flow across interfaces by means of a weir formulation. The procedure that is customary in the modeling of hydraulic structures is followed and any unsteady behavior across interfaces is neglected, i.e., the flow is locally assumed to be in equilibrium. Since the width of 1D-2D interfaces is very small compared to the length scales of typical 1D-2D applications, this is a valid assumption. Notice that, because the flow across interfaces is not discretized in detail (cf. Footnote 4), a meaningful time-dependent discretization of the flow equations at interfaces is not possible anyway.

Modeling the flow across the bank at an interface by means of a standard weir formulation, the conservation of mass and the transfer of normal momentum across interfaces are modeled by:

$$q_{\text{2D-1D}} = -q_{\text{1D-2D}} \ , \tag{1.242)a}$$

$$q_{\text{2D-1D}} = \begin{cases} 0 & \text{if } \zeta_{\text{2D},I} - z_s \leq 0 \\ & \text{and } \zeta_{\text{1D}} - z_s \leq 0 \ , \\ c_e c_w 2/3\sqrt{2g/3}(\zeta_{\text{2D},I} - z_s)^{3/2} & \text{if } \zeta_{\text{2D},I} - z_s > 0 \\ & \text{and } \zeta_{\text{2D},I} - z_s \geq 3/2(\zeta_{\text{1D}} - z_s) \ , \\ c_e c_w(\zeta_{\text{1D}} - z_s)\sqrt{2g(\zeta_{\text{2D},I} - \zeta_{\text{1D}})} & \text{if } 3/2(\zeta_{\text{1D}} - z_s) > \zeta_{\text{2D},I} - z_s \\ & \text{and } \zeta_{\text{2D},I} - z_s \geq \zeta_{\text{1D}} - z_s > 0 \ , \\ -c_e c_w 2/3\sqrt{2g/3}(\zeta_{\text{1D}} - z_s)^{3/2} & \text{if } \zeta_{\text{1D}} - z_s > 0 \\ & \text{and } \zeta_{\text{1D}} - z_s \geq 3/2(\zeta_{\text{2D},I} - z_s) \ , \\ -c_e c_w(\zeta_{\text{2D},I} - z_s)\sqrt{2g(\zeta_{\text{1D}} - \zeta_{\text{2D},I})} & \text{if } 3/2(\zeta_{\text{2D},I} - z_s) > \zeta_{\text{1D}} - z_s \\ & \text{and } \zeta_{\text{1D}} - z_s \geq \zeta_{\text{2D},I} - z_s > 0 \ . \end{cases} \tag{1.242)b}$$

[4] Because the variation of velocity and water level across interfaces is not included in the discretization, a reasonably accurate detailed modeling of the flow across an interface is not even possible, cf. section 1.4.2.

with c_e the discharge coefficient $[-]$ and c_w the lateral contraction coefficient $[-]$.

The second and fourth line in Equation (1.242)b specify the modeling of free-weir flow and the free-weir conditions when the flow is respectively from 2D to 1D and from 1D to 2D; the third and fifth line pertain to the drowned-weir flow regime. It has been verified that this weir formulation is a continuous and continuously differentiable function of ζ_{1D} and $\zeta_{\text{2D},I}$, *also* at the water levels where the weir flow changes regime or direction. Besides being essential for physically meaningful behavior, this is also essential for smooth numerical behavior.

If required, $\zeta_{\text{2D},I}$ in Equation (1.242)b could be corrected for the energy head normal to the interface by adding $u_{\text{2D},I}^2/(2g)$, with $u_{\text{2D},I}$ the normal velocity in the 2D domain at the interface. At present, this correction is assumed to be small enough to be negligible. Since 1D flow models do not include the modeling of the flow across 1D channels (and normal to 1D-2D interfaces), such a correction is not possible for ζ_{1D}. Notice that an energy-head correction based on the flow velocity tangential to the interface should not be applied, since flow parallel to a hydraulic structure has an at most limited effect on the dynamics of the flow across a structure.

As before, discharge $q_{\text{2D-1D}}$ (same for $q_{\text{1D-2D}}$) is defined *per unit length*, i.e., no multiplication of the expressions in Equation (1.242)b by the width W_s of the weir that here would be the length of a stretch of channel bank.

In view of the numerical implementation it is convenient to rewrite Equation (1.242)b in the form of a space-discretized normal momentum equation (cf. Equation (1.231)), adding a time derivative[5] to allow some (limited) modeling of the dynamic behavior of the flow at a 1D-2D interface or to improve its numerical performance (the convergence speed of the iterative solution procedure that will be applied to solve the interface equations, cf. section 1.4.4 below). Restricting the presentation to the *drowned-weir* flow regime for the moment, i.e., to an interface where $3/2(\zeta_{\text{1D}} - z_s) > \zeta_{\text{2D},I} - z_s > 2/3(\zeta_{\text{1D}} - z_s) > 0$ or, equivalently, $3/2(\zeta_{\text{2D},I} - z_s) > \zeta_{\text{1D}} - z_s > 2/3(\zeta_{\text{2D},I} - z_s) > 0$ (generalizations later), the equations Equation (1.242) become[6]:

$$A_I s_{c,I} u_{\text{2D},I} = -\Delta x_I q_{\text{1D-2D}} \ , \tag{1.243)a}$$

$$\Delta x_{\text{1D-2D}} s_{c,I} \frac{\partial u_{\text{2D},I}}{\partial t} + \alpha_{SF} g(\zeta_{\text{1D}} - \zeta_{\text{2D},I}) + \alpha_{SF} \Delta x_{u,I} b_{S,I} s_{c,I} u_{\text{2D},I} = 0 \ , \tag{1.243)b}$$

[5]Convection terms and viscosity terms are not present in Equation (1.243)b. Due to the geometry variations and hence flow variations at 1D-2D interfaces, cf. the green braces in the Figures 1.60, 1.61 and 1.62, these terms may locally be quite large. Integration of the normal momentum equation in conservative form across an interface shows that the change of normal momentum across an interface is determined by friction losses at the bottom (modeled by the last term in the left-hand side of Equation (1.243)b), by the overall water level difference (modeled by the second term), and by the balance of incoming and outgoing convection and viscous fluxes at the faces of the integration volume. The effect of the latter is assumed to be small enough to be negligible compared to the effect of the bottom friction at an interface. This is a reasonable assumption, since at the faces of an integration volume across an interface the convection and viscosity fluxes are relatively small. Note that this is in agreement with the applicability of hydraulic structure formulas and confirms the discussion in section 1.4.2.

[6]Kernkamp (2008) adds a time derivative to structure equations like Equation (1.242)b to obtain an equation of the form $\partial u_{\text{2D-1D}}/\partial t + |u_{\text{2D-1D}}|u_{\text{2D-1D}}/\Delta x_{u,I} = |u_{\text{2D-1D}}|/(\Delta x_{u,I} b_{S,I}) \times g(\zeta_{\text{2D},I} - \zeta_{\text{1D}})/\Delta x_{u,I}$. The balance between the time derivative and the water level gradient in this equation depends on the modeling of the friction loss across the structure, which is not in agreement with a momentum equation. In particular, for a very smooth weir modeled by a very small friction coefficient $b_{S,I}$, the effect of the time derivative in this equation vanishes. On the other hand, for a very large friction coefficient (and/or very small $|u_{\text{2D-1D}}|$) the balance between water level gradient and friction in Equation (1.242)b is replaced by a balance between time derivative and friction. This may result in structures reacting unrealistically slow to the flow dynamics. In contrast, Equation (1.243)b has the correct form of a momentum equation and hence the correct physical dependence on $b_{S,I}$.

Another important difference is that Equation (1.243)b has been designed such that the transition between drowned-weir flow and free-weir flow is continuous and smooth, just like in structure Equation (1.242)b (see the explanation on normalization coefficient α_{SF}). This is not the case in the Kernkamp formulation.

with $b_{S,I}$ the friction coefficient for a submerged weir at the interface, with A_I the value of an A_f at the 1D-2D interface, i.e., the surface of a cell face at the interface, with $u_{\text{2D},I}$ the velocity component u_f normal to that face (and hence at the interface) normal to the interface) directed from the 2D domain to the 1D domain if $s_{c,I} = 1$ and in the opposite direction if $s_{c,I} = -1$ (cf. section 1.4.2), with Δx_I the length of that cell face (the grid size *along* the interface), with $\Delta x_{u,I}$ the value of a Δx_u at the interface (the grid size *normal* to the interface), and with $\Delta x_{\text{1D-2D}}$ an additional model parameter that should normally be at most a fraction of the grid size near the interface, i.e., $\Delta x_{\text{1D-2D}} \ll \Delta x_{u,I}$. By definition, the $\Delta x_{u,I}$ are the distances between the points where the $\zeta_{\text{2D},i}$ and the $\zeta_{\text{2D},v}$ are located, i.e., the distances between the circumcenters of the 2D virtual grid cells outside the interface and those of the adjacent 2D grid cells inside the 2D domain, cf. Figure 1.62. Also by definition, i.e., because the 2D grid is orthogonal, the lines connecting $\zeta_{\text{2D},i}$ and $\zeta_{\text{2D},v}$ points are normal to the 1D-2D interface.

Normalization coefficient α_{SF} has been introduced in Equation (1.243)b to ensure a continuous formulation at the transition between drowned-weir flow and free-weir flow (at $\zeta_{\text{2D},I} - z_s = 3/2(\zeta_{\text{1D}} - z_s)$ and at $\zeta_{\text{1D}} - z_s = 3/2(\zeta_{\text{2D},I} - z_s)$) when the time derivative of $u_{\text{2D},I}$ is present. Since α_{SF} is included in both the second term and the third term of Equation (1.243)b, its value is irrelevant when $\Delta x_{\text{1D-2D}} = 0$.

The choice for $\Delta x_{u,I}$ in Equation (1.243)b is arbitrary. A typical length scale normal to the interface is needed to enable recasting Equation (1.242)b in the form of a space-discretized momentum equation like Equation (1.231) with a friction coefficient $b_{S,I}$ of dimension $[1/s]$.

Equation (1.243)a has been obtained by replacing $q_{\text{2D-1D}}$ in Equation (1.242)a by $(A_f/\Delta x_I)s_{c,f}u_f = (A_I/\Delta x_I)s_{c,I}u_{\text{2D},I}$ (cf. the mass fluxes in Equation (1.232) and the first expression in Equation (1.241)). The same substitution has been used to obtain Equation (1.243)b from the square of Equation (1.242)b. Parameter $\Delta x_{\text{1D-2D}}$ in Equation (1.243)b sort of represents the effective thickness of the interface, i.e., the size of the green braces in the Figures 1.60, 1.61 and 1.62. It has merely been introduced as a modeling parameter; by increasing or decreasing $\Delta x_{\text{1D-2D}}$, the relative importance of the time derivative in Equation (1.243)b can be increased or decreased. Apart from that time derivative, which provides a relaxation effect that may be physically realistic or that may be exaggerated for numerical purposes, Equation (1.243)b is equivalent with the third and fifth line in Equation (1.242)b (*drowned-weir* flow regime) when $\alpha_{SF} = 1$ and when $b_{S,I}$ is taken equal to:

$$b_{S,I} = \frac{A_I^2 \sqrt[k]{|u_{\text{2D},I}|^k + u_\varepsilon^k}}{2\Delta x_{u,I}\left(\Delta x_I c_e c_w(\zeta_{\text{1D/2D},I} - z_s)\right)^2}\ , \tag{1.244}$$

with $\zeta_{\text{1D/2D},I}$ equal to ζ_{1D} if $\zeta_{\text{2D},I} \geq \zeta_{\text{1D}}$ and equal to $\zeta_{\text{2D},I}$ if $\zeta_{\text{1D}} > \zeta_{\text{2D},I}$. Notice the division by $\Delta x_{u,I}$ in Equation (1.244) that compensates for the multiplication by $\Delta x_{u,I}$ in the last term of Equation (1.243)b.

Small positive parameter $u_\varepsilon > 0$ of dimension $[m/s]$ has been added in Equation (1.244) to stabilize the computations for small $|u_{\text{2D},I}| \lesssim u_\varepsilon$, when Equation (1.243)b would otherwise become highly nonlinear. The stabilization is required because of the explicit discretization in time of the coefficient $b_{S,I}$ in Equation (1.243)b (Picard linearization, see below). At present the stabilization is implemented for $k = 1$, i.e., the coefficient $|u_{\text{2D},I}|$ in Equation (1.244) has been replaced by $|u_{\text{2D},I}| + u_\varepsilon$, which gives a rather large perturbation of 1D-2D coupling equation Equation (1.243)b at larger $|u_{\text{2D},I}|$. Smaller perturbations are obtained for, e.g., $k = 2$ ($\Rightarrow \sqrt[k]{|u_{\text{2D},I}|^k + u_\varepsilon^k} = \sqrt{|u_{\text{2D},I}|^2 + u_\varepsilon^2}$) or $k \to \infty$ ($\Rightarrow \sqrt[k]{|u_{\text{2D},I}|^k + u_\varepsilon^k} \to \max(|u_{\text{2D},I}|, u_\varepsilon)$). These two options are currently investigated. As an alternative to stabilization by means of the introduction of u_ε, coefficient $b_{S,I}$ could be discretized implicitly in time. This would require an iterative solution procedure, which could

very well be combined with the iterative solution procedure for the 1D-2D coupling as a whole. It may be an option worth considering in the future.

For free-weir flow across an interface the same Equation (1.243)b may be used, but with ζ_{1D} replaced by z_s if $\zeta_{\text{2D},I} > z_s$ and $\zeta_{\text{2D},I} - z_s \geq 3/2(\zeta_{\text{1D}} - z_s)$, with $\zeta_{\text{2D},I}$ replaced by z_s if $\zeta_{\text{1D}} > z_s$ and $\zeta_{\text{1D}} - z_s \geq 3/2(\zeta_{\text{2D},I} - z_s)$, with $\alpha_{SF} = 1/3$, and with $b_{S,I}$ replaced by $b_{F,I}$, the friction coefficient for *free-weir* flow across the interface:

$$b_{F,I} = \frac{A_I^2 \sqrt[k]{|u_{\text{2D},I}|^k + u_\varepsilon^k}}{(2/3)^3 \Delta x_{u,I} \big(\Delta x_I c_e c_w (\zeta_{\text{1D/2D},I} - z_s)\big)^2} \ , \tag{1.245}$$

with $\zeta_{\text{1D/2D},I}$ equal to $\zeta_{\text{2D},I}$ if $\zeta_{\text{2D},I} \geq \zeta_{\text{1D}}$ and equal to ζ_{1D} if $\zeta_{\text{1D}} > \zeta_{\text{2D},I}$. This is *opposite* to the definition of $\zeta_{\text{1D/2D},I}$ in Equation (1.244), whence the coefficient $(2/3)^3$ instead of 2 in the denominator. This coefficient compensates for the sudden change in value of $\zeta_{\text{1D/2D},I} - z_s$ at the transition from drowned-weir flow to free-weir flow or vice versa at $\zeta_{\text{2D},I} - z_s = 3/2(\zeta_{\text{1D}} - z_s)$ or at $\zeta_{\text{1D}} - z_s = 3/2(\zeta_{\text{2D},I} - z_s)$.

Summarizing, Equation (1.243)a is always applied, in combination with:

IF $\zeta_{\text{2D},I} - z_s \leq 0$ AND $\zeta_{\text{1D}} - z_s \leq 0$ THEN
! **no flow** across interface
    $u_{\text{2D},I} = 0$ (or, equivalently, $q_{\text{1D-2D}} = 0$)
ELSE IF $\zeta_{\text{2D},I} - z_s \geq 3/2(\zeta_{\text{1D}} - z_s)$ THEN
! **free-weir flow** from 2D to 1D
    Equation (1.243)b with $\alpha_{SF} = 1/3$, with $\zeta_{\text{1D}} \Leftarrow z_s$, and with $b_{S,I} \Leftarrow b_{F,I}$ where $\zeta_{\text{1D/2D},I} \Leftarrow \zeta_{\text{2D},I}$
ELSE IF $\zeta_{\text{1D}} - z_s \geq 3/2(\zeta_{\text{2D},I} - z_s)$ THEN
! **free-weir flow** from 1D to 2D
    Equation (1.243)b with $\alpha_{SF} = 1/3$, with $\zeta_{\text{2D},I} \Leftarrow z_s$, and with $b_{S,I} \Leftarrow b_{F,I}$ where $\zeta_{\text{1D/2D},I} \Leftarrow \zeta_{\text{1D}}$
ELSE IF $\zeta_{\text{2D},I} \geq \zeta_{\text{1D}}$ THEN
! **drowned-weir flow** from 2D to 1D
    Equation (1.243)b with $\alpha_{SF} = 1$, and with $b_{S,I}$ where $\zeta_{\text{1D/2D},I} \Leftarrow \zeta_{\text{1D}}$
ELSE   ! i.e. IF $\zeta_{\text{1D}} > \zeta_{\text{2D},I}$
! **drowned-weir flow** from 1D to 2D
    Equation (1.243)b with $\alpha_{SF} = 1$, and with $b_{S,I}$ where $\zeta_{\text{1D/2D},I} \Leftarrow \zeta_{\text{2D},I}$
ENDIF

As mentioned before, first the case of a *drowned-weir* flow across an interface will be considered, which is modeled by Equation (1.243)a and Equation (1.243)b 'as is'. The free-weir flow case will be considered afterwards.

The discretization of Equation (1.243) in time must be in agreement with the discretizations Equation (1.232) and Equation (1.231), but must also be in agreement with Equation (1.242)b:

$$A_I^n s_{c,I} \big(\theta u_{\text{2D},I}^{n+1} + (1-\theta) u_{\text{2D},I}^n\big) = -\Delta x_I q_{\text{1D-2D}}^* \ , \tag{1.246)a}$$

$$\Delta x_{\text{1D-2D}} s_{c,I} \frac{u_{\text{2D},I}^{n+1} - u_{\text{2D},I}^n}{\Delta t} + \alpha_{SF} g (\zeta_{\text{1D}}^{n+1} - \zeta_{\text{2D},I}^{n+1}) + \alpha_{SF} \Delta x_{u,I} b_{S,I}^n s_{c,I} u_{\text{2D},I}^{n+1} = 0 \ . \tag{1.246)b}$$

The left-hand side of Equation (1.246)a is the same discretization as presented in the first expression of Equation (1.241). A backward Euler discretization ($\theta = 1$) has been applied in Equation (1.246)b to ensure that it is equivalent with the third and fifth line of Equation (1.242)b evaluated at the next time level in the limit of $\Delta x_{\text{1D-2D}} = 0$.

It has been decided to evaluate $b_{S,I}$ (and $b_{F,I}$) explicitly, whence the superscript n of $b^n_{S,I}$ in Equation (1.246)b. This makes the equation linear in the unknowns $u^{n+1}_{\text{2D},I}$, ζ^{n+1}_{1D} and $\zeta^{n+1}_{\text{2D},I}$, and therefore relatively easy to solve. It is also the procedure followed in SOBEK and D-Flow FM, cf. Kernkamp (2008). On the other hand, to ensure the best nonlinear performance (and, in consequence, the highest stability), the time-dependent variables $|u_{\text{2D},I}|$ and $\zeta_{\text{1D/2D}}$ in $b_{S,I}$ (and $b_{F,I}$) should be evaluated at the next time level $n+1$. Note that the time discretization of A_I in $b_{S,I}$ (and $b_{F,I}$) should be in agreement with that of the A_f in Equation (1.232), hence A_I in $b_{S,I}$ (and $b_{F,I}$) is always to be evaluated at previous time level n. Of course, if $b_{S,I}$ is (partially) evaluated at the next time level, Equation (1.246)b becomes a nonlinear equation that is to be solved iteratively.

For slow variation of the flow near the interface or when a small time step Δt is used, there will be hardly any difference in the results obtained with $b_{S,I}$ ($b_{F,I}$) evaluated at n and those obtained with $b_{S,I}$ ($b_{F,I}$) at $n+1$. At transient flow conditions near a 1D-2D interface, however, the differences may be large. This applies in particular to (free-weir!) flow at the onset of flooding, i.e., there may be noticeable temporary differences between results obtained with $b_{F,I}$ evaluated at n and those with $b_{F,I}$ at $n+1$.

Replacing next time level $n+1$ by iteration index p and separating terms with variables at the next iteration level p (that are to be determined) from terms that only depend on known information at previous time level n, Equation (1.246) becomes:

$$\begin{aligned}
&\theta A^n_I s_{c,I} u^p_{\text{2D},I} + \Delta x_I q^p_{\text{1D-2D}} = -(1-\theta) A^n_I s_{c,I} u^n_{\text{2D},I} \ , \\
&\Big(\frac{\Delta x_{\text{1D-2D}}}{\Delta t} + \alpha_{SF} \Delta x_{u,I} b^n_{S,I}\Big) s_{c,I} u^p_{\text{2D},I} + \alpha_{SF}\, g(\zeta^p_{\text{1D}} - \zeta^p_{\text{2D},I}) = \frac{\Delta x_{\text{1D-2D}}}{\Delta t} s_{c,I} u^n_{\text{2D},I} \ .
\end{aligned} \tag{1.247}$$

To get the coupling conditions formulated in the water level variables at the cell centers of the 2D domain, the linear interpolation $\zeta^p_{\text{2D},I} = (\zeta^p_{\text{2D},i} + \zeta^p_{\text{2D},v})/2$ is used and Equation (1.236) is used at the cell faces along the interface to replace $u^p_{\text{2D},I}$. The latter is obviously an equation very similar to Equation (1.240) and reads:

$$s_{c,I} u^p_{\text{2D},I} = s_{c,I} R^n_I - F^n_I (\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i}) \ , \tag{1.248}$$

with R^n_I and F^n_I as in Equation (1.237), and with $\zeta^p_{\text{2D},v}$ and $\zeta^p_{\text{2D},i}$ as in Figure 1.62. The result is:

$$-\theta A^n_I F^n_I (\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i}) + \Delta x_I q^p_{\text{1D-2D}} = -A^n_I s_{c,I} \Big(\theta R^n_I + (1-\theta) u^n_{\text{2D},I}\Big) \ , \tag{1.249)a}$$

$$\begin{aligned}
&-\Big(\frac{\Delta x_{\text{1D-2D}}}{\Delta t} + \alpha_{SF} \Delta x_{u,I} b^n_{S,I}\Big) F^n_I (\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i}) + \alpha_{SF}\, g\Big(\zeta^p_{\text{1D}} - \frac{\zeta^p_{\text{2D},i} + \zeta^p_{\text{2D},v}}{2}\Big) \\
&= \frac{\Delta x_{\text{1D-2D}}}{\Delta t} s_{c,I} u^n_{\text{2D},I} - \Big(\frac{\Delta x_{\text{1D-2D}}}{\Delta t} + \alpha_{SF} \Delta x_{u,I} b^n_{S,I}\Big) s_{c,I} R^n_I \ .
\end{aligned} \tag{1.249)b}$$

Notice that these equations define an *implicit* coupling across the 1D-2D interfaces, since the unknowns of the 2D domain ($\zeta_{\text{2D},i}$, $\zeta_{\text{2D},v}$) and those of the 1D domain ($q_{\text{1D-2D}}$, ζ_{1D}) are all at the next iteration level p. This may strongly complicate the implementation of the algorithm. It would require the construction of a system of equations composed of elements of both the 2D model and the 1D model, which is especially not obvious when dealing with separate implementations, as is the intention here (D-Flow FM as the 2D model, SOBEK as the 1D model). Nevertheless, as long as the 1D and 2D grids match at the interfaces[7] and the 1D and

[7] What is meant here is that the space discretizations should match. When the 1D models and 2D models use the same type of space discretizations (as is the case here) this is equivalent with the condition that the grids should match.

2D models all use the same time step[8], a direct implicit coupling, although still complex, might be feasible. However, a general and much more flexible 1D-2D coupling with non-matching grids at the interface and possibly the use of different time steps in the 2D and 1D model is virtually impossible with an implicit coupling; its implementation would be far too complex.

There is also a numerical reason for not applying an implicit coupling. When eliminating $q^p_{\text{1D-2D}}$ in Equation (1.249) (using Equation (1.238) for the 1D channel with lateral discharge q^p_c equal to $-q^p_{\text{1D-2D}}$[9], the resulting system of equations for the unknowns ζ^p will have a positive-definite symmetric matrix, because of the skew-symmetry of the discretization of the water level gradient in Equation (1.231) (and in Equation (1.236) and Equation (1.248)) and in Equation (1.243)b (and Equation (1.246)b). Systems of equations with a positive-definite symmetric matrix are guaranteed to be well posed and to be always solvable using a (preconditioned) CG iterative method, which is why this method is applied in D-Flow FM and SOBEK.

The skew-symmetry in Equation (1.243)b (and Equation (1.246)b) is *only* because of the drowned-weir case that is being considered for the moment. When the flow across the 1D-2D interface is to be modeled by a *free weir* an expression similar to Equation (1.243)b applies that, however, does *not* depend on either ζ_{1D} or $\zeta_{\text{2D},I}$, cf. the second and fourth line in Equation (1.242)b. Equations of the form Equation (1.243)b and Equation (1.246)b, and hence of the form Equation (1.249)b, still apply, but with $b^n_{S,I}$ replaced by $b^n_{F,I}$, with $\alpha_{SF} = 1/3$ instead of $\alpha_{SF} = 1$, and, most importantly, with the water level difference $\zeta_{\text{1D}} - \zeta_{\text{2D},I}$ replaced by either $\zeta_{\text{1D}} - z_s$ or $z_s - \zeta_{\text{2D},I}$, depending on whether the free-weir flow is from the 1D domain to the 2D domain or in the opposite direction. Such water level differences destroy the skew-symmetry of the equations and hence the symmetry of the system of ζ^p equations.

Non-symmetric systems of equations are avoided when the coupling equations Equation (1.249) are not applied implicitly, but *explicitly*. This also makes it feasible to develop and implement (in the future) a 1D-2D coupling that can handle non-matching grids at the interface and possibly the use of different time steps in the 2D and in the 1D model.

An *explicit* coupling involves a solution procedure where the implicit coupling equations Equation (1.249) are solved iteratively per nonlinear iteration p. This coupling iteration process is best combined with the nonlinear iteration loop. After all, the nonlinear iteration process per 1D and 2D model only needs to be solved up to (a fraction of) the convergence level of the coupling in between them. Solving the 1D and 2D models with a much higher precision than the coupling convergence error (a measure of $|\zeta^p_{\text{2D},I} - \zeta^{p-1}_{\text{2D},I}|$ and $|\zeta^p_{\text{1D}} - \zeta^{p-1}_{\text{1D}}|$) would be a waste of computational effort.

Although an explicit coupling can be realized by applying one of the coupling equations Equation (1.249) in one direction and the other one in the other direction (the naive coupling that is usually applied), a better performance is obtained by applying in each direction an optimized (and obviously different) combination. To determine this combination, first notice that for $\Delta x_{\text{1D-2D}} = 0$ and no friction losses across the 1D-2D interface (i.e., $b^n_{S,I} = 0$), Equation (1.249)b reduces to $\zeta^p_{\text{1D}} = (\zeta^p_{\text{2D},i} + \zeta^p_{\text{2D},v})/2$, i.e., coupling condition Equation (1.249)b essentially imposes a relation between the water levels at both sides of the interface (Dirichlet condition). The other coupling condition, Equation (1.249)a, obviously imposes a relation between the (lateral) velocities at both sides, cf. Equation (1.246)a where this condition originates from. It is a Neumann condition for $\zeta_{\text{2D},I}$, the water levels of the 2D domain.

[8]Not only the time steps, but the applied time integration methods as a whole should match. When the 1D models and 2D models use the same type of time integration methods (as is the case here) this is equivalent with the condition that the time steps should match.

[9]Recall that $q_{\text{1D-2D}}$ has been defined as the flux from the 1D domain to the 2D domain, cf. Equation (1.241), while q_c has been defined as an incoming flux, here incoming to the 1D domain, cf. Equation (1.232).

A suitable combination of coupling equations for the transfer of information to a 2D domain is one based on the concept of absorbing boundary conditions, see, e.g., Ye *et al.* (2011). The simplest form is the one that specifies the incoming Riemann invariant in normal direction $-h_{\text{2D},I}s_{c,I}u_{\text{2D-1D}} + \sqrt{gh_{\text{2D},I}}\zeta_{\text{2D},I}$[10], with $h_{\text{2D},I} = \zeta_{\text{2D},I} + d_{\text{2D},I}$ the total water depth of the 2D domain at the interface. This choice is equivalent with the Sommerfeld radiation condition. Formulated in the discretized water level this condition reads $(\zeta_{\text{2D},i} + \zeta_{\text{2D},v})/2 + \sqrt{gh_{\text{2D},I}}\Delta t/\Delta x_{u,I}(\zeta_{\text{2D},v} - \zeta_{\text{2D},i})$. Because all flow dynamics in lateral direction is neglected in a 1D model, the concept of Riemann invariants cannot be used at a 1D-2D interface in the other direction for the transfer of lateral information to a 1D domain, so here it is necessary to come up with something different.

To allow a full optimization of the coupling, it is proposed to use a parameterized combination of $-1/(\alpha_{SF}g)$ times Equation (1.249)b and $-1/(\theta A_I^n F_I^n)$ times Equation (1.249)a. The normalization is to get Equation (1.249)b in the form of (almost) a Dirichlet condition for the water level of the 2D domain at the interface, and Equation (1.249)a in the form of a Neumann condition. The combination is the Robin coupling condition:

$$\begin{aligned}
&\alpha_{\text{2D}}^n \frac{\zeta_{\text{2D},v}^p + \zeta_{\text{2D},i}^p}{2} + (\beta_{\text{2D}}^n + \alpha_{\text{2D}}^n f_{S,I}^n F_I^n)(\zeta_{\text{2D},v}^p - \zeta_{\text{2D},i}^p) \\
&= \alpha_{\text{2D}}^n \zeta_{\text{1D}}^{p-1} + \beta_{\text{2D}}^n \frac{\Delta x_I}{\theta A_I^n F_I^n} q_{\text{1D-2D}}^{p-1} \\
&+ \alpha_{\text{2D}}^n s_{c,I}\Big(f_{S,I}^n R_I^n - \frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF} g \Delta t} u_{\text{2D},I}^n\Big) + \beta_{\text{2D}}^n \frac{s_{c,I}}{\theta F_I^n}\Big(\theta R_I^n + (1-\theta) u_{\text{2D},I}^n\Big) \ ,
\end{aligned} \tag{1.250}$$

with (drowned-weir flow $\rightarrow$ substitute $\alpha_{SF} = 1$):

$$f_{S,I}^n = \Big(\frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF}\Delta x_{u,I}} + b_{S,I}^n \Delta t\Big)\frac{\Delta x_{u,I}}{g\Delta t} = \Big(\frac{\Delta x_{\text{1D-2D}}}{\Delta x_{u,I}} + b_{S,I}^n \Delta t\Big)\frac{\Delta x_{u,I}}{g\Delta t} \ . \tag{1.251}$$

The coupling parameters α_{2D}^n and β_{2D}^n (and the coupling parameters that will be introduced later) are allowed to vary in time, whence the superscript n. Like all other coefficients in the above equations (and like many terms and coefficients in the D-Flow FM and SOBEK time integration scheme) they are evaluated explicitly at the previous time level. For the moment the variation of coupling parameters inside the coupling iteration loop, evaluating them explicitly at the previous iteration level $p-1$, is not considered. In view of the strongly explicit nature of the solution algorithm as a whole, this is not expected to have a positive effect on the convergence properties of the 1D-2D coupling that is significant enough to be worth considering.

Obviously the scaling of the two coupling parameters α_{2D}^n and β_{2D}^n is irrelevant. The multiplication of α_{2D}^n and β_{2D}^n by any non-zero factor has no effect on the coupling, indicating that a single coupling parameter would have sufficed (to be obtained by replacing α_{2D}^n by, e.g., $1 - \beta_{\text{2D}}^n$). Nevertheless, two coupling parameters instead of one are considered, because it facilitates the implementation of the coupling. In particular, the use of two coupling parameters makes it easy to set either one of them to 0 or 1 and the other respectively to 1 or to some optimized value.

By choosing $\alpha_{\text{2D}}^n = 1$ and $\beta_{\text{2D}}^n = 1/2 - f_{S,I}^n F_I^n$, coupling Equation (1.250) only specifies $\zeta_{\text{2D},v}^p$, which variable can then immediately be eliminated from the system of equations for the ζ^p of the 2D domains. The simplicity and ease of implementation of this coupling seems to make this choice of β_{2D}^n (with $\alpha_{\text{2D}}^n = 1$) attractive. The main concern, however, should

[10] The minus sign of $s_{c,I}u_{\text{2D-1D}}^p$ is because the positive direction at cell faces has been defined as the direction pointing outward of a cell. This applies to the 2D faces at the boundaries of a 2D domain as well, and hence to the 2D faces at a 1D-2D interface.

be the convergence speed of the iterative explicit coupling per time step and the according optimization of β^n_{2D}.

For vanishing time derivative ($\Delta x_{\text{1D-2D}} = 0$) and vanishing friction ($b^n_{S,I} = 0$) at the 1D-2D interface ($\rightarrow f^n_{S,I} = 0$), coupling Equation (1.250) reduces to:

$$\begin{aligned}\alpha^n_{\text{2D}} \frac{\zeta^p_{\text{2D},v} + \zeta^p_{\text{2D},i}}{2} + \beta^n_{\text{2D}}(\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i}) = \alpha^n_{\text{2D}} \zeta^{p-1}_{\text{1D}} + \beta^n_{\text{2D}} \frac{\Delta x_I}{\theta A^n_I F^n_I} q^{p-1}_{\text{1D-2D}} \\ + \beta^n_{\text{2D}} \frac{s_{c,I}}{\theta F^n_I} \left(\theta R^n_I + (1-\theta) u^n_{\text{2D},I} \right) .\end{aligned} \tag{1.252}$$

In the second line only terms at the previous time level that do not affect the convergence properties of the explicit 1D-2D coupling.

For $\alpha^n_{\text{2D}} = 1$ and $\beta^n_{\text{2D}} = 0$, Equation (1.252) reduces to $(\zeta^p_{\text{2D},v} + \zeta^p_{\text{2D},i})/2 = \zeta^{p-1}_{\text{1D}}$, the discretization and iterative approximation of $\zeta^{n+1}_{\text{1D}} - \zeta^{n+1}_{\text{2D},I} = 0$. For $\alpha^n_{\text{2D}} = 0$ and $\beta^n_{\text{2D}} = 1$, one gets $\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i} = 1/(\theta F^n_I)\big(q^{p-1}_{\text{1D-2D}} \Delta x_I / A^n_I + s_{c,I}(\theta R^n_I + (1-\theta) u^n_{\text{2D-1D}})\big)$. Using Equation (1.248), this can be written as $s_{c,I}\big(\theta u^p_{\text{2D},I} + (1-\theta) u^n_{\text{2D},I}\big) = -q^{p-1}_{\text{1D-2D}} \Delta x_I / A^n_I$, showing that the other coupling contained in Equation (1.252) is of course Equation (1.246)a, the equation ensuring mass conservation across the interface.

Equation (1.250) defines the coupling from the 1D domain to the 2D domain. The coupling in the other direction, from the 2D domain to the 1D domain, is similar:

$$\begin{aligned}&\alpha^n_{\text{1D}} \zeta^p_{\text{1D}} - \beta^n_{\text{1D}} \frac{\Delta x_I}{\theta A^n_I F^n_I} q^p_{\text{1D-2D}} \\ &= \alpha^n_{\text{1D}} \frac{\zeta^{p-1}_{\text{2D},i} + \zeta^{p-1}_{\text{2D},v}}{2} + (\beta^n_{\text{1D}} - \alpha^n_{\text{1D}} f^n_{S,I} F^n_I)(\zeta^{p-1}_{\text{2D},i} - \zeta^{p-1}_{\text{2D},v}) \\ &- \alpha^n_{\text{1D}} s_{c,I} \left(f^n_{S,I} R^n_I - \frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF}\, g \Delta t} u^n_{\text{2D},I} \right) + \beta^n_{\text{1D}} \frac{s_{c,I}}{\theta F^n_I} \left(\theta R^n_I + (1-\theta) u^n_{\text{2D},I} \right) .\end{aligned} \tag{1.253}$$

The minus sign of the second term in the left-hand side is because $q^p_{\text{1D-2D}}$ is the *outgoing* flux from the 1D domain to the 2D domain, *not* a flux entering the 1D domain, like q^p_c in Equation (1.235) and Equation (1.238). NB, Equation (1.253) and Equation (1.250) are obviously equivalent. Apart from the iteration levels p and $p-1$ in the first and second line, the former can be obtained from the latter by setting $\alpha^n_{\text{1D}} = \alpha^n_{\text{2D}}$ and $\beta^n_{\text{1D}} = -\beta^n_{\text{2D}}$. The change of sign in the latter takes account of the fact that quantities at an interface that are normal to that interface and hence have a direction with respect to the interface (like velocity, mass flux and water level gradient across the interface) change sign when they are considered in opposite direction.

Equation (1.250) and Equation (1.253) are for the case that the flow across the 1D-2D interface is modeled as a *drowned weir*. If the flow is to be modeled by a *free weir*, one of the water levels in Equation (1.243)b (Equation (1.246)b) is to be replaced by z_s. (Also $b_{S,I}$ and $b^n_{S,I}$ are to be replaced by $b_{F,I}$ and $b^n_{F,I}$, while $\alpha_{SF} = 1/3$ instead of $\alpha_{SF} = 1$ is to be used.) For free-weir flow from the 2D domain to the 1D domain, this means replacement of ζ^p_{1D} in Equation (1.249)b by z_s, in which case the coupling equation becomes a boundary condition for the 2D domain:

$$\frac{\zeta^p_{\text{2D},v} + \zeta^p_{\text{2D},i}}{2} + f^n_{F,I} F^n_I (\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i}) = z_s + s_{c,I} \left(f^n_{F,I} R^n_I - \frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF}\, g \Delta t} u^n_{\text{2D},I} \right) , \tag{1.254}$$

with (free-weir flow $\rightarrow$ substitute $\alpha_{SF} = 1/3$):

$$f_{F,I}^n = \Big(\frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF}\Delta x_{u,I}} + b_{F,I}^n \Delta t\Big)\frac{\Delta x_{u,I}}{g\Delta t} = \Big(\frac{3\Delta x_{\text{1D-2D}}}{\Delta x_{u,I}} + b_{F,I}^n \Delta t\Big)\frac{\Delta x_{u,I}}{g\Delta t} \ . \tag{1.255}$$

The discharge determined at the boundary of the 2D domain is then simply imposed as *minus* a lateral discharge of the 1D domain (one-sided coupling from the 2D domain to the 1D domain). That is equivalent with applying Equation (1.253) for $\alpha_{\text{1D}}^n = 0$ and $\beta_{\text{1D}}^n = 1$.

Note that Equation (1.254) is obtained from Equation (1.250) by substituting $\zeta_{\text{1D}}^{p-1} = z_s$, $\alpha_{\text{2D}}^n = 1$ and $\beta_{\text{2D}}^n = 0$. In other words, coupling conditions Equation (1.250) and Equation (1.253) can be used for both the drowned-weir case and the free-weir case when the flow is from the 2D domain to the 1D domain.

Likewise, in the case of free-weir flow from the 1D domain to the 2D domain, $(\zeta_{\text{2D},i}^p + \zeta_{\text{2D},v}^p)/2$ in Equation (1.249)b is to be replaced by z_s. However, unlike the previous case, this does not lead to an equation with only variables defined in the 1D domain, because Equation (1.243)b has been formulated in terms of the velocity $u_{\text{2D},I}$ at the 2D side of the interface. An equation independent of variables at the next time level in the 2D domain is obtained by eliminating $u_{\text{2D},I}^p$ from the equations Equation (1.247) and substituting $\zeta_{\text{2D},I}^p = z_s$ in the result:

$$\zeta_{\text{1D}}^p - f_{F,I}^n \frac{\Delta x_I}{\theta A_I^n} q_{\text{1D-2D}}^p = z_s + s_{c,I}\Big(f_{F,I}^n \frac{1-\theta}{\theta} + \frac{3\Delta x_{\text{1D-2D}}}{g\Delta t}\Big) u_{\text{2D},I}^n \ . \tag{1.256}$$

The discharge determined at the boundary of the 1D domain is then simply imposed as discharge boundary condition to the 2D domain (one-sided coupling from the 1D domain to the 2D domain). That is equivalent with applying Equation (1.250) for $\alpha_{\text{2D}}^n = 0$ and $\beta_{\text{2D}}^n = 1$ (see interpretation of Equation (1.252) directly after that equation).

Similar as before with free-weir flow from the 2D domain to the 1D domain, Equation (1.256) is included in Equation (1.253). It is obtained by replacing $f_{S,I}^n$ by $f_{F,I}^n$ and setting $\alpha_{SF} = 1/3$ (for a free-weir formulation), and by substituting $(\zeta_{\text{2D},i}^{p-1} + \zeta_{\text{2D},v}^{p-1})/2 = z_s$, $\alpha_{\text{1D}}^n = 1$, and $\beta_{\text{1D}}^n = f_{F,I}^n F_I^n$. So coupling conditions Equation (1.250) and Equation (1.253) can also be used for both the drowned-weir case and the free-weir case when the flow is from the 1D domain to the 2D domain.

Care should be taken in case of extremely small $f_{F,I}^n$ (its value becomes zero for $\Delta x_{\text{1D-2D}} = 0$ and $b_{F,I}^n = 0$). For very small $f_{F,I}^n$ (which implies very small $\Delta x_{\text{1D-2D}}$) Equation (1.256) reduces to the condition $\zeta_{\text{1D}} = z_s$. This means that the water level in the 1D channel would become independent of the flow dynamics, which does not make sense. The same applies to Equation (1.254) by the way. That equation reduces for very small $f_{F,I}^n$ (and hence very small $\Delta x_{\text{1D-2D}}$) to the condition $(\zeta_{\text{2D},v}^p + \zeta_{\text{2D},i}^p)/2 = z_s$. Although physically incorrect, it does not pose a computational problem, so in that sense there is no reason for concern with a very small $f_{F,I}^n$ when considering a free-weir flow from 2D to 1D. NB, the situation is different for a drowned weir. As shown before, for $f_{S,I}^n$ very small that coupling reduces to $(\zeta_{\text{2D},v}^p + \zeta_{\text{2D},i}^p)/2 = \zeta_{\text{1D}}^{p-1}$, next to mass conservation Equation (1.243)a. This is the correct physical limit for a large water depth and hence very low friction losses at and across a 1D-2D interface.

1.4.4 Analysis of the lateral horizontal 1D-2D coupling

1.4.4.1 Preliminaries

To analyse and optimize the performance of the explicit solution procedure with strict model separation for the implicit horizontal 1D-2D coupling, rectangular domains, uniform conditions, and uniform rectangular grids are considered:

- ◇ a straight 1D-2D interface, arbitrarily set at the location $x = 0$;
- ◇ a straight 1D channel with uniform depth h_{1D}, width W_{1D} and friction coefficient b_{1D} at the side $x > 0$ of the interface, numerically modeled using a uniform grid with cells of size Δy;
- ◇ a 2D area with uniform depth h_{2D} and friction coefficient b_{2D} at the side $x < 0$ of the interface, numerically modeled using a uniform grid with rectangular cells of size Δx by Δy;
- ◇ uniform friction coefficient b_I at the interface of uniform thickness $\Delta x_{\text{1D-2D}}$, cf. equation Equation (1.243)b.

The uniform conditions make it possible to perform a normal-mode analysis.

> The underlying idea is that the modeling near 1D-2D interfaces will generally be such that in the optimization of the coupling parameters α^n_{2D}, β^n_{2D}, α^n_{1D} and β^n_{1D} in Equation (1.250) and Equation (1.253), the conditions are smooth enough for the coefficients to be assumed locally constant, for the 1D-2D interface to be assumed locally straight, and for the grid to be assumed locally uniform. If the conditions are not smooth (strong variations of water depth or grid size near a 1D-2D interface, strong bends in a 1D-2D interface), the optimization is assumed to give reasonable approximations of the optimal coupling parameters, especially when non-smooth conditions occur in only a small number of locations. Note that the normal-mode analysis presented below is the only feasible way to arrive at reasonable and simply to use estimations of the optimal coupling parameters that will perform better than no optimization.

In the straight and uniform 1D domain with uniform grid, water level equation Equation (1.238) becomes:

$$(1 + 2\text{CFL}^2_{\text{1D}})\zeta^p_j - \text{CFL}^2_{\text{1D}}(\zeta^p_{j-1} + \zeta^p_{j+1}) - q^p_{c,j}\Delta t/W_{\text{1D}} = \text{rhs}^n_{\text{1D},j} \ , \tag{1.257}$$

with j the control volume index in y-direction and with $\text{rhs}^n_{\text{1D},j}$ all the known terms at the previous time level n (note that the term $(S^{p-1}_\zeta \zeta^{p-1}_c - V^{p-1}_c)/\Delta t$ in Equation (1.238) vanishes for the constant-coefficient case that is considered here). Variable CFL_{1D} represents the coefficient $\Delta t A^n_f \theta F^n_u / S^{p-1}_\zeta$ in the y– or channel direction that for this uniform case becomes equal to:

$$\text{CFL}_{\text{1D}} = \frac{\theta \Delta t \sqrt{g h_{\text{1D}}}}{\Delta y \sqrt{1 + b_{\text{1D}} \Delta t}} \ . \tag{1.258}$$

In the uniform 2D domain with uniform rectangular grid, Equation (1.238) becomes:

$$\begin{aligned} &(1 + 2\text{CFL}^2_{\text{2D},x} + 2\text{CFL}^2_{\text{2D},y})\zeta^p_{i,j} \\ &- \text{CFL}^2_{\text{2D},x}(\zeta^p_{i-1,j} + \zeta^p_{i+1,j}) - \text{CFL}^2_{\text{2D},y}(\zeta^p_{i,j-1} + \zeta^p_{i,j+1}) = \text{rhs}^n_{\text{2D},i,j} \ , \end{aligned} \tag{1.259}$$

with i, j the control volume indices in x-, y-direction, $\text{rhs}^n_{\text{2D},i,j}$ all terms at time level n, and with $\text{CFL}_{\text{2D},x}$ and $\text{CFL}_{\text{2D},y}$ representing the coefficient $\Delta t A^n_f \theta F^n_u / S^{p-1}_\zeta$ in the two coordinate directions that here become equal to:

$$\text{CFL}_{\text{2D},x} = \frac{\theta \Delta t \sqrt{g h_{\text{2D}}}}{\Delta x \sqrt{1 + b_{\text{2D}} \Delta t}} \ , \ \text{CFL}_{\text{2D},y} = \frac{\theta \Delta t \sqrt{g h_{\text{2D}}}}{\Delta y \sqrt{1 + b_{\text{2D}} \Delta t}} \ . \tag{1.260}$$

Substituting $F_I^n = 1/(1+b_{\text{2D}}\Delta t)\times\theta g\Delta t/\Delta x$ (cf. Equation (1.237)) and $A_I^n = \Delta x_I h_{\text{2D},I} = \Delta x_I h_{\text{2D}}$ (depth across 2D domain is uniform), Equation (1.250) with Equation (1.251) becomes (coupling from 1D to 2D for *drowned-weir* flow):

$$\begin{aligned}&\alpha_{\text{2D}}^n \frac{\zeta_{\text{2D},v}^p + \zeta_{\text{2D},i}^p}{2} + \left(\beta_{\text{2D}}^n + \alpha_{\text{2D}}^n \frac{\theta(\Delta x_{\text{1D-2D}}/\Delta x + b_I\Delta t)}{1+b_{\text{2D}}\Delta t}\right)(\zeta_{\text{2D},v}^p - \zeta_{\text{2D},i}^p)\\ &= \alpha_{\text{2D}}^n \zeta_{\text{1D}}^{p-1} + \beta_{\text{2D}}^n \frac{\Delta x(1+b_{\text{2D}}\Delta t)}{\theta^2 g h_{\text{2D}}\Delta t} q_{\text{1D-2D}}^{p-1} + \text{rhs}_{\text{1D-2D}}^n\ ,\end{aligned} \tag{1.261}$$

with as before $\Delta x_{\text{1D-2D}} \ll \Delta x$ the effective thickness of the interface, and with all known terms at previous time level n collected in $\text{rhs}_{\text{1D-2D}}^n$. Introducing non-dimensional parameter:

$$b_I' = \frac{\theta(\Delta x_{\text{1D-2D}}/\Delta x + b_I\Delta t)}{1+b_{\text{2D}}\Delta t} \tag{1.262}$$

and substituting $\text{CFL}_{\text{2D},x}$ as defined in Equation (1.260), Equation (1.261) may be written as:

$$\begin{aligned}&\alpha_{\text{2D}}^n \frac{\zeta_{\text{2D},i}^p + \zeta_{\text{2D},v}^p}{2} + (\beta_{\text{2D}}^n + \alpha_{\text{2D}}^n b_I')(\zeta_{\text{2D},v}^p - \zeta_{\text{2D},i}^p)\\ &= \alpha_{\text{2D}}^n \zeta_{\text{1D}}^{p-1} + \beta_{\text{2D}}^n \frac{\Delta t}{\Delta x \text{CFL}_{\text{2D},x}^2} q_{\text{1D-2D}}^{p-1} + \text{rhs}_{\text{1D-2D}}^n\ .\end{aligned} \tag{1.263}$$

For $\alpha_{\text{2D}}^n = 1$ and $\beta_{\text{2D}}^n = 1/2 - b_I'$, the contribution of $\zeta_{\text{2D},i}^p$ in the left-hand side of Equation (1.263) vanishes and the coupling equation reduces to:

$$\zeta_{\text{2D},v}^p = \zeta_{\text{1D}}^{p-1} + (1/2 - b_I')\frac{\Delta t}{\Delta x \text{CFL}_{\text{2D},x}^2} q_{\text{1D-2D}}^{p-1} + (\text{rhs}_{\text{1D-2D}}^n)_{\alpha_{\text{2D}}^n=1,\beta_{\text{2D}}^n=1/2-b_I'}\ . \tag{1.264}$$

This equation is not to be applied as such. As mentioned before, better coupling performance is obtained with Equation (1.263) (Equation (1.261)) and an optimized β_{2D}^n. However, a parameterized combination of Equation (1.264) at different locations along the 1D-2D interface can be added to Equation (1.263) to enhance the possibilities for optimizing the coupling performance. Since Equation (1.263) and Equation (1.264) are both valid coupling equations, any linear combination is a valid coupling as well. This generalization of couplings between domains is well known in the field of domain decomposition.

An interesting extension of coupling Equation (1.263) applied along the interface at the indices j is to combine it with the coupling equation consisting of minus two times Equation (1.264) at j plus Equation (1.264) at $j-1$ and $j+1$. That addition represents the finite difference discretization of the second derivative of Equation (1.264) along the 1D-2D interface, which vanishes for a solution mode that is uniform along the interface (second-derivative discretization is zero for a constant) and is maximum for a highly oscillatory solution mode along the interface (second-derivative discretization is maximum for the wiggle mode). The extended coupling equation reads:

$$\begin{aligned}&\alpha_{\text{2D}}^n \frac{\zeta_{\text{2D},v,j}^p + \zeta_{\text{2D},i,j}^p}{2} + (\beta_{\text{2D}}^n + \alpha_{\text{2D}}^n b_I')(\zeta_{\text{2D},v,j}^p - \zeta_{\text{2D},i,j}^p)\\ &- \delta_{\text{2D}}^n(\zeta_{\text{2D},v,j-1}^p - 2\zeta_{\text{2D},v,j}^p + \zeta_{\text{2D},v,j+1}^p)\\ &= \alpha_{\text{2D}}^n \zeta_{\text{1D},j}^{p-1} + \beta_{\text{2D}}^n \frac{\Delta t}{\Delta x \text{CFL}_{\text{2D},x}^2} q_{\text{1D-2D},j}^{p-1}\\ &- \delta_{\text{2D}}^n\Big(\zeta_{\text{1D},j-1}^{p-1} - 2\zeta_{\text{1D},j}^{p-1} + \zeta_{\text{1D},j+1}^{p-1}\\ &\qquad + (1/2 - b_I')\frac{\Delta t}{\Delta x \text{CFL}_{\text{2D},x}^2}(q_{\text{1D-2D},j-1}^{p-1} - 2q_{\text{1D-2D},j}^{p-1} + q_{\text{1D-2D},j+1}^{p-1})\Big)\\ &+ \text{rhs}_{\text{1D-2D},j}^n - \delta_{\text{2D}}^n(\text{rhs}_{\text{1D-2D},j-1}^n - 2\text{rhs}_{\text{1D-2D},j}^n + \text{rhs}_{\text{1D-2D},j+1}^n)_{\alpha_{\text{2D}}^n=1,\beta_{\text{2D}}^n=1/2-b_I'}\ .\end{aligned} \tag{1.265}$$

Note the dependence of the $\text{rhs}^n_{\text{1D-2D}}$ on α^n_{2D} and β^n_{2D}.

Coupling Equation (1.265) uses the same computational stencil (minus one unknown) as Equation (1.238) and preserves (after a proper scaling) the symmetry of the system of the equations. For suitable choices of the coupling parameters ($\alpha^n_{\text{2D}} = 1$, $\beta^n_{\text{2D}} > -b'_I$, and $\delta^n_{\text{2D}} \geq 0$) the positive definiteness of the system is preserved as well.

Parameter δ^n_{2D} and accompanying terms have been introduced in Equation (1.265) to investigate the gain in performance that may be obtained with this extension of coupling Equation (1.263), with the intention to figure out if implementation of it (somewhere in the future) may be worth considering. NB, the combination of Equation (1.263) with the second difference of Equation (1.263) for two different values of β^n_{2D} (taking $\alpha^n_{\text{2D}} = 1$) will not be considered. Although in principle possible, it would destroy the symmetry of the matrix to be solved in 2D domains *and* would extend near 1D-2D interfaces the computational stencil of the ζ^p equations to be solved (the left-hand side of Equation (1.238)). Only the combination of Equation (1.263) with the second difference of Equation (1.263) for $\alpha^n_{\text{2D}} = 1$ and $\beta^n_{\text{2D}} = 1/2 - \theta b'_I$, i.e., with the second difference of Equation (1.264), is feasible enough to consider for implementation.

For $\alpha^n_{\text{2D}} = 1$ and $\beta^n_{\text{2D}} = -1/2 - b'_I$, the contribution of $\zeta^p_{\text{2D},v}$ in the left-hand side of Equation (1.263) vanishes. The result is an equation totally unsuitable for the coupling of the 1D domain to the 2D domain, among other things because combination with the equations Equation (1.238) destroys the positive definiteness of the system:

$$\begin{aligned}
&\zeta^p_{\text{2D},i} = \zeta^{p-1}_{\text{1D}} + (-1/2 - b'_I)\frac{\Delta t}{\Delta x \text{CFL}^2_{\text{2D},x}} q^{p-1}_{\text{1D-2D}} + (\text{rhs}^n_{\text{1D-2D}})_{\alpha^n_{\text{2D}}} = 1, \\
&\beta^n_{\text{2D}} = -1/2 - b'_I \ .
\end{aligned}$$

However, the equation might be useful in combination with Equation (1.263). The result is an equation similar to Equation (1.265):

$$\begin{aligned}
&\alpha^n_{\text{2D}} \frac{\zeta^p_{\text{2D},v,j} + \zeta^p_{\text{2D},i,j}}{2} + (\beta^n_{\text{2D}} + \alpha^n_{\text{2D}} b'_I)(\zeta^p_{\text{2D},v,j} - \zeta^p_{\text{2D},i,j}) \\
&- \delta^n_{\text{2D}}(\zeta^p_{\text{2D},i,j-1} - 2\zeta^p_{\text{2D},i,j} + \zeta^p_{\text{2D},i,j+1}) \\
&= \alpha^n_{\text{2D}} \zeta^{p-1}_{\text{1D},j} + \beta^n_{\text{2D}} \frac{\Delta t}{\Delta x \text{CFL}^2_{\text{2D},x}} q^{p-1}_{\text{1D-2D},j} \\
&- \delta^n_{\text{2D}} \Big(\zeta^{p-1}_{\text{1D},j-1} - 2\zeta^{p-1}_{\text{1D},j} + \zeta^{p-1}_{\text{1D},j+1} \\
&\qquad - (1/2 + b'_I)\frac{\Delta t}{\Delta x \text{CFL}^2_{\text{2D},x}} (q^{p-1}_{\text{1D-2D},j-1} - 2q^{p-1}_{\text{1D-2D},j} + q^{p-1}_{\text{1D-2D},j+1}) \Big) \\
&+ \text{rhs}^n_{\text{1D-2D},j} - \delta^n_{\text{2D}}(\text{rhs}^n_{\text{1D-2D},j-1} - 2\text{rhs}^n_{\text{1D-2D},j} + \text{rhs}^n_{\text{1D-2D},j+1})_{\alpha^n_{\text{2D}}=1, \beta^n_{\text{2D}}=-1/2-b'_I} \ .
\end{aligned} \tag{1.266}$$

The advantage of Equation (1.266) over Equation (1.265) is that it fits well within the current implementation of the solution procedure of Equation (1.238), in which the virtual unknowns $\zeta^p_{\text{2D},v}$ have been eliminated. Unlike Equation (1.265) that has several unknowns $\zeta^p_{\text{2D},v}$ in its left-hand side, Equation (1.266) with only $\zeta^p_{\text{2D},v,j}$ in its right-hand side does not require the extension of the current matrix solver with equations for the virtual unknowns. On the other hand, it is expected that the performance of Equation (1.266) will not be as good as that of Equation (1.265). In particular, it is expected that the demand for positive definiteness will impose a restriction on the choice of δ^n_{2D} that will hamper full optimization.

Exactly the same procedure as for the coupling from the 1D domain to the 2D domain is applied to construct the coupling in the other direction, from the 2D domain to the 1D domain.

Substitution of $F_I^n = 1/(1+b_{\text{2D}}\Delta t) \times \theta g \Delta t/\Delta x$ and $A_I^n = \Delta x_I h_{\text{2D},I} = \Delta x_I h_{\text{2D}}$ in Equation (1.253) yields, using Equation (1.251) (coupling from 2D to 1D for *drowned-weir* flow):

$$\begin{aligned}
&\alpha_{\text{1D}}^n \zeta_{\text{1D}}^p - \beta_{\text{1D}}^n \frac{\Delta x(1+b_{\text{2D}}\Delta t)}{\theta^2 g h_{\text{2D}} \Delta t} q_{\text{1D-2D}}^p \\
&= \alpha_{\text{1D}}^n \frac{\zeta_{\text{2D},i}^{p-1} + \zeta_{\text{2D},v}^{p-1}}{2} + \left(\beta_{\text{1D}}^n - \alpha_{\text{1D}}^n \frac{\theta(\Delta x_{\text{1D-2D}}/\Delta x + b_I \Delta t)}{1+b_{\text{2D}}\Delta t}\right)(\zeta_{\text{2D},i}^{p-1} - \zeta_{\text{2D},v}^{p-1}) + \text{rhs}_{\text{2D-1D}}^n \ .
\end{aligned} \tag{1.267}$$

Notice the similarity between this equation and Equation (1.261).

Upon substitution of b_I' as defined in Equation (1.262) and $\text{CFL}_{\text{2D},x}$ as defined in Equation (1.260), Equation (1.267) becomes:

$$\begin{aligned}
&\alpha_{\text{1D}}^n \zeta_{\text{1D}}^p - \beta_{\text{1D}}^n \frac{\Delta t}{\Delta x \text{CFL}_{\text{2D},x}^2} q_{\text{1D-2D}}^p \\
&= \alpha_{\text{1D}}^n \frac{\zeta_{\text{2D},i}^{p-1} + \zeta_{\text{2D},v}^{p-1}}{2} + (\beta_{\text{1D}}^n - \alpha_{\text{1D}}^n \theta b_I')(\zeta_{\text{2D},i}^{p-1} - \zeta_{\text{2D},v}^{p-1}) + \text{rhs}_{\text{2D-1D}}^n \ ,
\end{aligned} \tag{1.268}$$

which is similar to Equation (1.263).

The most useful simplified version of coupling Equation (1.268) is the one that is obtained for $\alpha_{\text{1D}}^n = 1$ and $\beta_{\text{1D}}^n = 0$:

$$\zeta_{\text{1D}}^p = \frac{\zeta_{\text{2D},i}^{p-1} + \zeta_{\text{2D},v}^{p-1}}{2} - b_I'(\zeta_{\text{2D},i}^{p-1} - \zeta_{\text{2D},v}^{p-1}) + (\text{rhs}_{\text{2D-1D}}^n)_{\alpha_{\text{1D}}^n=1,\beta_{\text{1D}}^n=0} \ . \tag{1.269}$$

The extended coupling obtained upon combining Equation (1.268) and the second difference of Equation (1.269) (minus two times Equation (1.269) at j plus Equation (1.269) at $j-1$ and $j+1$) reads:

$$\begin{aligned}
&\alpha_{\text{1D}}^n \zeta_{\text{1D},j}^p - \beta_{\text{1D}}^n \frac{\Delta t}{\Delta x \text{CFL}_{\text{2D},x}^2} q_{\text{1D-2D},j}^p - \delta_{\text{1D}}^n (\zeta_{\text{1D},j-1}^p - 2\zeta_{\text{1D},j}^p + \zeta_{\text{1D},j+1}^p) \\
&= \alpha_{\text{1D}}^n \frac{\zeta_{\text{2D},i,j}^{p-1} + \zeta_{\text{2D},v,j}^{p-1}}{2} + (\beta_{\text{1D}}^n - \alpha_{\text{1D}}^n b_I')(\zeta_{\text{2D},i,j}^{p-1} - \zeta_{\text{2D},v,j}^{p-1}) \\
&- \delta_{\text{1D}}^n \Big((\zeta_{\text{2D},i,j-1}^{p-1} + \zeta_{\text{2D},v,j-1}^{p-1})/2 - (\zeta_{\text{2D},i,j}^{p-1} + \zeta_{\text{2D},v,j}^{p-1}) + (\zeta_{\text{2D},i,j+1}^{p-1} + \zeta_{\text{2D},v,j+1}^{p-1})/2 \\
&\qquad + b_I' \big(\zeta_{\text{2D},v,j-1}^{p-1} - \zeta_{\text{2D},i,j-1}^{p-1} - 2(\zeta_{\text{2D},v,j}^{p-1} - \zeta_{\text{2D},i,j}^{p-1}) + \zeta_{\text{2D},v,j+1}^{p-1} - \zeta_{\text{2D},i,j+1}^{p-1} \big) \Big) \\
&+ \text{rhs}_{\text{2D-1D},j}^n - \delta_{\text{1D}}^n (\text{rhs}_{\text{2D-1D},j-1}^n - 2\text{rhs}_{\text{2D-1D},j}^n + \text{rhs}_{\text{2D-1D},j+1}^n)_{\alpha_{\text{1D}}^n=1,\beta_{\text{1D}}^n=0} \ .
\end{aligned} \tag{1.270}$$

Note again the dependence of the $\text{rhs}_{\text{2D-1D}}^n$ on α_{1D}^n and β_{1D}^n.

Equation (1.270) has only a single unknown $q_{\text{1D-2D}}^p$ in its left-hand side. This makes it relatively easy to combine it with the ζ^p equations Equation (1.238) to be solved in 1D domains. By combining both equations such that $q_{\text{1D-2D}}^p$ in the former and single variable q_c^p in the latter cancel (using $q_c = -q_{\text{1D-2D}}$), a system of equation in only the $\zeta_{\text{1D},j}^p$ is obtained that has the same tridiagonal structure as Equation (1.238). Implementation of Equation (1.270) should therefore be feasible. It is for this reason that Equation (1.269) has been considered for the extension of Equation (1.268).

Similar to the coupling from 1D to 2D, parameter δ_{1D}^n and accompanying terms have been introduced in Equation (1.270) to investigate the gain in performance that may be obtained

with this extension of coupling Equation (1.268) from 2D to 1D, with the intention to figure out if implementation of it (somewhere in the future) may be worth considering. NB, the combination of Equation (1.268) with the second difference of Equation (1.268) for two different values of β^n_{1D} (i.e., $\beta^n_{\text{1D}} = 0$ and a value $\beta^n_{\text{1D}} \neq 0$) will not be considered. Although in principle possible, with an extension based on $\beta^n_{\text{1D}} \neq 0$ one would get in the left-hand side of Equation (1.270) three coupled $q^p_{\text{1D-2D}}$ at the indices $j-1$, j and $j+1$ that cannot be easily eliminated from the equations. This would destroy the tridiagonal structure of the systems of equations to be solved in 1D domains. Only the combination of Equation (1.268) with the second difference of Equation (1.268) for $\alpha^n_{\text{1D}} = 1$ and $\beta^n_{\text{1D}} = 0$, i.e., with the second difference of Equation (1.269), is feasible enough to consider for implementation. NB, a proper choice of the coupling parameters β^n_{1D} and δ^n_{1D} (taking $\alpha^n_{\text{1D}} = 1$) is required to ensure the positive definiteness of the systems of equations.

1.4.4.2 Normal-mode analysis

Because of the simplifications introduced (straight 1D-2D interface, uniform depths, rectangular domains, uniform grids, cf. the beginning of Section 1.4.4.1) and ignoring the effect of the conditions (far) upstream and downstream along the 1D-2D interface, the solution in the direction tangential to the 1D-2D interface can be expanded in linearly independent Fourier modes. For the convergence analysis and the subsequent optimization of the coupling parameters, it is convenient to consider the (convergence) errors per time step $\Delta\zeta^p_j = \zeta^{n+1}_j - \zeta^p_j$ (in 1D) and $\Delta\zeta^p_{i,j} = \zeta^{n+1}_{i,j} - \zeta^p_{i,j}$ (in 2D), and to expand them in Fourier modes. So consider convergence-error modes of the form:

$$\Delta\zeta^p_j = \Delta Z^p_{\text{1D},k} \exp\left(iky_j\right) , \tag{1.271)a}$$

$$\Delta\zeta^p_{i,j} = \Delta Z^p_{\text{2D},k} \lambda^i_k \exp\left(iky_j\right) , \tag{1.271)b}$$

with $\Delta Z^p_{\text{1D},k}$ and $\Delta Z^p_{\text{2D},k}$ the amplitude of the mode with tangential wave number k ($0 \leq k\Delta y \leq \pi$) at iteration level p in respectively the 1D domain and the 2D domain, with $\exp(iky_j)$ the behavior of the modes in the direction tangential to the 1D-2D interface (in 1D and 2D the same, and assumed to be periodic), and with λ^i_k the exponential behavior of the error mode in 2D normal to the 1D-2D interface. The 1D error mode is uniform in that direction, since the solution in 1D channels has been assumed uniform in crosswise direction.

Note that the i inside the $\exp$ functions in Equation (1.271) denotes the unit imaginary number $\sqrt{-1}$ in the description of the oscillatory Fourier modes. The other occurrences of i (in Equation (1.271)b only) denote the index in x-direction (the direction normal to the 1D-2D interface) of the cell centers, with superscript i in λ^i_k the power in the exponential amplitude behavior of $\Delta\zeta^p_{i,j}$ normal to the interface.

It is easy to verify that, since ζ^{n+1}_j and $\zeta^{n+1}_{i,j}$ (and $q^{n+1}_{\text{1D-2D},j}$) are the converged solution of the equations Equation (1.257), Equation (1.259), Equation (1.263) (or Equation (1.265), or Equation (1.266)) and Equation (1.268) (or Equation (1.270)), $\Delta\zeta^p_j$ and $\Delta\zeta^p_{i,j}$ (and $\Delta q^p_{\text{1D-2D},j} = q^{n+1}_{\text{1D-2D},j} - q^p_{\text{1D-2D},j}$) satisfy the homogeneous version of these equations. The homogeneous equations are obtained by omitting all terms at the previous time level n, i.e., $\text{rhs}^n_{\text{1D},j}$, $\text{rhs}^n_{\text{2D},i,j}$, $\text{rhs}^n_{\text{1D-2D}}$ and $\text{rhs}^n_{\text{2D-1D}}$.

The first step in the analysis is the determination of the normal-mode convergence-error behavior in the 2D domain (no normal mode in the 1D domain where the convergence error normal to a 1D-2D interface is uniform). Inserting Equation (1.271)b in the homogenized Equation (1.259), one obtains for the 2D domain the quadratic equation:

$$1 + 2\text{CFL}^2_{\text{2D},x} + 4\sin^2(k\Delta y/2)\text{CFL}^2_{\text{2D},y} - \text{CFL}^2_{\text{2D},x}\left(\lambda^{-1}_k + \lambda_k\right) = 0 ,$$

whose two solutions are:

$$\lambda_{\pm k} = 1/(2\mathsf{CFL}_{2\mathsf{D},x}^2) + \Gamma + 1 \pm \sqrt{\big(1/(2\mathsf{CFL}_{2\mathsf{D},x}^2) + \Gamma\big)\big(1/(2\mathsf{CFL}_{2\mathsf{D},x}^2) + \Gamma + 2\big)}$$
$$\approx \frac{1}{\mathsf{CFL}_{2\mathsf{D},x}^{\pm 2}} \exp\big(\pm 2\mathsf{CFL}_{2\mathsf{D},x}^2 \pm 4\sin^2(k\Delta y/2)\mathsf{CFL}_{2\mathsf{D},y}^2\big) \ , \tag{1.272}$$

with $\Gamma = 2\sin^2(k\Delta y/2)\mathsf{CFL}_{2\mathsf{D},y}^2/\mathsf{CFL}_{2\mathsf{D},x}^2$. The approximate expansion in the second line is valid for small $\mathsf{CFL}_{2\mathsf{D},x}^2$ and $\mathsf{CFL}_{2\mathsf{D},y}^2$, when terms of $O(\mathsf{CFL}_{2\mathsf{D},x}^4)$, $O(\mathsf{CFL}_{2\mathsf{D},x}^2\mathsf{CFL}_{2\mathsf{D},y}^2)$ and $O(\mathsf{CFL}_{2\mathsf{D},y}^4)$ can be neglected. Note that for small k one can apply $\sin^2(k\Delta y/2) \approx (k\Delta y)^2/4$, neglecting terms of $O((k\Delta y)^4)$.

The two solutions Equation (1.272) describe the behavior in x-direction of two modes $\Delta\zeta_{i,j}^p$ per Fourier mode k that roughly speaking correspond with a left-going and a right-going error wave in the 2D domain. Because of the symmetry of Equation (1.259) it turns out that these modes are the inverse of each other, i.e., $\lambda_{-k}\lambda_{+k} = 1$.

The larger $\mathsf{CFL}_{2\mathsf{D},x}$, i.e., the larger the non-dimensional time step, the closer λ_{-k} and λ_{+k} are to 1 and the larger the penetration distance into the domain, while the larger k, i.e., the larger the non-dimensional angle of the error wave with respect to the normal direction, the more that penetration is tangential to the boundary. This behavior is recognized in Figure 1.63 that shows the normalized normal behavior of the two modes $\Delta\zeta_{i,j}^p$ per wave number k (one left-going, one right-going) (right-going mode λ_{-k} and left-going mode λ_{+k}) for a number of combinations of $\mathsf{CFL}_{2\mathsf{D}} = \mathsf{CFL}_{2\mathsf{D},x} = \mathsf{CFL}_{2\mathsf{D},y}$ and $k\Delta y$.

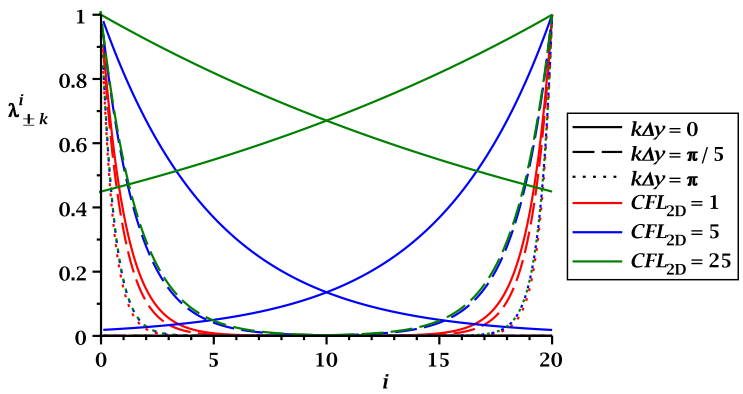

Figure 1.63: *Behavior of normal-mode solutions of Equation (1.259) for 3 different tangential modes $k\Delta y$ (solid, dashed and dotted lines) and 3 different values of $\mathsf{CFL}_{2\mathsf{D}} = \mathsf{CFL}_{2\mathsf{D},x} = \mathsf{CFL}_{2\mathsf{D},y}$ (red, blue and green lines).*

For $4\sin^2(k\Delta y/2)\mathsf{CFL}_{2\mathsf{D},y}^2 \ll 1$ and $\mathsf{CFL}_{2\mathsf{D},x} \gg 1$, expression Equation (1.272) can be approximated by $\lambda_{\pm k} \approx 1 \pm 1/\mathsf{CFL}_{2\mathsf{D},x}$, from which $\ln(\lambda_{\pm k}) \approx \pm 1/\mathsf{CFL}_{2\mathsf{D},x}$ can be obtained. This shows in particular that for the constant mode tangential to a 1D-2D interface ($k\Delta y = 0$) the exponential decay or increase in normal direction per grid cell $\ln(\lambda_{\pm k})$ is about inversely proportional to $\mathsf{CFL}_{2\mathsf{D},x}$ and hence can be rather small, as can also be observed in Figure 1.63: the larger $\mathsf{CFL}_{2\mathsf{D},x}$, the further a constant tangential-mode perturbation (convergence error) at a 1D-2D interface (which is a boundary for the 2D domain) penetrates into the 2D domain.

The situation is completely different for the shortest (wiggle) mode $k\Delta y = \pi$ along the 1D-2D interface. Assuming equal $\mathsf{CFL}_{2\mathsf{D},x}$ and $\mathsf{CFL}_{2\mathsf{D},y}$ for the moment, $\Gamma = 2$ and hence

$\lambda_{\pm k} \approx 3 \pm 2\sqrt{2}$ for $\text{CFL}_{2\text{D},x}$ sufficiently larger than 1. In fact, for $k\Delta y = \pi$, for all $\text{CFL}_{2\text{D},y} = \text{CFL}_{2\text{D},x} > 1$ the large exponential decay or increase per grid cell $\ln(\lambda_{\pm k}) \approx \ln(3 \pm 2\sqrt{2}) = \pm 1.763$. In other words, for the wiggle mode tangential to a 1D-2D interface ($k\Delta y = \pi$) the exponential decay or increase per grid cell $\ln(\lambda)$ is large and virtually independent of $\text{CFL}_{2\text{D},x}$, cf. Figure 1.63.

The large range in normal-mode behavior for large $\text{CFL}_{2\text{D},x}$ complicates the optimization of the interface coupling for maximum convergence speed, but also makes this optimization important.

It is now assumed that the optimization of the coupling at a 1D-2D interface can be considered as a problem that is independent of the coupling optimization at other interfaces. This assumption is obviously not true, but is expected to hold reasonably well. The reason for this is the exponential normal-mode decay. Figure 1.63 shows that when there are 20 grid cells between the 1D-2D interfaces present at both sides of a 2D domain, the effect of the solution at one interface on the solution at the other is less than $50\ \%$ at a $\text{CFL}_{2\text{D}}$ value as high as 25. The effect that the presence of an opposite interface at a distance of 20 grid cells has on the behavior of the coupling at a 1D-2D interface has twice that attenuation factor, i.e., it has an effect of about $20\ \%$ at $\text{CFL}_{2\text{D}} = 25$. Since 20 2D grid cells between 1D-2D interfaces is not a large number, but especially because in practice $\text{CFL}_{2\text{D}}$ will usually be (much) smaller than 25, the mutual influence between 1D-2D interfaces will generally be small.

An important exception is the situation that occurs when 1D channels are close together, such as in a braided river system as illustrated in Figure 1.58. The narrow 2D areas that are in between 1D channels may in that case be only a few grid cells wide. Also near nodes connecting branches at different angles there may be a mutual influence of the coupling at different 1D-2D interfaces. Because of the complexity of this problem it is not feasible to include the interdependence of couplings at different 1D-2D interfaces in the optimization, so this aspect will be ignored. Likewise, because of the simplifications introduced, the effect of (strongly) curved 1D-2D interfaces, non-uniform depths, varying grid sizes, varying 1D channel widths, and varying friction coefficients will be ignored.

Concerning the optimization strategy it is first noticed that, because 1D solutions are uniform in crosswise direction, all 1D normal convergence-error modes are constant. See also expression Equation (1.271)a that is independent of x, hence constant in x-direction. As a result, 1D convergence-error modes are 'global', in the sense that they depend equally on the convergence error at the 1D-2D interface at either side of the 1D channel. This creates an interdependency between the 1D-2D couplings at each of the two lateral boundaries of a 1D channel. Insight in the strength of that interdependency can be obtained from a normal-mode analysis where the 1D-2D couplings at both lateral boundaries of a 1D channel are considered simultaneously. Pending such an analysis, it is presently not known how strong that interdependency is.

In contrast to the 1D domains, the 2D domains have normal modes that decay exponentially with the distance from the boundaries, cf. Equation (1.271)b, Equation (1.272), and Figure 1.63. Assuming that this decay is large enough[11], the solution imposed at a boundary will within one time step hardly affect the solution at the opposity boundary.

[11] This requires a sufficiently large number of grid cells across the width of a 2D domain and a sufficiently small CFL number, cf. Figure 1.63.

1.4.5 Properties of the horizontal 1D-2D coupling

Expected properties of the SOBEK 3 1D-2D coupling:

- ◇ Works well for sufficiently small Courant number CFL or sufficiently wide 2D areas in between the 1D channel sections, hence may not work that well if large CFL *and* narrow 2D areas (as in between two parallel 1D channels that are close together, cf. Figure 1.58).
- ◇ Large CFL may be a problem because then large spreading in exponential behavior of normal modes, cf. Figure 1.63, while the optimization of the coupling is for a single 'average' mode. In consequence, the optimization is not optimal for the full range of modes.
  NB, a better optimization for the full range of modes requires consideration of the proposed extended couplings.
  IMPORTANT: large CFL are in particular a problem when the 1D grids and the 2D grids are conformal, i.e., match along an interface (the 1D and 2D grid size in tangential direction along the 1D-2D interfaces are equal). This is the current restriction; the presently available coupling has been developed with this grid restriction. Required is a coupling for 1D and 2D models as they are used in practice, where the grid size applied in the 1D channels is usually (much) larger than the grid size applied in the 2D areas. This (strongly) reduces the number of relevant Fourier modes, hence the spreading in normal-mode exponential behavior, and therefore improves the applicability of the applied single-mode optimization of the 1D-2D coupling. This is something to take into account when testing the coupling for the current conformal-grid limitation.
- ◇ Optimization of the coupling is based on the assumption of constant coefficients and straight 1D-2D interfaces, hence may not work that well in case of large variation in 2D depth or 2D friction coefficient along an interface, if large curvature of a 1D-2D interface, if corners in an interface at intersections of 1D channel sections, . . .

2 Numerical concepts (1D)

2.1 Staggered grid

The mathematical formulations of the shallow water flow and transport model D-Flow 1D were presented in the previous chapters. To solve these partial differential equations the equations should be transformed to the discrete space. In this chapter the space discretization and the time integrations methods in D-Flow 1D will be briefly presented.

For the numerical solution of the flow equations a so-called "staggered grid" is used (Stelling and Duinmeijer, 2003; Stelling and Verwey, 2006). A channel network is discretized in

- ◇ water level points (also called ζ-points, grid points or calculation points) and
- ◇ velocity points (also called u-points).

Figure 2.1 shows a schematic view of a staggered grid. Water level points form the basis of the computational grid, velocity points are located half between two ζ-points. At ζ-points the continuity equation is solved, while the momentum equation is solved at u-points. Consequently, on ζ-points a water level is computed, while on u-points the velocities are computed.

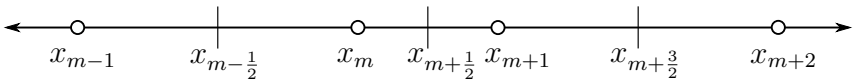

Figure 2.1: *Non-equidistant staggered grid for unsteady channel flow,*
ζ-points located at $x_m, m \in \mathbb{N}$,
u-points located at $x_{m+\frac{1}{2}}, m \in \mathbb{N}$

To understand the algorithms in D-Flow 1D it is important to know something about the history of the software package. The first version of D-Flow 1D was developed and programmed by G.S. Stelling for the simulation of unsteady flows in sewer systems. The hydraulics in such systems can be very complicated e.g. the flows can locally be supercritical. Therefore, numerical stability and robustness were major factors for the choice of a numerical method. It was therefore decided to apply in D-Flow 1D a **finite volume** approximation of the conservation laws on a **staggered grid** and for an implicit time integration of the hydrostatic pressure term. In D-Flow 1D, the final system of difference equations is solved with a Conjugate Gradient algorithm. The discretizations of the equations are made in such a way that the resulting system of equations is symmetric.

2.2 Computational grid and definitions

D-Flow 1D is a numerical model that is based on finite volumes. The arrangement of the flow variables is presented in Figure 2.2. Let x be the longitudinal coordinate, which follows the channel axis. The channel may be a branch of a river network. To discretise the balance equations in space, we have introduced a so-called staggered grid with m the grid index along the main axis, see Figure 2.2.

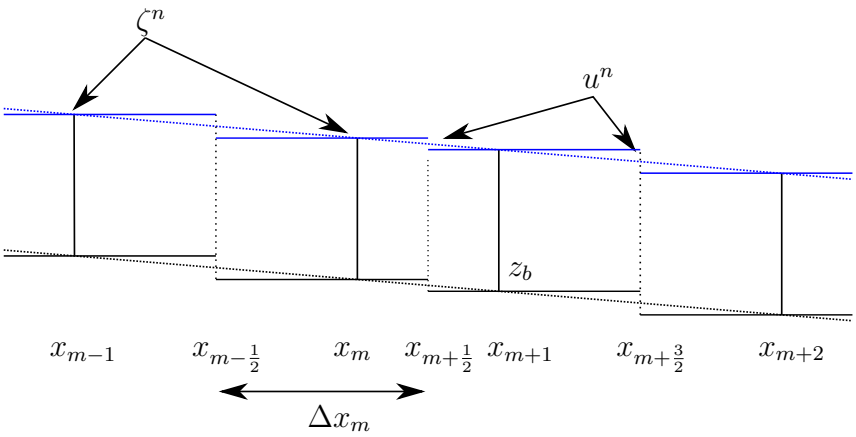

Figure 2.2: *Staggered flow grid*

The grid of the water level (pressure) points is given by $x_m, m \in \mathbb{N}$ and the velocity points by $x_{m+\frac{1}{2}}, m \in \mathbb{N}$. We approximate the river (channel or pipe) by finite volumes V_m around the water level points. In D-Flow 1D these finite volumes, at the water level points, for the mass balance are the total volume, i.e. the summation of the flow and the storage area.

D-Flow 1D, a one-dimensional flow model

We assume that the independent variables in our one-dimensional model the water level $\zeta\,(x,t)$, the velocity $u(x,t)$ and discharge $Q(x,t)$ are constant for a cross section (independent of y). The average flow velocity in the cross section $u\,(x,t)$ along the river axis, is defined as:

$$u\,(x,t) = \frac{\int_{z_b}^{\zeta(t)} \int_{y'} u(x,y',z',t) h(x,y')\, dy' dz'}{A_F\,(x,\zeta(t))} = \frac{Q}{A_F}\,(x,t) \tag{2.1}$$

River cross-sections often consist of a main channel through which the discharge is conveyed and shallow flood plains which are mainly functioning as storage areas, particular when groynes are present for channel regulation. Therfor the water in the storage area is assumed to be stagnant ($u(x,t) \equiv 0$). This assumption is important for the derivation of the advection terms in the following sections.

The length of a finite volume along the river axis is $\Delta x_m = \left[x_{m-\frac{1}{2}}, x_{m+\frac{1}{2}}\right]$ and the volume is defined by

$$V_{T,m}\,(\zeta_m(t)) = \int_{x_{m-\frac{1}{2}}}^{x_{m+\frac{1}{2}}} \int_{z_b(x')}^{\zeta(x',t)} W_T(x',z')\, dz' dx' \tag{2.2}$$

with

$V_{T,m}$	Total volume at water level point m,
W_T	Total width, including flow and storage area,
$\zeta_m(t)$	Time dependent water level at location m.

Note: that the vertical integral is from the bed level to the free surface with $W_T\,(x,z)$ the total width of the river. Let $h(x,y,t) = \zeta(x,t) - z_b(x)$ be the total water depth.

Instead of taking an integral in the vertical of the width over the total water depth h, we can also take an integral of the depth of the river, from the left bank $y = b_l(x)$ to the right bank of the river $y = b_r(x)$.

$$V_{T,m}\left(\zeta_m(t)\right) = \int_{x_{m-\frac{1}{2}}}^{x_{m+\frac{1}{2}}} \int_{z_b(x')}^{\zeta(x',t)} W_T(x', z')\, dz' dx' \tag{2.3}$$

$$= \int_{x_{m-\frac{1}{2}}}^{x_{m+\frac{1}{2}}} \int_{b_l(x')}^{b_r(x')} h(x', y', t)\, dy' dx' \tag{2.4}$$

The total area (summation of flow and storage area) of the cross section in the water level point is defined as:

$$A_T\left(\zeta_m(t)\right) = \int_{z_b(x_m)}^{\zeta_m(t)} W_T(x_m, z')\, dz' \tag{2.5}$$

$$= \int_{b_l(x_m)}^{b_r(x_m)} h(x_m, y', t)\, dy' \tag{2.6}$$

Because we assume that the average velocity component in the direction of the river axis for the storage area is zero (stagnant), we can split this integral over the total area of a cross section in a flowing part and a storage part.

$$A_T\left(x, \zeta(t)\right) = A_F\left(x, \zeta(t)\right) + A_S\left(x, \zeta(t)\right) \tag{2.7}$$

$$= \int_{y', u(x,t)} \left[\zeta(x,t) - z_b(x, y')\right] dy' \quad \text{(flow, time-dependent)}$$

$$+ \int_{y', u(x)\equiv 0} \left[\zeta(x,t) - z_b(x, y')\right] dy' \quad \text{(storage, stagnant)} \tag{2.8}$$

The flow area is defined as:

$$A_F\left(x, \zeta(t)\right) = \int_{y'} \left[\zeta(x,t) - z_b(x, y')\right] dy' \quad \text{velocity is time-dependent,} \tag{2.9}$$

and the storage area is defined as

$$A_S\left(x, \zeta(t)\right) = \int_{y'} \left[\zeta(x,t) - z_b(x, y')\right] dy' \quad \text{the water is stagnant.} \tag{2.10}$$

When the depths of the main channel and the flood plains are very different and the discharge through the flood plain can not be neglected, it is better to schematize the cross sections in a 1D-model into two parallel channels: one main channel or conveyance channel and one for the flood plain.

The total volume in a water level point is approximated by:

$$V_{T,m}\left(\zeta_m(t)\right) \approx A_T(\zeta_m(t))\Delta x_m \tag{2.11}$$

Note: Incase of morphology, the bed level is also a function of time, $z_b(t)$, and therefor the total depth is also dependent on the bed level ($h(t) = \zeta(t) - z_b(t)$) and not only on the water level, $\zeta(t)$.

2.3 Approximations on u-calculation points, upwinding

Due to the staggered grid the water levels and the bed levels are missing at the grid point $x_{m+\frac{1}{2}}$. These values, $\overline{\zeta}_{m+\frac{1}{2}}$ and $\overline{z}_{b,m+\frac{1}{2}}$, should be approximated by some kind of interpolation from the values at the water level points.

In D-Flow 1D, the flow area and the wet perimeter are specified by the user at the water level points. To determine the flow area in a velocity point an upwind approach is used to determine the total waterdepth h at the velocity point.

$$\overline{h}_{m+\frac{1}{2}}(t) = \begin{cases} \zeta_m(t) - z_{b_m} & u_{m+\frac{1}{2}} > 0 \\ \zeta_{m+1}(t) - z_{b_{m+1}} & u_{m+\frac{1}{2}} < 0 \\ \max\left(\zeta_m(t), \zeta_{m+1}(t)\right) - \max(z_{b_m}, z_{b_{m+1}}) & u_{m+\frac{1}{2}} = 0 \end{cases} \tag{2.12}$$

The approximation of the flow area in the cross section $x_{m+\frac{1}{2}}$ is given by:

$$A_F\left(x_{m+\frac{1}{2}}, \overline{\zeta}_{m+\frac{1}{2}}(t)\right) = \int_{y'} \overline{h}_{m+\frac{1}{2}}(x_{m+\frac{1}{2}}, y')\, dy' \quad \text{flow area} \tag{2.13}$$

Like the water levels we assume that the velocity in a one dimensional model is independent of the y-coordinate .

The water level and the velocity (discharge) characterise the flow in a one-dimensional network model.

2.4 Time integration

In this section the time integration of D-Flow 1D is explained. A subdivision is made between the time integration of the continuity equation and of the momentum equation.

2.4.1 Time integration of the continuity equation

The rate of change of the total mass in a finite volume approach at the water level point m ($V_{T,m}$) is the sum of the rate of change of the flow ($V_{F,m}$) and storage ($V_{S,m}$) volumes, and is given by:

$$\frac{dV_{T,m}}{dt} = \frac{dV_{F,m}}{dt} + \frac{dV_{S,m}}{dt} \tag{2.14}$$

The fluxes through the interfaces of the finite volume at the water level point m are given by

$$Q_{m+\frac{1}{2}} - Q_{m-\frac{1}{2}} \tag{2.15}$$

so the balance equation for the conservation of mass read

$$\frac{dV_{T,m}}{dt} + Q_{m+\frac{1}{2}} - Q_{m-\frac{1}{2}} = q_m \Delta x_m \tag{2.16}$$

with

q_m lateral discharges per unit length [m^2 s^{-1}]

These lateral discharges are volumes of water entering or leaving the finite volume. It may be a point source (pump, discharge) or an incoming flow across a specified branch (for instance rainfall or groundwater seepage). An incoming volume of water can also be supplied by a hydraulic structure.

In most of the 1D network branches, the volume of the finite volume for mass is a non-linear function of the water level $V_{T,m}\left(\zeta\left(t\right)\right)$.

The continuity and momentum equations are integrated in time with a so-called θ-scheme, leading to the discrete continuity equation:

$$\frac{V_{T,m}(\zeta^{n+1}) - V_{T,m}(\zeta^{n})}{\Delta t} + Q^{n+\theta}_{m+\frac{1}{2}} - Q^{n+\theta}_{m-\frac{1}{2}} = q_m \Delta x_m \tag{2.17}$$

with

$$\begin{aligned} Q^{n+\theta}_{m+\frac{1}{2}} &= \theta Q^{n+1}_{m+\frac{1}{2}} + (1-\theta) Q^{n}_{m+\frac{1}{2}} \\ &= \theta A_F\left(x_{m+\frac{1}{2}}, \overline{\zeta}^{n}_{m+\frac{1}{2}}\right) u^{n+1}_{m+\frac{1}{2}} + (1-\theta) A_F\left(x_{m+\frac{1}{2}}, \overline{\zeta}^{n}_{m+\frac{1}{2}}\right) u^{n}_{m+\frac{1}{2}} \end{aligned} \tag{2.18}$$

To solve this equation for the water level ζ^{n+1}_m at the new time level it is linearised, using:

$$V_{T,m}(\zeta^{n+1,p+1}) = V_{T,m}(\zeta^{n+1,p}) + \left(\zeta^{n+1,p+1}_m - \zeta^{n+1,p}_m\right) \frac{dV_{T,m}(\zeta^{n+1,p})}{d\zeta} \tag{2.19}$$

from Equation (2.3) it follows that the change of the finite volume with the water level is the total horizontal area of the finite volume:

$$V_{T,m}(\zeta^{n+1,p+1}) = V_{T,m}(\zeta^{n+1,p}) + \left(\zeta^{n+1,p+1}_m - \zeta^{n+1,p}_m\right) W^{n+1,p}_{T,m} \Delta x_m \tag{2.20}$$

p Iteration counter,

and $W_{T,m} = dA_{T,m}/d\zeta_m$.

If we take the limit to zero for the length of the finite volume the continuity equation becomes:

$$\frac{\partial A_T}{\partial t} + \frac{\partial Q}{\partial x} = q \tag{2.21}$$

after substitution of $A_T = A_F + A_S$ it read

$$\frac{\partial A_F}{\partial t} + \frac{\partial Q}{\partial x} = q - \frac{\partial A_S}{\partial t} \tag{2.22}$$

This equation is needed to substitute into the momentum equation (Equation (2.31)).

The total free surface area is related to the element length Δx_m according $A_{\zeta,m} = W_{T,m}\Delta x_m$. Substituting Equation (2.20) into Equation (2.17) and applying a so-called the θ-method, yields

$$\begin{aligned} A^n_{\zeta,m} \frac{\zeta^{n+1}_m - \zeta^n_m}{\Delta t} &+ \theta\left(A^n_{F,m+\frac{1}{2}} u^{n+1}_{m+\frac{1}{2}} - A^n_{F,m-\frac{1}{2}} u^{n+1}_{m-\frac{1}{2}}\right) \\ &+ (1-\theta)\left(A^n_{f,m+\frac{1}{2}} u^{n}_{m+\frac{1}{2}} - A^n_{f,m-\frac{1}{2}} u^{n}_{m-\frac{1}{2}}\right) = q_m \Delta x_m \end{aligned} \tag{2.23}$$

After separating the implicit and explicit part we arrived at:

$$\begin{aligned} &\frac{A^n_{\zeta,m}}{\Delta t}\zeta^{n+1}_m + \theta\left(A^n_F u^{n+1}\right)\Big|_{m+\frac{1}{2}} - \theta\left(A^n_F u^{n+1}\right)\Big|_{m-\frac{1}{2}} = \\ &\frac{A^n_{\zeta,m}}{\Delta t}\zeta^{n}_m - (1-\theta)\left(A^n_F u^n\right)\Big|_{m+\frac{1}{2}} + (1-\theta)\left(A^n_F u^n\right)\Big|_{m-\frac{1}{2}} + q_m \Delta x_m \end{aligned} \tag{2.24}$$

2.4.2 Time integration of the momentum equation

The momentum equation is derived from a momentum balance for a 3D Control Volume around a velocity point $x_{m+\frac{1}{2}}$. The volume of a control volume in a velocity point is defined as the integral over the flowing part of the cross section:

$$V_F\left(\zeta(t)\right)|_{m+\frac{1}{2}} = \int_{x_m}^{x_{m+1}} \int_{z_b(x')}^{\zeta(x',t)} W_f(x',z')\,dz'dx' \tag{2.25}$$

$$= \int_{x_m}^{x_{m+1}} \int_{b_l(x')}^{b_r(x')} h(x',y')\,dy'dx' \tag{2.26}$$

with $W_F\left(x,z\right)$ the width of the flow area at cross section x at level z. We assume that outside the flow area, W_F, the water is stagnant in the so-called storage area, so the horizontal velocity for that part of the cross section is zero and does not contribute to the momentum.

The volume of the control volume for the momentum balance, situated in a velocity point is approximated by:

$$V_{F,m+\frac{1}{2}}\left(\overline{\zeta}_{m+\frac{1}{2}}(t)\right) = A_F\left(\overline{\zeta}_{m+\frac{1}{2}}(t)\right)\Delta x_{m+\frac{1}{2}} \tag{2.27}$$

with $\Delta x_{m+\frac{1}{2}} = x_{m+1} - x_m$.

The rate of change of the momentum in a finite volume is

$$\left.\frac{d\left(\rho V u\right)}{dt}\right|_{m+\frac{1}{2}} \tag{2.28}$$

and should be the sum of in- and outflowing momentum fluxes through the cross sections at the water level points x_m and x_{m+1}. These convective momentum fluxes are given by:

$$\rho_{m+1}\overline{Q}_{m+1}\overline{u}_{m+1} - \rho_m\overline{Q}_m\overline{u}_m \tag{2.29}$$

Due to the staggered grid the discharge $\overline{Q}_m$, $\overline{Q}_{m+1}$, $\overline{u}_m$ and $\overline{u}_{m+1}$ should be approximated in the water level points x_m and x_{m+1}. In D-Flow 1D an average discharge at the water level points is used, defined as:

$$\overline{Q}_m = \frac{Q_{m+\frac{1}{2}} + Q_{m-\frac{1}{2}}}{2} \tag{2.30}$$

The transport term due to convection is in most rivers and channels of minor importance. Only in river sections with a strongly varying conveyance area, this term generates an energy loss, which should be taken into account accurately. For nodal points in 1D networks where more than two branches come together, the momentum equation is adjusted, see for a detailed description section 2.6.

The flow is accelerated due to the external forces, the pressure force related to the free surface gradient and decelerated by the bottom friction. For the bottom friction a quadratic friction law is assumed with friction coefficient c_f, and S_f the surface area where the bottom friction acts on the flowing fluid volume. We divide the momentum equation by the constant density ρ, we

get

$$\frac{d\left(V_{F,m+\frac{1}{2}}u_{m+\frac{1}{2}}\right)}{dt} + \overline{Q}_{m+1}\overline{u}_{m+1} - \overline{Q}_m\overline{u}_m + \overline{A}_f\left(\overline{\zeta}_{m+\frac{1}{2}}(t)\right)g\frac{\zeta_{m+1}-\zeta_m}{\Delta x} + c_f S_{F,m+\frac{1}{2}}u_{m+\frac{1}{2}}\left|u_{m+\frac{1}{2}}\right| = 0 \tag{2.31}$$

With $V_{T,m} = V_{F,m} + V_{S,m}$ the continuity equation can be written as (which shall be used to substitute into the momentum Equation (2.31))

$$\frac{dV_{F,m}}{dt} + Q_{m+\frac{1}{2}} - Q_{m-\frac{1}{2}} = q_m\Delta x - \frac{dV_{S,m}}{dt} \tag{2.32}$$

We use the chain rule to expand the first term:

$$\frac{d\left(V_{F,m+\frac{1}{2}}u_{m+\frac{1}{2}}\right)}{dt} = u_{m+\frac{1}{2}}\frac{dV_{F,m+\frac{1}{2}}}{dt} + V_{F,m+\frac{1}{2}}\frac{du_{m+\frac{1}{2}}}{dt} \tag{2.33}$$

so now it is possible to use the continuity equation for substitution of the term dV_F/dt, after we used the next approximation

$$V_{F,m+\frac{1}{2}} = \frac{V_{F,m}+V_{F,m+1}}{2} \tag{2.34}$$

Note: This approximation is only correct for equidistant grid sizes, because then

$$\Delta x_{m+\frac{1}{2}} = x_{m+1} - x_m = \frac{1}{2}(x_{m+\frac{1}{2}} - x_{m-\frac{1}{2}}) + \frac{1}{2}(x_{m+\frac{3}{2}} - x_{m+\frac{1}{2}}) \tag{2.35}$$

for non-equidistant grids $\frac{1}{2}(x_{m+\frac{1}{2}} - x_{m-\frac{1}{2}}) \neq x_m$ (the location of the water level point is not in the middle between two velocity points, see Figure 2.1), where the distances between the velocity points define the control volume for the continuity eqaution.

This separation of the controle volume into a flowing and stagnant part introduces an additional lateral discharge, the flow to and from the storage area. For rising water levels there is a flow from the flowing part to the storage area (withdrawal of momentum from the main steam) and for the falling water levels there is a flow from the storage area to the flowing area (with assumption: supplying no momentum to the main stream). For the momentum balance this introduces the following term at the right hand side of Equation (2.31):

$$\left(q_{m+\frac{1}{2}}\Delta x_{m+\frac{1}{2}} - \frac{dV_{S,m+\frac{1}{2}}}{dt}\right)u_b \tag{2.36}$$

with

u_b the velocity of the water that is exchanged between the flowing and storage area, defined by:

$$u_b = \begin{cases} u, & \text{if} \quad q_{m+\frac{1}{2}}\Delta x_{m+\frac{1}{2}} - \frac{dV_{S,m+\frac{1}{2}}}{dt} \leq 0 \quad \text{(withdrawal of flow area)} \\ 0, & \text{if} \quad q_{m+\frac{1}{2}}\Delta x_{m+\frac{1}{2}} - \frac{dV_{S,m+\frac{1}{2}}}{dt} > 0 \quad \text{(supply to flow area)} \end{cases} \tag{2.37}$$

$q_{m+\frac{1}{2}}$ Lateral discharge for a momentum control volume

If we multiply the continuity equation (Equation (2.32)) with the velocity $u^n_{m+\frac{1}{2}}$ and substract it from Equation (2.31) we get

$$\frac{du_{m+\frac{1}{2}}}{dt} + \frac{\overline{Q}_{m+1}}{V_{F,m+\frac{1}{2}}}\left(\overline{u}_{m+1} - u_{m+\frac{1}{2}}\right) - \frac{\overline{Q}_m}{V_{F,m+\frac{1}{2}}}\left(\overline{u}_m - u_{m+\frac{1}{2}}\right) + g\frac{\zeta_{m+1} - \zeta_m}{\Delta x} + \frac{c_f S_{F,m+\frac{1}{2}} u_{m+\frac{1}{2}} \left|u_{m+\frac{1}{2}}\right|}{V_{F,m+\frac{1}{2}}} = \left(q_{m+\frac{1}{2}} - \frac{dV_{S,m+\frac{1}{2}}}{dt}\right)\left(u_b - u_{m+\frac{1}{2}}\right) \quad (2.38)$$

Equation (2.38) is discretised using the θ-method as:

$$\frac{u^{n+1}_{m+\frac{1}{2}} - u^n_{m+\frac{1}{2}}}{\Delta t} + \frac{g}{\Delta x_{m+\frac{1}{2}}}\left\{\theta\left(\zeta^{n+1}_{m+1} - \zeta^{n+1}_m\right) + (1-\theta)\left(\zeta^n_{m+1} - \zeta^n_m\right)\right\} + B^n_{m+\frac{1}{2}} u^{n+1}_{m+\frac{1}{2}} = D^n_{m+\frac{1}{2}} \quad (2.39)$$

After separating the implict and explicit parts, we arrive at:

$$u^{n+1}_{m+\frac{1}{2}}\left(\frac{1}{\Delta t} + B^n_{m+\frac{1}{2}}\right) + \frac{g\theta}{\Delta x_{m+\frac{1}{2}}}\left(\zeta^{n+1}_{m+1} - \zeta^{n+1}_m\right) = \frac{u^n_{m+\frac{1}{2}}}{\Delta t} - \frac{g(1-\theta)}{\Delta x_{m+\frac{1}{2}}}\left(\zeta^n_{m+1} - \zeta^n_m\right) + D^n_{m+\frac{1}{2}} \quad (2.40)$$

A relation between velocity and water level terms is obtained:

$$u^{n+1}_{m+\frac{1}{2}} = R^n_{m+\frac{1}{2}} - F^n_{m+\frac{1}{2}}\left(\zeta^{n+1}_{m+1} - \zeta^{n+1}_m\right) \quad (2.41)$$

$$F^n_{m+\frac{1}{2}} = \frac{1}{\Delta t + B^n_{m+\frac{1}{2}}}\frac{g\theta}{\Delta x_{m+\frac{1}{2}}} \quad (2.42)$$

$$R^n_{m+\frac{1}{2}} = \frac{1}{\Delta t + B^n_{m+\frac{1}{2}}}\left(\frac{u^n_{m+\frac{1}{2}}}{\Delta t} - \frac{g(1-\theta)}{\Delta x_{m+\frac{1}{2}}}\left(\zeta^n_{m+1} - \zeta^n_m\right) + D^n_{m+\frac{1}{2}}\right) \quad (2.43)$$

In a numerical simulation, smaller high-frequency spurious modes may be generated, which depend on the size of the timestep. To prevent these modes from becoming large and imp[ortant one should use a value of θ that is larger than 0.5 to introduce a numerical dissipation mechanism. For most applications the recommended value of θ is 0.55. This gives usually sufficient dissipation without decreasing the accuracy too much.

2.4.3 Time integration of the combined continuity and momentum equation

The discretised continuity Equation (2.24) and momentum Equation (2.17) form a linear system with the two unknowns, water level and flow velocity. From these equations the velocity may be eliminated. The velocities are copied from Equation (2.17):

$$u^{n+1}_{m-\frac{1}{2}} = R^n_{m-\frac{1}{2}} - F^n_{m-\frac{1}{2}}\left(\zeta^{n+1}_m - \zeta^{n+1}_{m-1}\right) \quad (2.44)$$

$$u^{n+1}_{m+\frac{1}{2}} = R^n_{m+\frac{1}{2}} - F^n_{m+\frac{1}{2}}\left(\zeta^{n+1}_{m+1} - \zeta^{n+1}_m\right) \quad (2.45)$$

Substitution into Equation (2.24) leads to:

$$\begin{aligned}\frac{A^n_{\zeta;m}}{\Delta t}\zeta^{n+1}_m &+ \theta A^n_{F;m+\frac{1}{2}}\left\{R^n_{m+\frac{1}{2}} - F^n_{m+\frac{1}{2}}\left(\zeta^{n+1}_{m+1} - \zeta^{n+1}_m\right)\right\}\\ &- \theta A^n_{F;m-\frac{1}{2}}\left\{R^n_{m-\frac{1}{2}} - F^n_{m-\frac{1}{2}}\left(\zeta^{n+1}_m - \zeta^{n+1}_{m-1}\right)\right\}\\ &= \frac{A^n_{\zeta;m}}{\Delta t}\zeta^n_m - (1-\theta)\,A^n_{F;m+\frac{1}{2}}u^n_{m+\frac{1}{2}} + (1-\theta)\,A^n_{F;m-\frac{1}{2}}u^n_{m-\frac{1}{2}} + Q^n_m\end{aligned} \tag{2.46}$$

Or, after moving the known terms to the right side:

$$\begin{aligned}\frac{A^n_{\zeta;m}}{\Delta t}\zeta^{n+1}_m &- \theta A^n_{F;m+\frac{1}{2}}F^n_{m+\frac{1}{2}}\left(\zeta^{n+1}_{m+1} - \zeta^{n+1}_m\right)\\ &+ \theta A^n_{F;m-\frac{1}{2}}F^n_{m-\frac{1}{2}}\left(\zeta^{n+1}_m - \zeta^{n+1}_{m-1}\right)\\ &= \frac{A^n_{\zeta;m}}{\Delta t}\zeta^n_m - (1-\theta)\,A^n_{F;m+\frac{1}{2}}u^n_{m+\frac{1}{2}} - \theta A^n_{F;m+\frac{1}{2}}R^n_{m+\frac{1}{2}}\\ &+ (1-\theta)\,A^n_{F;m-\frac{1}{2}}u^n_{m-\frac{1}{2}} + \theta A^n_{F;m-\frac{1}{2}}R^n_{m-\frac{1}{2}} + Q^n_m\end{aligned} \tag{2.47}$$

This is a tridiagonal system of equations for the water level, which is solved by forward and backward elimination.

2.5 Numerical method for the transport equation

The advection term for the time integration of the transport equation for salinity is denoted by F^n_i and is calculated for the previous time step (explicit). The dispersion term is solved implicitely. As a result the formula for the discretization of the advection diffusion equation can be written into:

$$\begin{aligned}V^{n+1}_i s^{n+1}_i &= V^n_i s^n_i + F^n_i\\ &+\Delta t\left((1-\theta)\sum_{k\in K}A^n_k D^n_k\frac{s^n_{i_k} - s^n_i}{\Delta x_k} + \theta\sum_{k\in K}A^n_k D^n_k\frac{s^{n+1}_{i_k} - s^{n+1}_i}{\Delta x_k}\right)\end{aligned} \tag{2.48}$$

Where F^n_i denotes the advection.

Which can be rewritten in

$$\left(V^{n+1}_i + \theta\Delta t\sum_{k\in K}\frac{A^n_k D^n_k}{\Delta x_k}\right)s^{n+1}_i - \theta\Delta t\sum_{k\in K}\frac{A^n_k D^n_k s^{n+1}_{i_k}}{\Delta x_k} = V^n_i s^n_i \tag{2.49}$$

$$+F^n_i + \Delta t\left((1-\theta)\sum_{k\in K}A^n_k D^n_k\frac{s^n_{i_k} - s^n_i}{\Delta x_k}\right) \tag{2.50}$$

The matrix coefficients for SOBEK will now be:

$$b_i^n = V_i^{n+1} + \theta \Delta t \sum_{k \in K} \frac{A_k^n D_k^n}{\Delta x_k} \tag{2.51}$$

$$c_{i_k}^n = -\theta \Delta t \frac{A_k^n D_k^n}{\Delta x_k} \tag{2.52}$$

$$d_i^n = V_i^n s_i^n + F_i^n + \Delta t \left((1-\theta) \sum_{k \in K} A_k^n D_k^n \frac{s_{i_k}^n - s_i^n}{\Delta x_k} \right) \tag{2.53}$$

In which:

K	the collection of all links of computational cell i
K_{in}	the collection of link with inflow
K_{out}	$= K - K_{in}$ and therefore the collection of links with outflow.
b_i	Diagonal element (i, i) in matrix
c_{i_k}	Off diagonal element (i, j) and (j, i)
d_i	right hand side element i
s_{i_k}	salinity of computational cell j
V_i	> 0, which is de volume of cell i (at different time levels)
A_k	≥ 0, is the wetted cross-section of link k
D_k^n	≥ 0, is the dispersion coefficient at link k
Δx_k	> 0, mesh size over link k

As a result we can write:

$$b_i^n s_i^{n+1} + \sum_{k \in K} c_{i_k}^n s_{i_k}^{n+1} = d_i^n \tag{2.54}$$

For the advection term two schemes are available, first the upwind scheme:

$$F_i = \Delta t \left(\sum_{k \in K_{in}} |q_k^n| \, s_{i_k}^n - \sum_{k \in K_{out}} |q_k^n| \, s_i^n \right) \tag{2.55}$$

Second the "Van Leer-2" advection scheme:

For $u_k \geq 0$:

$$F_k = u_k A_k \left(c_i + \alpha \left(1 - CFL_k\right) \left(c_i - c_{i-1}\right) \frac{c_{i+1} - c_i}{c_{i+1} - c_{i-1}} \right) \tag{2.56}$$

where:

$$CFL_k = \frac{\Delta t \, |u_k|}{\Delta x} \tag{2.57}$$

$$\alpha = \begin{cases} 0, \text{ when } \left| \frac{c_{i+1} - 2c_i + c_{i-1}}{c_{i+1} - c_{i-1}} \right| > 1, \text{ (local max. or min.)}, \\ 1, \text{ when } \left| \frac{c_{i+1} - 2c_i + c_{i-1}}{c_{i+1} - c_{i-1}} \right| \leq 1, \text{ (monotone)}. \end{cases} \tag{2.58}$$

For $u_k < 0$:

$$F_k = u_k A_k \left(c_{i+1} + \alpha \left(1 - CFL_k\right) \left(c_i - c_{i+1}\right) \frac{c_{i+1} - c_{i+2}}{c_i - c_{i+2}} \right) \tag{2.59}$$

where:

$$CFL_k = \frac{\Delta t\,|u_k|}{\Delta x} \tag{2.60}$$

$$\alpha = \begin{cases} 0, \text{ when } \left|\frac{c_{i+2}-2c_{i+1}+c_i}{c_{i+2}-c_i}\right| > 1, \text{ (local max. or min.)}, \\ 1, \text{ when } \left|\frac{c_{i+2}-2c_{i+1}+c_i}{c_{i+2}-c_i}\right| \leq 1, \text{ (monotone)}. \end{cases} \tag{2.61}$$

2.5.1 Open boundary conditions for the transport equation

As described before, boundaries limit the computational area. At a boundary a boundary condition is required. At a closed boundary we have a no flow condition, as a result the salinity flux is set to 0 as well. At an open boundary during outflow (ebb) the concentration must be free. During inflow (flood) a boundary condition is needed. At inflow we have to specify the concentration which may be determined by the concentration at outflow of the former ebb period. Because usually only the background concentration is known from measurements or from a larger model area, a special boundary condition based on the concentration in the interior area in combination with a return time is used, which does not completely fix the concentration at the background value. This is the so-called Thatcher-Harleman boundary condition (Thatcher and Harleman, 1972b), which is discussed in the next section.

The transport of dissolved substances such as salt is described by the advection-diffusion equation. The horizontal transport is advection dominated and the equation is of hyperbolic type. At inflow, one boundary condition is needed and the concentration is specified. At outflow, no boundary condition is allowed. The concentration is determined by pure advection from the interior area:

$$\frac{\partial C}{\partial t} + U\frac{\partial C}{\partial x} = 0 \tag{2.62}$$

In D-Flow 1D the dispersive fluxes through the open boundaries at both inflow and outflow are switched off. therefore we have:

$$D_H\frac{\partial^2 C}{\partial x^2} = 0 \tag{2.63}$$

In case a branch k coincides with a boundary condition, and the flow is directed into the model, we have:

$$b_i s_i^{n+1} + \sum_{k\in K_{intern}} c_{i_k} s_{i_k}^{n+1} = d_i - c_{i_{boun}} s_{i_{boun}}^{n+1} \tag{2.64}$$

K_{intern} is the collection of internal interfaces that do not coincide with a boundary node. And $s_{i_{boun}}^{n+1}$ is the salinity in the boundary point at the new time level.

In case an interface k coincides with a boundary condition and the flow is directed outward of the model. We assume the dispersion flux is equal to 0. As a result:

$$s_{i_{boun}}^{n+1} = s_{i_{boun}}^{n} - \frac{\Delta t}{\Delta x}|u_k^n|\left(s_i^n - s_{i_{boun}}^n\right) \tag{2.65}$$

$$b_i s_i^{n+1} + \sum_{k\in K_{intern}} c_{i_k} s_{i_k}^{n+1} = d_i - c_{i_{boun}} s_{i_{boun}}^{n+1} \tag{2.66}$$

2.5.2 Thatcher Harleman time lag

If the concentration at outflow differs from the boundary condition at inflow, there is a discontinuity in the concentration at the turn of the flow. The transition of the concentration at the boundary from the outflow value to the inflow value may take some time. This depends on the refreshment of the water in the boundary region. The transition time is called the return time. The functional relationship describing the variation in concentration from the slack-water value to the present value is arbitrary. In D-Flow 1D a half-cosine variation is used. After the return time, the boundary value remains constant until outward flow begins (Thatcher and Harleman, 1972b). The mathematical formulation of this memory effect is given as follows:

$$C\left(t\right) = C^{out} + \frac{1}{2}(C^{bnd} - C^{out}) \left[\cos\left\{\pi \frac{T_{ret} - t_{out}}{T_{ret}}\right\} + 1\right] \ , \qquad 0 \leq t_{out} \leq T_{ret} \tag{2.67}$$

where

C^{out}	is the computed concentration at the open boundary at the last time of outward flow,
C^{bnd}	is the background concentration described by you,
t_{out}	is the elapsed time since the last outflow and
T_{ret}	is the constituent return period.

When the flow turns inward ($t_{out} = 0$), the concentration is set equal to C^{out}. During the interval $0 \leq t_{out} \leq T_{ret}$ the concentration will return to the background concentration C^{bnd}. After that period, the concentration will remain C^{bnd}. The mechanism is illustrated in Figure 2.3.

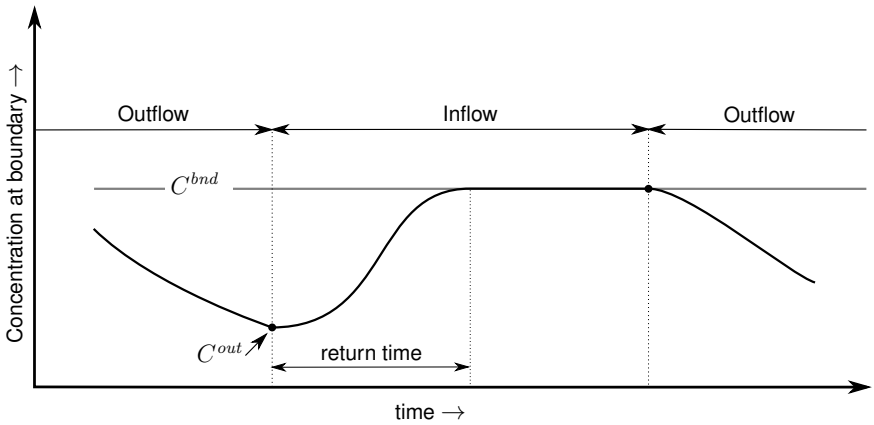

Figure 2.3: *Illustration of memory effect for open boundary*

2.5.2.1 Equation of state

The density of water ρ is a function of salinity (s) and temperature (t). In D-Flow 1D you can choose between three different formulations for the equation of state (Eckart, UNESCO or Eckart_modified).

When salinity and/or temperature transport is turned of the initial or default salinity and/or temperature is used for calculating the density of the water.

The first empirical relation derived by Eckart (1958) is based on a limited number of measurements dating from 1910 (only two salinities at 5 temperatures). In the original equation the pressure is present, but at low pressures the effect on density can be neglected.

Eckart formulation

The Eckart formulation is given by (Eckart, 1958):

Range: $0 < T < 40$ °C, $0 < s < 40$ ppt

$$\rho = \frac{1000 P_0}{\lambda + \alpha_0 P_0}, \tag{2.68}$$

where:

$$\lambda = 1779.5 + 11.25T - 0.0745T^2 - (3.80 + 0.01T)\, s, \tag{2.69}$$
$$\alpha_0 = 0.6980, \tag{2.70}$$
$$P_0 = 5890 + 38T - 0.375T^2 + 3s. \tag{2.71}$$

with the salinity s in ppt and the water temperature T in °C.

For fresh water the Eckart formulation does not give the maximum density at the temperature of 4 degrees Celsius. This is an important physical aspect when simulating the thermocline formation in lakes or stagnant basins. Therefore also the so-called UNESCO formulation is implemented in D-Flow 1D.

UNESCO formulation

The UNESCO formulation is given by (UNESCO, 1981a):

Range: $0 < T < 40$ °C, $0.5 < s < 43$ ppt

$$\rho = \rho_0 + As + Bs^{3/2} + Cs^2 \tag{2.72}$$

where

$$\rho_0 = 999.842\,594 + 6.793\,952 \cdot 10^{-2}T - 9.095\,290 \cdot 10^{-3}T^2 + \\ + 1.001\,685 \cdot 10^{-4}T^3 - 1.120\,083 \cdot 10^{-6}T^4 + 6.536332 \cdot 10^{-9}T^5 \tag{2.73}$$
$$A = 8.244\,93 \cdot 10^{-1} - 4.0899 \cdot 10^{-3}T + 7.6438 \cdot 10^{-5}T^2 + \\ -8.2467 \cdot 10^{-7}T^3 + 5.3875 \cdot 10^{-9}T^4 \tag{2.74}$$
$$B = -5.724\,66 \cdot 10^{-3} + 1.0227 \cdot 10^{-4}T - 1.6546 \cdot 10^{-6}T^2 \tag{2.75}$$
$$C = 4.8314 \cdot 10^{-4} \tag{2.76}$$

Remarks:

- Equation (2.74) is known as the International Equation of State for Seawater (EOS80) and is based on 467 data points. The standard error of the equation is $3.6 \cdot 10^{-3}$ kg/m^3 (Millero and Poisson, 1981).
- The Practical Salinity Scale (UNESCO, 1981b, Par. 3.2) and the International Equation of State are meant for use in all oceanic waters. However, these equations should be used with caution in waters that have a chemical composition different from standard seawater. In such waters, densities derived with the methods based on practical salinity measurements and the International Equation of State may deviate measurably from the true densities. However, in water masses different in composition from standard seawater the *differences* in densities derived by the new equations involve only very small errors.

Recommendation

The UNESCO formulae serve as an international standard. Further, the UNESCO formulae show the correct temperature of 4 degrees Celsius where fresh water has its maximum density. The latter is of importance for thermal stratification in deep lakes in moderate climate zones. Therefore we recommend the application of the UNESCO formulae and to select the Eckart formulation only for consistence with previous projects in which it has been used. At this moment the default is UNESCO.

Nevertheless, the UNESCO formulae have their limitations as they are based on the general properties of seawater mixed with fresh water. Particularly in cases where marginal density differences play a role, typically lakes, a variable mineral content of the water may create density differences not detected by just temperature and salinity. For such dedicated cases, you are warranted to check the accuracy of the UNESCO formulae against experimental density-relations derived from the (lake) water. In case of deviations or other constituents determining the water density, we then recommend to contact the Helpdesk for further assistance such as the implementation of a more dedicated density formulation.

2.5.3 Boundary conditions

The 3D and 2D depth-averaged shallow water or long-wave equations applied in Delft3D-FLOW represent a hyperbolic (inviscid case) or parabolic (viscid case) set of partial differential equations. To get a well-posed mathematical problem with a unique solution, a set of initial and boundary conditions for water levels and horizontal velocities must be specified. The contour of the model domain consists of parts along “land-water” lines (river banks, coast lines) which are called closed boundaries and parts across the flow field which are called open boundaries. Closed boundaries are natural boundaries. The velocities normal to a closed boundary are set to zero. Open boundaries are always artificial “water-water” boundaries. In a numerical model open boundaries are introduced to restrict the computational area and so the computational effort. They are situated as far away as possible from the area of interest. Long waves propagating out of the model area should not be hampered by the open boundaries. The reflection should be minimal.

The solution of the shallow water equations can be split in a steady state solution and a transient solution. The steady state solution depends on the boundary conditions and the forcing terms while the transient solution follows from the deviation between the initial condition and the steady state solution at the start of the simulation, the reflection at the open boundaries and the amount of dissipation.

Usually it is assumed that initially the model is at rest (“cold start”). The start of the simulation will generate short progressive waves with a wavelength coupled with the length of the model area (eigen oscillations). When disturbances are reflected at the open boundaries and are trapped in the model area, a standing wave may be generated. A water level boundary acts as a nodal point and a velocity or closed boundary acts as an anti-nodal point. The transient solution may propagate out of the model area through the open boundaries (Riemann boundary condition) or be reduced at the open boundaries (low pass filter) by use of the α-coefficient, Eqs. **??** and **??**, or die out due to bottom friction and viscosity. The spin-up time of a model is dependent on the time-scale with which the transient solution dies out.

The steady state solution is completely dependent on the boundary conditions. It is reached after the transient solution died out. The number of boundary conditions specified at any particular point of the boundary should be equal to the number of characteristics entering the region at that point, see Courant and Hilbert (1962) or Vreugdenhil (1989). Delft3D-FLOW assumes that the flow at the open boundaries is sub-critical, which means that the magnitude

of the flow is smaller than the velocity of wave propagation. Sub-critical flow means that the Froude number, which is defined by:

$$Fr = \frac{|U|}{\sqrt{gH}}, \tag{2.77}$$

is smaller than one. For sub-critical flow we distinguish two situations, viz. inflow and outflow. At inflow, we have to specify two boundary conditions and at outflow we have to specify one boundary condition. For tidal flow the number of required boundary conditions varies between ebb and flood. The first boundary condition is an external forcing by the water level, the normal velocity, the discharge rate or the Riemann invariant specified by you. The second boundary condition is a built-in boundary condition. At inflow the velocity component along the open boundary is set to zero. The influence of this built-in boundary condition is in most cases restricted to only a few grid cells near the open boundary. It would be better to specify the tangential velocity component. The tangential velocity may be determined from measurements or from model results of a larger domain (nesting of models). In the present implementation of the open boundary conditions in Delft3D-FLOW it is not possible yet to specify the tangential velocity component at input. To get a realistic flow pattern near the open boundary, it is suggested to define the model boundaries at locations where the grid lines of the boundary are perpendicular to the flow.

For modelling the region outside the computational domain, incident waves should be known and are prescribed at the open boundaries. To reduce the reflections at the open boundaries a so-called weakly reflecting boundary condition may be applied. This boundary condition is derived using the Riemann invariants of the linearised shallow water equations without Coriolis and bottom friction (Verboom and Slob, 1984). The boundary condition is non-reflective for waves, which pass normal to the boundary (wave front parallel with open boundary). If we do not prescribe exactly the incoming Riemann invariants at an open boundary, the outgoing wave will reflect at the boundary and propagate back as a disturbance into the model area. Usually information on both water level and velocity is not available and consequently we cannot specify the Riemann invariant. A water level or velocity boundary condition can be made weakly reflective for short wave components originating from the initial conditions or the eigen frequencies by adding the time derivative of the Riemann invariant see Eqs. **??** and **??**. If these Riemann invariants are not used then these short wave components may disturb the solution for a long time.

In the vertical direction the momentum equations are parabolic. The vertical velocity profile is determined by the vertical eddy viscosity and the boundary conditions at the bed (bed stress) and at the free surface (wind stress). The spin-up time of the vertical profiles is dependent on the vertical eddy viscosity.

When the grid size near the boundaries is larger than the thickness of the boundary layers occurring in the flow, the shear-stresses along the lateral boundaries can be neglected. A so-called free slip boundary condition is applied at all lateral boundaries. For the simulation of laboratory flumes the effect of side wall friction can be added to the system of equations in Delft3D-FLOW by a so-called partial slip boundary condition.

2.5.4 Dispersion coefficient for salinity

Several dispersion formulations for salinity have been implemented and validated using D-Flow 1D (Daniels, 2016; Huismans *et al.*, 2017). Most formulations were modified to be used in a numerical package for dynamic situations, whereas they were defined for simple cases that can be solved analytically. The Thatcher and Harleman (1972a) formulation was modified by changing a number of things, amongst others by adding a c/c_0 term and by using the tidal averaged gradient of the salt concentration, which gave better results for real estuaries

(Rijkswaterstaat, 1984). The Kuijper and van Rijn (Kuijper and Van Rijn, 2011) formulations were simplified by omitting the damping term, since these contributions are small compared to others and cannot easily be determined for branched estuaries. In the implementation the dispersion coefficient is determined from tidal averaged variables, whereas it is defined in the theory at high water slack, which is generally a small difference.

The validation with respect to flume experiments of a prismatic estuary showed that the Kuijper and van Rijn- prismatic formulation performed best (Daniels, 2016; Huismans *et al.*, 2017). Another advantage is that Kuijper and Van Rijn (2011) show that their dispersion formula should give the same result no matter where in the estuary the mouth is defined, in contrast with the tidal dispersion term of the Thatcher-Harleman formulation that is determined in the estuary mouth. For these reasons, the Kuijper and van Rijn-prismatic formulation is the default in D-Flow 1D for the tidal dispersion term.

The dispersion coefficient in D-Flow 1D consists of three contributions:

1 A time-constant background dispersion coefficient:

$$D = f_1(x) \tag{2.78}$$

with:

$f_1(x)$	Background dispersion coefficient [m^2/s]

f_1 covers mixing effects that are not in the tidal dispersion or shear dispersion. Ideally, $f_1 = 0$, which is the default value.

2 The Taylor dispersion coefficient due to shear in the flow and variation in salt concentration, which may vary both in space and time:

$$D = f_3(x)|u(x,t)|d(x,t)\frac{\sqrt{g}}{C(x)} \tag{2.79}$$

with:

$f_3(x)$	Shear dispersion coefficient [-]
$C(x)$	$= \frac{1}{n(x)}R(x)^{1/6}$ and $d \approx R$

Thatcher and Harleman (1972a) derived a value for this shear dispersion coefficient of 77 m$^{1/2}/s$, which is 25 [-] when divided by $\sqrt{g}$.
The default value for f_3 in D-Flow 1D is 25 [-].

3 Various formulations exist for the tidal dispersion. To make it possible to use the various formulations, a general formulation is implemented in D-Flow 1D:

$$D = f_4 u_0^{c_3} E_0^{c_4} N_r^{c_5} \left(\frac{C}{\sqrt{g}}\right)^{c_6} \left(\frac{d_0}{a}\right)^{c_7} \left(\frac{d_0}{E_0}\right)^{c_8} \left(1 + c_9\left(\frac{B_0}{E_0}\right)^2\right) \left(\frac{\langle s\rangle}{\langle s_0\rangle}\right)^{c_{10}} \left|\langle\frac{\partial s}{\partial x}\rangle\right|^{c_{11}} \tag{2.80}$$

with:

$f_4(x)$	Tidal dispersion coefficient [-],
E_0	tidal excursion length [m],
B_0	width in the mouth [m],
$\langle s_k\rangle$	The tidal averaged salt concentration at location k [kg/m^3],
$c_3 \ldots c_{11}$	coefficients that lead to the selected tidal dispersion formulation.

$$N_r = \frac{\Delta\rho}{\rho_w}\frac{gdQ_fT}{u_0^2 P_e} \tag{2.81}$$

Each of the formulations can be set by setting the constants f_3, f_4 and c_3 until c_{11} to the correct values. The modified Thatcher-Harleman tidal dispersion term can be obtained indirectly. The estuary length is not included in Equation (2.80) this formulation, since it was not clear whether it was meant as the convergence length or the length in which the tide has effect. As a work around, this formulation van be used by incorporating the value of a definition for the estuary length in the f_4 value.
The default formulation is the Kuijper and Van Rijn-prismatic formulation. The complete set for this tidal dispersion term is given in Table 2.1:

Table 2.1: *Default coefficients for the Kuijper and Van Rijn-prismatic formulation*

c_3	c_4	c_5	c_6	c_7	c_8	c_9	c_{10}	c_{11}
1	1	0.5	1	0	1	0	0.5	0

Using this set of coefficients results in the formulation of Kuijper and Van Rijn (2011) for prismatic channels:

$$D = f_4(x)u_0(x,t)N_r(x,t)^{1/2}d_0\frac{\sqrt{C}(x)}{g}\frac{\langle s_k\rangle}{\langle s_{0_{mouth}}\rangle} \tag{2.82}$$

Based on physical considerations a value of $f_4 = 6 \pm 30\%$ should be used (Kuijper and Van Rijn, 2011). Note that differences with the modified Thatcher-Harleman formulation include a higher power of N_r, direct dependencies with Chézy roughness and depth and the use of the salinity in the mouth versus the gradient of salinity.

2.5.4.1 Branch mouth relation

Usually dispersion coefficients are defined in the estuary mouth. Since the ratio of fresh water discharge over the tidal effects may vary in the estuary, the dispersion coefficient is calculated at the mouth of each branch in D-Flow 1D. In this way the characteristics in a branch determine the mixing and therewith salt transport.

This is implemented in a way that the user does not need to specify at which end of the branch the mouth is locatied. The mouth for a branch is the node with the highest average concentration over two tidal periods. As a result, the possibility exists that during a computation the location of the mouth changes.

2.5.4.2 Calculating parameters

The shear dispersion coefficient is determined in each calculation point according Equation (2.79). The background dispersion coefficient (f_1) is not calculated, but user defined.

The tidal dispersion coefficient (Equation (2.80)) contains tidally averaged values, and is calculated after each complete tidal period. The tidal period starts and ends at the moment that at the previous time step the flow was directed out of the sea boundary of the model, while at the current time step the flow is directed into the sea boundary of model. In order to be able to determine the tidal period the location of a the main sea boundary needs to be given by the user.

The minimal time of a tidal period must be 12 hours. At the completion of each tidal period all tidal parameters are averaged over two consecutive tidal periods. The timestep may vary. This is accounted for in the averaging. The tidally averaged values are calculated at the beginning and the end of each branch. From the parameters at the side with the highest salinity the dispersion coefficient is derived.

Note:

- All discharges directed into the branch are positive (and outflowing discharges are negative).
- The local mouth is at the beginning or the end of a branch. The actual location of the mouth is determined after a full tidal period.
  The node with the highest average salinity is taken as the mouth.
- Q_m denotes the discharge in point m.
- Δt is the current time step. The time step can vary during the simulation.
- A_{u_m} is the flow area in point m.
- d_m is the water depth in point m.
- s_m is the salinity in point m.
- u_m is the flow velocity in point m.

These parameters are determined in the local mouths during computation:

- $P_{e_{mouth}}$: flood volume

$$P_{e_{mouth}} = \frac{1}{4} \sum_{2T_p} |Q_{k_{mouth}}| \, \Delta t \tag{2.83}$$

  The flood volume is determined by integrating the absolute value of the discharge. In this way the flood volume is corrected for the additional fresh water discharge. More details can be found in Huismans *et al.* (2017).
- $Q_{f_{mouth}}$: fresh water discharge

$$Q_{f_{mouth}} = \left| \frac{1}{2T_p} \sum_{2T_p} Q_{k_{mouth}} \Delta t \right| \tag{2.84}$$

- A_{mouth}: Flow area

$$A_{mouth} = \frac{1}{2T_p} \sum_{2T_p} A_{u,k_{mouth}} \Delta t \tag{2.85}$$

- ρ_w: density of fresh water, using a default temperature of 14 °C, when temperature is not modelled,
- $\rho_{sea_{mouth}}$: average density of the water in the mouth, using a default temperature (T) of 14 °C, when temperature is not modelled.

$$\rho_{sea_{mouth}} = \rho(\langle s_0 \rangle, \langle T_0 \rangle) \tag{2.86}$$

- h_0: average water depth in a branch.

$$h_0 = \frac{1}{2T_p} \sum_{2T_p} \left(\frac{1}{L_{branch}} \sum_{m \in branch} \frac{h_{m-\frac{1}{2}} + h_{m+\frac{1}{2}}}{2} \Delta x \right) \Delta t \tag{2.87}$$

  The average water depth on a branch is computed by averaging the water depth in between each grid point weighted for the distance in between the grid points on that branch. Subsequently, the average water depth over a tidal period is computed.
- u_0: maximal flood velocity

$$u_0 = \max_{2T_p} (u_{mouth}) + \frac{Q_{f_{mouth}}}{A_{mouth}} \tag{2.88}$$

The maximal flood velocity is determined as the maximal value during flood or the minimal value when flow is negative during flood. The mean river flow velocity is added to correct for the reduction of the flood velocity due to the river discharge. More details can be found in Huismans *et al.* (2017)

◇ $\langle s_m \rangle$: average salinity concentration in grid point m

$$\langle s_m \rangle = \frac{1}{2T_p} \sum_{2T_p} s_m \Delta t \tag{2.89}$$

◇ $\langle s_{0_{mouth}} \rangle$: average salinity in mouth

$$\langle s_{0_{mouth}} \rangle = \langle s_{m_{mouth}} \rangle \tag{2.90}$$

◇ E_0: the tidal excursion, which is used in only some formulations It is estimated as

$$E_0 = u_0 \frac{T}{\pi} \tag{2.91}$$

2.5.5 Baroclinic term

When one of the boundaries of a river or estuary is in contact with the sea, salt water may enter the system. This effect is important for the water flow, because salt water has a higher density than fresh water. Salt intrusion into a river or estuary is truly a two- or even three-dimensional phenomenon. The dense sea water flows under the fresh water in a so-called salt wedge (see Figure 2.4).

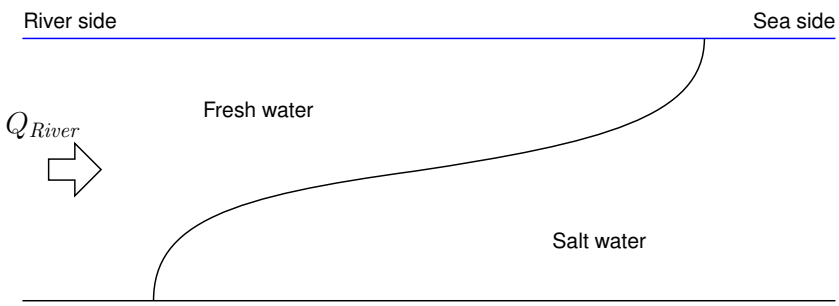

Figure 2.4: *Salt wedge*

In D-Flow 1D, the density in a cross-section is assumed uniform. The effect on the water flow is represented in the momentum equation through the density term:

$$-\frac{1}{2}\frac{g}{\rho_0}\frac{\partial \rho}{\partial x} h^2 \tag{2.92}$$

in which:

g	acceleration due to gravity [m/s^2]
ρ_0	reference density [kg/m^3]
ρ	density of salt water [kg/m^3]
h	water depth [m]

This so-called baroclinic pressure term is added to the momentum equation. We will now explain how this term has been derived.

Starting with the 3D Navier Stokes equations we have:

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho_0}\frac{\partial p}{\partial x} \tag{2.93}$$

$$\frac{\partial p}{\partial z} = -g\rho_0 \tag{2.94}$$

By integration from the bottom $z = z_b(x, y)$ to the free water surface $z = \zeta(x, y, t)$ we obtain:

$$\int_{z_b}^{\zeta} \left(\frac{\partial u}{\partial t}\right) dz' = -\frac{1}{\rho_0}\int_{z_b}^{\zeta} \left(\frac{\partial p}{\partial x}\right) dz' \tag{2.95}$$

By integration at the free water surface we obtain for the pressure: $z' = z$ to $z' = \zeta(x, y, t)$ (with z-axis positive upward):

$$p\,(x, y, z, t) = P_{atm} + \rho_0 g\,(\,\zeta(x, y, t) - z) \tag{2.96}$$

As boundary condition we assume that the pressure at the free water surface is equal to the atmospheric pressure P_{atm}, which we assume to be 0. Subtitution of Equation (2.96) into Equation (2.95) yields:

$$\frac{\partial\,(hU)}{\partial t} - U\frac{\partial\,h}{\partial t} = -\frac{g}{\rho_0}\int_{z_b}^{\zeta} \left(\frac{\partial \rho}{\partial x}\right) (\zeta(x, y, t) - z)\,dz' - g\int_{z_b}^{\zeta} \left(\frac{\partial \zeta}{\partial x}\right) dz' \tag{2.97}$$

$$= -\frac{g}{\rho_0}\left(\frac{\partial \rho}{\partial x}\right)\int_{z_b}^{\zeta} (\zeta(x, y, t) - z)\;dz' - gh\left(\frac{\partial \zeta}{\partial x}\right) \tag{2.98}$$

$$= -\frac{g}{\rho_0}\left(\frac{\partial \rho}{\partial x}\right)\left[\zeta z - \tfrac{1}{2}z^2\right]_{z_b}^{\zeta} - gh\left(\frac{\partial \zeta}{\partial x}\right) \tag{2.99}$$

$$= -\frac{g}{\rho_0}\left(\frac{\partial \rho}{\partial x}\right)\left(\tfrac{1}{2}\zeta^2 + \tfrac{1}{2}z_b^2 - \zeta z_b\right) - gh\left(\frac{\partial \zeta}{\partial x}\right) \tag{2.100}$$

$$= -\tfrac{1}{2}\frac{g}{\rho_0}\left(\frac{\partial \rho}{\partial x}\right)(\zeta - z_b)^2 - gh\left(\frac{\partial \zeta}{\partial x}\right) \tag{2.101}$$

$$= -\tfrac{1}{2}\frac{g}{\rho_0}\left(\frac{\partial \rho}{\partial x}\right)h^2 - gh\left(\frac{\partial \zeta}{\partial x}\right) \tag{2.102}$$

The first term in the right-hand side corresponds to the additional baroclinic term.

2.6 Advection at junctions

In this section the numerical aspects of advection at junctions is explained. Junctions are nodes where more than two branches join. Water flows from one or more branches into one branch or divides between two or more branches.

D-Flow 1D solves the continuity and momentum equations. In this section we introduce index k to designate separate branches. In the previous chapters this was not necessary.

Note: in this chapter we use a slightly different notation:

- ◇ in superscript: indices that indicate time level and iteration level per time stepe
- ◇ in subscript: indices that indicate location
- ◇ the extra index k indicating a branch

2.6.1 Introduction

See Figure 2.5 for the definition of the discrete variables at and near a junction.

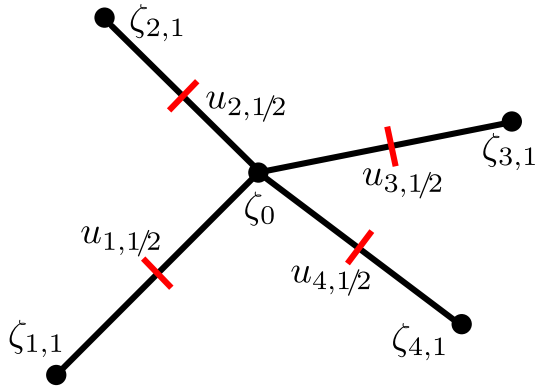

Figure 2.5: *A junction connecting 4 branches.*

A junction connects k branches, with $k > 2$. The connection node at the junction gets the index 0, hence the water level at the junction is denoted by ζ_0. The next node in each branch gets the index 1, hence the water level at those nodes is denoted by $\zeta_{k,1}$, $\forall k$. The flow velocity in the first cell center is denoted by $u_{k,1/2}$, $\forall k$. The direction of the flow in a branch is indicated by s_k:

$$s_k = \begin{cases} 1 & \text{if } u_k > 0 \text{ is directed from junction } 0 \ , \\ -1 & \text{if } u_k > 0 \text{ is directed toward junction } 0 \ . \end{cases} \tag{2.103}$$

Discharges in m^3 s^{-1} are computed at velocity points:

$$Q_{k,m+1/2} = A_{F,k,m+1/2} u_{k,m+1/2} \ , \tag{2.104}$$

with $A_F = A_F(\zeta)$ the cross-sectional *flow* area that is a function of the water level. We also have $A_T = A_T(\zeta)$ the *total* wetted cross-sectional area as a function of the water level. The difference between both is the wetted cross-sectional *storage* area $A_S = A_S(\zeta) = A_T(\zeta) - A_F(\zeta)$.

Note: about the grid size. Regardless of whether the grid points correspond with water-level points m or velocity points $m + 1/2$, the basic grid size *must* be defined at *water-level points* and *not* at velocity points! This is required for the procedure below where the space discretization of continuity equation and momentum equation in conservative form are combined to a conservative space discretization of the momentum equation in non-conservative form. So either we have $\Delta x_m = x_{m+1/2} - x_{m-1/2}$ (grid points at velocity points), or we have $\Delta x_m = (x_{m+1} - x_{m-1})/2$ (grid points at water-level points). At the end/beginning of branches (boundaries, junctions) we either have $\Delta x_0 = 2(x_{1/2} - x_0)$ or $\Delta x_0 = x_1 - x_0$ (or something like that). To obtain the grid size at velocity points, we average the grid size at water-level points: $\overline{\Delta x}_{m+1/2} = (\Delta x_m + \Delta x_{m+1})/2$.

2.6.2 Continuity equation

The continuity equation reads:

$$\frac{\partial A_T}{\partial t} + \frac{\partial Q}{\partial x} = q \ , \tag{2.105}$$

with lateral discharge q in m^2 s^{-1}.

2.6.2.1 Branch discretization

Discretization of Equation (2.105) in space inside a branch k reads:

$$\frac{dV_{T,k,m}}{dt} + Q_{k,m+1/2} - Q_{k,m-1/2} = \Delta x_{k,m} q_{k,m} \ , \tag{2.106}$$

while its discretization in both space and time reads:

$$\frac{V_{T,k,m}^{n+1} - V_{T,k,m}^{n}}{\Delta t} + Q_{k,m+1/2}^{n+\theta} - Q_{k,m-1/2}^{n+\theta} = \Delta x_{k,m} q_{k,m}^{n+\theta} \ , \tag{2.107}$$

with $V_{T,k,m} = V_{T,k,m}(\zeta_{k,m})$ the *total* water volume that is a function of the water level and probably computed by means of (likewise for $V_{F,k,m} = V_{F,k,m}(\zeta_{k,m})$ and $V_{S,k,m} = V_{S,k,m}(\zeta_{k,m})$, the *flowing* and *storage* water volume):

$$\begin{aligned} &V_{T,k,m} = A_{T,k,m}\Delta x_{k,m} \ , \ V_{F,k,m} = A_{F,k,m}\Delta x_{k,m} \ , \\ &V_{S,k,m} = V_{T,k,m} - V_{F,k,m} = (A_{T,k,m} - A_{F,k,m})\Delta x_k, m = A_{S,k,m}\Delta x_{k,m} \ , \end{aligned} \tag{2.108}$$

with $A_{T,k,m} = A_{T,k,m}(\zeta_{k,m})$, etc.

Discretization Equation (2.106) is nonlinear in ζ^{n+1}. It is linearized by evaluating the discharges partially explicit by means of $Q_{k,m+1/2}^{n+\theta} = A_{F,k,m+1/2}^{n} u_{k,m+1/2}^{n+\theta}$ and by the linearization:

$$V_{T,k,m}^{n+1,p+1} \approx V_{T,k,m}^{n+1,p} + W_{T,k,m}^{n+1,p}\Delta x_{k,m}\big(\zeta_{k,m}^{n+1,p+1} - \zeta_{k,m}^{n+1,p}\big) \ , \tag{2.109}$$

with p the iteration index and $W_{T,k,m} = dA_{T,k,m}/d\zeta_{k,m}$. Strict mass conservation in time is obtained, independent of the total number of iterations P, provided that the total water volume at the previous time level is taken equal to the last approximation of the next total water volume at the previous time step:

$$V_{T,k,m}^{n} = V_{T,k,m}^{n,P} + W_{T,k,m}^{n,P}\Delta x_{k,m}\big(\zeta_{k,m}^{n,P+1} - \zeta_{k,m}^{n,P}\big) \ . \tag{2.110}$$

2.6.2.2 Boundary discretization

For completeness sake, we briefly mention how the continuity equation may have been discretized at boundaries. Discretization of Equation (2.105) in space at a boundary $m = M$ of a branch k is:

$$\frac{1}{2}\frac{dV_{T,M}}{dt} + Q_M - Q_{M-\frac{1}{2}} = q_M \frac{\Delta x_M}{2} \ , \tag{2.111}$$

with Q_M the discharge at the boundary that is either imposed as a boundary condition or . . . ?

IMPORTANT: we *always* have the $V_{T,m}$ representing volumes pertaining to a *full* grid cell, *also* at the beginning and end of branches, where half of that grid cell is 'virtual' (outside the

branch). This avoids having to make some exceptions later on, which is convenient. Because Equation (2.111) is a discretization over half a grid cell, the time derivative is over a total water volume pertaining to half a grid cell, whence the time derivative of $V_{T,k,M}/2$.

Instead of Equation (2.111), the discretization of the continuity equation at the boundary may also have been obtained by applying Equation (2.106) for $m = M$, with $Q_{M+\frac{1}{2}}$ (i.e., $A_{F,M+\frac{1}{2}}$ and $u_{M+\frac{1}{2}}$, cf. Equation (2.104)) obtained from (linear) extrapolation.

2.6.2.3 Junction discretization

The space discretization of the continuity equation at a junction is in fact the sum of discretizations like Equation (2.111) at the end (or beginning) of a number of branches, where the discharges at the end points ($Q_{k,M}$ in Equation (2.111)) cancel against each other. It reads:

$$\frac{d}{dt}\Big(\sum_k V_{T,k,0}/2\Big) + \sum_k s_k Q_{k,1/2} = \sum_k \Delta x_{k,0}/2\, q_{k,0} \ , \tag{2.112}$$

with the sums taken over the discharges in the nearest velocity points and over the lateral discharges across the nearest half grid cells of all branches. Discretization in both space and time is obtained in the same way as in the step from Equation (2.106) to Equation (2.107).

2.6.3 Momentum equation

The momentum equation is:

$$\frac{\partial A_F u}{\partial t} + \frac{\partial Q u}{\partial x} + g A_F \frac{\partial \zeta}{\partial x} + c_f S_F u|u| = \Big(q - \frac{\partial A_S}{\partial t}\Big) u_b \ , \tag{2.113}$$

with:

$$u_b = \begin{cases} u & \text{if } q - \partial A_S/\partial t < 0 \text{ (flow from flow area to storage area and beyond)} \ , \\ 0 & \text{if } q - \partial A_S/\partial t > 0 \text{ (flow from storage area and beyond to flow area)} \ . \end{cases} \tag{2.114}$$

The right-hand side of Equation (2.113) can then be written as $\min(0, q - \partial A_S/\partial t)u$. That is a term consisting of u times a negative definite coefficient and hence guaranteed to dissipate/remove momentum.

2.6.3.1 Branch discretization

The space discretization of the momentum equation in conservative form Equation (2.113) inside a branch k reads:

$$\begin{aligned} &\frac{d(\overline{V}_{F,m+1/2} u_{m+1/2})}{dt} + \overline{Q}_{m+1}\overline{u}_{m+1} - \overline{Q}_m \overline{u}_m \\ &+ g\frac{\overline{V}_{F,m+1/2}}{\overline{\Delta x}_{m+1/2}}\big(\zeta_{m+1} - \zeta_m\big) + \Delta x\, c_f S_{F,m+1/2} u_{m+1/2}|u_{m+1/2}| \\ &= \Big(\overline{\Delta x\, q}|_{m+1/2} - \frac{d\overline{V}_{S,m+1/2}}{dt}\Big) u_{b,m+1/2} \ , \end{aligned} \tag{2.115}$$

with

$$\begin{aligned}
\overline{V}_{F,m+1/2} &= (V_{F,m} + V_{F,m+1})/2, \\
\overline{\Delta x}_{m+1/2} &= (\Delta x_m + \Delta x_{m+1})/2, \\
\overline{Q}_m &= (Q_{m+1/2} + Q_{m-1/2})/2, \\
\overline{u}_m &\quad \text{determined by means of upwind, and} \\
\overline{\Delta x\, q}|_{m+1/2} &= (\Delta x_m q_m + \Delta x_{m+1} q_{m+1})/2.
\end{aligned}$$

Expanding the time derivative in the left-hand side of Equation (2.115) (using $\partial \overline{V}_F u/\partial dt = \overline{V}_F \partial u/\partial t + u \partial \overline{V}_F/\partial t$), subtracting $u_{m+1/2}$ times $d\overline{V}_{T,m+1/2}/dt = (dV_{T,m}/dt + dV_{T,m+1}/dt)/2$ using Equation (2.106), and dividing the result by $\overline{V}_{F,m+1/2}$, we get that special discretization of the momentum equation in non-conservative form Equation (2.116) that nevertheless ensures conservation of momentum in space because of the way it has been constructed:

$$\begin{aligned}
&\frac{du_{m+1/2}}{dt} + \frac{\overline{Q}_{m+1}}{\overline{V}_{F,m+1/2}}(\overline{u}_{m+1} - u_{m+1/2}) - \frac{\overline{Q}_m}{\overline{V}_{F,m+1/2}}(\overline{u}_m - u_{m+1/2}) \\
&+ g\frac{\zeta_{m+1} - \zeta_m}{\overline{\Delta x}_{m+1/2}} + \Delta x\, c_f \frac{S_{F,m+1/2}}{\overline{V}_{F,m+1/2}} u_{m+1/2}|u_{m+1/2}| \\
&= \left(\overline{\Delta x\, q}|_{m+1/2} - \frac{d\overline{V}_{S,m+1/2}}{dt}\right)\frac{u_{b,m+1/2} - u_{m+1/2}}{\overline{V}_{F,m+1/2}} \ .
\end{aligned} \tag{2.116}$$

This equation is equal to Equation (2.38).

2.6.3.2 Junction discretization

If we would find a way to come up with reliable expressions for the $\overline{Q}_{k,0}$ and the $\overline{u}_{k,0}$ (the discharge and upwind velocity per branch k at the junction where the branches are connected), then discretization Equation (2.115) can also be applied at junctions.

A suitable expression for $\overline{Q}_{k,0}$ is:

$$\overline{Q}_{k,0} = \frac{1}{K}\left((K-1)Q_{k,1/2} - s_k \sum_{l \neq k} s_l Q_{l,1/2}\right) \ , \tag{2.117}$$

with K the number of branches connected at junction 0. This is a weighting of $(K-1)/K$ times the discharge in branch k and $1/K$ times the discharges in the $K-1$ other branches connected at junction 0.

Using this expression, a discretization of the continuity equation per half grid cell next to junction 0 can be defined:

$$\frac{dV_{T,k,0}/2}{dt} + s_k(Q_{k,1/2} - \overline{Q}_{k,0}) = \Delta x_{k,0}/2\, q_{k,0} \ . \tag{2.118}$$

It is easy to see that for $K = 2$ (node connecting two branches) this expression is equal to Equation (2.106), as it should be. It is also easy to verify that $\sum_k s_k \overline{Q}_{k,0} = 0$, i.e., the sum of Equation (2.118) over all branches k equals Equation (2.106), showing that Equation (2.117) ensures mass conservation.

Next the $\overline{u}_{k,0}$. If the flow is *toward* the junction, application of upwind is straightforward:

$$\overline{u}_{k,0} = u_{k,1/2} \ \text{ if } \ s_k \overline{Q}_{k,0} \leq 0 \ . \tag{2.119}$$

To ensure momentum conservation, for the flow *from* the junction we should have . . . (Discretizations of momentum equation per grid cell near junction cannot simply be added together, because we have to take momentum direction into account. Hence take sum of momentum fluxes toward junction and demand that to be equal to sum of momentum fluxes from junction. Leads to correct procedure in case of $K = 2$):

$$\sum_{s_k \overline{Q}_{k,0} > 0} \overline{u}_{k,0} \overline{Q}_{k,0} = \sum_{s_k \overline{Q}_{k,0} \leq 0} \overline{u}_{k,0} \overline{Q}_{k,0} \ . \tag{2.120}$$

By definition, the terms in both left-hand side and right-hand side are all nearly always positive. In a very dynamic situation, where some $\overline{Q}_{k,0}$ and the corresponding $\overline{u}_{k,0}$ may temporarily have opposite signs, terms in left- and/or right-hand side may become slightly negative.

It may be reasonable to take the $\overline{u}_{k,0}$ in the left-hand side of Equation (2.120) proportional to the flow speed in branch k:

$$\overline{u}_{k,0} = \alpha \overline{Q}_{k,0} / V_{F,k,0} \ \text{ if } \ s_k \overline{Q}_{k,0} > 0 \ . \tag{2.121}$$

Put this expression in Equation (2.120) and solve for α. Done! To avoid division by zero, adapt Equation (2.120) to something like:

$$\sum_{s_k \overline{Q}_{k,0} > 0} \overline{u}_{k,0} \overline{Q}_{k,0} = \max \left(A_{F,\mathrm{typ}} u_\varepsilon^2, \sum_{s_k \overline{Q}_{k,0} \leq 0} \overline{u}_{k,0} \overline{Q}_{k,0} \right) \ ,$$

with u_ε some very small velocity and $A_{F,\mathrm{typ}}$ some typical (very small) cross-sectional flow area. Strict momentum conservation at junctions is then lost, but only when flow velocities are very small (and may change sign).

3 Numerical concepts (1D2D)

3.1 Lateral 1D-2D coupling

3.1.1 General

The implementation of the 1D-2D coupling is based on the equations derived in section 1.4.

Equation (1.250) for the transfer of information from 1D to 2D:

$$\begin{aligned}
&\alpha^n_{\text{2D}}\frac{\zeta^p_{\text{2D},v}+\zeta^p_{\text{2D},i}}{2}+(\beta^n_{\text{2D}}+\alpha^n_{\text{2D}}f^n_I F^n_I)(\zeta^p_{\text{2D},v}-\zeta^p_{\text{2D},i})\\
&=\alpha^n_{\text{2D}}\zeta^{p-1}_{\text{1D}}+\beta^n_{\text{2D}}\frac{\Delta x_I}{\theta A^n_I F^n_I}q^{p-1}_{\text{1D-2D}}\\
&+\alpha^n_{\text{2D}}s_{c,I}\Big(f^n_I R^n_I-\frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF}g\Delta t}u^n_{\text{2D},I}\Big)+\beta^n_{\text{2D}}\frac{s_{c,I}}{\theta F^n_I}\Big(\theta R^n_I+(1-\theta)u^n_{\text{2D},I}\Big)\ ,
\end{aligned}\tag{3.1}$$

with n denoting variables at the previous time level t^n, and p and $p-1$ denoting variables at the current and previous iteration level of the solution at the next time level t^{n+1}. Variables without a superscript are time independent.

Equation (1.253) for the transfer of information from 2D to 1D:

$$\begin{aligned}
&\alpha^n_{\text{1D}}\zeta^p_{\text{1D}}-\beta^n_{\text{1D}}\frac{\Delta x_I}{\theta A^n_I F^n_I}q^p_{\text{1D-2D}}\\
&=\alpha^n_{\text{1D}}\frac{\zeta^{p-1}_{\text{2D},i}+\zeta^{p-1}_{\text{2D},v}}{2}+(\beta^n_{\text{1D}}-\alpha^n_{\text{1D}}f^n_I F^n_I)(\zeta^{p-1}_{\text{2D},i}-\zeta^{p-1}_{\text{2D},v})\\
&-\alpha^n_{\text{1D}}s_{c,I}\Big(f^n_I R^n_I-\frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF}g\Delta t}u^n_{\text{2D},I}\Big)+\beta^n_{\text{1D}}\frac{s_{c,I}}{\theta F^n_I}\Big(\theta R^n_I+(1-\theta)u^n_{\text{2D},I}\Big)\ .
\end{aligned}\tag{3.2}$$

The different flow regimes across the interface, the conditions at which they apply, and the way they are modeled, are (pseudo code, with $\zeta^n_{\text{2D},I}=(\zeta^n_{\text{2D},i}+\zeta^n_{\text{2D},v})/2$):

IF $\zeta^n_{\text{2D},I}-z_s\le 0$ AND $\zeta^n_{\text{1D}}-z_s\le 0$ THEN
! **no flow** across interface

$u^n_{\text{2D},I}=0$ and $q^n_{\text{1D-2D}}=0$

ELSE IF $\zeta^n_{\text{2D},I}-z_s\ge 3/2(\zeta^n_{\text{1D}}-z_s)$ THEN
! **free-weir flow** from 2D to 1D

Equation (3.1) with $\alpha_{SF}=1/3$, $\zeta^{p-1}_{\text{1D}}\Leftarrow z_s$, $\alpha^n_{\text{2D}}=1$, $\beta^n_{\text{2D}}=0$,
Equation (3.2) with $\alpha^n_{\text{1D}}=0$ ($\Rightarrow$ value of α_{SF} and ζ^p_{1D} irrelevant) , $\beta^n_{\text{1D}}=1$,
and with $f^n_I=f^n_{F,I}=\Big(\frac{3\Delta x_{\text{1D-2D}}}{\Delta x_{u,I}}+\frac{(A^n_I)^2|u^n_{\text{2D},I}|\Delta t}{(2/3)^3\Delta x_{u,I}\big(\Delta x_I c_e c_w(\zeta^n_{\text{2D},I}-z_s)\big)^2}\Big)\frac{\Delta x_{u,I}}{g\Delta t}$.

ELSE IF $\zeta^n_{\text{1D}}-z_s\ge 3/2(\zeta^n_{\text{2D},I}-z_s)$ THEN
! **free-weir flow** from 1D to 2D

Equation (3.2) with $\alpha_{SF} = 1/3$, $\dfrac{\zeta_{\text{2D},i}^{p-1} + \zeta_{\text{2D},v}^{p-1}}{2} \Leftarrow z_s$, $\alpha_{\text{1D}}^n = 1$, $\beta_{\text{1D}}^n = f_I^n F_I^n$,

Equation (3.1) with $\alpha_{\text{2D}}^n = 0$ ($\Rightarrow$ value of α_{SF} and ζ_{1D}^{p-1} irrelevant) , $\beta_{\text{2D}}^n = 1$,

and with $f_I^n = f_{F,I}^n = \Big(\dfrac{3\Delta x_{\text{1D-2D}}}{\Delta x_{u,I}} + \dfrac{(A_I^n)^2 |u_{\text{2D},I}^n| \Delta t}{(2/3)^3 \Delta x_{u,I} \big(\Delta x_I c_e c_w (\zeta_{\text{1D}}^n - z_s)\big)^2}\Big) \dfrac{\Delta x_{u,I}}{g \Delta t}$.

ELSE

! **drowned-weir flow** across interface

Equation (3.1) with $\alpha_{SF} = 1$, $\alpha_{\text{2D}}^n = 1$, $\beta_{\text{2D}}^n = \ldots$ (optimized value, see below)

Equation (3.2) with $\alpha_{SF} = 1$, $\alpha_{\text{1D}}^n = 1$, $\beta_{\text{1D}}^n = \ldots$ (optimized value, see below)

and with $f_I^n = f_{S,I}^n = \Big(\dfrac{\Delta x_{\text{1D-2D}}}{\Delta x_{u,I}} + \dfrac{(A_I^n)^2 |u_{\text{2D},I}^n| \Delta t}{2\Delta x_{u,I} \big(\Delta x_I c_e c_w (\zeta_{\text{1D/2D},I}^n - z_s)\big)^2}\Big) \dfrac{\Delta x_{u,I}}{g \Delta t}$,

where IF $\zeta_{\text{2D},I}^n \geq \zeta_{\text{1D}}^n$ THEN $\zeta_{\text{1D/2D},I}^n \Leftarrow \zeta_{\text{1D}}^n$ ELSE $\zeta_{\text{1D/2D},I}^n \Leftarrow \zeta_{\text{2D},I}^n$ ENDIF

ENDIF

with:

$F_I^n =$ first order term in linearization of momentum equation at the interface
$R_I^n =$ zeroth order term in linearization of momentum equation at the interface
$\Delta x_I =$ length of the 2D cell face (along the interface)
$\Delta x_{u,I} =$ grid size of the 2D cell at the interface (normal to the interface)
$\Delta x_{\text{1D-2D}} =$ sort of represents the effective thickness of the interface
$A_I^n =$ Flow area at the 1D-2D interface
$Q_{\text{2D-1D}}^p = -Q_{\text{1D-2D}}^p = s_{c,I} A_I^n u_I^p$
$s_{c,I} u_{\text{2D-1D}}^p = s_{c,I} R_I^n - F_I^n (\zeta_{\text{2D},v}^p - \zeta_{\text{2D},i}^p)$
$z_s =$ crest level of the interface

3.1.2 Optimization of the coupling parameters

A normal-mode analysis has been used to analyse and optimize the performance of the explicit solution procedure for the implicit horizontal 1D-2D coupling by considering rectangular domains, uniform conditions, and uniform rectangular grids:

- ◇ a straight 1D-2D interface of uniform thickness and with a uniform friction coefficient between
- ◇ a straight 1D channel with uniform depth, width, friction coefficient and grid at one side, and
- ◇ a 2D area with uniform depth, friction coefficient and grid consisting of rectangular cells at the other side.

This implies that all coefficients in Equation (3.1) and Equation (3.2) relevant to the coupling, i.e., f_I^n, F_I^n, Δx_I, and A_I^n, are considered to be constant, both in time and in space.

The uniform conditions are required to perform a normal-mode analysis. It is a reasonable assumption when the coefficients can locally be assumed constant, when the 1D-2D interface can locally be assumed straight, and when the grids can locally be assumed uniform. If the conditions are not smooth (e.g., strong variations in water depth or grid size near a 1D-2D interface, strong bends in a 1D-2D interface), the optimization is assumed to give a reasonable

approximation of the optimal coupling parameters, especially when non-smooth conditions occur in only a small number of locations.

In addition to the uniform conditions, it has been assumed for the time being that the solution procedure of each 1D-2D coupling can be considered independently. In reality, the iterative performance of the 1D-2D coupling at one side of a 1D channel (only relevant if drowned-weir flow) can be *strongly* affected by the coupling at the other side. It is easy to see that this applies in particular when we have drowned-weir flow or free-weir flow from 1D to 2D at that other side[1]. Taking the effect of a coupling at the other side of a channel into account in the optimization is p.m. There may also be some mutual influence between the 1D-2D couplings at either side of a 2D area, especially if the distance between the two interfaces is small (small 2D strip between two 1D channels). Taking this effect into account in the optimization of the couplings is not feasible.

A normal-mode analysis of the 1D-2D coupling shows that the optimal value of β^n_{2D} and β^n_{1D} for a Fourier-mode perturbation with wave number k, $0 \leq k \leq \pi/\Delta x_I$, are:

$$\beta^{n,opt}_{\text{2D}} = \frac{\Delta x_{u,I}\text{CFL}^2_{\text{2D},n}}{W\big(1+4(\sin(k\Delta x_I/2)\text{CFL}^n_{\text{1D}})^2\big)} \ , \tag{3.3}$$

$$\beta^{n,opt}_{\text{1D}} = f^n_I F^n_I + \frac{\sqrt{1+4(\sin(k\Delta x_I/2)\text{CFL}^n_{\text{2D},s})^2+4(\text{CFL}^n_{\text{2D},n})^2}}{2\sqrt{1+4(\sin(k\Delta x_I/2)\text{CFL}^n_{\text{2D},s})^2}} \ , \tag{3.4}$$

with W the width of the 1D channel, with CFL_{1D} the CFL number in the 1D domain (along the interface), and with $\text{CFL}_{\text{2D},n}$ and $\text{CFL}_{\text{2D},s}$ the CFL number in the 2D domain normal to and along the interface.

Note: when Equation (3.3) is inserted in Equation (3.1) with CFL as specified below in Equation (3.6), the coefficient of $q^{p-1}_{\text{1D-2D}}$ in the right-hand side would become:

$$\beta^{n,opt}_{\text{2D}}\frac{\Delta x_I}{\theta A^n_I F^n_I} = \frac{\Delta t}{W\big(1+4(\sin(k\Delta x_I/2)\text{CFL}^n_{\text{1D}})^2\big)} \ .$$

This shows that in an *optimal* 1D-2D coupling the terms in Equation (3.1) at the previous iteration level $p-1$ (first and second term in the right-hand side) *only* depend on variables of the 1D domain. Likewise, inserting Equation (3.4) in Equation (3.2) (with $\alpha^n_{\text{1D}} = 1$) to obtain:

$$\beta^{n,opt}_{\text{1D}} - f^n_I F^n_I = +\frac{\sqrt{1+4(\sin(k\Delta x_I/2)\text{CFL}^n_{\text{2D},s})^2+4(\text{CFL}^n_{\text{2D},n})^2}}{2\sqrt{1+4(\sin(k\Delta x_I/2)\text{CFL}^n_{\text{2D},s})^2}} \ ,$$

we see that in an optimal coupling in the other direction, the terms at the previous iteration level $p-1$ (again the first and second term in the right-hand side) only depend on variables of the 2D domain. The part in the coefficient of $\zeta^p_{\text{2D},v} - \zeta^p_{\text{2D},i}$ that depends on f^n_I, a factor depending on parameters and variables pertaining to the weir formulation applied at the 1D-2D interface, vanishes.

The above is true in general. It has been verified that the convergence speed of an implicit-coupling solver consisting of the iterative explicit update of coupling information is always

[1] In the case of free-weir flow from 2D to 1D at that other side, an additional lateral discharge is imposed to the 1D channel that does not depend on the 1D flow solution and hence will have little or no effect on the convergence speed of the 1D-2D coupling at this side of the 1D channel.

maximized by minimizing the transfer of information. Since information to be transfered (by definition at a previous iteration level) is determined by the numerics of the domain it is coming from, it is these numerics that determine the optimal information. In consequence, optimal information that is transfered can only depend on variables and parameters defined in a single domain. This fact may be used to simplify the implementation of the coupling.

The general expression of the CFL number in a certain direction is obtained from Equation (1.238):

$$\text{CFL}^n = \sqrt{\theta \Delta t A_f^n F_u^n / S_\zeta^n} \ . \tag{3.5}$$

This expression can be used to estimate the value of all three CFL numbers[2] $\text{CFL}_{1\text{D}}$, $\text{CFL}_{2\text{D},n}$ and $\text{CFL}_{2\text{D},s}$. In case of a (fairly regular) 2D unstructured grid, $\text{CFL}_{2\text{D},s}$ can be taken equal to $\text{CFL}_{2\text{D},n}$. Alternatively, the values obtained from the cell faces touching the 1D-2D interface can be used. To reduce a possibly strong variation of $\text{CFL}_{1\text{D}}$, $\text{CFL}_{2\text{D},n}$ and $\text{CFL}_{2\text{D},s}$ along the interface (after all, in the optimization these values are assumed constant), the application of a simple explicit space filter is recommended, e.g., a $1/4 - 1/2 - 1/4$ weighting.

Expressions Equation (3.3) and Equation (3.4) are for a single Fourier mode only. To obtain expressions that can be used for all modes, the value of Equation (3.3) and Equation (3.4) for some 'average' k can be used, or some average of Equation (3.3) and Equation (3.4) evaluated at the smallest and largest wave number $k = 0$ and $k = \pi/\Delta x_I$ can be used. A combination of these options, e.g., the use of weighted averages of Equation (3.3) and Equation (3.4) evaluated at different k, is possible as well. There is not much difference between these options. The analysis indicates that a good convergence performance of the 1D-2D coupling for a wide variety of conditions can be obtained by using Equation (3.3) and Equation (3.4) evaluated at well-selected k-values. So we propose to use (for the time being):

$$\beta_{2\text{D}}^n = \beta_{2\text{D}}^{n,opt}(k\Delta x_I = \delta_{2\text{D}}^n \pi) \ , \ \beta_{1\text{D}}^n = \beta_{1\text{D}}^{n,opt}(k\Delta x_I = \delta_{1\text{D}}^n \pi) \ , \tag{3.7}$$

with:

$$\delta_{2\text{D}}^n = 1/16 \ , \ \delta_{1\text{D}}^n = 3/8 \ . \tag{3.8}$$

Figure 3.1 shows the convergence speed that is obtained with this setting, as predicted by the analysis. Note that the solution procedure is 'exact' (full convergence in a finite number of iterations) for $k\Delta x_I = \pi/16$ and $k\Delta x_I = 3\pi/8$. These values are set by the choice of $\delta_{2\text{D}}^n$ and $\delta_{1\text{D}}^n$. The values specified in Equation (3.8) have been chosen such that a high convergence speed is obtained over the full range of Fourier modes.

We recall that Equation (3.7) with Equation (3.8) is just a rough optimization 'by hand', given the limitations of the current iterative 1D-2D solution procedure and its analysis. The assumption underlying the analysis that locally the conditions are smooth enough to be considered

[2] If the bottom of a 2D grid cell is fully submerged, its surface S_ζ is constant and can, for a quadrilateral 2D cell near a 1D-2D interface, be approximated by $S_{\zeta,I} \approx \Delta x_I \Delta x_{u,I}$. So for quadrilateral cells we have:

$$\text{CFL}_{2\text{D},n}^n \approx \sqrt{\theta \Delta t A_I^n F_I^n / \Delta x_I / \Delta x_{u,I}} \ , \tag{3.6}$$

as well as $\text{CFL}_{2\text{D},s}^n \approx \text{CFL}_{2\text{D},n}^n \Delta x_{u,I} / \Delta x_I$. These may be more suitable expressions for $\text{CFL}_{2\text{D},n}^n$ and $\text{CFL}_{2\text{D},s}^n$ than Equation (3.5) in case of quadrilateral 2D cells near the interface. However, these expressions do not seem to be suitable for triangular 2D grid cells, where we have $S_{\zeta,I} \approx 1/2 \Delta x_I \Delta x_{u,I}$. Moreover, because of the use of $S_{\zeta,I}^n$, Equation (3.5) automatically includes the effect of the shape of the grid cell (not included in the analysis where the optimized expressions of $\beta_{2\text{D}}^n$ and $\beta_{1\text{D}}^n$ were obtained), which may be advantageous. On the other hand, expression Equation (3.6) is more convenient for implementation.

Since the variables that are required are already computed in SOBEK, the evaluation of $\text{CFL}_{1\text{D}}$ should be straightforward. The same applies to $\text{CFL}_{2\text{D},n}$ and $\text{CFL}_{2\text{D},s}$, whether computed using Equation (3.5) or using Equation (3.6).

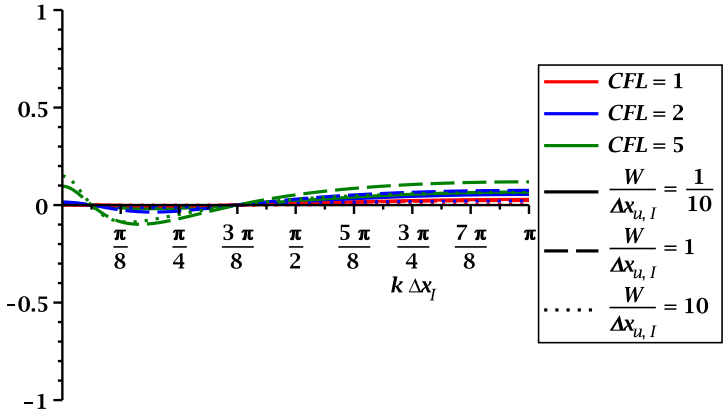

Figure 3.1: *Predicted reduction factor per 1D-2D iteration as a function of dimensionless wave number $k\Delta x_I$ for several ratios between channel width W and normal 2D grid size $\Delta x_{u,I}$ and several CFL-values, taking all CFL numbers equal ($CFL_{1D} = CFL_{2D,s} = CFL_{2D,n} = CFL$).*

uniform is obviously not valid in practice, but may be acceptable. A more significant drawback of the current analysis is that it does not take into account the presence of a 1D-2D coupling at the other side of a 1D channel. The good news is that it seems to be possible to extend the analysis of the 1D-2D coupling for this case. Extensions of the coupling for better overall performance across the entire range of Fourier modes are possible as well. Pending such developments, one must be aware of a 1D-2D coupling not yet performing well in difficult applications. This is largely a temporary situation.

3.1.3 Implementation of the 1D–to–2D coupling into the 2D system of equations

Rewrite Equation (3.1) as:

$$b^n_{\text{2D},v}\zeta^p_{\text{2D},v} + b^n_{\text{2D},i}\zeta^p_{\text{2D},i} = d^{p-1}_{\text{2D}} \ , \tag{3.9}$$

with

$$\begin{aligned}
b^n_{\text{2D},v} &= \frac{\alpha^n_{\text{2D}}}{2} + (\beta^n_{\text{2D}} + \alpha^n_{\text{2D}} f^n_I F^n_I) \ , \\
b^n_{\text{2D},i} &= \frac{\alpha^n_{\text{2D}}}{2} - (\beta^n_{\text{2D}} + \alpha^n_{\text{2D}} f^n_I F^n_I) \ , \\
d^{p-1}_{\text{2D}} &= \alpha^n_{\text{2D}}\zeta^{p-1}_{\text{1D}} + \beta^n_{\text{2D}}\frac{\Delta x_I}{\theta A^n_I F^n_I} q^{p-1}_{\text{1D-2D}} \\
&\quad + \alpha^n_{\text{2D}} s_{c,I}\left(f^n_I R^n_I - \frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF} g \Delta t} u^n_{\text{2D},I}\right) + \beta^n_{\text{2D}}\frac{s_{c,I}}{\theta F^n_I}\left(\theta R^n_I + (1-\theta) u^n_{\text{2D},I}\right) \ .
\end{aligned}$$

In general we have for each control volume the equation:

$$\begin{aligned}
&\left(\frac{S^{p-1}_\zeta}{\Delta t} + \sum_{\text{cell faces}} A^n_f \theta F^n_u\right)\zeta^p_c - \sum_{\text{cell faces}} A^n_f \theta F^n_u \zeta^p_a - Q^p_{\text{lat},c} \\
&= \frac{S^{p-1}_\zeta \zeta^{p-1}_c - V^{p-1}_c}{\Delta t} + \frac{V^n_c}{\Delta t} - \sum_{\text{cell faces}} s_{c,f} A^n_f (1-\theta) u^n_f - \sum_{\text{cell faces}} s_{c,f} A^n_f \theta R^n_u \ ,
\end{aligned} \tag{3.10}$$

with $Q_{\text{lat},c}$ the discharge *into* the grid cell.

Rewrite Equation (3.10) as:

$$\Big(\frac{S_\zeta^{p-1}}{\Delta t} + bb_c^n\Big)\zeta_c^p - \sum_{\text{cell faces}} cc_{c,f}^n \zeta_a^p - Q_{\text{lat},c}^p = \frac{S_\zeta^{p-1}\zeta_c^{p-1} - V_c^{p-1}}{\Delta t} + \frac{V_c^n}{\Delta t} + dd_c^n \ , \quad (3.11)$$

with:

$$\begin{aligned} bb_c^n &= \sum_{\text{cell faces}} \theta A_f^n F_u^n \ , \\ cc_{c,f}^n &= -\theta A_f^n F_u^n \ , \\ dd_c^n &= -\sum_{\text{cell faces}} s_{c,f}\big(\theta A_f^n R_u^n + (1-\theta)Q_f^n\big) \ . \end{aligned} \quad (3.12)$$

The coefficients Equation (3.12) are used in SOBEK and D-Flow FM to define the system of equations.

At the 1D2D interface we have the equations Equation (3.11) and Equation (3.9). Rewriting Equation (3.9) in such a way that the system is symmetrical results in:

$$cc_{c,\text{2D},I}^n \frac{b_{\text{2D},v}^n}{b_{\text{2D},i}^n}\zeta_{\text{2D},v}^p + cc_{c,\text{2D},I}^n \zeta_{\text{2D},i}^p = cc_{c,\text{2D},I}^n \frac{d_{\text{2D}}^{p-1}}{b_{\text{2D},i}^n} \ ,$$

Using this equation in the matrix coefficients, we get:

$$\begin{aligned} bb_{v,\text{2D},I}^n &= cc_{c,\text{2D},I}^n \frac{b_{\text{2D},i}^n}{b_{\text{2D},i}^n} \\ dd_{v,\text{2D},I}^{p-1} &= cc_{c,\text{2D},I}^n \frac{d_{\text{2D}}^{p-1}}{b_{\text{2D},i}^n} \ . \end{aligned} \quad (3.13)$$

3.1.4 Implementation of the 2D–to–1D coupling into the 1D system of equations

Equation (3.2) is rewritten as:

$$b_{\text{1D},s}^n \zeta_{\text{1D}}^p + b_{\text{1D},Q}^n Q_{\text{1D-2D}}^p = d_{\text{1D}}^{p-1} \ ,$$

with

$$\begin{aligned} b_{\text{1D},s}^n &= \alpha_{\text{1D}}^n \ , \\ b_{\text{1D},Q}^n &= -\beta_{\text{1D}}^n \frac{1}{\theta A_I^n F_I^n} \ , \\ d_{\text{1D}}^{p-1} &= \alpha_{\text{1D}}^n \frac{\zeta_{\text{2D},i}^{p-1} + \zeta_{\text{2D},v}^{p-1}}{2} + (\beta_{\text{1D}}^n - \alpha_{\text{1D}}^n f_I^n F_I^n)(\zeta_{\text{2D},i}^{p-1} - \zeta_{\text{2D},v}^{p-1}) \\ &\quad - \alpha_{\text{1D}}^n s_{c,I}\Big(f_I^n R_I^n - \frac{\Delta x_{\text{1D-2D}}}{\alpha_{SF} g \Delta t} u_{\text{2D},I}^n\Big) + \beta_{\text{1D}}^n \frac{s_{c,I}}{\theta F_I^n}\Big(\theta R_I^n + (1-\theta)u_{\text{2D},I}^n\Big) \ , \end{aligned}$$

which results in:

$$Q_{\text{1D-2D}}^p = -\frac{b_{\text{1D},s}^n}{b_{\text{1D},Q}^n}\zeta_{\text{1D}}^p + \frac{d_{\text{1D}}^{p-1}}{b_{\text{1D},Q}^n} \ , \quad (3.14)$$

where $-Q_{\text{1D-2D}}^p = -\Delta x_I q_{\text{1D-2D}}^p$ (notice the *minus* signs) is the lateral discharge *into* the 1D cell over the current time step.

Substituting Equation (3.14) into Equation (3.11) results in:

$$
\begin{aligned}
bb^n_{c,\mathsf{1D},I} &= -\frac{b^n_{\mathsf{1D},s}}{b^n_{\mathsf{1D},Q}} + \sum_{\text{cell faces}} \theta A^n_f F^n_u \ , \\
cc^n_{c,\mathsf{1D},I} &= -\theta A^n_f F^n_u \ , \\
dd^{p-1}_{c,\mathsf{1D},I} &= -\frac{d^{p-1}_{\mathsf{1D}}}{b^n_{\mathsf{1D},Q}} - \sum_{\text{cell faces}} s_{c,f}\left(\theta A^n_f R^n_u + (1-\theta) Q^n_f\right) \ .
\end{aligned}
\tag{3.15}
$$

Notice that $cc^n_{c,\mathsf{1D},I} = cc^n_{c,f}$, i.e., this coefficient remains unchanged.

3.2 Incorporation of boundary conditions in SOBEK

For a simple 1D channel with a discharge boundary condition at one end and a water level boundary condition at the other end, the symmetric system of equations looks like:

$$
\begin{pmatrix}
1 & 0 & & & & & \\
0 & b_2 & c_2 & & & & \\
 & c_2 & b_3 & c_3 & & & \\
 & & c_3 & \ddots & \ddots & & \\
 & & & \ddots & b_{n-2} & c_{n-2} & \\
 & & & & c_{n-2} & \tilde{b}_{n-1} & 0 \\
 & & & & & 0 & 1
\end{pmatrix}
\cdot
\begin{pmatrix}
\zeta_1 \\ \zeta_2 \\ \zeta_3 \\ \vdots \\ \zeta_{n-2} \\ \zeta_{n-1} \\ \zeta_n
\end{pmatrix}
=
\begin{pmatrix}
d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{n-2} \\ \tilde{d}_{n-1} \\ d_n
\end{pmatrix}
\tag{3.16}
$$

A boundary condition at grid point b with interior point i connected by link l

1. All coefficients on row n and column b in the system matrix are set to 0, except for the diagonal coefficient b_b.
2. The diagonal coefficient b_b is set to 1.
3. For a water level boundary condition: the right-hand side coefficient d_b is set to the boundary value ζ_{bnd}:

$$d_b = \zeta_{bnd} \tag{3.17}$$

4. For a discharge boundary condition:

$$d_b = \zeta^n_i \tag{3.18}$$

$$\tilde{b}_i = b_i + dir \cdot Q_{bnd} \tag{3.19}$$

   where dir denotes the direction of the flow at the boundary with respect to the orientation of the link.
5. The momentum equation is then substituted into the equation for the interior point:

$$\tilde{d}_i = d_i + A_u(l) \cdot (dir \cdot R_u(l) + F_u(l) \times d_b) \tag{3.20}$$

$$\tilde{b}_i = b_i + A_u(l) F_u(l) \tag{3.21}$$

3.3 Interface requirements for SOBEK and D-Flow FM

In D-Flow 1D and D-Flow FM the user-defined time step can be reduced. In this chapter we observe the inner loop for the computation of one sub time step.

For the computation of a coupled of D-Flow 1D (1D flow) and D-Flow FM (2D/3D flow), we suggest the following work flow (see Figure 3.2).

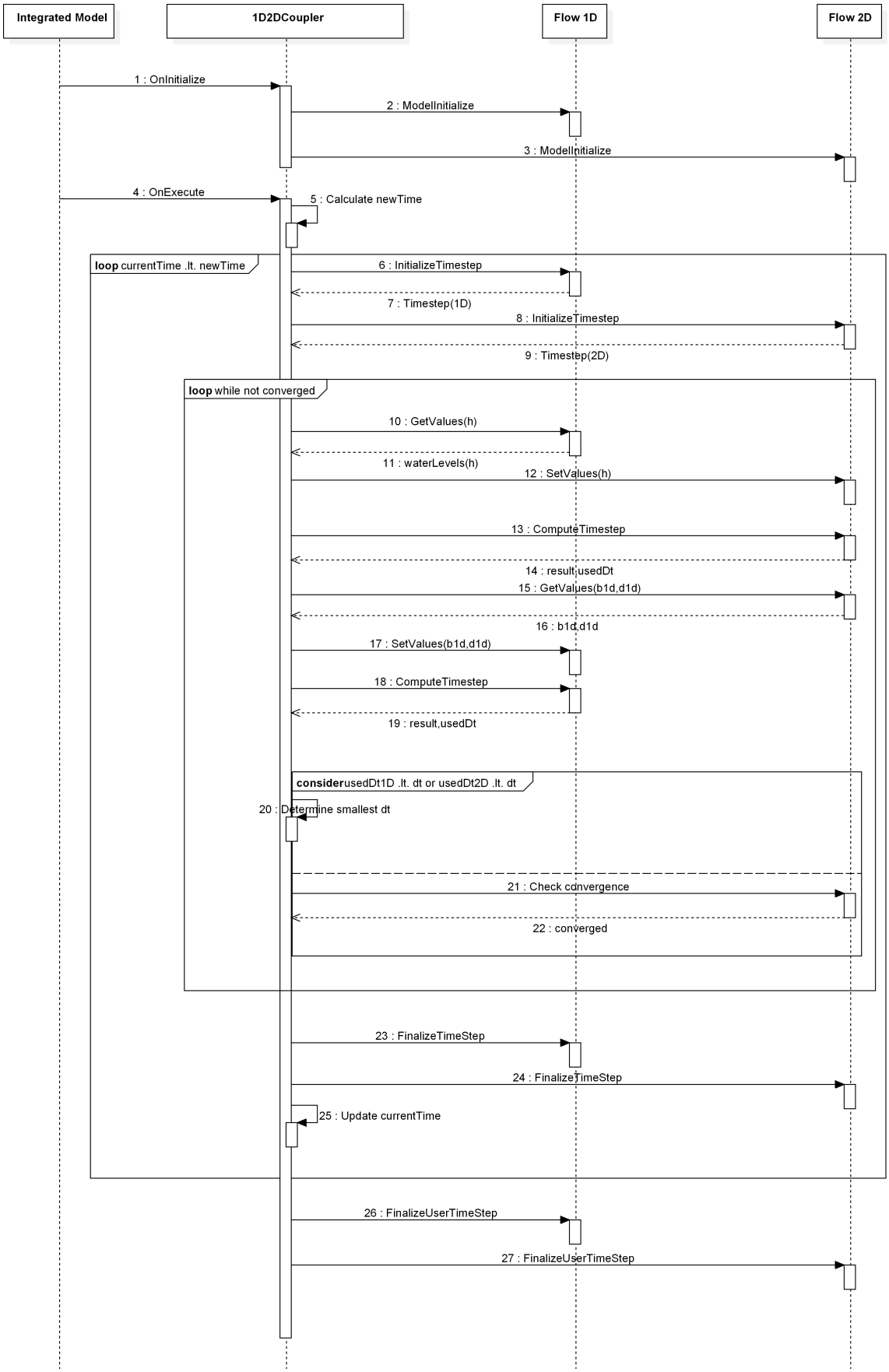

Figure 3.2: *Sequence diagram of coupled time step computation.*

In the computational core of both D-Flow 1D and D-Flow FM the computation of a time step has to be split up in parts, in order to allow for iterative computation. As a result the following interface is required:

- *InitializeTimestep*
- *ComputeTimeStep*
- *FinalizeTimestep*

For now we are restricting ourselves to 1D-2D interfaces where the D-Flow FM grids and D-Flow 1D grids match, i.e., to 1D-2D interfaces where there is a single connection that is (close to) normal to the interface between consecutive D-Flow FM water levels and consecutive D-Flow 1D water levels along the interface. We also impose for now the restriction that the time step in both models is to be identical. However, we anticipate the development of a 1D-2D coupling for non-matching spatial discretizations and time steps in the future. So although the transfer of data from 1D to 2D and vice versa is trivial at present, it will not be trivial in the future. For this reason we have defined the two separate subroutines `MapDataFromSOBEK2DFlowFM()` and `MapDataFromDFlowFM2SOBEK()` for the implementation of the required mappings in space and in time from 1D to 2D and from 2D to 1D.

In addition to this:

- The computation of the 1D-2D interface formulation for the 2D side has to be implemented in D-Flow FM.
- D-Flow FM requires the 1D water levels and the D-Flow 1D lateral discharge at the 1D-2D interfaces.
- The computation of the 1D-2D interface formulation for the 1D side has to be implemented in D-Flow 1D.
- D-Flow 1D requires the water levels in the 2D virtual grid cells and 2D inner grid cells at the 1D-2D interfaces.
- A number of coefficients that are computed in D-Flow FM have to be transferred to D-Flow 1D.
- A 1D-2D interface convergence check needs to be implemented.
- Each time step a final D-Flow 1D computation with imposed lateral discharge from the 2D domains has to be performed to enforce mass conservation.

4 File description for D-Flow1D

4.1 Introduction

The input files for the computational core are described in this chapter.

The main model definition input file is the <∗.md1d>-file. In this file the overall input parameters are set, such as start time, stop time, time step, constant initial values and numerical parameters.

Furthermore the name and location of the other input files are listed, such as the input files for the network, structures, cross-sections etc. All files are described below.

4.2 Model definition file for D-Flow 1D (<∗.md1d>)

Note: Some long keywords are split up by means of a dash '-'.
The two parts of those keywords should be concatenated WITHOUT the dash.

Table 4.1: *Branch mouth relations.*

Keyword	Type	Default	Description
`[NumericalOptions]`			
`teta`	double	`1.0`	Θ value in implicit transport solver (1 is fully implicit, 0 is fully explicit)
`tidalPeriod`	double	`12.417`	Tidal period (hours)
`maxMouthsPerBranch`	integer	`1`	maximum of mouths related to one branch
`advectionScheme`	string	`vanLeer-2`	possible values: ◇ `vanLeer-2`: van Leer 2 scheme ◇ `upwind`: standard upwind scheme,
`[Mouth]`			
`nodeId`	string		id of the node to which the mouth is related to
`estuaryLength`	double		Length of the estuary (m)
`refWaterDepth`	double		Reference water depth in mouth (m)
`refSalinity-Concentration`	double		Reference salinity in mouth (kg/m3)
`charFloodVelocity`	double		Characteristic flood velocity (m/s)
`freshWaterDischarge`	double	Optional	Fresh water discharge (m3/s)
`floodVolume`	double	Optional	Flood volume entering mouth during flood (m3)
`rhoSeaWater`	double		ρ sea water (kg/m3)
`rhoFreshWater`	double		ρ fresh water (kg/m3)
`branchesCount`	integer		number of branches in branches
`branches`	strings		id's of the branches attached to mouth

Table 4.2: *The D-Flow 1D file definition (<∗.md1d>).*

Keyword	**Type**	**Default**	**Description**
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`modelDef`	File type. Do not edit this.
`[Files]`			

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
networkFile			Name of the network file
networkUgridFile			Name of the ugrid-network file
crossLocFile			File with cross-section locations
crossDefFile			File with cross-section definitions
structureFile			Name of the structure file
obsPointsFile			Name of the observation points file
initialWaterLevelFile			File with initial water level
initialWaterDepthFile			File with initial water depth
initialDischargeFile			File with initial discharge
retentionFile			File with retention areas
boundLocFile			File containing boundary locations
latDischargeLocFile			File containing lateral discharge locations
boundCondFile			File containing boundary conditions and lateral discharges
wlevStateFileIn		wlevStateFileIn.xyz	Water level state input file
wlevStateFileOut		wlevStateFileOut.xyz	Water level state output file
roughnessFilePath		Roughness	Location of the files with roughness data
roughnessFiles			Files containing the roughness data (; delimited)
sobekSimIniFile			Name of the sobeksim-ini file. Will be phased out later
initialSalinityFile			File with initial salinity
dispersionFile			Name of the file containing the Thatcher Harleman F1-Dispersion term
f1File			Name of the file containing the Thatcher Harleman F1-Dispersion term
f3File			Name of the file containing the Thatcher Harleman F3-Dispersion term
f4File			Name of the file containing the Thatcher Harleman F4-Dispersion term
salinityParametersFile			Name of the file containing the Salinity Parameters
ConvLengthFile			Name of the file containing the convergence length of estuary
windShieldingFile			Name of the file containing the windshielding factor
[GlobalValues]			
UseInitialWaterDepth	Int	0	Use initial water depth instead of water level. 0=false, 1=true
InitialWaterLevel	Double	0.0	Initial water level for locations where no spatial varying data is defined
InitialWaterDepth	Double	0.0	Initial water depth for locations where no spatial varying data is defined
InitialDischarge	Double	0.0	Initial discharge for branches where no spatial varying data is defined
InitialSalinity	Double	0.0	Initial salinity level for locations where no spatial varying data is defined
InitialTemperature	Double	15.0	Initial salinity level for locations where no spatial varying data is defined
Dispersion	Double	0.0	Dispersion (TH-1) for locations where no spatial varying data is defined
F1	Double	0.0	Dispersion (TH-1) for locations where no spatial varying data is defined
F3	Double	0.0	Thatcher-Harleman F3 for locations where no spatial varying data is defined
F4	Double	0.0	Thatcher-Harleman F4 for locations where no spatial varying data is defined

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
ConvLength	Double	0.0	Convergence length of estuary for locations where no spatial varying data is defined
[InitialConditions]			
InitialEmptyWells	Int	0	0=false, 1=true
[Time]			
StartTime	DateTime	0	yyyy-MM-dd HH:mm:ss
StopTime	DateTime	0	yyyy-MM-dd HH:mm:ss
TimeStep	Double	0.0	in seconds
MapOutputTimeStep	Double	0.0	in seconds for gridpoints.nc and reachsegments.nc
HisOutputTimeStep	Double	0.0	in seconds for all other output files
[Restart]			
WriteRestart			Write restart files (=1) or not (=0). If equal to 1, restart files <sobek_yyyyMMdd_HHmmss>.nda and <sobek_yyyyMMdd_HHmmss>.ndf are written for each point-in-time (yyyy-MM-dd HH:mm:ss) specified by RestartStartTime, RestartStopTime and RestartTimeStep.
UseRestart			Simulation is initialized using restart files (=1) or not (=0).
RestartStartTime	DateTime		Restart start time (yyyy-MM-dd HH:mm:ss)
RestartStopTime	DateTime		Restart stop time (yyyy-MM-dd HH:mm:ss) Whatever stop time has been specified, if write restart is on there is always a restart file written at the end of the simulation.
RestartTimeStep	Double		Interval to write restart files in seconds
RestartFile			Name of the restart (e.g. <name>.nda and <name>.ndf) files, used to initialize the simulation. When no name is specified, default names sobek.nda and sobek.ndf are used.
[ResultsNodes]			0, 1, current, true, minimum, maximum, average
AdaptedCrossSec	Int	0	Adapted Cross-Sections (-), 1.0 or 0.0
BedLevel	Int	0	Bed Level (m AD), only relevant when Morphology switched on
ChangeArea	Int	0	Change in area per morphologic step (m2)
CumChangeArea	Int	0	Cumulative change in area (m2)
Density	Int	0	Density
EffectiveBackRad	Int	0	Effective back radiation
GrainSizeD50	Int	0	Grain Size D50 (mm)
GrainSizeD90	Int	0	Grain Size D90 (mm)
HeatLossConv	Int	0	Heat loss due to convection
HeatLossEvap	Int	0	Heat loss due to evaporation
HeatLossForcedConv	Int	0	Heat loss due to forced convection
HeatLossForcedEvap	Int	0	Heat loss due to forced evaporation
HeatLossFreeConv	Int	0	Heat loss due to free convection
HeatLossFreeEvap	Int	0	Heat loss due to free evaporation
IntSediTrans	Int	0	Integrated Sediment Transport (kg)
Lateral1d2d	Int	0	Lateral Discharge from 2D to 1D
LateralOnNodes	Int	0	0=false, 1=true
MeanBedLevelMain	Int	0	Mean Bed Level Main (m AD)
MorDepth	Int	0	Morphology Depth (m)
MorVelocity	Int	0	Morphology Velocity (m/s)
MorWaterLevel	Int	0	Morphology Water Level (m AD)
MorWidth	Int	0	Morphology Width (m)
NegativeDepth	Int	0	Negative depth

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
NetSolarRad	Int	0	Net incident solar radiation
NoIteration	Int	0	Number of iterations
RadFluxClearSky	Int	0	Radiation flux for clear sky condition
Salinity	Int	0	Salt concentration
SedimentTransport	Int	0	Sediment Transport (kg/s)
SedimentTransport-Left	Int	0	Sediment Transport Left (kg/s)
SedimentTransport-Right	Int	0	Sediment Transport Right (kg/s)
ShieldsParameter	Int	0	Shields Parameter (-)
TotalArea	Int	0	Total area
Temperature	Int	0	Water temperature in degrees Celcius
TotalHeatFlux	Int	0	Total heat flux
Totalwidth	Int	0	Total width
Volume	Int	0	Water volume
WaterDepth	Int	0	Water depth
WaterLevel	Int	1	Water level
[ResultsBranches]			0, 1, current, true, minimum, maximum, average
AreaFP1	Int	0	FloodPlain1 Flow area
AreaFP2	Int	0	FloodPlain2 Flow area
AreaMain	Int	0	Main Flow area
ChezyFP1	Int	0	FloodPlain1 Chezy values
ChezyFP2	Int	0	FloodPlain2 Chezy values
ChezyMain	Int	0	Main Chezy values
Discharge	Int	1	Discharge
DischargeFP1	Int	0	FloodPlain1 Discharge
DischargeFP2	Int	0	FloodPlain2 Discharge
DischargeMain	Int	0	Main Discharge
Dispersion	Int	1	0=false, 1=true
EnergyLevels	Int	0	Energy head
F1	Int	0	F1-term (m2/s)
F3	Int	0	F3-term (m2/s)
F4	Int	0	F4-term (m2/s)
FlowArea	Int	0	Flow area
FlowChezy	Int	0	Chezy values
FlowConv	Int	0	Conveyance
FlowHydrad	Int	0	Hydraulic radius
FlowWidth	Int	0	Flow Width
Froude	Int	0	Froude number
HydradFP1	Int	0	FloodPlain1 Hydraulic radius
HydradFP2	Int	0	FloodPlain2 Hydraulic radius
HydradMain	Int	0	Main Hydraulic radius
MomAcceleration	Int	None	Resulting acceleration term in momentum equation
MomAdvection	Int	None	Advection term in momentum equation
MomBedStress	Int	None	Bed stress term in momentum equation
MomLateralCorrection	Int	None	lateral/storage loss term in momentum equation
MomLosses	Int	None	Various losses term in momentum equation
MomPressure	Int	None	Pressure term in momentum equation
MomWindStress	Int	None	wind stress term in momentum equation
TimeStepEstimation	Int	0	Time step estimation
Velocity	Int	0	Velocity
WaterLevelGradient	Int	0	Water level gradient
WidthFP1	Int	0	FloodPlain1 Flow width
WidthFP2	Int	0	FloodPlain2 Flow width
WidthMain	Int	0	Main Flow width
WindVelocity	Int	0	0=false, 1=true

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Keyword	Type	Default	Description
			(continued from previous page)
WindDirection	Int	0	0=false, 1=true
[ResultsStructures]			0, 1, current, true, minimum, maximum, average
CrestLevel	Int	0	0=false, 1=true
CrestWidth	Int	0	0=false, 1=true
Discharge	Int	0	Discharge (s)
FlowArea	Int	0	Flow area (s)
GateLowerEdgeLevel	Int	0	0=false, 1=true
GateOpeningHeight	Int	0	0=false, 1=true
Head	Int	0	Head Difference (s)
PressureDifference	Int	0	Pressure difference (s)
State	Int	-1	Classification of structure flow: < 0 = Not Applicable, 0 = No flow, 1 = Free Weir Flow, 2 = Drowned Weir Flow, 3 = Free Gate Flow, 4 = Drowned Gate Flow, 5 = Free Flow, 6 = Drowned Flow.
Velocity	Int	0	Velocity (s)
WaterLevelAtCrest	Int	0	Water level at crest (s)
WaterLevelDown	Int	0	Water level down (s)
WaterLevelUp	Int	0	Water level up (s)
[ResultsPumps]			0, 1, current, true, minimum, maximum, average
DeliverySideLevel	Int	0	Water Level Delivery Side
PumpCapacity	Int	0	Pump Capacity
PumpDischarge	Int	0	Pump Discharge
PumpHead	Int	0	Pump Head
ReductionFactor	Int	0	Pump Reduction Factor
SuctionSideLevel	Int	0	Water Level Suction Side
[ResultsObservationpoints]			0, 1, current, true, minimum, maximum, average
Discharge	Int	1	Discharge (op)
Salinity	Int	0	Salt concentration (op)
Velocity	Int	0	Velocity (op)
WaterDepth	Int	1	Water depth (op)
WaterLevel	Int	1	Water level (op)
[ResultsRetentions]			
Volume	Int	0	Volume (rt)
WaterLevel	Int	0	Water level (rt)
[ResultsLaterals]			
ActualDischarge	Int	0	Actual Lateral Discharge (l)
DefinedDischarge	Int	0	Defined Lateral Discharge (l)
LateralDifference	Int	0	Difference between Actual and Defined Discharge (l)
WaterLevel	Int	0	Water Level at Lateral Location (l)
[ResultsWaterBalance]			
Bal2d1dIn	Int	0	Lateral1D2DDischarge In
Bal2d1dOut	Int	0	Lateral1D2DDischarge Out
Bal2d1dTot	Int	0	Lateral1D2DDischarge Total
BalBoundariesIn	Int	0	Boundaries In
BalBoundariesOut	Int	0	Boundaries Out
BalBoundariesTot	Int	0	Boundaries Total
BalError	Int	0	VolumeError

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Keyword	Type	Default	Description
			(continued from previous page)
BalLatIn	Int	0	LateralDischarge In
BalLatOut	Int	0	LateralDischarge Out
BalLatTot	Int	0	LateralDischarge Total
BalStorage	Int	0	Storage
BalVolume	Int	0	TotalVolume
[ResultsGeneral]			Not exported by Delta Shell
ActualValue	Int	1	0=false, 1=true
DelwaqNoStaggered-Grid	Int	0	0=false, 1=true
FlowAnalysis-TimeSeries	Int	0	0=false, 1=true
MeanValue	Int	0	0=false, 1=true
MaximumValue	Int	0	0=false, 1=true
SobeksimStamp	Int	1	0=false, 1=true
[Sediment]			
D50	Double	0.0005	
D90	Double	0.001	
DepthUsedForSediment	Double	0.3	
[Specials]			
DesignFactorDLG	Double	1.0	
[NumericalParameters]			
AccelerationTerm-Factor	Double	1.0	
AccurateVersusSpeed	Int	3	
CourantNumber	Double	1.0	
DtMinimum	Double	0.001	
EpsilonValueVolume	Double	0.0001	
EpsilonValueWater-Depth	Double	0.0001	
FloodingDivided-ByDrying	Double	10.0	
Gravity	Double	9.81	
MaxDegree	Int	6	
MaxIterations	Int	8	
MaxTimeStep	Double	dt_sobek	
MinimumLength	Double	1.0	
MinimumSurface-atStreet	Double	0.1	
MinimumSurfaceinNode	Double	0.1	
RelaxationFactor	Double	1.0	
Rho	Double	1000	
StructureInertia-DampingFactor	Double	1.0	
Theta	Double	1.0	
ThresholdValueDrying	Double	0.001	Default is: ThresholdValueFlooding / FloodingDividedByDrying
ThresholdValue-Flooding	Double	0.01	
UseOmp	Int	0	0=false, 1=true
UseTimeStepReducer-Structures	Int	0	0=false, 1=true
[SimulationOptions]			
allowablelarger-timestep	Int	0	
			(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
allowabletimestep-limiter	Int	30	
AllowableVolume-Error	Double	10.0	
AllowCrestLevel-BelowBottom	Int	0	0=false, 1=true
Cflcheckalllinks	Int	0	0=false, 1=true
Channel	Int	1	0=false, 1=true Only relevant for licenses, can be removed later
DebugTime	Int	0	0=false, 1=true
DepthsBelowBobs	Int	0	0=false, 1=true
DispMaxFactor	Double	0.45	
DumpInput	Int	0	0=false, 1=true
Iadvec1D	Int	2	Possible values: 1 Conservation of Momentum 2 Weighted Average of Conservation of Momentum and Conservation of Energy in Contraction and Expansion 3 Weighted Average of Conservation of Momentum and Conservation of Energy in Contraction Only 4 Weighted Average of Conservation of Momentum and Conservation of Energy in Expansion Only 5 Conservation of Energy
Jchecknans	Int	0	0=false, 1=true, Check for NANs
Junctionadvection	Int	0	Possible values: 0 None 1 Open 1D Links not in Line 2 Also in Line
LaboratoryTest	Int	0	0=false, 1=true Special test
LaboratoryTimeStep	Double	current dt_sobek	Relevant when LaboratoryTest is true
LaboratoryTotalStep	Int	current nstep	Relevant when LaboratoryTest is true
Limtyphu1D	Int	1	Possible values: 1 Upwind 2 Central in Cross-Sections 3 Central in Water Levels
LoggingLevel	Char	info	Possible values: ◇ info ◇ debug ◇ warning ◇ error ◇ fatal ◇ none
Manhloss	Int	0	0=false, 1=true
ManholeLosses	Int	0	0=false, 1=true Two different flags with different actions for apparently the same thing. Needs investigation.
MissingValue	Real	-999.999	
NoSuperCritical-Inflow	Int	0	0=false, 1=true
			(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
Onlineplot	Int	0	0=false, 1=true
PercentAllowable-VolumeError	Double	1.0	
PreissmannMin-ClosedManholes	Double	0.001	
ReadNetworkFromUgrid	Int	0	0=false, 1=true
River	Int	1	0=false, 1=true Only relevant for licenses, can be removed later
Sewer	Int	0	0=false, 1=true Only relevant for licenses, can be removed later
SiphonUpstream-ThresholdSwitchOff	Double	0.1	
StrucAlfa	Double	0.9	
StrucFlowDirection-AccuracyFactor	Double	1.0	
StructureDynamics-Factor	Double	1.0	
StructureStability-Factor	Int	0	
ThresholdFor-SummerDike	Double	0.4	
TimersOutput-Frequency	Int	0	Only relevant when UseTimers is true
use1d2dcoupling	Int	0	0=false, 1=true
UseEnergyHead-Structures	Int	0	0=false, 1=true
UseTimers	Int	0	0=false, 1=true
Usevariableteta	Int	0	0=false, 1=true
UseWlevStateFile	Int	0	0=false, 1=true
VolumeCheck	Int	0	0=false, 1=true
VolumeCorrection	Int	0	0=false, 1=true
WaterQualityInUse	Int	0	0=false, 1=true
WriteNetCDF	Int	0	0=false, 1=true
[AdvancedOptions]			
CacheMode	Char	none	Possible values: ◇ none ◇ write ◇ read
CalculateDelwaq-Output	Int	0	0=false, 1=true
FillCulvertsWithGL	Int	0	0=false, 1=true If true this overrules MaxLoweringCrossAtCulvert
ExtraResistance-GeneralStructure	Double	0.0	
FillCulvertsWithGL	Int	0	0=false, 1=true
LateralLocation	Int	0	Possible values: 0 Nearest Bed Level Point 1 Lowest Bed Level Point
Latitude	double	52.00667	Latitude of the model
Longitude	double	4.35556	Longitude of the model
MaxLoweringCross-AtCulvert	Double	0.0	

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
MaxVolFact	Double	0.9	Relevant when NoNegativeQlatWhenThereIsNoWater is true
NoNegativeQlatWhenThereIsNoWater	Int	1	0=false, 1=true
TransitionHeightSD	Double	1.0	
[Salinity]			
SaltComputation	Int	0	0=false, 1=true
DiffusionAtBoundaries	Int	0	0=false, 1=true
[TransportComputation]			
Temperature	Int	0	0=false, 1=true
Density	Char	eckhart_modified	Possible values: ◇ eckhart_modified ◇ eckhart ◇ Unesco
HeatTransferModel	Char	transport	Possible values: ◇ transport ◇ excess ◇ composite
[Temperature]			
SurfaceArea	double	10^6	Exposed surface area (used in Excess temperature model) (m^2)
AtmosphericPressure	double	10^5	Atmospheric pressure (N/m^2)
DaltonNumber	double	0.0015	Dalton number
HeatCapacityWater	double	3930	Heat capacity of water (J/kgK)
StantonNumber	double	0.00145	Stanton number
[Morphology]			
CalculateMorphology	Int	0	0=false, 1=true
SedimentInputFile	char	-	name of sediment input file
MorphologyInputFile	char	-	name of morphology description file
[ObservationPoints]			
interpolationType	Char	OBS_NEAREST	Possible values: ◇ OBS_NEAREST ◇ OBS_LINEAR
[StorageTable]			
WriteStorageTables	Int	0	Write storage table file (0=false, 1=true)
StorageOutputFile	Char	-	Name of storage table file (e.g. output/storage.nc)
StorageTableIncrement	double	0.1	Desired increment to write storage tables
ExtraHeight	double	0.0	Extend the storage tables with an additional height

4.3 Network file

Table 4.3: *Network definition.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`network`	File type. Do not edit this.
`[Map]`			
`coordinateSystem`			Used coordinate system for locations
`topoGeomFile`			Name of the file used for the topology and geometry definition of the network
`[Node]`			
`id`	String		Unique network id
`name`	String		Given name in the user interface
`x`	DOUBLE		X-coordinate of the node (m)
`y`	DOUBLE		Y-coordinate of the node (m)
`[Branch]`			
`id`	String		Unique network id
`name`	String		Given name in the user interface
`fromNode`	String		Node id of node at start of branch
`toNode`	String		Node id of node at end of branch
`order`	INTEGER		Order number will be used for interpolation over branches. A chain of branches with the same order number (> 0) will be treated as one.
`geometry`	String		**(GUI ONLY)** The geometry of how the branch should go.
`gridPointsCount`	INT		Number of grid points on a branch
`gridPointX`	DOUBLE[]		X-coordinates of the grid points (m)
`gridPointY`	DOUBLE[]		Y-coordinates of the grid points (m)
`gridPointOffsets`	DOUBLE[]		Chainage of the grid points on the branch (m)
`gridPointID`	String[]		ID's of the grid points separated by ';'

Notes

1. the number of values in xValuesAtGridPoints, yValuesAtGridPoints and chainages must be identical. A mismatch in length will result in an error.
2. The first point in xValuesAtGridPoints, yValuesAtGridPoints and chainages corresponds to the grid point coinciding with the node, defined in "toNode"
3. The last point in xValuesAtGridPoints, yValuesAtGridPoints and chainages corresponds to the grid point coinciding with the node, defined in "fromNode"
4. The topoGeomFile is not required for step one of this project.

4.4 Cross section files

Cross sections are defined in two files. The first file contains the cross section definitions, containing the geometrical parameters. And the location file, containing the location of the actual cross section in the network, including the bed level (or a shift).

4.4.1 Cross section definition file

The definition file contains the geometrical parameters for each different cross section.

Table 4.4: *Cross section definition file.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`		`1`	Major file version. Do not edit this.
`minorVersion`		`0`	Minor file version. Do not edit this.
`fileType`		`crossDef`	File type. Do not edit this.
`[Definition]`			
`id`	String		Unique cross-section definition id
`type`	String		Cross section type. Possible values are: ◇ rectangle ◇ tabulated ◇ circle ◇ egg ◇ yz ◇ xyz ◇ ellipse ◇ arch ◇ cunette ◇ steelcunette ◇ trapezium
`isShared`	String		(Optional, default=false=0) if is shared set to true=1
`thalweg`	Double		**(GUI ONLY)**
`...`			type specific items

The type specific items can be found in the subsequent subsections.

4.4.1.1 Tabulated cross section definition

Additional items for the tabulated cross section definition.

Table 4.5: *Tabulated cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = tabulated`	String		
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Heights / Levels (m AD)
`flowWidths`	Double[]		Flow widths at heights (m)
`totalWidths`	Double[]		Total widths at heights (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`sd_crest`		0.0	Summer dike crest level (m AD)
`sd_baseLevel`		0.0	Summer dike base level (m AD)

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
sd_flowArea		0.0	Flow area behind summer dike (m^2)
sd_totalArea		0.0	Total area behind summer dike (m^2)
main			Width of main section (m)
floodPlain1			Width of Floodplain 1 (m)

Default for main: max(FlowWidth)
Default for Floodplain1: max(FlowWidth) - Main
FloodPlain2 is defined as max(FlowWidth) - Main - Floodplain1

4.4.1.2 YZ cross section definition

Additional items for the YZ cross section definition.

Table 4.6: *YZ cross section definition.*

Keyword	Type	Default	Description
[Definition]			
type = yz	String		
yzCount	Int		Number of YZ-coordinates
yValues	Double[]		Y-coordinates (m)
zValues	Double[]		Z-coordinates (m AD)
sectionCount	Int	1	Number of friction sections
roughnessNames	String[]		Names of the roughness sections
roughnessPositions	Double[]		Locations where the roughness sections start and end. Always one more than sectionCount
deltaZStorage	Double[]		**(GUI ONLY)**

4.4.1.3 XYZ cross section definition

Additional items for the XYZ cross section definition.

Table 4.7: *XYZ cross section definition.*

Keyword	Type	Default	Description
[Definition]			
type = xyz	String		
xyzCount	Int		Number of XYZ-coordinates
xCoors	Double[]		X-coordinates (m)
yCoors	Double[]		Y-coordinates (m)
zCoors	Double[]		Z-coordinates (m AD)
yValues	Double[]		Y-values of the YZ-converted cross section (m)
zValues	Double[]		Z-values of the YZ-converted cross section (m AD)
sectionCount	Int	1	Number of friction sections
roughnessNames	String[]		Names of the roughness sections
roughnessPositions	Double[]		Locations where the roughness sections start and end. Always one more than sectionCount
roughnessTypePos	Int[]		Roughness types for positive flow
roughnessValuesPos	Double[]		Roughness values for positive flow
roughnessTypeNeg	Int[]		Roughness types for negative flow
roughnessValuesNeg	Double[]		Roughness values for negative flow
deltaZStorage	Double[]		**(GUI ONLY)**

The values in xCoors, yCoors and zCoors defines the xyz cross section as it is defined in the user

interface. The translation of this profile into a YZ-cross section, that can be used in the computational core is given in yValues, zValues.

4.4.1.4 Circle cross section definition

Additional items for the circular shaped cross section definition.

Table 4.8: *Circle cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = circle`	CString20		
`diameter`	double		Diameter of the circle (m)
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Levels (m AD)
`flowWidths`	Double[]		Flow widths at levels (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapezium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.5 Egg cross section definition

Additional items for the egg shaped cross section definition.

Table 4.9: *Egg cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = egg`	String		
`width`	Double		Width of the egg profile (m) Height is calculated: 1.5 * Width
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Levels (m AD)
`flowWidths`	Double[]		Flow widths at levels (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapezium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.6 Ellipse cross section definition

Additional items for the ellipse shaped cross section definition.

Table 4.10: *Ellipse cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
`type = ellipse`	String		
`height`	Double		Height of the ellipse (m)
`width`	Double		Width of the ellipse (m)
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Levels (m AD)
`flowWidths`	Double[]		Flow widths at levels (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapezium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.7 Rectangle cross section definition

Additional items for the rectangular cross section definition.

Table 4.11: *Rectangular cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = rectangle`	String		
`width`	Double		Width of the rectangle (m)
`height`	Double		Height of the rectangle (m)
`closed`	Int	1	1 = closed channel, 0 = open
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapezium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.8 Arch cross section definition

Additional items for the arch cross section definition.

Table 4.12: *Arch cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = arch`	String		
`width`	Double		Width of the cross section (m)
`height`	Double		Height of cross section (m)
`arcHeight`	Double		Height of arch (m)
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Levels (m AD)
`flowWidths`	Double[]		Flow widths at levels (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapezium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.9 Cunette cross section definition

Additional items for the cunette cross section definition.

Table 4.13: *Cunette cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = cunette`	String		
`width`	Double		Width of the cunette (m). Height is calculated: 0.634 * Width
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Levels (m AD)
`flowWidths`	Double[]		Flow widths at levels (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapezium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.10 Steel cunette cross section definition

Additional items for the steel cunette cross section definition.

Table 4.14: *Steel cunette cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = steelcunette`	String		
`height`	Double		Height of cunette (m)
`r`	Double		Radius r (m)
`r1`	Double		Radius r1 (m)
`r2`	Double		Radius r2 (m)
`r3`	Double		Radius r3 (m)
`a`	Double		Angle a (degrees)
`a1`	Double		Angle a1 (degrees)
`numLevels`	Int		Number of levels in table
`levels`	Double[]		Levels (m AD)
`flowWidths`	Double[]		Flow widths at levels (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapazium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.1.11 Trapezium cross section definition

Additional items for the trapezium cross section definition.

Table 4.15: *Trapezium cross section definition.*

Keyword	Type	Default	Description
`[Definition]`			
`type = trapezium`	String		
`slope`	Double		Slope of trapezium (m)
`maximumFlowWidth`	Double		Maximum floww idth of trapezium (m)
`bottomWidth`	Double		bottom width of trapezium (m)
`groundlayerUsed`	Int	0	Flag for ground layer (0=None, <>0=Yes)
`groundlayer`	Double	0.0	Thickness of ground layer (m)
`roughnessNames`	String		Name of the roughness section of this definition.

Note: In SOBEK 3.4 the cross sections of type circle, egg, ellipse, rectangle, arch, cunette, steel cunette and trapazium are transformed in Delta Shell into a tabulated cross section. For compatibility reasons, the arrays levels and flowWidths are added to the description of the cross section. And subsequently these are the arrays that will be used in the computational core.

4.4.2 Cross section location file

Table 4.16: *Cross section location.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`crossLoc`	File type. Do not edit this.
`[CrossSection]`			
`id`	String		Unique network id
`name`	String		Given name in the user interface
`branchId`	String		Branch on which the cross section is located
`chainage`	Double		Location on the branch (m)
`shift`	Double		Level shift of the cross section definition (m)
`definition`	String		Id of cross section definition

4.5 Structure file

Table 4.17: *Structure definition.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`structure`	File type. Do not edit this.
`[Structure]`			
`id`	String		Unique definition id
`name`	String		Given name in the user interface
`branchId`	String		Branch on which the structure is located
`chainage`	Double		Chainage on the branch (m)
`compound`	Int		All structures with the same compound id are considered to be elements of the same compound structure.
`type`	String		Structure type. Possible values: ◇ weir ◇ universalWeir ◇ riverWeir ◇ advancedWeir ◇ culvert (use for (inverted) siphon too) ◇ bridge ◇ pump ◇ orifice ◇ generalStructure ◇ extraResistance
`...`			Structure type specific data

NOTE: the elements of the compound structure must be placed on the same branch and chainage. Otherwise an error message is given by the computational core.

4.5.1 Weir

Structure type specific data for a weir structure.

Table 4.18: *Weir definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = weir`	String		
`crestlevel`	Double		Crest level of weir (m AD)
`crestwidth`	Double		Width of weir (m)
`dischargecoeff`	Double		Discharge coefficient (-)
`latdiscoeff`	Double		Lateral discharge coefficient (-)
`allowedflowdir`	Int		0 = Both, 1=Positive, 2=Negative, 3=None

4.5.2 Universal Weir

Structure type specific data for a universal weir structure.

Table 4.19: *Universal Weir definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = universalWeir`	String		
`allowedflowdir`	Int		0 = Both, 1=Positive, 2=Negative, 3=None
`levelscount`	Int		Number of YZ-Values
`yValues`	Double[]		y-values of the cross section (m)
`zValues`	Double[]		z-values of the cross section (m)
`crestlevel`	Double		Crest level of weir (m AD)
`dischargecoeff`	Double		Discharge coefficient (-)
`freesubmergedfactor`	Double	0.667	Normally 0.667 (2/3) (-)

4.5.3 River Weir

Structure type specific data for a river weir structure.

Table 4.20: *River Weir definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = riverWeir`	String		
`crestlevel`	Double		Crest level (m AD)
`crestwidth`	Double		Crest width (m)
`poscwcoef`	Double		Coefficient for positive direction (-)
`posslimlimit`	Double		Submergence limit for positive direction (-)
`negcwcoef`	Double		Coefficient for negative flow direction (-)
`negslimlimit`	Double		Submergence limit for negative direction (-)
			Reduction table for positive flow direction
`possfcount`	Int		Number of table rows
`possf`	Double[]		Values for (h2 - z) / (h1 - z) (-)
`posred`	Double[]		Reduction factors (-)
			Reduction table for negative flow direction
`negsfcount`	Int		Number of table rows
`negsf`	Double[]		Values for (h2 - z) / (h1 - z) (-)
`negred`	Double[]		Reduction factors (-)

4.5.4 Advanced Weir

Structure type specific data for an advanced weir structure.

Table 4.21: *Advanced Weir definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = advancedWeir`	String		
`crestlevel`	Double		Crest level of weir (m AD)
`crestwidth`	Double		Crest width (m)
`npiers`	Int		Number of piers
`posheight`	Double		Upstream face height for positive flow direction (m)
`posdesignhead`	Double		Weir design head for positive flow direction (m)
`pospiercontractcoef`	Double		Pier contraction coefficient for positive flow direction (-)

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
posabutcontractcoef	Double		Abutment contraction coefficient for positive flow direction (-)
negheight	Double		Upstream face height for negative flow direction (m)
negdesignhead	Double		Weir design head for negative flow direction (m)
negpiercontractcoef	Double		Pier contraction coefficient for negative flow direction (-)
negabutcontractcoef	Double		Abutment contraction coefficient for negative flow direction (-)

4.5.5 Culvert, Siphon or Inverted Siphon

Structure type specific data for a culvert, siphon or inverted siphon structure.

Table 4.22: *Culvert, Siphon or Inverted Siphon definition.*

Keyword	Type	Default	Description
[Structure]			
type = culvert	String		Alt. type = siphon, type = invertedSiphon
allowedFlowdir	Int		0 = Both, 1=Positive, 2=Negative, 3=None
leftLevel	Double		Left bed level (m AD)
rightLevel	Double		Right bed level (m AD)
csDefId	String		Id of Cross-Section Definition
length	Double		Length (m)
inletLossCoeff	Double		Inlet loss coefficient (-)
outletLossCoeff	Double		Outlet loss coefficient (-)
valveOnOff	Int		Flag for having valve or not (0=no valve, 1=valve)
iniValveOpen	Double		Initial valve opening height (m)
			Table for loss coefficients of valve
lossCoeffCount	Int		Number of rows in table
relOpening	Double[]		Relative valve opening (0.0 — 1.0)
lossCoeff	Double[]		Loss coefficients (-)
bedFrictionType	Int		Friction type, possible values are: ◇ Chezy = 1 ◇ Manning = 4 ◇ Nikuradse = 5 ◇ Strickler = 6 ◇ WhiteColebrook = 7 ◇ BosBijkerk = 9
bedFriction	Double		Friction Value
groundFrictionType	Int		Friction type for ground layer
groundFriction	Double		Friction value for ground layer
Siphon and Inverted Siphon:			
bendLossCoeff	Double		Bend loss coefficient (-)
Siphon Only:			
turnOnLevel	Double		Start level of operation of siphon (m AD)
turnOffLevel	Double		Stop level of operation of siphon (m AD)

4.5.6 Bridge

Structure type specific data for a bridge structure.

Table 4.23: *Bridge Structure definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = bridge`	String		Alt. type = bridgePillar
`allowedFlowdir`	Int		0 = Both, 1=Positive, 2=Negative, 3=None
`bedLevel`	Double		Lowest point of bridge profile (crest) (m AD)
`Pillar Bridge Only:`			
`pillarWidth`	Double		Total width of pillars in flow direction (m)
`formFactor`	Double		Shape factor (-)
`Abutment, Fixed Bed or Soil Bed Bridge Only:  (standard bridge)`			
`csDefId`	C40		Id of Cross-Section Definition
`length`	Double		Length (m)
`inletLossCoeff`	Double		Inlet loss coefficient (-)
`outletLossCoeff`	Double		Outlet loss coefficient (-)
`bedFrictionType`	Int		Friction type
`bedFriction`	Double		Friction Value
`groundFrictionType`	Int		Friction type for ground layer
`groundFriction`	Double		Friction value for ground layer

4.5.7 Pump

Structure type specific data for a pumping station structure.

Table 4.24: *Pump definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = pump`	String		
`direction`	Int	1	Pump direction (>0 = positive, <0 = negative) ABS(direction): 1: Suction side control 2: Delivery side control 3: Suction and Delivery side econtrol
`nrstages`	Int	1	Number of stages in pump
`capacity`	Double[]		Pump capacity (m3/s)
`startlevelsuctionside`	Double[]		Start level suction side (m AD)
`stoplevelsuctionside`	Double[]		Stop level suction side (m AD)
`startleveldeliveryside`	Double[]		Start level at delivery side (m AD)
`stopleveldeliveryside`	Double[]		Stop level at delivery side (m AD)
`reductionfactorlevels`	Int	0	Number of levels in reduction table
`head`	Double[]		Head, the difference in waterlevel (delivery side - suction side) (m)
`reductionfactor`	Double[]		Reduction factor (-)

4.5.8 Orifice

Structure type specific data for an orifice structure.

Table 4.25: *Orifice definition.*

Keyword	Type	Default	Description
`[Structure]`			
`type = orifice`	String		
`allowedflowdir`	Int		0 = Both, 1=Positive, 2=Negative, 3=None
`crestlevel`	Double		Crest level (m AD)

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Keyword		Default	Description
			(continued from previous page)
crestwidth	Double		Crest width (m)
openlevel	Double		Gate height (m)
contractioncoeff	Double		Contraction coefficient (-)
latcontrcoeff	Double		Lateral contraction coefficient (-)
uselimitflowpos	int		0 = unlimited, 1 = limited
limitflowpos	Double		Maximum positive flow (m3/s)
uselimitflowneg	int		0 = unlimited, 1 = limited
limitflowneg	Double		Maximum negative flow (m3/s)

4.5.9 General Structure

Structure type specific data for a general structure.

Table 4.26: *General Structure definition.*

Keyword	Type	Default	Description
[Structure]			
type = generalStructure	String		
widthleftW1	Double		Width left side of structure (m)
widthleftWsdl	Double		Width structure left side (m)
widthcenter	Double		Width structure centre (m)
widthrightWsdr	Double		Width structure right side (m)
widthrightW2	Double		Width right side of structure (m)
levelleftZb1	Double		Bed level left side of structure (m AD)
levelleftZbsl	Double		Bed level left side structure (m AD)
levelcenter	Double		Bed level at centre of structure (m AD)
levelrightZbsr	Double		Bed level right side structure (m AD)
levelrightZb2	Double		Bed level right side of structure (m AD)
gateheight	Double		Gate lower edge level (m AD)
posfreegateflowcoeff	Double		Positive free gate flow (-)
posdrowngateflowcoeff	Double		Positive drowned gate flow (-)
posfreeweirflowcoeff	Double		Positive free weir flow (-)
posdrownweirflowcoeff	Double		Positive drowned weir flow (-)
poscontrcoeffreegate	Double		Positive flow contraction coefficient (-)
negfreegateflowcoeff	Double		Negative free gate flow (-)
negdrowngateflowcoeff	Double		Negative drowned gate flow (-)
negfreeweirflowcoeff	Double		Negative free weir flow (-)
negdrownweirflowcoeff	Double		Negative drowned weir flow (-)
negcontrcoeffreegate	Double		Negative flow contraction coefficient (-)
extraresistance	Double	0	Extra resistance (-)

4.5.10 Extra Resistance

Structure type specific data for extra resistance.

Table 4.27: *Extra Resistance definition.*

Keyword	Type	Default	Description
[Structure]			
type = extraResistance	String		
numValues	Int		Number of values
levels	Double[]		Water levels (m AD)
ksi	Double[]		KSI-values (s2/m5)

4.6 Spatial Data

4.6.1 General

Spatial Data refers to all kinds of input data that can be defined on the network. The following keywords refer to "Spatial Data" files:

- initialWaterLevelFile
- initialWaterDepthFile
- initialDischargeFile
- initialSalinityFile
- dispersionFile (or F1)
- roughnessFiles
- F3
- F4
- windShielding

Table 4.28: *Spatial data definitions.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`spatialData`	File type. Do not edit this.
`[Content]`			
`quantity`			Possible values: ◇ InitialWaterLevel ◇ InitialWaterDepth ◇ InitialDischarge ◇ InitialSalinity ◇ Dispersion ◇ WindShielding ◇ ThatcherHarleman_F3 ◇ ThatcherHarleman_F4
`interpolate`	Int	0	0=false, 1=true
`[Definition]`			
`branchid`			
`chainage`			
`value`			

The roughness files are an extension on the regular "Spatial Data" files.

4.6.2 Roughness

Table 4.29: *Roughness definition.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`	String	`roughness`	File type. Do not edit this.
`[Content]`			
`sectionId`	String		The name of the roughness section
`flowDirection`	Int	0	Type of flow direction, possible value : ◇ 0=false=normal ◇ 1=true=reversed
`In case of (normal) Roughness Section only :`			
`globalType`	Int	1	The global default type for this section, possible values: ◇ Chezy = 1 ◇ Manning = 4 ◇ Nikuradse = 5 ◇ Strickler = 6 ◇ WhiteColebrook = 7 ◇ BosBijkerk = 9
`globalValue`	Double		The global default value for this section
`[BranchProperties]`			
`branchid`	String		The name of the branch
`roughnessType`	Int	1	Type of roughness definition, possible values: ◇ Chezy = 1 ◇ Manning = 4 ◇ Nikuradse = 5 ◇ Strickler = 6 ◇ WhiteColebrook = 7 ◇ BosBijkerk = 9
`functionType`	Int		Function type for the calculation of the value, possible values ◇ Constant = 0 ◇ FunctionOfDischarge = 1 ◇ FunctionOfWaterLevel = 2
`In case of function of discharge or water level:`			
`numLevels`	Int		Number of levels in table
`levels`	Double[]		

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Keyword	Type	Default	Description
			(continued from previous page)
`[Definition]`			
`branchid`			
`chainage`			
`In case of constant discharge or water leve`			
`value`	Double		
`In case of table of discharge or water level`			
`values`	Double[]		

Filename: roughness-{sectionid}-{flowdirection}.ini

4.7 Boundary locations file

For ease of implementation, the Boundary locations are defined in two files; The first containing the Boundary locations (nodes), and the second containing the Lateral discharge locations.

4.7.1 Boundary locations (on nodes)

Table 4.30: *Definition of boundary nodes.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`boundLocs`	File type. Do not edit this.
`[Boundary]`			
`nodeId`	String		Node on which the boundary is located
`type`	Int		Boundary type Possible values are: ◇ waterLevel = 1 ◇ discharge = 2
`thatcher-harlemancoeff`	Double	0.0	Thatcher-Harleman return time in seconds (Only present for nodes with a salinity boundary condition)

4.7.2 Lateral Discharge locations

Table 4.31: *Definition of lateral discharges.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`latLocs`	File type. Do not edit this.
`[LateralDischarge]`			
`id`	String		Unique lateral discharge id
`name`	String		Long name in the user interface
`branchId`	String		Branch on which the lateral discharge is located
`chainage`	Double		If length=0, the location on the branch (m) of the point-lateral discharge. If length > 0, the begin-point on the branch (m) of the diffusive lateral discharge.
`length`	Double		(Optional, default=0) If length=0, a point-lateral discharge is defined. If length > 0, the length of the diffusive lateral discharge. (m)

4.8 Boundary conditions file

The format for Sobek-3's boundary conditions file based on the BC-format for D-FlowFM's boundary signal data. It can be used to combine multiple quantities for multiple locations in a single (ASCII) file. Each location will be defined in a separate 'BC block', which consists of a column-wise table, (the data), and a header specifiying how this table should be interpreted.

Table 4.32: *Definition of boundary conditions.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`boundConds`	File type. Do not edit this.
`[Boundary] or [LateralDischarge]`			
`name`	String		Name of the boundary location (node id)
`function`			Possible values: ◇ Constant ◇ TimeSeries ◇ QHTable
`time-interpolation`			Possible values: ◇ linear ◇ linear-extrapolate ◇ block-from (value holds from specified time step) ◇ block-to (value holds until next specified time step)
`periodic`	Bool	false	Repeat table after last time step? 0=false, 1=true
Repeated quantity/time:			
`quantity`			Possible values: ◇ time ◇ water_level ◇ water_discharge ◇ sea_water_salinity ◇ wind_speed ◇ wind_from_direction ◇ humidity ◇ water_temperature ◇ air_temperature ◇ cloudiness
`unit`			Possible values (time): ◇ seconds since *begintime* ◇ minutes since *begintime* ◇ hours since *begintime* format: yyyy-MM-dd hh:mm:ss +00:00 (+00:00: time zone)

(continued on next page)

Keyword	Type	Default	Description
			(continued from previous page)
			Possible values (other than time): ◇ m^3/s (discharge) ◇ m/s (velocity) ◇ m (water level) ◇ m/s (wind speed) ◇ ppt (salt, parts per thousand) ◇ kg/s (salt, mass per second) ◇ degrees C (air_emperature) ◇ degrees C (water_emperature) ◇ percentage (humidity and cloudiness)
`{data}` `(a column per quant/unit)`			

4.9 Observation points file

Table 4.33: *Definition of observation points.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`obsPoints`	File type. Do not edit this.
`[ObservationPoint]`			
`id`	String		Unique observation point id
`name`	String		Long name in the user interface
`branchId`	String		Branch on which the observation point is located
`chainage`	Double		Location on the branch (m)

4.10 Retentions file

Table 4.34: *Definition of Retentions.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`retentions`	File type. Do not edit this.
`[Retention]`			
`id`	String		Unique retention id
`name`	String		Long name in the user interface
`branchId`	String		Branch on which the observation point is located
`chainage`	Double		Location on the branch (m)
`storageType`	string	Reservoir	possible values: ◇ Reservoir ◇ Closed ◇ Loss
`useTable`*	Int		0=false, 1=true
`If` **`NOT`** `UseTable:`			
`bedLevel`	Double		Bed level of the retention area
`area`	Double		Area at the bed level
`streetLevel`	Double		street level of the retention area
`streetStorageArea`	Double		Area at street level
`If UseTable:`			
`numLevels`	Int		Number of levels in storage area table
`levels`	Double []		Levels in storage area table
`storageArea`	Double []		Areas in storage area table
`interpolate`	Int	0	interpolation type 0 = linear, 1 = block function

Note: * If UseTable is false than the interpolation type is always a block function.

4.11 Thatcher Harleman coefficients

Table 4.35: *spatial distributed Thatcher Harleman F4 term.*

Keyword	Type	Default	Description
`[Files]`			
`salinityParametersFile`			Name of the file containing the salinity parameters
`f4`			Name of the file containing the Thatcher Harleman F4-Dispersion term

4.12 Salinity Parameters File

Table 4.36: *Branch mouth relations.*

Keyword	Type	Default	Description
`[NumericalOptions]`			
`teta`	double	`1.0`	Θ value in implicit transport solver (1 is fully implicit, 0 is fully explicit)
`tidalPeriod`	double	`12.417`	Tidal period (hours)
`maxMouthsPerBranch`	integer	`1`	maximum of mouths related to one branch
`advectionScheme`	string	`vanLeer-2`	possible values: ◇ `vanLeer-2`: van Leer 2 scheme ◇ `upwind`: standard upwind scheme,
`[Mouth]`			
`nodeId`	string		id of the node to which the mouth is related to
`estuaryLength`	double		Length of the estuary (m)
`refWaterDepth`	double		Reference water depth in mouth (m)
`refSalinityConcentration`	double		Reference salinity in mouth (kg/m3)
`charFloodVelocity`	double		Characteristic flood velocity (m/s)
`freshWaterDischarge`	double	Optional	Fresh water discharge (m3/s)
`floodVolume`	double	Optional	Flood volume entering mouth during flood (m3)
`rhoSeaWater`	double		ρ sea water (kg/m3)
`rhoFreshWater`	double		ρ fresh water (kg/m3)
`branchesCount`	integer		number of branches in branches
`branches`	strings		id's of the branches attached to mouth

5 File description for D-Flow1D2D

Table 5.1: *The D-Flow1D2D file definition (<1d2d.ini>).*

Keyword	Type	Default	Description
`[Model]`			
`type`	string		modeltype, possible values are: Flow1D or FlowFM
`name`	string		name of the model
`directory`	string		location of the modeldefinition file. Can be relative to the location of this file.
`modelDefinitionFile`	string		name of the modeldefinition file.
`[Files]`			
`mappingFile`	string		name of the 1d2d mapping file
`logFile`	string		name of the log file
`[Parameters]`			
`maximumIterations`	int	3	maximum number of iterations
`maximumError`	double	0.0001	maximum allowable error
`minimumWidth`	double	10	minimum width of 1d channel

Table 5.2: *1D2D mapping file.*

Keyword	Type	Default	Description
`[1d2dLink]`			
`XY_1D`	double∗2		(X,Y)-coordinates of the 1D coupling location
`XY_2D`	double∗2		(X,Y)-coordinates of the 2D coupling location

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